

Analytic Property Testing: Directed Isoperimetry and Monotonicity

by

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A thesis
presented to the University of Waterloo
in fulfillment of the
thesis requirement for the degree of
Doctor of Philosophy
in
Computer Science

Waterloo, Ontario, Canada, 2025

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Author's Declaration

This thesis consists of material all of which I authored or co-authored: see Statement of Contributions included in the thesis. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners.

I understand that my thesis may be made electronically available to the public.

Statement of Contributions

The results in this thesis have appeared in the following works:

1. The results in [Chapter 4](#) appeared in the paper [\[Fer23\]](#), which was solely authored by me.
2. The results in [Chapters 5](#) and [7](#) appeared in the paper [\[Fer24\]](#), which was solely authored by me.
3. The results in [Chapter 6](#) were developed in collaboration with Lap Chi Lau, and appear in the manuscript [\[FF25\]](#), authored by Yumou Fei and myself (in submission at the time of this writing). Those results should be considered as equal collaboration between the co-authors.

Abstract

Property testing is a computational paradigm that aims to design algorithms for extremely fast decision-making about massive inputs. Property testing has been studied for nearly three decades, primarily with a focus on testing properties about discrete objects such as Boolean functions and graphs. In this thesis, we study property testing for inherently continuous objects, namely functions with real-valued domain and range—which we call the *analytic setting*. We study the central problem of *monotonicity testing* in this setting, where the input is a continuous function $f : [0, 1]^d \rightarrow \mathbb{R}$ and the algorithm must decide whether f is monotone with respect to its input coordinates, or *far* from monotone in the appropriate sense (namely with respect to the L^p distance).

The central theme of this thesis is a connection between monotonicity testing and *directed isoperimetric inequalities*, which are analogues of classical isoperimetric inequalities that have been shown to be intimately related to monotonicity testing in discrete settings. Indeed, many algorithmic advances in monotonicity testing over the last decade have been obtained via new directed isoperimetric inequalities.

We show that the connection between directed isoperimetry and monotonicity also holds in the analytic setting, and indeed reveals new relationships between property testing and areas of mathematics such as partial differential equations and optimal transport theory. The main results in this thesis are the *directed Poincaré inequality* $\text{dist}_2^{\text{mono}}(f)^2 \lesssim \mathbb{E}[|\nabla^- f|^2]$ for Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ defined over the solid unit cube, where $\text{dist}_2^{\text{mono}}(f)$ denotes the L^2 distance to monotonicity of f and the “directed gradient” operator $\nabla^- f$ measures the local violations of monotonicity of f ; and a monotonicity tester for this setting with query complexity $\tilde{O}(\sqrt{d})$. We obtain the directed Poincaré inequality by studying a new partial differential equation called the *directed heat equation*.

In our study of monotonicity testing and its connection to directed isoperimetry, we also systematize classical and directed isoperimetric inequalities in continuous and discrete settings; obtain a variety of upper and lower bounds for monotonicity testing of Lipschitz functions in other settings of interest such as the one-dimensional line and the hypergrid; and develop directed Poincaré inequalities for *directed graphs* by studying a dynamical process called the *directed heat flow* via directed analogues of classical spectral theory.

Acknowledgements

This thesis would not exist if not for a number of important people in my life.

First and foremost, I am indebted to my advisor, Eric Blais, who one day graciously accepted my cold call and promptly welcomed me to his research group. Eric has unfailingly taken me seriously and entertained my ideas—even more than I would, at times—and provided me with a friendly and supportive environment for growth and self-discovery in my research career.

I am also thankful to all the members of my thesis committee for taking the time to serve on my committee, and giving me the opportunity to hear their invaluable perspectives. I am grateful to Lap Chi Lau for his supportive presence since my first year, and for taking the time to digest my work and inviting me for a series of stimulating research discussions that led to the development of [Chapter 6](#) of this thesis. I also thank Sepehr Assadi, who has always been willing to offer feedback and advice on my work and career.

I am ever grateful to Nathan Harms, who is not only a close friend but also a mentor and inspiring presence. Nathan taught me much of what I know about how to do research, and his taste and intuition have been a gift to our fun collaborations.

I am also grateful to Vahid Asadi, Mohammad Hossein Ebtehaj, Parth Mittal, Argyris Mouzakis, Cameron Seth, Vihan Shah, Janani Sundaresan, and all the members of our Algorithms and Complexity group for their support and companionship throughout these years. I also thank Michael Kapralov for hosting me for a visit to EPFL, and for the Theory Group at EPFL for welcoming me during that time.

I was fortunate to have the opportunity to visit Sofya Raskhodnikova at Boston University for a term, and I am grateful for her welcoming attitude, advice, and insight in our collaboration. A number of people made my time in Boston special and productive, including Mark Bun, Rathin Desai, Yumou Fei, Esty Kelman, Ephraim Linder, Cassandra Marcussen, Elchanan Mossel, Shivam Nadimpalli, Diptaksho Palit, Ronitt Rubinfeld, and Adam Smith.

Finally, I must warmly acknowledge my family and my partner, who have always been by my side and who make everything else worthwhile. You have made the best moments all the better, and helped me get through the most difficult ones. Thank you.

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Chapter 1

Introduction

“But equations belong to everybody. We’re going to roll up our sleeves and give it our best shot.”

Cédric Villani, *Birth of a Theorem: A Mathematical Adventure*

When is it possible to make reliable decisions using an extremely small amount of information? Can we derive knowledge about a system purely from local observations of it? In philosophy, this kind of question dates back to Hume’s treatment of the problem of induction [Hum08], i.e. the problem of deducing laws about the natural world from experience. In that context, Hume posited that any such process must be predicated on some form of *regularity* of nature: the expectation – which cannot be ascertained from experience alone – that local observations made today reflect general principles that extend through time and space, thus enabling the acquisition of knowledge.

This thesis considers a computational instantiation of this general question: how can algorithms make reliable decisions about extremely large inputs using only a small number of local observations? Specifically, we study the ability of algorithms to draw conclusions about *continuous* objects, whose infinite-dimensional nature appears to pose a formidable challenge to any finite observer. Nevertheless, under reasonable assumptions of *mathematical* regularity, we will uncover *local to global* phenomena which algorithms can exploit to achieve the foregoing goal.

The computational paradigm of making decisions by looking at only a small portion of the input is called *property testing* [RS96; GGR98]. The setting is the following. Suppose we have access to an input object x from a universe $\mathcal{X}$, and we would like to check whether

x satisfies some property $\mathcal{P} \subset \mathcal{X}$ while reading only a small portion of x . This goal is too ambitious as stated: if x does not satisfy property $\mathcal{P}$ but only differs from some $y \in \mathcal{P}$ in a few bits, then no algorithm can hope to catch this difference without inspecting most of x . However, as we will see, the following two relaxations suffice to arrive at a tractable problem:

1. *Distance*: we only require to distinguish between objects $x \in \mathcal{P}$ and those that are *far* from $\mathcal{P}$, i.e. they are substantially different from every $y \in \mathcal{P}$ (in a precise sense specified by the problem formulation).
2. *Randomness*: we allow ourselves to make randomized decisions, and admit a small probability of error.

Algorithms that accomplish this relaxed task are called *property testers*.

1.1 Property testing in the analytic setting

Property testing has been an explicit topic of study for about three decades [RS96; GGR98] (see also the books [Gol17; BY22]), with origins tracing back to program checking [BLR90] and probabilistically checkable proofs [BFLS91]. At a high level, most existing work in property testing can be classified under one or more of the following broad themes:¹

1. *Algebraic*: the input is a function f taking values in a field, and the goal is to check whether f satisfies some algebraic property such as linearity or being a low-degree function; see e.g. [BLR90; RS96; KS08; RS97; FY20; ABFKY23] and the survey [KMS00].
2. *Combinatorial*: here we may distinguish between two salient areas with combinatorial flavour. In one line of study, the input is a combinatorial object such as a graph or hypergraph, and the goal is to test properties such as bipartiteness, k -colorability, existence of a large clique, connectivity, and planarity or minor-freeness; see e.g. [Gol97; GGR98; GR02; FN05; ELR24; BS24; SS24; Set25]. In another line of work, the input is a Boolean or real-valued function on a discrete domain such as the hypercube, and properties of interest include being a junta [FKRSS04; Bla09] (i.e. completely

¹It is impractical to give a full account of past works under all of these themes, so we give a small sample of pointers to the literature instead.

determined by only a few input variables), a decision list or tree [DLMORSW07], or a monotone function [EKKRV98; Ras99; DGLRRS99; GGLRS00; CS13; CS16; KMS18; BCS23a].

3. *Geometric*: the input is typically a Boolean function f , and the goal is to check whether f can be captured by a “simple” geometric concept such as a halfspace, an intersection of halfspaces, or a low-degree polynomial threshold function [MORS10; BBY12; DMN19; Har19; BFH21]. Closely related perspectives ask whether the set $f^{-1}(1)$ satisfies properties such as convexity [RV04; CFSS17; BB20; HY22; BBH24; CDNSW25] or connectedness [Ras03; BMRR24], or whether real-valued f satisfies a property such as convexity [BBB20].
4. *Statistical*: here, the input is a *probability distribution* p , and the algorithm typically only has access to independent samples from p (or perhaps certain kinds of conditional samples). We may ask about properties such as uniformity [GR11; DK16; VV17a], independence [BFFKRW01] or support size [VV17b; WY19; AFL24; FH25], or versions of the properties mentioned above applied to probability distributions, e.g. monotonicity [AGPRY19; RV20; CCRSW25]. See also the survey [Can22].

Of course, the lines between these categories are not sharp, and some problems (and associated mathematical techniques) do straddle them. For example, in *distribution-free testing* [HK07], distance between the input f and the property $\mathcal{P}$ is measured against an unknown input probability distribution from which the algorithm can only take samples, so statistical considerations play a key role in this setting along with the other themes. Property testing of Boolean functions is also intimately connected with Fourier analysis, which lends an analytic flavour to this area of study.

One common feature of the themes above, and indeed of most prior works in property testing, is some form of *discreteness* – either of the underlying domain, e.g. a graph or the Boolean hypercube; or of the range, e.g. Boolean-valued functions (but see Chapter 3 for a discussion of related work, including research initiated by [FY20] on algebraic property testing in the real-valued setting). In this thesis, we ask: what kinds of decisions are algorithms able to make in the *fully continuous* setting, where the input function f has real-valued domain and range? Can we obtain query-efficient property testers, despite the fact that any finite number of queries only captures a measure-zero portion of any continuous object?

This thesis explores an instantiation of the question above which we call *analytic* property testing. This setting is characterized by two features of the problem description.

First, the input is a real-valued function defined on Euclidean space, for example a function $f : [0, 1]^d \rightarrow \mathbb{R}$ over the solid unit cube; typically, we require some regularity from f such as Lipschitz continuity. Second, we measure distance between the input f and the property $\mathcal{P}$ according to L^p distance (for some appropriate choice of p). In contrast with the Hamming distance, which is more common in discrete settings, the L^p distances are a natural fit for the continuous nature of the analytic setting, since they are sensitive not only to the presence of errors but also to their magnitude². These features imply a third characteristic of the analytic setting, which is that new *mathematical techniques* must be brought to bear compared to previous work in the other settings. Indeed, this thesis obtains its computational results (new algorithms) by recruiting tools from real and functional analysis, ordinary and partial differential equations, and optimal transport theory.

We initiate research into analytic property testing with a study of *monotonicity testing*, which we introduce next.

1.2 Monotonicity testing

Monotonicity testing has been a central topic of study in property testing for over 25 years [Ras99; DGLRRS99; GGLRS00], and continues to be a prominent object of recent research. Very informally, the problem asks: given a function f , is it true that the value $f(x)$ can only grow larger when we make the input x larger? We may motivate the study of this problem from at least three perspectives:

1. As a simple instantiation of property testing. In theoretical research, it is often sensible and productive to understand the simple cases of a theory as well as possible as a first step toward more elaborate settings. Monotonicity testing, as we will see below, is a natural problem with a simple statement, and therefore provides an excellent testbed for property testing research more generally.
2. As a natural problem connected to real-world concepts. The question of whether one or more variates are positively associated with a response variable is fundamental to much of empirical science, and parametric forms of this question have been extensively studied in statistics [BJG98; Che19]; property testing offers a complementary angle on this class of problems. From a more contemporary standpoint, monotonicity has also been proposed and studied as a desirable characteristic of machine learning models due to fairness, interpretability, and quality assurance considerations [SW20].

²Indeed, in the statistical setting of distribution testing, the standard notion of distance is the total variation or L^1 distance.

3. As a source of theoretical insights and innovation. Many mathematical tools have been developed, or found novel applications, in the course of research in monotonicity testing. A non-exhaustive list includes routing on the hypercube [LR01; BCGM12], isoperimetric inequalities [CS16; KMS18; BCS23b; BKKM23], analysis of random walks [BCS23a], domain reduction techniques [BCS20; HY22], Ramsey theory [Fis04], and constructions of Ruzsa-Szemerédi graphs [FLNRRS02]. This thesis adds to this list topics such as partial differential equations (PDEs) and optimal transport theory.

Here, we briefly summarize the landscape of monotonicity testing, and then provide a more systematic review in [Chapter 3](#). The general formulation of the problem is as follows. Let $(X, \preceq)$ be a partially-ordered set, i.e. X is a ground set and $\preceq$ is a partial order on X . Let R be a totally ordered set (which is typically either $\{0, 1\}$ or $\mathbb{R}$). Then, given an input function $f : X \rightarrow R$, the goal of a monotonicity testing algorithm is to distinguish between the following two cases:

1. f is monotone, that is, for every $x, y \in X$ satisfying $x \preceq y$, we have $f(x) \leq f(y)$.
2. f is *far* from monotone, meaning that $\text{dist}(f, g) > \varepsilon$ for all monotone $g : X \rightarrow R$, where dist is a fixed distance metric on functions from X to R and $\varepsilon > 0$ is an input parameter. (Most often, dist has been the relative Hamming distance, i.e. the fraction of inputs on which f and g disagree.)

The algorithm may access f via queries to an *oracle*, which typically takes an arbitrary element $x \in X$ as input and provides the value $f(x)$ back to the algorithm. The goal of the algorithm is to reach a correct decision about f using as few queries as possible, i.e. with the smallest possible *query complexity*. Specifically, the algorithm should *accept* when f is monotone and *reject* when f is far from monotone, both with probability³ at least $2/3$. (When f is neither monotone nor far from monotone, either outcome is acceptable. As already remarked, this flexibility is key to making efficient property testing possible.)

One fundamental setting for monotonicity testing is the fully Boolean regime, where 1) the partially-ordered set is the Boolean hypercube $\{0, 1\}^d$, under the natural partial order whereby $x \preceq y$ if $x_i \leq y_i$ for all $i \in [d]$; and 2) we take Boolean range $R = \{0, 1\}$. The first works on monotonicity testing [Ras99; DGLRS99; GGLRS00] studied this setting, as well as the generalizations where the domain is allowed to be the hypergrid $[n]^d$ and the range is allowed to be any set R . A hallmark result established that, in the fully Boolean case, monotonicity can be tested using $O(d/\varepsilon)$ queries [Ras99; GGLRS00].

³This probability may be amplified to any desired level $1 - \delta$ by repeating the test $O(\log(1/\delta))$ times.

Since then, a vast array of works has continued to drive progress on this problem. A series of works on the Boolean case has pushed the query complexity down to $\tilde{O}(\sqrt{d}/\varepsilon^2)$ [KMS18], and more recent works have obtained similar results for the case of Boolean-valued functions on the hypergrid [BCS23a]; monotonicity testing of real-valued functions on the hypercube and hypergrid has also been studied [CS13; BKR24]; and the challenging setting of general partially-ordered domains was studied in early work and recently revisited [FLNRRS02; LRV22; LV23].

1.2.1 Monotonicity testing in the analytic setting.

As already mentioned, in this thesis we study monotonicity testing in the analytic setting. Specifically, we consider functions $f : [0, 1]^d \rightarrow \mathbb{R}$ defined on the solid unit cube, and make the following modelling choices.

L^p distance. First, we let the distance metric **dist** be an L^p metric, such as L^1 or L^2 . This is a more natural choice for our continuous setting than the Hamming metric because it captures not only whether two functions disagree at a point x , but also by *how much* they disagree. For example, by slightly perturbing a monotone function f , we may obtain a function $\tilde{f}$ which is $\Omega(1)$ -far from monotone in Hamming distance even though any algorithm will observe, via its queries, an output that is consistent with a monotone function.⁴ In this example, the perturbed function $\tilde{f}$ is indeed arbitrarily close to f in L^p distance, in line with our intuition that this notion of distance is a better fit to our setting. This choice follows the work of [BRY14a], who introduced L^p testing with a similar motivation and indeed studied monotonicity testing (for functions with discrete domain), along with other problems (we review this and other prior work in Chapter 3).

Lipschitz regularity. Even after choosing an appropriate distance metric, it turns out that continuous space is too unwieldy to permit testability without further assumptions on f . For example, we may perturb f in an arbitrarily small region by an arbitrarily large amount to obtain a function $\tilde{f}$ that is very far from monotone, yet indistinguishable from f . We consider Lipschitz regularity to be a natural enough constraint on f which, as we will see, is also sufficient to enable efficient testers as well as mediate connections between

⁴Formally, for any *nonadaptive* algorithm $\mathcal{A}$, i.e. one that determines all of its queries in advance, we may make the perturbation to f fine enough so that $\mathcal{A}$ will not observe any violations of monotonicity. We may extend this argument to adaptive testers under reasonable assumptions, e.g. that $\mathcal{A}$ only uses a bounded number of bits of precision from each output before making its next query.

monotonicity and well-established areas of mathematical analysis such as isoperimetric inequalities, rearrangements, partial differential equations, and optimal transport.

Therefore, our problem is the following:

Given oracle access to an M -Lipschitz function $f : [0, 1]^d \rightarrow \mathbb{R}$, distinguish between 1) f is monotone; and 2) f is far from every monotone $g : [0, 1]^d \rightarrow \mathbb{R}$ in L^p distance.

To complete the specification of the problem, we must define the oracle access in this model. We consider two types of oracle:

Value oracle: Given point x , this oracle outputs the value $f(x)$.

Directional derivative oracle: Given point x and vector $v \in \mathbb{R}^d$, this oracle outputs the derivative of f along v at point x , given by $\nabla_v f(x) = v \cdot \nabla f(x)$, as long as f is differentiable at x . Otherwise,⁵ it outputs a special symbol $\perp$.

A directional derivative oracle is weaker than a full first-order oracle, which would return the entire gradient [BV04], and it seems like a reasonable model for the high-dimensional setting; for example, obtaining the full gradient costs d queries, rather than a single query. This type of oracle has also been studied in optimization research, see e.g. [CWZ21]. Finally, we remark (in Chapter 5) that, under reasonable assumptions, these oracle queries may be simulated by value queries in our applications, so we consider directional derivative queries to be a primitive that is both natural and reasonable in our model.

1.3 Directed isoperimetry

Since the introduction of monotonicity testing and especially over the last decade, a series of works has revealed striking connections between monotonicity testing and directed analogues of well-studied and ubiquitous isoperimetric inequalities such as Poincaré, Margulis, and Talagrand inequalities. Directed isoperimetric inequalities and their role in monotonicity testing were first recognized and formalized in the breakthrough work of [CS16], and lie at the core of a number of subsequent advances [KMS18; BCS23b; BKKM23; BKR24]. This thesis builds upon these ideas by showing that continuous functions also enjoy directed isoperimetric phenomena, which underlie algorithms for monotonicity testing in the analytic setting.

⁵By Rademacher’s theorem, a Lipschitz function is differentiable almost everywhere, so non-differentiability will never be a concern for us.

Before summarizing the connections between directed isoperimetry and monotonicity and previewing our contributions, let us start with an example that will already motivate the overall picture by returning to the classic monotonicity tester of [GGLRS00] for the fully Boolean case of functions $f : \{0, 1\}^d \rightarrow \{0, 1\}$.

The tester of [GGLRS00] is remarkably simple: first, sample a vertex $x \in \{0, 1\}^d$ and another vertex y directly “above” x , i.e. y is obtained from x by flipping a single bit of x from 0 to 1; we call (x, y) an *edge*. If $f(x) > f(y)$, i.e. f violates monotonicity along this edge, reject. If no violating edge is found after repeating this test $O(d/\varepsilon)$ times, accept. This algorithm is called the *edge tester*.

Why should this algorithm work? Certainly it accepts whenever f is monotone, so the point is to argue that this algorithm samples a *violating edge* with good probability when f is far from monotone. A key result in [GGLRS00] shows exactly that:

Theorem 1.1 ([Ras99; GGLRS00]). *Suppose $f : \{0, 1\}^d \rightarrow \{0, 1\}$ is ε -far from monotone in relative Hamming distance. Let $V(f)$ denote the number of violating edges in f . Then*

$$V(f) \geq \varepsilon \cdot 2^d.$$

Thus, when f is ε -far from monotone, at least an $\Omega(\varepsilon/d)$ -fraction of the edges violates monotonicity. Hence $O(d/\varepsilon)$ queries suffice to detect one of them and successfully reject.

Broadly speaking, isoperimetric phenomena are relationships between the volume and surface area of sets or, under suitable interpretation, functions. As first observed by [CS16], statements such as Theorem 1.1 may be seen as *directed analogues* of classical isoperimetry. Indeed, if we think of “distance to monotonicity” as a notion of volume (e.g. the volume of the smallest set where f needs to change to reach any monotone function), and “violating edges” as a notion of surface area, then Theorem 1.1 is a lower bound on (directed) surface area in terms of (directed) volume.

In this thesis, we observe that the family of *Poincaré inequalities* provides a suitable framework for discussing, systematizing, and extending this connection between classical isoperimetry, directed analogues, and monotonicity testing, as we now preview.

Let $p, q \geq 1$. We say a (classical) (L^p, ℓ^q) -*Poincaré inequality* is an inequality of the form

$$\text{dist}_p^{\text{const}}(f)^p \leq C \mathbb{E} [\|\nabla f\|_q^p]$$

for, say, all functions $f : \{0, 1\}^d \rightarrow \mathbb{R}$ on the Boolean cube, or all functions $f : [0, 1]^d \rightarrow \mathbb{R}$ on the unit cube, or perhaps only the Boolean-valued functions on these domains. Here $\text{dist}_p^{\text{const}}(f)$ denotes the L^p distance of f to the closest constant function when the domain

is given uniform probability measure. Then the following isoperimetric results are classical Poincaré inequalities:

1. for functions $f : \{0, 1\}^d \rightarrow \mathbb{R}$, the classical Poincaré inequality $\text{Var}[f] \leq C \text{Inf}[f]$, where $\text{Inf}[f]$ denotes the total influence of f , is an (L^2, ℓ^2) -Poincaré inequality (see e.g. [O’D14]);
2. in the same setting, the Talagrand inequality [Tal93] is an (L^1, ℓ^2) inequality;
3. for smooth functions $f : [0, 1]^d \rightarrow \mathbb{R}$, the (L^2, ℓ^2) inequality is often called *the* Poincaré inequality, especially in mathematical analysis (see e.g. [BGL14]); and
4. in the same setting, the (L^1, ℓ^2) inequality was proved by Bobkov & Houdré [BH97].

Now, let $\text{dist}_p^{\text{mono}}(f)$ denote the L^p distance of f to the closest monotone function. Then a *directed* (L^p, ℓ^q) -Poincaré inequality is an inequality of the form

$$\text{dist}_p^{\text{mono}}(f)^p \leq C \mathbb{E} [\|\nabla^- f\|_q^p] , \quad (1.1)$$

where $\nabla^- f := \min\{0, \nabla f\}$, the *directed gradient* of f , captures the local violations of monotonicity of f . Note that both the classical and directed inequalities relate a local property (violations of the “constant” or “monotone” property, captured by the gradient) to a global one (distance to a constant or monotone function).

As observed by [CS16] and mentioned above, it turns out that many classical isoperimetric inequalities enjoy directed analogues. For example,

1. Theorem 1.1 above is a directed (L^1, ℓ^1) -Poincaré inequality for functions $f : \{0, 1\}^d \rightarrow \{0, 1\}$;
2. [KMS18] proved the stronger (L^1, ℓ^2) inequality for functions $f : \{0, 1\}^d \rightarrow \{0, 1\}$, in analogy with Talagrand’s result in the classical setting; and
3. Related directed isoperimetric statements include a directed analogue of the Margulis inequality by [CS16], and isoperimetric inequalities for real-valued functions on the Boolean cube [BKR24] and for Boolean functions on the hypergrid [BCS23b; BKKM23]. We review these in Chapter 3.

What makes the story above especially compelling is that each new directed isoperimetric inequality also enabled an *algorithmic result*, namely a monotonicity tester with

improved query complexity in the same setting. While we defer a more detailed review to [Chapter 3](#), one central example is that, for functions $f : \{0, 1\}^d \rightarrow \{0, 1\}$, the celebrated work of [\[KMS18\]](#) essentially resolved the question for nonadaptive testers by giving an $\tilde{O}(\sqrt{d}/\varepsilon^2)$ query tester.

Previously, directed isoperimetric inequalities and their applications to monotonicity testing were only known in discrete domains. On the other hand, ever since the original “isoperimetric problem” about shapes in Euclidean space, isoperimetric inequalities have enjoyed a rich and fruitful history with connections to mathematical physics, geometry, probability theory, diffusion processes, optimal transport, and so on; see e.g. [\[Vil09; BGL14\]](#). Closest to our subject, the Poincaré inequality itself first appeared in the study of partial differential equations arising in mathematical physics by Poincaré [\[Poi90\]](#). Thus, if there are unifying principles underlying both classical and directed isoperimetric phenomena, as well as the connection to monotonicity, it seems reasonable to expect such a principle to manifest itself in the analytic setting.

Accordingly, this thesis strengthens the connection between classical isoperimetry, directed isoperimetry, and monotonicity testing by extending it to the analytic setting. We show that sufficiently regular (e.g. Lipschitz) functions $f : [0, 1]^d \rightarrow \mathbb{R}$ satisfy directed Poincaré inequalities, and that this implies query-efficient monotonicity testers for such functions. Toward this goal, we extend the aforementioned classical connections between isoperimetry and other areas of mathematical analysis to the directed setting. For example, in [Chapter 7](#) we introduce a new nonlinear partial differential equation, the *directed heat equation*, whose convergence properties we use to prove a directed (L^2, ℓ^2) -Poincaré inequality for functions $f : [0, 1]^d \rightarrow \mathbb{R}$ via optimal transport theory.

1.4 Main results and organization

In [Chapter 2](#), we introduce notation and technical preliminaries used throughout the thesis, and formally define the problem of monotonicity testing in the analytic setting. [Chapter 3](#) summarizes prior work relevant to this thesis. After that, the main body of the thesis can be roughly divided into three parts, which proceed as follows.

Part I: Monotonicity testing in the analytic setting. This part encompasses [Chapters 4](#) and [5](#). In [Chapter 4](#), we begin our study of directed isoperimetry by giving directed (L^1, ℓ^1) -Poincaré inequalities in discrete and continuous settings, and study their implications to monotonicity testing. We give upper and lower bounds for testing monotonicity of

functions over the unit cube $[0, 1]^d$ and the hypergrid $[n]^d$, including nearly-tight bounds for the one-dimension cases of the discrete and continuous line. The main result from [Chapter 4](#) is an $O(d)$ -query tester for monotonicity of Lipschitz functions on the unit cube:

Theorem 1.2 (Testing monotonicity with $O(d)$ queries). *There is a nonadaptive L^1 monotonicity tester for M -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ with query complexity $O\left(\frac{dM}{\varepsilon}\right)$ and one-sided error.*

[Theorem 1.2](#) is shown via a directed (L^1, ℓ^1) -Poincaré inequality, which is conceptually similar to how [Theorem 1.1](#) is used to show the classical $O(d/\varepsilon)$ bound for monotonicity testing of Boolean functions. Specifically, the tester from [Theorem 1.2](#) works by sampling points x and coordinates $i \in [d]$ uniformly at random, and using directional derivative queries to reject if f is decreasing in direction i at point x . Correctness is shown using the directed Poincaré inequality, which says that if f is far from monotone, then the total magnitude of its negative partial derivatives must be large – and since each partial derivative is bounded thanks to the Lipschitz assumption, the partial derivatives must be strictly negative in a set of large measure, which the tester stands good chance of hitting with the given query complexity.

[Theorem 1.2](#) suggests a natural follow-up question: can we obtain a $o(d)$ -query tester? A complementary lower bound result of [Chapter 4](#) shows that any tester that works as above, i.e. by sampling directional derivatives aligned with a single direction i one at a time, cannot do better than [Theorem 1.2](#). Hence any tester with sublinear query complexity, if it is based on directional derivatives, must make use of directional derivative queries that are not aligned with any single direction i – intuitively, the tester would need to “probe” many directions at once.

[Chapter 5](#) shows that this strategy can indeed work, *if* we show a different directed Poincaré inequality, namely an (L^2, ℓ^2) inequality (which is indeed more standard in the classical, undirected case). Specifically, we show that assuming a directed (L^2, ℓ^2) -Poincaré inequality, monotonicity of M -Lipschitz functions can be tested using $\tilde{O}(\sqrt{d}M^2/\varepsilon^2)$ queries. Intuitively, note that directional derivative queries are inner products between a query vector v and a (directed) gradient vector $\nabla^- f(x)$, so it may not be entirely unexpected that a guarantee in terms of ℓ^2 geometry is better suited to proving bounds on such inner products. The main task left to the rest of the thesis, then, is to prove such an (L^2, ℓ^2) inequality.

Interlude: Directed isoperimetry on graphs via the heat equation. Our approach to proving the (L^2, ℓ^2) inequality will be to analyze a certain dynamical system –

the *directed heat equation* – and show that its convergence properties imply the desired inequality. This idea draws from the original motivation of the Poincaré inequality itself, which, as we already mentioned, arose from Poincaré’s work on partial differential equations in mathematical physics [Poi90], including the *classical* heat equation. The novel aspect of our approach, then, is to instantiate and study the appropriate directed analogues of these classical connections. At a high level, we will study dynamical systems which take an initial state f (our input function) and converge, over time, to an equilibrium state f^* which is monotone. Then, by characterizing the rate of convergence of this process, we may place an upper bound on the distance between f and f^* in terms of an “energy functional” which, it turns out, is precisely the right-hand side in our desired Poincaré inequality (1.1).

However, before moving to the setting of nonlinear PDEs – whose rigorous study brings about a host of technical considerations – we first study, in Chapter 6, a version of this approach for *discrete* domains, with the goal of first showing a directed (L^2, ℓ^2) -Poincaré inequality for real-valued functions $f : \{0, 1\}^d \rightarrow \mathbb{R}$ defined on the Boolean cube.

In fact, it is convenient and informative to consider the more general setting of *directed graphs*. In this setting, the appropriate dynamical system is given by a directed version of the classical Laplacian operator from spectral graph theory, whose study was initiated by [Yos16] in the context of network analysis. We show that the dynamical system (heat flow) generated by the directed Laplacian yields a directed Poincaré inequality for any directed graph; this inequality is made quantitative by the *dynamical spectral gap* of the graph, which characterizes the convergence rate of the dynamical system. Chapter 6 concludes by exactly characterizing the dynamical spectral gap of the (directed) Boolean cube, which implies the directed (L^2, ℓ^2) -Poincaré inequality for that domain, and by showing a family of *reverse* Poincaré inequalities quantifying the extent to which our directed isoperimetric inequalities are tight.

Remark 1.3. In the manuscript [FF25], we combined the directed Poincaré inequalities from Chapter 6 with an approximate version of the FKG inequality [FKG71] to obtain improved bounds on the mixing time of random walks on monotone subsets of the hypercube. That application falls outside the scope of this thesis, so we refer to [FF25] for details.

Part II: The continuous setting via PDEs and optimal transport. The final Chapter 7 executes the plan outlined above to obtain a directed (L^2, ℓ^2) -Poincaré inequality for Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ on the unit cube, and therefore a monotonicity tester for this setting using roughly $\sqrt{d}$ queries. The following two theorems summarize the main results that follow from this chapter.

Theorem 1.4 (Directed Poincaré inequality). *There exists a universal constant $C > 0$ such that, for all Lipschitz $f : [0, 1]^d \rightarrow \mathbb{R}$,*

$$\text{dist}_2^{\text{mono}}(f)^2 \leq C \mathbb{E} [\|\nabla^- f\|_2^2] . \quad (1.2)$$

Theorem 1.5 (Testing monotonicity with $\tilde{O}(\sqrt{d})$ queries). *There exists a nonadaptive, directional derivative L^2 monotonicity tester for M -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ with query complexity $\tilde{O}(\sqrt{d}M^2/\varepsilon^2)$ and one-sided error.*

(In comparison with [Theorem 1.2](#), we note that an L^2 tester is also an L^1 tester, because L^2 -distance over a probability space is larger than L^1 -distance. See [Chapter 2](#) for formal definitions, and [Chapter 5](#) for further discussion.)

As mentioned above, the proof of [Theorem 1.4](#) proceeds via the analysis of a dynamical system which takes f into a *monotone equilibrium* f^* ; informally, we show that this process converges exponentially fast, which implies that the initial “energy” of f – namely the right-hand side of (1.2) – yields an upper bound on the distance from f to f^* , and hence on $\text{dist}_2^{\text{mono}}(f)$. However, the analysis of the continuous case is more intricate than in the discrete setting of the Boolean cube from [Chapter 6](#), as we outline next.

The first step is to define and analyze a nonlinear PDE, the directed heat equation, whose behaviour matches the description above. However, even in one dimension, the nonlinear aspect of this process proves surprisingly resistant to standard arguments, and showing basic results such as existence, uniqueness and regularity of solutions already requires a number of targeted technical arguments. Therefore, we constrain our PDE analysis to the one-dimensional case. We analyze the directed heat equation in the formalism of gradient flows and maximal monotone operators [[Bre73](#); [AC84](#)]. To handle the nonlinear nature of our system, we introduce a canonical decomposition $u = u\uparrow + u\downarrow$ of a function $u : [0, 1] \rightarrow \mathbb{R}$ into a nondecreasing $u\uparrow$ and a nonincreasing $u\downarrow$, which allows us to isolate the phenomena we can control in $u\downarrow$ (and in fact recover a bit of linearity), and deal with the less well-behaved $u\uparrow$ only when necessary.

To promote the conclusions of this study to our multidimensional setting of interest, we draw from another area of mathematics – *optimal transport*. At a high level, optimal transport is concerned with the “cost” of transforming a source probability measure μ into a target probability measure ν , both over an underlying metric space Ω , when we measure cost as a combination of 1) how much probability mass is moved and 2) how far each “particle” of mass moves in Ω . The connection to our PDE study is natural: a PDE stipulates, at each moment in time, how mass must be flowing at each point in the system. By tracking this flow over time, we recover an optimal transport picture

of the directed heat equation; then, by sequentially applying our one-dimensional results to each direction $i = 1, 2, \dots, d$ in turn, we can reason – in optimal transport terms – about the total distance travelled by each “particle” at the end of the entire process. This study yields a Poincaré-type inequality that is indeed multidimensional, but which bounds optimal transport distance as opposed to L^2 distance. Fortunately, there exist relatively standard analytical arguments to translate back and forth between these two flavours of inequalities, and we conclude the proof of [Theorem 1.4](#) by showing that the appropriate directed analogue of such an argument does hold in our setting.

Conclusion. Finally, [Chapter 8](#) concludes this thesis with some reflections on the results obtained, concrete open questions suggested by our results, and broader directions for future research.

Chapter 2

Preliminaries

2.1 General notation

In this thesis, $\mathbb{N}$ denotes the set of strictly positive integers $\{1, 2, \dots\}$. Throughout this thesis, $d \in \mathbb{N}$ denotes the dimension of the ambient space $\mathbb{R}^d$ unless otherwise specified. For $m \in \mathbb{N}$, we write $[n]$ to denote the set $\{i \in \mathbb{N} : i \leq n\}$.

For any $x \in \mathbb{R}$, we write x^+ for $\max\{0, x\}$ and x^- for $\max\{0, -x\}$, and we extend this notation to vectors $u \in \mathbb{R}^d$ in the natural way: $u^+, u^- \in \mathbb{R}^d$ are given by $u_i^+ := (u_i)^+$ and $u_i^- := (u_i)^-$. For real numbers a and b , we use the notation $a \wedge b := \min(a, b)$ and $a \vee b := \max(a, b)$.

We write $\mathbf{1}[\cdot]$ for the indicator function, which takes value in $\{0, 1\}$. We write e_i for the i -th basis vector, i.e. $e_i \in \{0, 1\}^d$ satisfies $e_i(j) = \mathbf{1}[i = j]$ for all j . Given a vector $u \in \mathbb{R}^d$ and $q \geq 1$, we write $\|u\|_q$ for its ℓ^q -norm and $|u| := \|u\|_2$ for its Euclidean norm.

For a vector $u \in \mathbb{R}^d$, we let $\text{supp}(u) \subseteq [d]$ denote the set of indices where u is non-zero, and write $\|u\|_0 := |\text{supp}(u)|$.

For two points $x, y \in \mathbb{R}^d$, we write $x \preceq y$ if $x_i \leq y_i$ for every $i \in [d]$, and $y \succeq x$ if $x \preceq y$. Given set $\Omega \subset \mathbb{R}^d$ and function $f : \Omega \rightarrow \mathbb{R}$, we say f is *monotone* if, for all $x, y \in \Omega$ satisfying $x \preceq y$, we have $f(x) \leq f(y)$.

We use the notation $a \stackrel{?}{\leq} b$, $a \stackrel{?}{=} b$, etc. within a proof to denote (in)equalities that have not yet been established. We write $a \lesssim b$ to denote that $a \leq c \cdot b$ for some absolute constant $c > 0$, and similarly for $a \gtrsim b$. We write $a \asymp b$ to denote that $a \lesssim b$ and $b \lesssim a$.

2.2 Sets and measures

We denote the closure of a set $D \subset \mathbb{R}^d$ by $\overline{D}$.

We write dx for the “natural” measure for integration in a particular context, namely the Lebesgue measure or discrete measure when we are working with continuous or discrete domains, respectively, as follows. In continuous contexts, where $\Omega \subset \mathbb{R}^d$ is a Lebesgue-measurable set (e.g. $\Omega = [0, 1]^d$), and for Lebesgue integrable $f : \Omega \rightarrow \mathbb{R}$, we write $\int_{\Omega} f dx$ for the Lebesgue integral of f over Ω . In discrete contexts, where $\Omega \subset \mathbb{Z}^d$ is a finite set (e.g. $\Omega = [n]^d$), and for function $f : \Omega \rightarrow \mathbb{R}$, we write $\int_{\Omega} f dx := \sum_{x \in \Omega} f(x)$. It will always be clear from context whether we are working with the continuous or discrete setting. We will occasionally employ similar notation with other variable names, such as dy or dt , which should be understood in the same way.

When we need to refer to the Lebesgue measure of a set Ω explicitly, we write $\mathcal{L}(\Omega)$.

Throughout this thesis, all measures on $\mathbb{R}^d$ are Borel measures. We say that measure μ over $\Omega \subset \mathbb{R}^d$ is *absolutely continuous* if it is absolutely continuous with respect to the Lebesgue measure, i.e. if there exists a measurable function $f : \Omega \rightarrow \mathbb{R}_{\geq 0}$ satisfying $d\mu = f dx$. In this case, f is called the Radon-Nikodym derivative, or *density* of μ .

2.3 L^p spaces and Lipschitz functions

For a measure space (Ω, Σ, μ) and measurable function $f : \Omega \rightarrow \mathbb{R}$, we write $\int_{\Omega} f d\mu$ for the Lebesgue integral of f over this space when this integral exists. Then for $1 \leq p < +\infty$, the space $L^p(\Omega, \mu)$ is the set of measurable functions f such that $|f|^p$ is Lebesgue integrable with respect to μ , i.e. $\int_{\Omega} |f|^p d\mu < +\infty$, and we write the L^p norm of such functions as $\|f\|_{L^p(\Omega, \mu)} = \left(\int_{\Omega} |f|^p d\mu\right)^{1/p}$. When μ is the Lebesgue or counting measure, we also shorten this notation and simply write $L^p(\Omega)$.

Let $q \geq 1$, and let $\Omega \subset \mathbb{R}^d$ be a set endowed with the ℓ^q distance metric (note that Ω could be a discrete set). Given function $f : \Omega \rightarrow \mathbb{R}$ and parameter $M > 0$, we say f is (ℓ^q, M) -Lipschitz, or *M-Lipschitz with respect to the ℓ^q geometry* if, for every $x, y \in \Omega$, we have $|f(x) - f(y)| \leq M\|x - y\|_q$. We say that f is Lipschitz if it is (ℓ^q, M) -Lipschitz for some choice of M (in which case this also holds for any other choice of geometry ℓ^q and appropriate choice of M'). When we are considering the standard ℓ^2 (Euclidean) geometry, we also omit q from the notation and simply say that f is M -Lipschitz (which is the standard notion in the continuous setting).

Remark 2.1. If $p \leq q$ and f is (ℓ^q, M) -Lipschitz, then f is also (ℓ^p, M) -Lipschitz.

Remark 2.2. If $\Omega \subset \mathbb{R}^d$ is a bounded, convex open set, then every Lipschitz function $f : \Omega \rightarrow \mathbb{R}$ is bounded, and hence Lebesgue integrable and in space $L^p(\Omega)$ for every $p \geq 1$. If $\Omega \subset \mathbb{Z}^d$ is a finite set, then every function $f : \Omega \rightarrow \mathbb{R}$ is Lipschitz.

We also recall that Lipschitz functions over an open set are differentiable almost everywhere, i.e. outside of a set of measure zero:

Theorem 2.3 (Rademacher's theorem). *If $\Omega \subset \mathbb{R}^d$ is an open set and $f : \Omega \rightarrow \mathbb{R}$ is Lipschitz, then f is differentiable almost everywhere in Ω .*

2.4 Distance to constant and monotonicity

Let Ω be either a bounded Lebesgue-measurable set endowed with Lebesgue measure, or a finite discrete set endowed with the counting measure, and let ν denote the Lebesgue or counting measure, respectively. Let μ be the probability measure on Ω obtained by rescaling the measure ν , that is, $\mu(S) := \frac{\nu(S)}{\nu(\Omega)}$ for all measurable $S \subseteq \Omega$. We call μ the *uniform measure over Ω* .

Given $p \geq 1$ and a function $f \in L^p(\Omega)$, we define the L^p distance to monotonicity of f as

$$\text{dist}_p^{\text{mono}}(f) := \inf \left\{ \|f - g\|_{L^p(\Omega, \mu)} : g \in L^p(\Omega, \mu) \text{ monotone} \right\},$$

where $\|f - g\|_{L^p(\Omega, \mu)} = \left(\int_{\Omega} |f - g|^p d\mu \right)^{1/p} = \mathbb{E}_{x \sim \mu} [|f(x) - g(x)|^p]^{1/p}$ by definition. Similarly, we define the L^p distance from f to a constant function by

$$\text{dist}_p^{\text{const}}(f) := \inf \left\{ \|f - g\|_{L^p(\Omega, \mu)} : g \in L^p(\Omega, \mu) \text{ constant} \right\}.$$

2.5 Directed partial derivatives and gradients

Continuous setting. Let $\Omega \subset \mathbb{R}^d$ be an open set, and let $f : \Omega \rightarrow \mathbb{R}$ be Lipschitz. Then by Rademacher's theorem f is differentiable almost everywhere in Ω . For each $x \in \Omega$ where f is differentiable, let $\nabla f(x) = (\partial_1 f(x), \dots, \partial_d f(x))$ denote its gradient, where $\partial_i f(x)$ is the partial derivative of f along the i -th coordinate at x . Then, let $\partial_i^- := 0 \wedge \partial_i$, i.e. for every x where f is differentiable we have $\partial_i^- f(x) = -(\partial_i f(x))^-$. We call ∂_i^- the *directed*

partial derivative operator in direction i . Then we define the *directed gradient* operator by $\nabla^- := (\partial_1^-, \dots, \partial_d^-)$, again defined on every x where f is differentiable. Note that $\nabla^- f \preceq 0$. We also similarly define $\partial_i^+ := 0 \vee \partial_i$ and $\nabla_i^+ := (\partial_1^+, \dots, \partial_d^+)$.

Discrete setting. Now considering the hypergrid domains, let $f : [n]^d \rightarrow \mathbb{R}$. Fix $x \in [n]^d$ and $i \in [d]$. We then define the (discrete) partial derivative of f along the i -th coordinate at x by $\partial_i f(x) := f(x + e_i) - f(x)$ if $x_i < n$, and $\partial_i f(x) := 0$ if $x_i = n$. We then define its discrete gradient by $\nabla := (\partial_1, \dots, \partial_n)$. Their directed counterparts are defined as above: $\partial_i^- := \min\{0, \partial_i\}$ and $\nabla^- := (\partial_1^-, \dots, \partial_n^-)$, and similarly for ∂_i^+ and ∇^+ .

2.6 Monotonicity testing

Here, we define monotonicity testing of Lipschitz functions against the L^p distance, which is the main problem studied in this thesis. Our definition follows the original definition of L^p testing by [BRY14a], with the differences that 1) [BRY14a] focused on functions with bounded range, whereas we focus on Lipschitz functions whose range can grow with the dimension d ; and 2) [BRY14a] focused on the discrete setting, whereas we will cover both discrete and continuous settings (and focus on the latter).

Definition 2.4 (Monotonicity tester). Let $p, q \geq 1$. Let Ω be either the unit cube $[0, 1]^d$ (continuous setting) or the hypergrid $[n]^d$ for some $n \in \mathbb{N}$ (discrete setting). Given parameters $M, \varepsilon > 0$, we say a randomized algorithm $\mathcal{A}$ is an L^p *monotonicity tester* for (ℓ^q, M) -Lipschitz functions with query complexity $m = m(\Omega, M, \varepsilon)$ if, for every (ℓ^q, M) -Lipschitz input function $f : \Omega \rightarrow \mathbb{R}$, algorithm $\mathcal{A}$ makes m oracle queries to f and 1) accepts with probability at least $2/3$ if f is monotone; 2) rejects with probability at least $2/3$ if $\text{dist}_p^{\text{mono}}(f) > \varepsilon$.

As stated in Chapter 1, oracle queries may be either value queries or directional derivative queries:

Value query Given point x , this oracle outputs the value $f(x)$.

Directional derivative query Given point x and vector $v \in \mathbb{R}^d$, this oracle outputs the derivative of f along v at point x , given by $\nabla_v f(x) = v \cdot \nabla f(x)$, as long as f is differentiable at x . Otherwise, it outputs a special symbol $\perp$.

A monotonicity tester is *nonadaptive* if it decides all its queries in advance before seeing any output from the oracle; otherwise, it is *adaptive*. A monotonicity tester has *one-sided error* if it accepts monotone functions with probability 1; otherwise, it has *two-sided error*.

2.7 Classical and directed Poincaré inequalities

Let Ω be either a bounded open subset of $\mathbb{R}^d$ endowed with Lebesgue measure, or the hypergrid $[n]^d$ endowed with the counting measure. Let μ be the uniform measure over Ω .

Given $p, q \geq 1$, we say a (classical) (L^p, ℓ^q) -Poincaré inequality is an inequality of the form

$$\text{dist}_p^{\text{const}}(f)^p \leq C \mathbb{E} [\|\nabla f\|_q^p]$$

for all Lipschitz¹ functions $f : \Omega \rightarrow \mathbb{R}$. A *directed* (L^p, ℓ^q) -Poincaré inequality is an inequality of the form

$$\text{dist}_p^{\text{mono}}(f)^p \leq C \mathbb{E} [\|\nabla^- f\|_q^p]$$

for all Lipschitz functions $f : \Omega \rightarrow \mathbb{R}$.

Note that the parameter p determines the *order of integration*, whereas q determines the *geometry* in which we measure the norms of gradient vectors. These are orthogonal concerns, and we may vary each parameter independently to obtain stronger, weaker, or incomparable inequalities. In general, making q larger yields stronger inequalities for a fixed p , while varying p yields incomparable inequalities for a fixed q .

¹Our results hold under a relaxed condition, but we state the Lipschitz condition here for simplicity.

Chapter 3

Prior work

In this chapter, we summarize the lines of research that are most closely related to the work in this thesis. We start with an overview of the landscape of monotonicity testing; give some background on classical and directed isoperimetric inequalities in continuous and discrete domains; describe prior work on property testing with respect to L^p distances; and conclude by outlining a closely related line of works on testing algebraic properties of functions in the real-valued setting, and other recent work on testing properties of functions in real-valued settings.

3.1 Monotonicity testing

We first summarize results on testing monotonicity with respect to the Hamming distance, which is the most prevalent notion of distance in prior works in property testing. We also point the reader to the recent thesis of Black [Bla23] as an excellent resource.

Boolean-valued functions. Goldreich et al. [GGLRS00] formally introduced the problem of monotonicity testing, and that work and the Master’s thesis [Ras99] gave testers for functions on the hypergrid $[n]^d$ with query complexities $O(d \log(n)/\varepsilon)$ and $O((d/\varepsilon)^2)$; note that the latter bound is independent of n , and the query complexity of testers with this property was subsequently improved to $O((d/\varepsilon) \log^2(d/\varepsilon))$ by Dodis et al. [DGLRRS99] and to $O((d/\varepsilon) \log(d/\varepsilon))$ by Berman, Raskhodnikova and Yaroslavl’tsev [BRY14a]. For functions on the Boolean cube $\{0, 1\}^d$, Chakrabarty and Seshadhri [CS16] gave the first $o(d)$ tester, subsequently improved by Chen, Servedio and Tan [CST14], culminating in

the $\tilde{O}(\sqrt{d}/\varepsilon^2)$ tester of Khot, Minzer and Safra [KMS18], which essentially resolved the question for nonadaptive testers. Whether adaptivity helps in monotonicity testing is still an open question; see the lower bounds below, and also work of Chakrabarty and Seshadhri [CS19] on a connection between the query complexity of monotonicity testing and the *total influence* of the input function.

Returning to hypergrid domains $[n]^d$, Black, Chakrabarty and Seshadhri [BCS18; BCS20] established first testers with $o(d)$ query complexity and, via a domain reduction technique, also obtained $o(d)$ testers for product distributions on $\mathbb{R}^d$ (and the alternative proof of Harms and Yoshida [HY22] improves the number of *samples* drawn by the tester when the distribution is unknown). Subsequent works of Black, Chakrabarty and Seshadhri [BCS23b] and Braverman et al. [BKKM23] attained the optimal dependence on d at the cost of a dependence on n , with upper bounds of the form $\tilde{O}(\sqrt{d} \text{poly}(n))$. Most recently, Black, Chakrabarty and Seshadhri [BCS23a] gave a tester with query complexity $O(d^{1/2+o(1)}/\varepsilon^2)$, which is almost optimal for nonadaptive algorithms, and again extends to product measures on $\mathbb{R}^d$.

Real-valued functions. Ergün et al. [EKKR98] gave a tester with query complexity $O(\log(n)/\varepsilon)$ for real-valued functions on the line $[n]$; the tight query complexity of this problem was more recently shown to be $\Theta(\log(\varepsilon n)/\varepsilon)$ by Belovs [Bel18]. As for functions on the hypergrid $[n]^d$, the early works on monotonicity testing [GGLRS00; DGLRRS99; Ras99] also gave testers for larger ranges, but the query complexity depends on the size of the range. Then, Chakrabarty and Seshadhri [CS13] gave a nonadaptive tester with one-sided error and (optimal) query complexity $O(n \log(d)/\varepsilon)$. On the Boolean cube, Black, Kalemaj and Raskhodnikova [BKR24] gave a tester with query complexity $\tilde{O}\left(\min\left\{r\sqrt{d}/\varepsilon^2, d/\varepsilon\right\}\right)$ for real-valued functions f with image size r , and showed that this is optimal (for constant ε) for nonadaptive testers with one-sided error.

Lower bounds. We briefly summarize the known lower bounds for these problems; all lower bounds listed are for testers with two-sided error unless noted otherwise. For Boolean functions on the Boolean cube $\{0, 1\}^d$, there is a near-optimal lower bound of $\tilde{\Omega}(\sqrt{d})$ for nonadaptive testers by Chen, Waingarten and Xie [CWX17], which improves on a line of prior results [FLNRRS02; CST14; CDST15]. For adaptive testers, Belovs and Blais [BB16] gave the first polynomial lower bound of $\tilde{\Omega}(d^{1/4})$, since improved to $\tilde{\Omega}(d^{1/3})$ by Chen, Waingarten and Xie [CWX17].

Turning to real-valued functions, Fischer [Fis04] combined Ramsey theory arguments with a result of Ergün et al. [EKKR98] to show a $\Omega(\log n)$ lower bound for adaptive testers

on the line $[n]$. On the Boolean cube, Briët et al. [BCGM12] gave a $\Omega(d/\varepsilon)$ nonadaptive one-sided lower bound, and Blais, Brody and Matulef [BBM12] gave an adaptive lower bound of $\Omega(d)$. On the hypergrid, Blais, Raskhodnikova and Yaroslavtsev [BRY14b] gave a non-adaptive lower bound of $\Omega(d \log n)$ by communication complexity arguments, Chakrabarty and Seshadhri [CS14] showed the optimal lower bound of $\Omega(d \log(n)/\varepsilon - \log(1/\varepsilon)/\varepsilon)$ for adaptive testers using Ramsey theory (which involves functions with large range), and Belovs [Bel18] gave an alternative proof of this bound that does not use Ramsey theory.

Other settings and models of testing. This thesis focuses on monotonicity testing with respect to the uniform distribution over product domains (the Boolean cube, hypergrids, and the solid unit cube), when the algorithm is allowed to query the input function at arbitrary points. Each of these conditions may be altered to obtain different, usually more challenging settings. While we refrain from a full review here, we give some pointers to the literature. If we drop the assumption that the distribution over the input domain is uniform, but rather allow it to be an arbitrary unknown distribution $\mathcal{D}$ to which the algorithm gets sample access, we obtain the challenging *distribution-free testing* setting, which was defined in the early work of [GGR98] and directly studied by Halevy and Kushilevitz [HK07], who already showed that monotonicity testing requires $\exp(\Theta(d))$ queries in this setting. We may instead keep the uniform distribution but require that our access to the input f come from uniformly sampled points x only, rather than arbitrary points x . This setting is called *sample-based testing* [GR16], and in this case, the number of samples required is again known to be exponential in $\Theta(d)$ [GGLRS00; Bla24]. Finally, early work [FLNRRS02] studied the challenging setting where the domain of f is a general partially-ordered set, and recent work on *proper learning* of monotone functions has also obtained progress in this setting [LRV22; LV23].

3.2 Classical and directed isoperimetric inequalities

The central object of study of this thesis is the connection between monotonicity and directed analogues of classical isoperimetric inequalities. Let us trace here an extremely brief history of developments that are most relevant to our study. We start with the original work of Poincaré [Poi90], which yields inequalities of the type $\|f - \mathbb{E}[f]\|_{L^p(\Omega)} \leq C(\Omega) \|\nabla f\|_{L^p(\Omega)}$, i.e. (L^p, ℓ^2) -inequalities in our notation, for sufficiently smooth domains Ω and sufficiently integrable¹ functions f . The optimal constant $C(\Omega)$, also called the

¹More precisely, for f in the appropriate Sobolev space.

Poincaré constant of Ω , depends on properties of this domain², and often the goal is to establish the sharp constant for families of domains Ω . Payne and Weinberger [PW57] and Acosta and Durán [AD04] (see also [Beb03]) showed that, for convex domains in the cases $p = 2$ and $p = 1$ respectively, $C(\Omega)$ is essentially upper bounded by the diameter of Ω , and this bound is tight in general. However, for specific structured domains such as the product domain $[0, 1]^d$, the diameter characterization falls short of yielding a dimension-free inequality (see also the literature on logarithmic Sobolev inequalities [Gro75]).

Making progress on this front in the discrete setting, the landmark work of Talagrand [Tal93] gave an (L^1, ℓ^2) -inequality for domain $\Omega = \{0, 1\}^d$, with $C = C(\Omega)$ independent of d , and established connections with earlier works of Margulis on graph connectivity [Mar74] and Pisier on probability in Banach spaces [Pis86]. (More recently, Fourier-analytic proofs of Talagrand’s inequality have also been given [KKKMS21; EKLM24].) In continuous settings, similar results were first established for the Gaussian measure in connection with the Gaussian isoperimetric inequality [Bob97; BL96; ST78; Bor75; Lat03]. Tying back to our present settings of interest, Bobkov and Houdré [BH97] showed that a dimension-independent Poincaré inequality also holds for product measures in $\mathbb{R}^n$, implying in particular an (L^1, ℓ^2) -inequality (analogously to Talagrand’s result) for domain $[0, 1]^d$.

As outlined in Chapter 1, it is these dimension-free isoperimetric inequalities for discrete and continuous product measures whose directed analogues have implications for the structure of monotone functions and therefore for monotonicity testing. This connection was first explicitly observed by Chakrabarty and Seshadhri [CS16], who obtained the first $o(d)$ query tester for monotonicity of functions $f : \{0, 1\}^d \rightarrow \{0, 1\}$ by showing a directed analogue of the *Margulis inequality* for such functions. That inequality is similar to the Poincaré inequalities we have discussed so far, but its guarantee is a combination of an “edge boundary size” and a “vertex boundary size”, as follows. For each vertex x , write³ $\Gamma_f(x) := \mathbb{1} [\nabla^- f(x) \neq 0]$. Chakrabarty and Seshadhri showed that, for Boolean functions f , we have

$$\text{dist}_2^{\text{mono}}(f)^2 \lesssim \mathbb{E} [\|\nabla^- f\|_2^2] \cdot \mathbb{E} [\Gamma_f(x)] . \quad (3.1)$$

Later on, Khot, Minzer and Safra [KMS18] obtained their $\tilde{O}(\sqrt{d}/\varepsilon^2)$ tester for Boolean functions by showing the directed (L^1, ℓ^2) -Poincaré inequality (up to a logarithmic factor,

²For example, it is characterized by the first nontrivial eigenvalue of the Laplacian operator on smooth bounded Ω . There are additional considerations relating the assumptions made of f on the boundary $\partial\Omega$ and the (Dirichlet or Neumann) boundary conditions associated with the Laplacian; see [KN15] for a survey.

³We could in fact make this stronger by assigning $\Gamma_f(x) = 0$ whenever $f(x) = 0$, but we omit this detail for simplicity. See also the remark below on robust inequalities.

which was later removed by Pallavoor, Raskhodnikova and Waingarten [PRW22])

$$\text{dist}_2^{\text{mono}}(f) \lesssim \mathbb{E} [\|\nabla^- f\|_2] . \quad (3.2)$$

By writing $\|\nabla^- f(x)\|_2 = \|\nabla^- f(x)\|_2 \cdot \Gamma_f(x)$ and applying the Cauchy-Schwarz inequality, we see that (3.2) implies (3.1).

Recent work [BCS23b; BKKM23; BKR24] has also explored close analogues of these inequalities to tackle the problems of testing monotonicity of Boolean functions on the hypergrid and real-valued functions on the Boolean cube (with respect to Hamming distance), as we now briefly summarize.

Define the vector-valued operators ϕ and ϕ^- on functions $f : [n]^d \rightarrow \mathbb{R}$ as follows: for each $x \in [n]^d$ and $i \in [d]$,

$$\begin{aligned} (\phi f(x))_i &:= \mathbb{1} [\exists y : (x \preceq_i y \text{ or } y \preceq_i x) \text{ and } f(x) \neq f(y)] , \\ (\phi^- f(x))_i &:= \mathbb{1} [(\exists y : x \preceq_i y, f(x) > f(y)) \text{ or } (\exists y : y \preceq_i x, f(y) > f(x))] , \end{aligned}$$

where we write $x \preceq_i y$ if $x_j = y_j$ for every $j \neq i$, and $x_i \leq y_i$. Compared to the gradient, these operators 1) are only sensitive to the order relation between function values (which suits the setting of Hamming testing); and 2) capture “long range” violations of monotonicity (accordingly, the corresponding hypergrid testers are not edge testers). Then, Poincaré-type inequalities where the gradient ∇ is replaced with ϕ (and similarly for their directed counterparts) were given by Black, Chakrabarty and Seshadhri [BCS23b] and Braverman et al. [BKKM23] for Boolean functions on the hypergrid, and by Black, Kalemaj and Raskhodnikova [BKR24] for real-valued functions on the Boolean cube; and these works gave monotonicity testers in the respective settings.

One important aspect omitted in the discussion above is the notion of *robustness* for isoperimetric inequalities in discrete settings, introduced by Khot, Minzer and Safra [KMS18] and subsequently featuring in most other monotonicity testing results as well. At a high level, the idea is that a directed isoperimetric inequality can be used to obtain an efficient monotonicity tester by establishing the existence of a “good subgraph” of violating edges, which is a generalization of a matching. However, even under a directed Poincaré inequality, it could informally be the case that a few vertices are responsible for many violating edges, which seems insufficient to yield the existence of such a good subgraph (indeed, a similar difficulty was already tackled in [CS16]). The idea of robustness, then, is to assign each edge (x, y) a *colour*, red or blue, and then break up the directed gradient ∇^- into two components – a “red component” ∇_{red}^- , which counts red edges but only at the vertices x with $f(x) = 1$, and a “blue component” ∇_{blue}^- , which counts blue edges but only at the

Inequality	Setting	Discrete		Continuous
		$\{0, 1\}^d \rightarrow \{0, 1\}$	$\{0, 1\}^d \rightarrow \mathbb{R}$	$[0, 1]^d \rightarrow \mathbb{R}$
(L^1, ℓ^1) -Poincaré	$d_1^{\text{const}}(f) \lesssim \mathbb{E} [\ \nabla f\ _1]$	* [Tal93]	* [Tal93]	* [BH97]
	$d_1^{\text{mono}}(f) \lesssim \mathbb{E} [\ \nabla^- f\ _1]$	[GGLRS00]	Theorem 4.2	Theorem 4.1
(L^1, ℓ^2) -Poincaré	$d_1^{\text{const}}(f) \lesssim \mathbb{E} [\ \nabla f\ _2]$	* [Tal93]	[Tal93]	[BH97]
	$d_1^{\text{mono}}(f) \lesssim \mathbb{E} [\ \nabla^- f\ _2]$	[KMS18; PRW22]	[CCRSW25]	Conjecture 4.43
(L^2, ℓ^2) -Poincaré	$d_2^{\text{const}}(f)^2 \lesssim \mathbb{E} [\ \nabla f\ _2^2]$	*	Classical, see [O'D14]	Classical [Poi90]
	$d_2^{\text{mono}}(f)^2 \lesssim \mathbb{E} [\ \nabla^- f\ _2^2]$	[GGLRS00]	Theorem 6.25	Theorem 7.1
Related inequalities	$d_H^{\text{mono}}(f) \lesssim \mathbb{E} [\ \phi^- f\ _2]$	For $f : \{0, 1\}^d \rightarrow \mathbb{R}$ [BKR24]		
	$d_H^{\text{mono}}(f) \lesssim \mathbb{E} [\ \phi^- f\ _2]$	For $f : [n]^d \rightarrow \{0, 1\}$ [BCS23b; BKKM23]		

Table 3.1: Classical and directed isoperimetric inequalities on discrete and continuous domains. Cells marked with * indicate inequalities that follow from another entry in the table. For simplicity, logarithmic factors in the inequalities are ignored. We write d for dist , and write d_H to denote relative Hamming distance.

vertices y with $f(y) = 0$. A robust inequality holds if $\mathbb{E} [\|\nabla_{\text{red}}^- \|_2] + \mathbb{E} [\|\nabla_{\text{blue}}^- \|_2]$ is large for *every* colouring of the edges, and this yields the desired stronger structural guarantees. We refer to [KMS18] for further details on robust isoperimetric inequalities.

Finally, in a recent development subsequent to the work in this thesis, Chakrabarty et al. [CCRSW25] extended the (L^1, ℓ^2) -inequality of Khot, Minzer and Safra [KMS18] from Boolean-valued functions $f : \{0, 1\}^d \rightarrow \{0, 1\}$ to all real-valued functions $f : \{0, 1\}^d \rightarrow \mathbb{R}$, using a thresholding argument of Berman, Raskhodnikova and Yaroslavtsev [BRY14a]. This answered (up to a logarithmic factor) a question left open by the work of Chapter 4 [Fer23]. The authors used this inequality to establish an exciting new connection to monotonicity testing of *probability distributions*, rather than functions, in a certain “sub-cube conditioning” sampling model.

Table 3.1 summarizes the situation described in this section by showing, for each of the settings discussed above, the best-known classical and directed Poincaré-type inequalities, including the results given in this thesis.

3.3 L^p testing

We have mentioned that the analytic setting of property testing, which we study in this thesis, departs from most previously-studied settings by making two complementary choices: 1) the input functions have real-valued domain and range; and 2) we measure distance with respect to the L^p distance, rather than Hamming distance. The latter choice puts us in the field of “ L^p testing”, which was first systematically investigated by Berman, Raskhodnikova and Yaroslavtsev [BRY14a]. We briefly summarize that work here, and then discuss the connection with our results in more detail in Chapter 4.

In [BRY14a], the authors initiated the systematic study of L^p testing of real-valued functions defined on hypergrid domains, i.e. $f : [n]^d \rightarrow \mathbb{R}$. Some type of regularity must be assumed if one wishes to perform L^p testing of real-valued functions, and in [BRY14a] the authors assumed that f has bounded range: their results are stated for functions taking values in $[0, 1]$, and other ranges $[0, r]$ may be handled by appropriately rescaling both the input function f and the parameter ε . They studied the properties of monotonicity, Lipschitzness, convexity and submodularity.

For monotonicity specifically, the authors gave nonadaptive L^p monotonicity testers with one-sided error on the hypergrid $[n]^d$ with query complexity $O((d/\varepsilon^p) \log(d/\varepsilon^p))$ —note this is independent of n , bypassing the Hamming testing lower bound—and a lower bound of $\Omega((1/\varepsilon^p) \log(1/\varepsilon^p))$ for nonadaptive testers with one-sided error; on the line $[n]$, they showed there is an $O(1/\varepsilon^p)$ nonadaptive tester with one-sided error and a matching lower bound for adaptive testers with two-sided error. They also gave a reduction, based on a thresholding argument, from L^1 monotonicity testing to Hamming testing of Boolean functions for nonadaptive one-sided testers, so in particular L^1 testing functions with range $[0, 1]$ is no harder than Hamming testing functions with Boolean range. Finally, they studied the problem of approximating the L^1 distance of f to monotonicity (i.e. the *tolerant testing* framework of Parnas, Ron and Rubinfeld [PRR06]), and gave approximation algorithms up to additive error δ with $\text{poly}(1/\delta)$ query complexity for the line ($d = 1$) and two-dimensional grid ($d = 2$).

Parameterized testing. We also remark that our problem, which is parameterized by the upper bound M on the Lipschitz constant of input functions, lies under the umbrella of *parameterized property testing*, formally introduced by Pallavoor, Raskhodnikova and Varma [PRV17], and refer to [PRV17] for an introduction to, and results on this type of tester.

3.4 Algebraic property testing over the reals

Another recent line of research [FY20; ABFKY23; AKM25], which is closely related to the topic of this thesis, has revisited classic problems from algebraic property testing in the fully real-valued setting: given query access to a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, the goal is to test whether f satisfies a property such as *linearity*, or being a *low-degree* multivariate polynomial.

Specifically, Fleming and Yoshida [FY20] initiated this line of investigation by giving a linearity tester for functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ using $\tilde{O}(1/\varepsilon)$ queries. Then, Arora et al. [ABFKY23] obtained a tester for degree- k functions using $\text{poly}(k/\varepsilon)$ queries, and Arora, Kelman and Meir [AKM25] simplified some of the techniques from the previous works while eliminating logarithmic factors from the query complexity.

The main difference between those works and ours lies in the notion of distance in the formulation of the problem: the works [FY20; ABFKY23; AKM25] considered Hamming distance with respect to an underlying distribution $\mathcal{D}$ over $\mathbb{R}^d$, that is, the distance between two functions $f, g : \mathbb{R}^d \rightarrow \mathbb{R}$ is

$$\text{dist}(f, g) = \mathbb{P}_{x \sim \mathcal{D}} [f(x) \neq g(x)] .$$

In fact, those works studied the *distribution-free testing* setting studied by Halevy and Kushilevitz [HK07], where the distribution $\mathcal{D}$ with respect to which distances are measured is an unknown input, to which the algorithm only has sample access. Building upon the “self-correct and test” approach of Halevy and Kushilevitz, they augmented central ideas from algebraic property testing with a host of arguments and tools to handle the challenges of the real-valued setting.

In contrast, in this thesis we build upon earlier work of Berman, Raskhodnikova and Yaroslavtsev [BRY14a] and measure distance according to the L^p norm, which contemplates not only *whether* functions f and g disagree at a point x , but also by *how much*. While we make this choice primarily for conceptual reasons – L^p testing is a natural computational task – it is also interesting to observe the technical ramifications of such a choice: Hamming testing and L^p testing seem to call for different mathematical tools that reveal different aspects of our problems of interest, with Hamming distance being a more natural fit for combinatorial and algebraic arguments, and L^p distance showing better affinity with analytic arguments.⁴

⁴Nevertheless, in Chapter 4 we will also see that the Lipschitz condition provides a bridge of sorts between these two scenarios, indeed leading to a nearly-tight characterization of the query complexity of L^1 monotonicity testing on the line, by a reduction to the classic problem of testing sortedness [EKKR98].

3.5 Testing properties of parametric models

Motivated by linear regression and *sparse recovery*, [CDS20] studied the problem of deciding whether a noisy linear model $y = w \cdot x + \eta$ is k -sparse, meaning that the unknown weight vector $w \in \mathbb{R}^d$ is supported on at most k elements, or whether w is ε -far from every k -sparse vector in normalized ℓ^2 distance. The algorithm receives samples $(x, y = w \cdot x + \eta)$, where x is distributed according to a fixed product distribution $\mathcal{D} = X^d$ and the noise η is sampled independently at random from a fixed distribution, and must make its decision using only these samples (rather than arbitrary queries). The main result of [CDS20] is an algorithm whose sample complexity depends only on the one-dimensional marginal X of $\mathcal{D}$ and on the distribution of the noise η , but not on d .

The main similarity between the work of [CDS20] and this thesis is that [CDS20] also applied the framework of property testing in a fully real-valued setting, i.e. the “noisy function” $x \in \mathbb{R}^d \mapsto w \cdot x + \eta \in \mathbb{R}$. Since the input is assumed to come from a linear model with added noise, we may see their result as an application of property testing to a *parametric* setting in the language of statistics. Accordingly, their notion of distance for testing is defined in terms of the parameter vector $w \in \mathbb{R}^d$. In contrast, in this thesis we apply property testing to Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ with no parametric specification, and in this setting the natural notion of distance is in terms of the *outputs* of f instead. One interesting connection between these two settings is that, in certain parametric situations, the Lipschitz condition assumed in our work could be derived a priori or otherwise approximately tested by using the model specification.

Part I: Monotonicity testing in the analytic setting

Chapter 4

Testing with $O(d)$ queries via rearrangements

In this chapter, we begin our investigation of directed (L^p, ℓ^q) -Poincaré inequalities by studying the (L^1, ℓ^1) setting, which we use to derive upper bounds for L^1 monotonicity testing of Lipschitz functions defined on the hypergrid $[n]^d$ and solid unit cube $[0, 1]^d$. The main results in this chapter are a directed (L^1, ℓ^1) -Poincaré inequality for Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ ([Theorem 4.1](#)), and an L^1 monotonicity tester for M -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ using $O(dM/\varepsilon)$ queries ([Theorem 4.3](#)). We also give lower bounds for these problems, derive tighter results for the special case of the discrete and continuous lines (i.e. $d = 1$), and compare our results to what can be derived from prior work on L^p testing by Berman, Raskhodnikova and Yaroslavtsev [[BRY14a](#)].

4.1 Results and main ideas

4.1.1 Directed Poincaré-type inequalities

Our first result is a directed (L^1, ℓ^1) -Poincaré inequality for Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$. This result may be seen as a continuous analogue of [Theorem 1.1](#), which says that Boolean functions which are far from monotone have many violating edges. Indeed, our argument is inspired by the proof in [[Ras99](#); [GGLRS00](#)], and the monotonicity tester we will obtain from our inequality is a continuous analogue of the $O(d)$ monotonicity tester for Boolean functions.

Theorem 4.1. *Let $f : [0, 1]^d \rightarrow \mathbb{R}$ be a Lipschitz function with monotone rearrangement f^* . Then*

$$\text{dist}_1^{\text{mono}}(f) \asymp \mathbb{E} [|f - f^*|] \lesssim \mathbb{E} [\|\nabla^- f\|_1] . \quad (4.1)$$

As hinted in the statement, a crucial tool for this result is the *monotone rearrangement* f^* of f . We construct f^* by a sequence of axis-aligned rearrangements $R_1, \dots, R_n$; each R_i is the *non-symmetric monotone rearrangement* operator along dimension i , which naturally generalizes the *sorting* operator of [GGLRS00] to the continuous case. For each coordinate $i \in [d]$, the operator R_i takes f into an equimeasurable function $R_i f$ that is monotone in the i -th coordinate, at a “cost” $\mathbb{E} [|f - R_i f|]$ that is upper bounded by $\mathbb{E} [|\partial_i^- f|]$. We show that each application R_i can only decrease the “cost” associated with further applications R_j , so that the total cost of obtaining f^* (i.e. the middle expression in (4.1)) may be upper bounded, via the triangle inequality, by the sum of all directed partial derivatives, i.e. the right-hand side of (4.1).

A technically simpler version of this argument also yields a directed (L^1, ℓ^1) -Poincaré inequality for real-valued functions on the hypergrid. We also note that Theorems 4.1 and 4.2 are both tight up to constant factors.

Theorem 4.2. *Let $f : [n]^d \rightarrow \mathbb{R}$ and let f^* be its monotone rearrangement. Then*

$$\text{dist}_1^{\text{mono}}(f) \asymp \mathbb{E} [|f - f^*|] \lesssim n \mathbb{E} [\|\nabla^- f\|_1] .$$

Table 3.1 in Chapter 3 shows these two results in the broader context of classical and directed isoperimetric inequalities for continuous and discrete settings.

4.1.2 Testing monotonicity on the unit cube and hypergrid

Equipped with the results above, we give a monotonicity tester for Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$, and the same technique yields a tester for functions on the hypergrid as well. The testers are parameterized by an upper bound M on the Lipschitz constant of f with respect to the ℓ^1 geometry, in the terminology from Chapter 2.

Both of our testers are *partial derivative testers*. These are algorithms which only have access to a directional derivative oracle and, moreover, their queries are promised to be axis-aligned vectors. In the discrete case, these are usually called *edge testers* [GGLRS00; CS13].

Theorem 4.3. *There is a nonadaptive partial derivative L^1 monotonicity tester for (ℓ^1, M) -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ with query complexity $O\left(\frac{dM}{\varepsilon}\right)$ and one-sided error.*

Similarly, there is a nonadaptive partial derivative L^1 monotonicity tester for (ℓ^1, M) -Lipschitz functions $f : [n]^d$ with query complexity $O\left(\frac{dnM}{\varepsilon}\right)$ and one-sided error.

The testers work by sampling points x and coordinates $i \in [d]$ uniformly at random, and using directional derivative queries to reject if $\partial_i^- f(x) < 0$. Their correctness is shown using [Theorems 4.1](#) and [4.2](#), which imply that, when f is ε -far from monotone in L^1 -distance, the total magnitude of its negative partial derivatives must be large—and since each partial derivative is at most M by assumption, the values $\partial_i^- f(x)$ must be strictly negative in a set of large measure, which the tester stands good chance of hitting with the given query complexity.

4.1.3 Testing monotonicity on the line

The results above, linking a directed Poincaré inequality with a monotonicity tester that uses partial derivative queries and has linear dependence on d , paint a close parallel between our analytic setting and the case of the edge tester on the Boolean cube [\[GGLRS00\]](#). On the other hand, we also show a strong separation between Hamming and L^1 testing. Focusing on the simpler problem of monotonicity testing *on the line*, we show that the tight query complexity of L^1 monotonicity testing Lipschitz functions grows with the square root of the size of the (continuous or discrete) domain.

Theorem 4.4. *There exist nonadaptive L^1 monotonicity testers for M -Lipschitz functions $f : [0, n] \rightarrow \mathbb{R}$ and $f : [n] \rightarrow \mathbb{R}$ with query complexity $\tilde{O}\left(\sqrt{nM/\varepsilon}\right)$. The testers use value queries and have one-sided error.*

This result (along with the near-tight lower bounds in [Section 4.1.4](#)) stands in contrast with the case of Hamming testing functions $f : [n] \rightarrow \mathbb{R}$, which has sample complexity $\Theta(\log n)$ [\[EKKRV98; Fis04; BRY14b; Bel18\]](#). Intuitively, this difference arises because an M -Lipschitz function may violate monotonicity with rate of change M , so the area under the curve may grow quadratically in violating regions. The proof is in fact a reduction to the Hamming case, using the Lipschitz assumption to establish a connection between the L^1 and Hamming distances to monotonicity.

4.1.4 Lower bounds

We give two types of lower bounds: under no assumptions about the tester and for constant d , we show that the dependence of [Theorem 4.3](#) on M/ε is close to optimal¹. We give stronger bounds for the special case of partial derivative testers (such as the ones from [Theorem 4.3](#)), essentially showing that our analysis of the partial derivative tester is tight.

We first state our general lower bound for constant dimension d .

Theorem 4.5. *Let d be a constant. Any L^1 monotonicity tester (with two-sided error, and adaptive value and directional derivative queries) for (ℓ^1, M) -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ requires at least $\Omega\left((M/\varepsilon)^{\frac{d}{d+1}}\right)$ queries.*

Similarly, any L^1 monotonicity tester (with two-sided error and adaptive queries) for (ℓ^1, M) -Lipschitz functions $f : [n]^d \rightarrow \mathbb{R}$ requires at least $\Omega\left(\min\left\{(nM/\varepsilon)^{\frac{d}{d+1}}, n^d\right\}\right)$ queries.

Note that the bounds above cannot be improved beyond logarithmic factors, because they nearly attain the upper bounds for the line from [Theorem 4.4](#). It also follows that adaptivity (essentially) does not help with L^1 monotonicity testing on the line, matching the situation for Hamming testing [[Fis04](#); [CS14](#); [Bel18](#)].

[Theorem 4.5](#) is obtained via a “hole” construction, which hides a non-monotone region of f inside an ℓ^1 -ball B of radius r . We choose r such the violations of monotonicity inside B are large enough to make f ε -far from monotone, but at the same time, the ball B is hard to find using few queries. However, this construction has poor dependence on d .

To lower bound the query complexity of partial derivative testers with better dependence on d , we employ a simpler “step” construction, which essentially chooses a coordinate i and hides a small negative-slope region on every line along coordinate i . These functions are far from monotone, but a partial derivative tester must correctly guess both i and the negative-slope region to detect them. We conclude that [Theorem 4.3](#) is optimal for partial derivative testers on the unit cube, and optimal for edge testers on the hypergrid for constant ε and M .

Theorem 4.6. *Any partial derivative L^1 monotonicity tester for (ℓ^1, M) -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ (with two-sided error and adaptive queries) requires at least $\Omega(dM/\varepsilon)$ queries.*

¹Note that one may always multiply the input values by $1/M$ to reduce the problem to the case with Lipschitz constant 1 and proximity parameter ε/M , so this is the right ratio to look at.

Domain	Hamming testing $f : \Omega \rightarrow \mathbb{R}$	L^1 -testing (prior works) f is (ℓ^1, M) -Lipschitz	L^1 -testing (this work) f is (ℓ^1, M) -Lipschitz
$\Omega = [0, 1]^d$	Infeasible	$\tilde{O}\left(\frac{d^2 M}{\varepsilon}\right)$ (*) [BRY14a]	$O\left(\frac{dM}{\varepsilon}\right)$ p.d.t.
		—	$\Omega\left(\left(\frac{M}{\varepsilon}\right)^{\frac{d}{d+1}}\right)$ const. d $\Omega\left(\frac{dM}{\varepsilon}\right)$ p.d.t.
$\Omega = [n]^d$	$O\left(\frac{d \log n}{\varepsilon}\right)$ [CS13]	$\tilde{O}\left(\frac{d^2 n M}{\varepsilon}\right)$ (*) [BRY14a]	$O\left(\frac{dnM}{\varepsilon}\right)$ p.d.t.
	$\Omega\left(\frac{d \log(n) - \log(1/\varepsilon)}{\varepsilon}\right)$ [CS14]	$\tilde{\Omega}\left(\frac{M}{\varepsilon}\right)$ n.a. 1-s. [BRY14a] $\Omega(d \log n)$ n.a. [BRY14b]	$\Omega\left(\left(\frac{nM}{\varepsilon}\right)^{\frac{d}{d+1}}\right)$ const. d $\Omega(dn)$ p.d.t.

Table 4.1: Query complexity bounds for testing monotonicity on the unit cube and hypergrid. Upper bounds are for nonadaptive (n.a.) algorithms with one-sided error (1-s.), and lower bounds are for adaptive algorithms with two-sided error, unless stated otherwise. For L^1 -testing, the upper bounds derived from prior works (*) are specialized to the Lipschitz case by us; see the text for details. Our lower bounds hold either for constant (const.) d , or for partial derivative testers (p.d.t.).

For sufficiently small constant ε and constant M , any partial derivative L^1 monotonicity tester for (ℓ^1, M) -Lipschitz functions $f : [m]^n \rightarrow \mathbb{R}$ (with two-sided error and adaptive queries) requires at least $\Omega(dn)$ queries.

Table 4.1 summarizes our upper and lower bounds for testing monotonicity on the unit cube and hypergrid, along with the analogous Hamming testing results for intuition and bounds for L^1 testing from prior works. See Section 4.6 for a comparison of our upper bounds and those given in [BRY14a] for L^1 testing of monotonicity, and an explanation of how we derive the upper bounds shown in the table from their results (which are stated under incomparable assumptions, namely bounded range instead of Lipschitz continuity). We sketch in Appendix A how to obtain the $\Omega(d \log n)$ nonadaptive lower bound shown in the table via the construction of Blais, Raskhodnikova and Yaroslavl'tsev [BRY14b] for the Hamming setting.

Organization. The rest of this chapter is organized as follows. In Section 4.2, we prove our directed Poincaré inequalities on the unit cube (Theorem 4.1) and hypergrid (Theo-

rem 4.2), and in Section 4.3 we give our L^1 monotonicity testers for these domains (Theorem 4.3). Section 4.4 gives the upper bound for testing functions on the line (Theorem 4.4), and in Section 4.5 we prove our lower bounds (Theorems 4.5 and 4.6). In Section 4.6, we discuss the relation between our results and those from prior work on L^p testing [BRY14a] and sketch the upper bounds shown in Table 4.1 derived from their results. Section 4.7 concludes this chapter with a brief discussion of the results and some open questions.

4.2 Directed Poincaré inequalities

In this section, we establish Theorems 4.1 and 4.2. We start with the one-dimensional case, i.e. functions on the line, and then generalize to higher dimensions. In each subsection, we focus our presentation on the setting where the domain is continuous (corresponding to our results for the unit cube), and then show how the same proof strategy (more easily) yields analogous results for discrete domains (corresponding to our results for the hypergrid).

4.2.1 One-dimensional case

Let $n > 0$, let $I := (0, n)$, and let $f : \bar{I} \rightarrow \mathbb{R}$ be a measurable function. We wish to show that $\|f - f^*\|_{L^1(I)} \lesssim n \|\partial^- f\|_{L^1(I)}$, where f^* is the monotone rearrangement of f . We first introduce the monotone rearrangement, and then show this inequality using an elementary calculus argument.

Monotone rearrangement

Here, we introduce the (non-symmetric, non-decreasing) monotone rearrangement of a one-dimensional function. We follow the definition of [Kaw85], with the slight modification that we are interested in the *non-decreasing* rearrangement, whereas most of the literature usually favours the non-increasing rearrangement. The difference is purely syntactic, and our choice more conveniently matches the convention in the monotonicity testing literature. Up to this choice, our definition also agrees with that of [BS88, Chapter 2], and we refer the reader to these two texts for a comprehensive treatment.

We define the (lower) *level sets* of $f : \bar{I} \rightarrow \mathbb{R}$ as the sets

$$\bar{I}_c := \{x \in \bar{I} : f(x) \leq c\}$$

for all $c \in \mathbb{R}$. For nonempty measurable $S \subset \mathbb{R}$ of finite measure, the *rearrangement* of S is the set

$$S^* := [0, \mathcal{L}(S)]$$

(recall that $\mathcal{L}$ denotes the Lebesgue measure), and we define $\emptyset^* := \emptyset$. For a level set $\bar{I}_c$, we write $\bar{I}_c^*$ to mean $(\bar{I}_c)^*$.

Definition 4.7. The *monotone rearrangement* of f is the function $f^* : \bar{I} \rightarrow \mathbb{R}$ given by

$$f^*(x) := \inf \left\{ c \in \mathbb{R} : x \in \bar{I}_c^* \right\}. \quad (4.2)$$

Note that f^* is always a non-decreasing function.

We note two well-known properties of the monotone rearrangement: equimeasurability and order preservation. Two functions f, g are called *equimeasurable* if $\mathcal{L}\{f \geq c\} = \mathcal{L}\{g \geq c\}$ for every $c \in \mathbb{R}$. A mapping $u \mapsto u^*$ is called *order preserving* if $f(x) \leq g(x)$ for all $x \in \bar{I}$ implies $f^*(x) \leq g^*(x)$ for all $x \in \bar{I}$. See [BS88, Chapter 2, Proposition 1.7] for a proof of the following facts.

Fact 4.8. *Let $f : \bar{I} \rightarrow \mathbb{R}$ be a measurable function. Then f and f^* are equimeasurable.*

Fact 4.9. *The mapping $f \mapsto f^*$ is order preserving.*

Absolutely continuous functions and the one-dimensional Poincaré inequality

Let $f : \bar{I} \rightarrow \mathbb{R}$ be absolutely continuous. It follows that f has a derivative ∂f almost everywhere (i.e. outside a set of measure zero), $\partial f \in L^1(I)$ (i.e. its derivative is Lebesgue integrable), and

$$f(x) = f(0) + \int_0^x \partial f(t) dt$$

for all $x \in \bar{I}$. It also follows that $\partial^- f \in L^1(I)$.

We may now show our one-dimensional inequality.

Lemma 4.10. *Let $f : \bar{I} \rightarrow \mathbb{R}$ be absolutely continuous. Then $\|f - f^*\|_{L^1(I)} \leq 2n \|\partial^- f\|_{L^1(I)}$.*

Proof. Let $S := \{x \in \bar{I} : f^*(x) > f(x)\}$, and note that S is a measurable set because f, f^* are measurable functions (the latter by Fact 4.8). Moreover, since f and f^* are

equimeasurable (by the same result), we have $\int f \, dx = \int f^* \, dx$ and therefore

$$\begin{aligned} \|f - f^*\|_{L^1(I)} &= \int_I |f - f^*| \, dx = \int_S (f^* - f) \, dx + \int_{I \setminus S} (f - f^*) \, dx \\ &= \int_S (f^* - f) \, dx + \left(\int_I (f - f^*) \, dx - \int_S (f - f^*) \, dx \right) = 2 \int_S (f^* - f) \, dx. \end{aligned}$$

Hence our goal is to show that

$$\int_S (f^* - f) \, dx \leq n \|\partial^- f\|_{L^1(I)}.$$

Let $x \in \bar{I}$. We claim that there exists $x' \in [0, x]$ such that $f(x') \geq f^*(x)$. Suppose this is not the case. Then since f is continuous on $[0, x]$, by the extreme value theorem it attains its maximum and therefore there exists $c < f^*(x)$ such that $f(y) \leq c$ for all $y \in [0, x]$. Thus $[0, x] \subseteq \bar{I}_c$, so $\mathcal{L}(\bar{I}_c) \geq x$ and hence $x \in \bar{I}_c^*$. Then, by [Definition 4.7](#), $f^*(x) \leq c < f^*(x)$, a contradiction. Thus the claim is proved.

Now, let $x \in S$ and fix some $x' \in [0, x]$ such that $f(x') \geq f^*(x)$. Since f is absolutely continuous, we have

$$f^*(x) - f(x) \leq f(x') - f(x) = - \int_{x'}^x \partial f(t) \, dt \leq - \int_0^x \partial^- f(t) \, dt = \|\partial^- f\|_{L^1(I)}.$$

The result follows by applying this estimate to all x :

$$\int_S (f^* - f) \, dx \leq \int_S \|\partial^- f\|_{L^1(I)} \, dx = \mathcal{L}(S) \|\partial^- f\|_{L^1(I)} \leq n \|\partial^- f\|_{L^1(I)}. \quad \square$$

Discrete case

Let $n \in \mathbb{N}$ and let $I := [n]$. We may define the monotone rearrangement $f^* : I \rightarrow \mathbb{R}$ of $f : I \rightarrow \mathbb{R}$ as in [Definition 4.7](#) by identifying $\bar{I}$ with I and writing $S^* := [|S|]$ for each finite $S \subset \mathbb{N}$. More directly, f^* is the function such that $f^*(1) \leq f^*(2) \leq \dots \leq f^*(n)$ is the *sorted sequence* of the values $f(1), f(2), \dots, f(n)$. It is easy to show that the directed version of [Lemma 4.10](#) holds, and in fact one may simply repeat the proof of that lemma.

Lemma 4.11. *Let $f : [n] \rightarrow \mathbb{R}$. Then $\|f - f^*\|_{L^1(I)} \leq 2n \|\partial^- f\|_{L^1(I)}$.*

4.2.2 Multidimensional case

In the continuous case, we ultimately only require an inequality on the unit cube $[0, 1]^n$. However, we will first work in slightly more generality and consider functions defined on a *box* in $\mathbb{R}^d$, defined below. This approach makes some of the steps more transparent, and also gives intuition for the discrete case of the hypergrid.

Definition 4.12. Let $a \in \mathbb{R}_{>0}^d$. The *box of size a* is the closure $\overline{B} \subset \mathbb{R}^d$ of $B = (0, a_1) \times \cdots \times (0, a_d)$.

For the rest of this section, $\overline{B} \subset \mathbb{R}^d$ will always denote such a box.

Notation. For $x \in \mathbb{R}^d$, $y \in \mathbb{R}$ and $i \in [d]$, we will use the notation x^{-i} to denote the vector in $\mathbb{R}^{[d] \setminus \{i\}}$ obtained by removing the i -th coordinate from x (note that the indexing is not changed), and we will write (x^{-i}, y) as a shorthand for the vector $(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_d) \in \mathbb{R}^d$. We will also write x^{-i} directly to denote any vector in $\mathbb{R}^{[d] \setminus \{i\}}$. For function $f : \overline{B} \rightarrow \mathbb{R}$ and $x^{-i} \in \mathbb{R}^{[d] \setminus \{i\}}$, we will write $f_{x^{-i}}$ for the function given by $f_{x^{-i}}(y) = f(x^{-i}, y)$ for all $(x^{-i}, y) \in \overline{B}$. For any set $D \in \mathbb{R}^d$, we will denote by D^{-i} the projection $\{x^{-i} : x \in D\}$, and extend this notation in the natural way to more indices, e.g. D^{-i-j} .

Definition 4.13 (Rearrangement in direction i). Let $f : \overline{B} \rightarrow \mathbb{R}$ be a measurable function and let $i \in [d]$. The *rearrangement of f in direction i* is the function $R_i f : \overline{B} \rightarrow \mathbb{R}$ given by

$$(R_i f)_{x^{-i}} := (f_{x^{-i}})^* \quad (4.3)$$

for all $x^{-i} \in (\overline{B})^{-i}$. We call each R_i the *rearrangement operator in direction i* .

We may put (4.3) in words as follows: on each line in direction i determined by point x^{-i} , the restriction of $R_i f$ to that line is the monotone rearrangement of the restriction of f to that line.

Proposition 4.14. Let $\overline{B}$ be the box of size $a \in \mathbb{R}^d$, and let $f : \overline{B} \rightarrow \mathbb{R}$ be Lipschitz. Then for each $i \in [d]$,

$$\|f - R_i f\|_{L^1(B)} \leq 2a_i \|\partial_i^- f\|_{L^1(B)} .$$

Proof. Since f is Lipschitz, each $f_{x^{-i}} : [0, a_i] \rightarrow \mathbb{R}$ is Lipschitz and *a fortiori* absolutely continuous. The result follows from Lemma 4.10, using Tonelli's theorem to choose the

order of integration. □

A key ingredient in our multi-dimensional argument is that the rearrangement operator preserves Lipschitz continuity.

Lemma 4.15 ([Kaw85, Lemma 2.12]). *If $f : \overline{B} \rightarrow \mathbb{R}$ is Lipschitz (with Lipschitz constant L), then $R_i f$ is Lipschitz (with Lipschitz constant $2L$).*

We are now ready to define the (multidimensional) monotone rearrangement f^* of f .

Definition 4.16. Let $f : \overline{B} \rightarrow \mathbb{R}$ be Lipschitz. The *monotone rearrangement* of f is the function

$$f^* := R_d R_{d-1} \cdots R_1 f.$$

We first show that f^* is indeed a monotone function.

Proposition 4.17. *Let $f : \overline{B} \rightarrow \mathbb{R}$ be Lipschitz. Then f^* is monotone.*

Proof. Say that $g : \overline{B} \rightarrow \mathbb{R}$ is *monotone in direction i* if $g_{x^{-i}}$ is non-decreasing for all $x^{-i} \in (\overline{B})^{-i}$. Then g is monotone if and only if it is monotone in direction i for every $i \in [d]$. Note that $R_i f$ is monotone in direction i by definition of monotone rearrangement. Therefore, it suffices to prove that if f is monotone in direction j , then $R_i f$ is also monotone in direction j .

Suppose f is monotone in direction j , and suppose $i < j$ without loss of generality. Let $a \in \mathbb{R}^d$ be the size of B . Let $x^{-j} \in (\overline{B})^{-j}$ and $0 \leq y_1 < y_2 \leq a_j$, so that $(x^{-j}, y_1), (x^{-j}, y_2) \in \overline{B}$. We need to show that $(R_i f)(x^{-j}, y_1) \leq (R_i f)(x^{-j}, y_2)$. Let $\overline{I}_i := [0, a_i]$. For each $k \in \{1, 2\}$, let $g_k : \overline{I}_i \rightarrow \mathbb{R}$ be given by

$$g_k(z) := f(x_1, \dots, x_{i-1}, z, x_{i+1}, \dots, x_{j-1}, y_k, x_{j+1}, \dots, x_n).$$

Note that

$$g_k^*(z) = (R_i f)(x_1, \dots, x_{i-1}, z, x_{i+1}, \dots, x_{j-1}, y_k, x_{j+1}, \dots, x_n)$$

for every $z \in \overline{I}_i$, and therefore our goal is to show that $g_1^*(x_i) \leq g_2^*(x_i)$. But f being monotone in direction j means that $g_1(z) \leq g_2(z)$ for all $z \in \overline{I}_i$, so by the order preserving property (Fact 4.9) of the monotone rearrangement we get that $g_1^*(x_i) \leq g_2^*(x_i)$, which concludes the proof. □

It is well-known that the monotone rearrangement is a nonexpansive operator. Actually a stronger fact holds, as we note below.

Proposition 4.18 ([CT80]). *Let $n > 0$ and let $f, g \in L^1[0, n]$. Then f^*, g^* satisfy*

$$\int_{[0, n]} (f^* - g^*)^- dx \leq \int_{[0, n]} (f - g)^- dx$$

and

$$\int_{[0, n]} |f^* - g^*| dx \leq \int_{[0, n]} |f - g| dx.$$

The result above is stated for functions on the interval. Taking the integral over the box B and repeating for each operator R_i yields the nonexpansiveness of our monotone rearrangement operator, as also noted by [Kaw85].

Corollary 4.19. *Let $f, g \in L^1(\overline{B})$. Then $\|f^* - g^*\|_{L^1(B)} \leq \|f - g\|_{L^1(B)}$.*

We show that the rearrangement operator can only make the norm of the directed partial derivatives smaller, i.e. decrease the violations of monotonicity, which is the key step in this proof. This result may be seen as a directed version of nonexpansiveness.

Proposition 4.20. *Let $f : \overline{B} \rightarrow \mathbb{R}$ be Lipschitz and let $i, j \in [d]$. Then $\|\partial_j^-(R_i f)\|_{L^1(B)} \leq \|\partial_j^- f\|_{L^1(B)}$.*

Proof. We may assume that $i \neq j$, since otherwise the left-hand side is zero. We will use the following convention for variables names: $w \in \mathbb{R}^d$ will denote points in B ; $z \in \mathbb{R}^{[d] \setminus \{i, j\}}$ will denote points in B^{-i-j} ; $x \in \mathbb{R}$ will denote points in $(0, a_i)$ (indexing the i -th dimension); and $y \in \mathbb{R}$ will denote points in $(0, a_j)$ (indexing the j -th dimension). Recall that, for each $i \in [d]$, e_i denotes the i -th basis vector.

Since f is Lipschitz, so is $R_i f$ by Lemma 4.15. By Rademacher's theorem, these functions are differentiable almost everywhere. Therefore, let $D \subseteq B$ be a measurable

set such that f and $R_i f$ are differentiable in D and $\mathcal{L}(D) = \mathcal{L}(B)$. We have

$$\begin{aligned}
\|\partial_j^-(R_i f)\|_{L^1(B)} &= \int_D |\partial_j^-(R_i f)| \, dw \\
&= \int_D \left[\lim_{h \rightarrow 0} \left(\frac{(R_i f)(w + h e_j) - (R_i f)(w)}{h} \right)^- \right] dw \\
&\stackrel{(BC1)}{=} \lim_{h \rightarrow 0} \int_D \left(\frac{(R_i f)(w + h e_j) - (R_i f)(w)}{h} \right)^- dw \\
&\stackrel{(D1)}{=} \lim_{h \rightarrow 0} \int_B \left(\frac{(R_i f)(w + h e_j) - (R_i f)(w)}{h} \right)^- dw \\
&\stackrel{(T1)}{=} \lim_{h \rightarrow 0} \int_{B^{-i-j}} \int_{(0, a_j)} \int_{(0, a_i)} \left(\frac{(R_i f)(z, y + h, x) - (R_i f)(z, y, x)}{h} \right)^- dx \, dy \, dz \\
&\leq \lim_{h \rightarrow 0} \int_{B^{-i-j}} \int_{(0, a_j)} \int_{(0, a_i)} \left(\frac{f(z, y + h, x) - f(z, y, x)}{h} \right)^- dx \, dy \, dz \\
&\stackrel{(T2)}{=} \lim_{h \rightarrow 0} \int_B \left(\frac{f(w + h e_j) - f(w)}{h} \right)^- dw \\
&\stackrel{(D2)}{=} \lim_{h \rightarrow 0} \int_D \left(\frac{f(w + h e_j) - f(w)}{h} \right)^- dw \\
&\stackrel{(BC2)}{=} \int_D \left[\lim_{h \rightarrow 0} \left(\frac{f(w + h e_j) - f(w)}{h} \right)^- \right] dw \\
&= \int_D |\partial_j^- f| \, dw \\
&= \|\partial_j^- f\|_{L^1(B)} .
\end{aligned}$$

Equalities (BC1) and (BC2) hold by the bounded convergence theorem, which applies because the difference quotients are uniformly bounded by the Lipschitz constants of $R_i f$ and f (respectively), and because $R_i f$ and f are differentiable in D (which gives pointwise convergence of the limits). Equalities (D1) and (D2) hold again by the uniform boundedness of the difference quotients, along with the fact that $\mathcal{L}(B \setminus D) = 0$. Equalities (T1) and (T2) hold by Tonelli's theorem. Finally, the inequality holds by [Proposition 4.18](#), since $(R_i f)(z, y + h, \cdot)$ is the monotone rearrangement of $f(z, y + h, \cdot)$ and $(R_i f)(z, y, \cdot)$ is the monotone rearrangement of $f(z, y, \cdot)$. $\square$

We are now ready to prove our directed (L^1, ℓ^1) -Poincaré inequality.

Theorem 4.21. *Let B be the box of size $a \in \mathbb{R}^d$ and let $f : \overline{B} \rightarrow \mathbb{R}$ be Lipschitz. Then*

$$\|f - f^*\|_{L^1(B)} \leq 2 \sum_{i=1}^d a_i \|\partial_i^- f\|_{L^1(B)} .$$

Proof. We have

$$\begin{aligned} \|f - f^*\|_{L^1(B)} &\leq \sum_{i=1}^d \|R_{i-1} \cdots R_1 f - R_i \cdots R_1 f\|_{L^1(B)} && \text{(Triangle inequality)} \\ &\leq 2 \sum_{i=1}^d a_i \|\partial_i^-(R_{i-1} \cdots R_1 f)\|_{L^1(B)} && \text{(Lemma 4.15 and Proposition 4.14)} \\ &\leq 2 \sum_{i=1}^d a_i \|\partial_i^- f\|_{L^1(B)} && \text{(Proposition 4.20)} . \end{aligned}$$

□

Setting $B = (0, 1)^d$ yields the inequality portion of [Theorem 4.1](#).

Corollary 4.22. *Let $B = (0, 1)^d$ and let $f : \overline{B} \rightarrow \mathbb{R}$ be Lipschitz. Then*

$$\mathbb{E} [|f - f^*|] = \|f - f^*\|_{L^1(B)} \leq 2 \int_B \|\nabla^- f\|_1 dx = 2\mathbb{E} [\|\nabla^- f\|_1] .$$

To complete the proof of [Theorem 4.1](#), we need to show that $\text{dist}_1^{\text{mono}}(f) \asymp \mathbb{E} [|f - f^*|]$, i.e. that the monotone rearrangement is “essentially optimal” as a target monotone function for f . The inequality $\text{dist}_1^{\text{mono}}(f) \leq \mathbb{E} [|f - f^*|]$ is immediate from the fact that f^* is monotone. The inequality in the other direction follows from the nonexpansiveness of the rearrangement operator, with essentially the same proof as that of [\[KMS18\]](#) for the Boolean cube, as we now show.

Proposition 4.23. *Let $f : [0, 1]^d \rightarrow \mathbb{R}$ be Lipschitz. Then $\mathbb{E} [|f - f^*|] \leq 2\text{dist}_1^{\text{mono}}(f)$.*

Proof. Let $B := (0, 1)^d$ and let $g \in L^1(B)$ be any monotone function. It follows that $g^* = g$ almost everywhere (that is, except perhaps at jump discontinuities of g). By [Corollary 4.19](#), we have that $\|f^* - g^*\|_{L^1(B)} \leq \|f - g\|_{L^1(B)}$. Using the triangle inequality, we obtain

$$\begin{aligned} \|f - f^*\|_{L^1(B)} &\leq \|f - g\|_{L^1(B)} + \|g - f^*\|_{L^1(B)} \\ &= \|f - g\|_{L^1(B)} + \|f^* - g^*\|_{L^1(B)} \leq 2 \|f - g\|_{L^1(B)} . \end{aligned}$$

The claim follows by taking the infimum over the choice of g . □

Tightness of the inequality. To check that [Corollary 4.22](#) is tight up to constant factors, it suffices to take the linear function $f : [0, 1]^d \rightarrow \mathbb{R}$ given by $f(x) = 1 - x_1$ for all $x \in [0, 1]^d$. Then f^* is given by $f^*(x) = x_1$, so $\mathbb{E} [|f - f^*|] = 1/2$ while $\mathbb{E} [\|\nabla^- f\|_1] = 1$, as needed.

Discrete case

The proof above carries over to the case of the hypergrid almost unmodified, as we now outline. We now consider functions $f : [n]^d \rightarrow \mathbb{R}$, so the box B is replaced with $[n]^d$ and its dimensions a_i are all replaced with the length n of the hypergrid. We define the rearrangement in direction i , $R_i f$, as in [Definition 4.13](#) by sorting the restrictions of f to each line along direction i . We also define f^* as in [Definition 4.16](#) by subsequent applications of each operator R_i . Then [Proposition 4.14](#) carries over by applying the one-dimensional [Lemma 4.11](#), and the proof of [Proposition 4.17](#) carries over unmodified.

The nonexpansiveness properties [Proposition 4.18](#) and [Corollary 4.19](#) also carry over unmodified, and the key [Proposition 4.20](#) carries over with a more immediate proof: the use of [Proposition 4.18](#) remains the same, but rather than expanding the definition of derivative and reasoning about the limit, the discrete argument boils down to showing the inequality

$$\int_{[n]^d} ((R_i f)(w + e_j) - (R_i f)(w))^- dw \leq \int_{[n]^d} (f(w + e_j) - f(w))^- dw,$$

which follows immediately from the discrete version of [Proposition 4.18](#) by summing over all lines in direction i . Then, the hypergrid version of [Theorem 4.21](#) follows by the same application of the triangle inequality, and we conclude the inequality portion of [Theorem 4.2](#).

Theorem 4.24. *Let $f : [n]^d \rightarrow \mathbb{R}$. Then $\mathbb{E} [|f - f^*|] \leq 2n \mathbb{E} [\|\nabla^- f\|_1]$.*

The discrete version of [Proposition 4.23](#) follows identically, and we state it here for convenience.

Proposition 4.25. *Let $f : [n]^d \rightarrow \mathbb{R}$. Then $\mathbb{E} [|f - f^*|] \leq 2 \text{dist}_1^{\text{mono}}(f)$.*

Finally, the tightness of [Theorem 4.24](#) is mostly easily verified via the following step function: letting n be even for simplicity, define $f : [n]^d \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x_1 \leq n/2, \\ 0 & \text{if } x_1 > n/2. \end{cases}$$

Then f^* is obtained by flipping this function along the first coordinate, or equivalently swapping the values 1 and 0 in the definition above. Thus $\mathbb{E} [|f - f^*|] = 1$. On the other hand, $\|\nabla^- f\|_1$ takes value 1 on exactly one point in each line along the first coordinate, and 0 elsewhere. Hence $\mathbb{E} [\|\nabla^- f\|_1] = 1/n$, as needed.

4.3 Monotonicity testers

In this section, we use the directed Poincaré inequalities on the unit cube and hypergrid to show that the natural partial derivative tester (or edge tester) attains the upper bounds from [Theorem 4.3](#).

Let Ω denote either $[0, 1]^d$ or $[n]^d$, and let $q(\Omega, M, \varepsilon)$ denote the query complexity of our testers for (ℓ^1, M) -Lipschitz functions on these domains, as follows:

$$q([0, 1]^d, M, \varepsilon) := \Theta\left(\frac{dM}{\varepsilon}\right) \quad \text{and} \quad q([n]^d, M, \varepsilon) := \Theta\left(\frac{dnM}{\varepsilon}\right).$$

The tester is given in [Algorithm 1](#). It is clear that this algorithm is a nonadaptive partial derivative tester, and that it always accepts monotone functions. It suffices to show that it rejects with good probability when $\text{dist}_1^{\text{mono}}(f) > \varepsilon$.

Algorithm 1 L^1 monotonicity tester for Lipschitz functions using partial derivative queries

Input: Partial derivative oracle access to Lipschitz function $f : \Omega \rightarrow \mathbb{R}$.

Output: Accept if f is monotone, reject if $\text{dist}_1^{\text{mono}}(f) > \varepsilon$.

Requirement: f is (ℓ^1, M) -Lipschitz.

procedure PARTIALDERIVATIVETESTER($f, \Omega, M, \varepsilon$)

repeat $q(\Omega, M, \varepsilon)$ **times**

 Sample $x \in \Omega$ uniformly at random.

 Sample $i \in [d]$ uniformly at random.

Reject if $\partial_i f(x) < 0$.

end repeat

Accept.

Lemma 4.26. *Let Ω be one of $[0, 1]^d$ or $[n]^d$, and let $f : \Omega \rightarrow \mathbb{R}$ be an (ℓ^1, M) -Lipschitz function. Suppose $\text{dist}_1^{\text{mono}}(f) > \varepsilon$. Then [Algorithm 1](#) rejects with probability at least $2/3$.*

Proof. Write μ for the uniform measure over Ω , which is the Lebesgue measure if $\Omega = [0, 1]^d$ or the normalized counting measure if $\Omega = [n]^d$.

Continuous case. Suppose $\Omega = [0, 1]^d$. Let $D \subseteq [0, 1]^d$ be a measurable set such that f is differentiable on D and $\mu(D) = 1$, which exists by Rademacher's theorem. For each $i \in [d]$, let $S_i := \{x \in D : \partial_i f(x) < 0\}$. A standard argument gives that each $S_i \subset \mathbb{R}^n$ is a measurable set. We claim that

$$\sum_{i=1}^d \mu(S_i) > \frac{\varepsilon}{2M}.$$

Suppose this is not the case. By the Lipschitz continuity of f , we have that $|\partial_i f(x)| \leq M$ for every $x \in D$ and $i \in [d]$, and therefore

$$2 \sum_{i=1}^d \mathbb{E} [|\partial_i^- f|] \leq 2M \sum_{i=1}^d \mu(S_i) \leq \varepsilon.$$

On the other hand, the assumption that $\text{dist}_1^{\text{mono}}(f) > \varepsilon$ and [Corollary 4.22](#) yield

$$\varepsilon < \mathbb{E} [|f - f^*|] \leq 2\mathbb{E} [\|\nabla^- f\|_1] = 2 \sum_{i=1}^d \mathbb{E} [|\partial_i^- f|],$$

a contradiction. Therefore the claim holds.

Now, the probability that one iteration of the tester rejects is the probability that $x \in S_i$ when x and i are sampled uniformly at random. This probability is

$$\mathbb{P}[\text{Iteration rejects}] = \sum_{j=1}^d \mathbb{P}_i [i = j] \mathbb{P}_x [x \in S_j] = \sum_{j=1}^d \frac{1}{d} \cdot \mu(S_j) > \frac{\varepsilon}{2dM}.$$

Thus $\Theta\left(\frac{dM}{\varepsilon}\right)$ iterations suffice to reject with high constant probability.

Discrete case. Suppose $\Omega = [n]^d$. The proof proceeds the same way, but we give it explicitly for completeness. For each $i \in [d]$, let $S_i := \{x \in [n]^d : \partial_i f(x) < 0\}$. We then claim that

$$\sum_{i=1}^d \mu(S_i) > \frac{\varepsilon}{2nM}.$$

Indeed, if this is not the case, then since $|\partial_i f(x)| \leq M$ for every i and x , we get that

$$2 \sum_{i=1}^d \mathbb{E} [|\partial_i^- f|] \leq 2M \sum_{i=1}^d \mu(S_i) \leq \frac{\varepsilon}{n}.$$

On the other hand, the assumption that $\text{dist}_1^{\text{mono}}(f) > \varepsilon$ and [Theorem 4.24](#) yield

$$\frac{\varepsilon}{n} < \frac{1}{n} \cdot \mathbb{E} [|f - f^*|] \leq \frac{1}{n} \cdot 2n \mathbb{E} [\|\nabla^- f\|_1] = 2 \sum_{i=1}^d \mathbb{E} [|\partial_i^- f|] ,$$

a contradiction. Thus the claim holds, and the probability that one iteration of the tester rejects is

$$\mathbb{P} [\text{Iteration rejects}] = \sum_{j=1}^d \mathbb{P}_i [i = j] \mathbb{P}_x [x \in S_j] = \sum_{j=1}^d \frac{1}{d} \cdot \mu(S_j) > \frac{\varepsilon}{2dnM} .$$

Thus $\Theta\left(\frac{dnM}{\varepsilon}\right)$ iterations suffice to reject with high constant probability. $\square$

4.4 L^1 -testing monotonicity on the line

In this section, we show the upper bounds for L^1 monotonicity testing on the line from [Theorem 4.4](#). The main idea is to reduce from L^1 testing to Hamming testing by using the Lipschitz constant to show that, if the L^1 distance to monotonicity is large, then the Hamming distance to monotonicity must be somewhat large as well; combined with the Hamming testers [\[EKKRV98; Bel18\]](#), this yields an L^1 tester for the discrete line $[n]$.

To obtain a tester for the continuous line $[0, n]$, we furthermore apply a discretization strategy inspired by domain reduction and downsampling ideas from previous works on property testing [\[BCS20; HY22\]](#). The idea is that, given ε and M , we may impose a fine enough grid on $[0, n]$ such that the function defined on that grid preserves the L^1 distance to monotonicity compared to the continuous function; again, the Lipschitz assumption is essential for this step.

In this section, we will follow the convention of denoting functions on continuous domains by f, g , and those on discrete domains by $\bar{f}, \bar{g}$. Depending on the context, it will be clear whether $\bar{f}$ is an arbitrary function or one obtained by discretizing a particular function f . We will also write “ f is M -Lipschitz” without specifying the ℓ^q geometry, since all choices are equivalent in one dimension.

Lemma 4.27 (Discretization preserves distance to monotonicity). *Let $n, M, \varepsilon > 0$ and let $f : [0, n] \rightarrow \mathbb{R}$ be an M -Lipschitz function. Let the discretized function $\bar{f} : [n'] \rightarrow \mathbb{R}$, for suitable choice of $n' = \Theta(nM/\varepsilon)$, be given by $\bar{f}(i) = f(\delta i)$ for each $i \in [n']$, where $\delta := n/n'$. Then if $\text{dist}_1^{\text{mono}}(f) > \varepsilon$, we have $\text{dist}_1^{\text{mono}}(\bar{f}) > \varepsilon/4$.*

Proof. Let $n' \in [cnM/\varepsilon, 2cnM/\varepsilon]$ be an integer, where c is a sufficiently large universal constant.² Let $\bar{f} : [n'] \rightarrow \mathbb{R}$ be the function given in the statement, and suppose $\text{dist}_1^{\text{mono}}(f) > \varepsilon$.

Let $\bar{g} : [n'] \rightarrow \mathbb{R}$ be the monotone rearrangement of $\bar{f}$. It is easy to check that $\bar{g}$ is Lipschitz with at most the Lipschitz constant of $\bar{f}$. Let $g : [0, n] \rightarrow \mathbb{R}$ be the following piecewise linear function whose discretization is $\bar{g}$: for each $i \in [n']$ we set $g(\delta i) = \bar{g}(i)$, and g is the linear spline induced by these points elsewhere (and constant in the segment $[0, \delta]$). Then clearly g is monotone, and thus $\mathbb{E}[|f - g|] > \varepsilon$. Moreover, g is M -Lipschitz, since its steepest slope is the same as that of $\bar{g}$ up to the coordinate change.³ Hence, we have

$$\begin{aligned}
\varepsilon &< \mathbb{E}[|f - g|] = \frac{1}{n} \int_0^n |f(x) - g(x)| dx = \frac{1}{n} \sum_{i=1}^{n'} \int_{(i-1)\delta}^{i\delta} |f(x) - g(x)| dx \\
&= \frac{1}{n} \sum_{i=1}^{n'} \int_{(i-1)\delta}^{i\delta} |(f(i\delta) \pm M\delta) - (g(i\delta) \pm M\delta)| dx && \text{(Lipschitz property)} \\
&\leq \frac{1}{n} \sum_{i=1}^{n'} \int_{(i-1)\delta}^{i\delta} (|\bar{f}(i) - \bar{g}(i)| + 2M\delta) dx \\
&= \frac{1}{n} \left(2n'M\delta^2 + \delta \sum_{i=1}^{n'} |\bar{f}(i) - \bar{g}(i)| \right) = \frac{2nM}{n'} + \frac{1}{n'} \sum_{i=1}^{n'} |\bar{f}(i) - \bar{g}(i)| \leq \frac{2\varepsilon}{c} + \mathbb{E}[\bar{f} - \bar{g}] ,
\end{aligned}$$

where we used the notation $a \pm b$ to denote any number in the interval $[a - b, a + b]$.

We may set $c \geq 4$ so that $2\varepsilon/c \leq \varepsilon/2$. Therefore, we obtain $\mathbb{E}[\bar{f} - \bar{g}] > \varepsilon/2$. Since $\bar{g}$ is the monotone rearrangement of $\bar{f}$, [Proposition 4.25](#) implies that $\mathbb{E}[\bar{f} - \bar{g}] \leq 2\text{dist}_1^{\text{mono}}(\bar{f})$. We conclude that $\text{dist}_1^{\text{mono}}(\bar{f}) > \varepsilon/4$, as desired. $\square$

Observation 4.28. *The function $\bar{f}$ defined in [Lemma 4.27](#) is ε -Lipschitz. Indeed, since $n' \geq nM/\varepsilon$, we have*

$$|\bar{f}(i) - \bar{f}(i+1)| = |f(\delta i) - f(\delta(i+1))| \leq M\delta = Mn/n' \leq \varepsilon .$$

²We may assume that $nM/\varepsilon > 1$, otherwise the problem is trivial: the maximum L^1 distance from monotonicity attainable by an M -Lipschitz function is $\frac{1}{n} \cdot \frac{n \cdot nM}{2} = nM/2$. Thus the given interval indeed contains an integer.

³Formally, if f is M -Lipschitz, then $\bar{f}$ is M' -Lipschitz for $M' = Mn/n'$, hence so is its monotone rearrangement $\bar{g}$. Then since the steepest slope of g must come from two vertices of the spline, g is Lipschitz with Lipschitz constant $M'n'/n = M$.

Write $\text{dist}_H(f, g) := \mathbb{P}[f \neq g]$ for the relative Hamming distance between f and g , and write $\text{dist}_H^{\text{mono}}(f) := \inf_{\text{monotone } g} \text{dist}_H(f, g)$ for the relative Hamming distance to monotonicity of f .

Lemma 4.29 (Far in L^1 distance implies far in Hamming distance). *Let $\bar{f} : [n'] \rightarrow \mathbb{R}$ be an M' -Lipschitz function. Then $\text{dist}_H^{\text{mono}}(\bar{f}) \geq \sqrt{\frac{\text{dist}_1^{\text{mono}}(\bar{f})}{n' M'}}$.*

Proof. Let $S \subseteq [n']$ be a set such that 1) $|S| = \text{dist}_H^{\text{mono}}(\bar{f}) \cdot n'$; and 2) it suffices to change $\bar{f}$ on inputs in S to obtain a monotone function; note that S exists by definition of Hamming distance. Write S as the union of maximal, pairwise disjoint contiguous intervals, $S = I_1 \cup \dots \cup I_k$.

We define a monotone function $\bar{g} : [n'] \rightarrow \mathbb{R}$ as follows. For each $i \in S$, set $i^* \in [n'] \setminus S$ as follows: if there exists $j \in [n'] \setminus S$ such that $j > i$, pick the smallest such j ; otherwise, pick the largest $j \in [n'] \setminus S$. In other words, i^* is obtained by picking a direction (right if possible, otherwise left) and choosing the first point outside the interval I_ℓ that contains i . Now, define $\bar{g}$ by

$$\bar{g}(i) = \begin{cases} \bar{f}(i) & \text{if } i \notin S \\ \bar{f}(i^*) & \text{if } i \in S. \end{cases}$$

We first claim that $\bar{g}$ is monotone. Indeed the sequence of values $(\bar{f}(i))_{i \in [n'] \setminus S}$ (taken in order of increasing i) is monotone by the choice of S , and since $\bar{g}$ is obtained by extending some of these values into flat regions, the resulting function is also monotone. Therefore we can upper bound the L^1 distance of $\bar{f}$ to monotonicity by

$$\begin{aligned} \text{dist}_1^{\text{mono}}(\bar{f}) &\leq \mathbb{E}[|\bar{f} - \bar{g}|] = \frac{1}{n'} \sum_{i=1}^{n'} |\bar{f}(i) - \bar{g}(i)| = \frac{1}{n'} \sum_{j=1}^k \sum_{i \in I_j} |\bar{f}(i) - \bar{f}(i^*)| \\ &= \frac{1}{n'} \sum_{j=1}^k \sum_{i \in I_j} |(\bar{f}(i^*) \pm M'|i - i^*|) - \bar{f}(i^*)| \quad (\text{Lipschitz property}) \\ &\leq \frac{M'}{n'} \sum_{j=1}^k \sum_{i \in I_j} |i - i^*| \leq \frac{M'}{n'} \sum_{j=1}^k \sum_{i \in I_j} |I_j| = \frac{M'}{n'} \sum_{j=1}^k |I_j|^2 \\ &\leq \frac{M'}{n'} \cdot |S|^2 \quad (\text{Since } |I_1| + \dots + |I_k| = |S|) \\ &= M' \cdot \text{dist}_H^{\text{mono}}(\bar{f})^2 \cdot n'. \end{aligned}$$

The claim follows. □

Combining the two lemmas with the classical Hamming monotonicity tester of [EKKRV98], the following theorem establishes Theorem 4.4 for the continuous domain $[0, n]$.

Theorem 4.30. *There exists a nonadaptive one-sided L^1 monotonicity tester for M -Lipschitz functions $f : [0, n] \rightarrow \mathbb{R}$ with query complexity $O\left(\sqrt{\frac{nM}{\varepsilon}} \log\left(\frac{nM}{\varepsilon}\right)\right)$.*

Proof. The tester works as follows. It first fixes $n' = \Theta(nM/\varepsilon)$ as given by Lemma 4.27. Let $\bar{f} : [n'] \rightarrow \mathbb{R}$ be the discretization defined therein ($\bar{f}$ is not explicitly computed upfront, but rather queried as needed). The algorithm then simulates the (nonadaptive, one-sided) monotonicity tester of [EKKRV98] on the function $\bar{f}$ with proximity parameter $\varepsilon' = \Theta\left(\sqrt{\frac{\varepsilon}{nM}}\right)$ (the constant may easily be made explicit), producing $f(\delta i) = f(in/n')$ whenever the simulation queries $\bar{f}(i)$. The algorithm returns the result produced by the simulated tester. The query complexity claim follows from the fact that the tester of [EKKRV98] has query complexity $O\left(\frac{1}{\varepsilon'} \log n'\right)$.

We now show correctness. When f is monotone, so is $\bar{f}$, so the algorithm accepts since the tester of [EKKRV98] has one-sided error. Now, suppose $\text{dist}_1^{\text{mono}}(f) > \varepsilon$. Then $\text{dist}_1^{\text{mono}}(\bar{f}) > \varepsilon/4$ by Lemma 4.27. Moreover, since $\bar{f}$ is ε -Lipschitz by Observation 4.28, Lemma 4.29 implies that

$$\text{dist}_H^{\text{mono}}(\bar{f}) \geq \sqrt{\frac{\text{dist}_1^{\text{mono}}(\bar{f})}{n'\varepsilon}} > \sqrt{\frac{1}{4n'}} = \Omega\left(\sqrt{\frac{\varepsilon}{nM}}\right).$$

Since this is the proximity parameter ε' used to instantiate the tester of [EKKRV98], the algorithm rejects with high constant probability, as needed. $\square$

Lemma 4.29 itself also implies Theorem 4.4 for the discrete domain $[n]$. This time, we use the Hamming tester of [Bel18] to obtain a slightly more precise query complexity bound⁴.

Theorem 4.31. *There exists a nonadaptive one-sided L^1 monotonicity tester for M -Lipschitz functions $\bar{f} : [n] \rightarrow \mathbb{R}$ with query complexity $O\left(\sqrt{\frac{nM}{\varepsilon}} \log\left(\frac{n\varepsilon}{M}\right)\right)$ when $\varepsilon/M \geq 4/n$, and $O(n)$ otherwise.*

Proof. The tester sets $\varepsilon' := \sqrt{\frac{\varepsilon}{nM}}$, and then runs the (nonadaptive, one-sided) Hamming monotonicity tester of [Bel18] on the line $[n]$ with proximity parameter ε' . That tester has

⁴One may check that this refinement would have no effect in Theorem 4.30

query complexity $O\left(\frac{1}{\varepsilon'} \log(\varepsilon' n)\right)$ when $\varepsilon' \geq 2/n$ and (trivially) $O(n)$ otherwise, which gives the claimed upper bound. It remains to show correctness.

When $\bar{f}$ is monotone, the algorithm accepts since the tester of [Bel18] has one-sided error. Now, suppose $\text{dist}_1^{\text{mono}}(\bar{f}) > \varepsilon$. Then Lemma 4.29 yields

$$\text{dist}_H^{\text{mono}}(\bar{f}) > \sqrt{\frac{\varepsilon}{nM}} = \varepsilon',$$

so the tester rejects with high constant probability. $\square$

4.5 Lower bounds

In this section, we prove our lower bounds for testing monotonicity on the unit cube and on the hypergrid. We first show our general lower bounds based on a “hole” construction, which hides a monotonicity violating region inside a randomly placed ℓ^1 -ball; these bounds imply near tightness of our upper bounds for testing on the line from Section 4.4. Then we give our lower bounds for partial derivative testers, which show that the analysis of our tester in Section 4.3 is tight.

Definition 4.32 (ℓ^1 -ball). Let Ω be one of $\mathbb{R}^d$ or $\mathbb{Z}^d$, let $x \in \Omega$ and let $r > 0$ be a real number. The ℓ^1 -ball of radius r centered at x is the set $B_1^d(r, x) := \{y \in \Omega : \|x - y\|_1 \leq r\}$. We also write $B_1^d(r) := B_1^d(r, 0)$.

It will be clear from context whether the domain should be taken to be continuous or discrete, i.e. whether $B_1^d(r, c)$ should be understood under $\Omega = \mathbb{R}^d$ or $\Omega = \mathbb{Z}^d$. Let ν denote the Lebesgue measure when $\Omega = \mathbb{R}^d$, and the counting measure over $\mathbb{Z}^d$ when $\Omega = \mathbb{Z}^d$.

We give the following simple bounds on the volume of continuous and discrete ℓ^1 -balls. Since we do not require particularly tight bounds, we opt for a simple formulation and elementary proof.

Proposition 4.33. *There exist functions $c_1, c_2 : \mathbb{N} \rightarrow \mathbb{R}_{>0}$ satisfying the following. Let $d \in \mathbb{N}$. Let Ω be one of $\mathbb{R}^d$ or $\mathbb{Z}^d$. Let $r \in \mathbb{R}$ satisfy $r > 0$ if $\Omega = \mathbb{R}^d$, and $r \geq 1$ if $\Omega = \mathbb{Z}^d$. Then*

$$c_1(d)r^d \leq \nu(B_1^d(r)) \leq c_2(d)r^d.$$

Proof. First suppose $\Omega = \mathbb{R}^d$. Then we have the following formula for the area of the ℓ^1 -ball of radius r (see e.g. [Wan05]):

$$\nu(B_1^d(r)) = \frac{(2r)^d}{d!}.$$

The result follows as long as $c_1(d) \leq 2^d/d!$ and $c_2(d) \geq 2^d/d!$.

Now, suppose $\Omega = \mathbb{Z}^n$, and suppose r is an integer without loss of generality (because since $r \geq 1$, there exist integers within a factor of 2 above and below r). We proceed by an inductive argument. For $d = 1$, the volume is

$$\nu(B_1^1) = 1 + 2 \sum_{k=1}^r 1 = 1 + 2r,$$

so the claim holds as long as $c_1(1) \leq 2$ and $c_2(1) \geq 3$. Assuming the claim for some $d \in \mathbb{N}$, we have

$$\begin{aligned} \nu(B_1^{d+1}(r)) &= |\{x = (x_1, \dots, x_d, x_{d+1}) : \|x\|_1 \leq r\}| = \sum_{y=-r}^r |\{x' = (x'_1, \dots, x'_d) : \|x'\|_1 \leq r - |y|\}| \\ &= \nu(B_1^d(r)) + 2 \sum_{k=1}^r \nu(B_1^d(r - k)). \end{aligned}$$

Since the last expression is at most $3r \cdot \nu(B_1^d(r))$, using the inductive hypothesis we conclude that

$$\nu(B_1^{d+1}(r)) \leq 3r \cdot c_2(d)r^d = 3c_2(d)r^{d+1} \leq c_2(d+1)r^{d+1},$$

the last inequality as long as $c_2(n+1) \geq 3c_2(n)$.

For the lower bound, we consider two cases. Note that $r - k \geq r/2$ for at least $\lfloor r/2 \rfloor$ values of k . When $r \geq 2$, we have $\lfloor r/2 \rfloor \geq r/3$, and then

$$\nu(B_1^{d+1}(r)) \geq c_1(d)r^d + 2 \sum_{k=1}^r c_1(d)(r - k)^d > \frac{2r}{3} \cdot c_1(d) \left(\frac{r}{2}\right)^d \geq c_1(d+1)r^{d+1},$$

the last inequality as long as $c_1(n+1) \leq \frac{2}{3} \cdot 2^{-d} \cdot c_1(n)$. On the other hand, if $r < 2$, the bound follows easily for small enough $c_1(d+1)$, since

$$\nu(B_1^{d+1}(r)) \geq c_1(d)r^d + 2 \sum_{k=1}^r c_1(d)(r - k)^d > \frac{c_1(d)r^{d+1}}{r} > \frac{c_1(d)}{2}r^{d+1}. \quad \square$$

Remark 4.34. The constants $c_1(d)$ and $c_2(d)$ in [Proposition 4.33](#) have poor dependence on d , and this is unavoidable at least in the continuous case. This fact is essentially the reason why our general lower bound construction is only efficient for constant d .

We now prove our tester-independent lower bounds. Note that there exists a tester for (ℓ^1, M) -Lipschitz functions with proximity parameter ε if and only if there exists a tester for $(\ell^1, 1)$ -Lipschitz functions with proximity parameter ε/M (by rescaling the input). Therefore it suffices to prove the theorems for the case $M = 1$. The following two theorems establish the continuous and discrete cases of [Theorem 4.5](#).

Theorem 4.35 (Lower bound for constant d on the unit cube). *Let $d \in \mathbb{N}$ be a constant. Any L^1 monotonicity tester (with two-sided error, and adaptive value and directional derivative queries) for $(\ell^1, 1)$ -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ requires at least $\Omega\left((1/\varepsilon)^{\frac{d}{d+1}}\right)$ queries.*

Proof. We construct a family of functions that are ε -far from monotone in L^1 distance such that any deterministic algorithm cannot distinguish between a function chosen uniformly at random from this family and the constant-0 function with fewer than the announced number of queries; then, the claim will follow from Yao's principle.

Each such function f is constructed as follows. Let $c \in [0, 1]^d$ be a point such that the ball $B_1^d(r, c)$ is completely inside $[0, 1]^d$, for radius r to be chosen below. Then f takes value 0 everywhere outside $B_1^d(r, c)$, and inside this ball, it takes value

$$f(x) = -r + \|x - c\|_1$$

for each $x \in B_1^d(r, c)$. Then f is $(\ell^1, 1)$ -Lipschitz. We now lower bound $\text{dist}_1^{\text{mono}}(f)$. Fix any $x' \in [0, 1]^{d-1}$ and consider the line of points (y, x') for $y \in [0, 1]$, i.e. the line along the first coordinate with remaining coordinates set to x' . Suppose this line intersects $B_1^d(r, c)$. Then this intersection occurs on some interval $[a, b]$ of y -values, and on this interval, f first decreases from $f(a, x') = 0$ to $f\left(\frac{a+b}{2}, x'\right) = -\frac{b-a}{2}$ with rate 1, and then increases with rate 1 back to $f(b, x') = 0$. Any monotone function g is in particular monotone over this line, and it is easy to see that this requires total change to f proportional to the area under this curve:

$$\int_0^1 |f(y, x') - g(y, x')| dy \gtrsim \int_0^1 |f(y, x')| dy.$$

Now, since this holds for each line intersecting $B_1^d(r, c)$, and the collection of such lines gives a partition of $B_1^d(r, c)$, the total distance between f and any monotone function g is

lower bounded (up to a constant) by the L^1 -norm of f :

$$\int_{[0,1]^d} |f - g| \, dx \gtrsim \int_{[0,1]^d} |f| \, dx ,$$

and since this holds for every choice of g , we conclude that

$$\text{dist}_1^{\text{mono}}(f) \gtrsim \int_{[0,1]^d} |f| \, dx .$$

We now note that this last expression is half the volume of an ℓ^1 -ball in dimension $d + 1$: for each point $x \in B_1^d(r, c)$, the contribution to the integrand is $|f(x)| = r - \|x - c\|_1$, corresponding to the measure of points (x, z') for all $0 \leq z' \leq z$ where $z = r - \|x - c\|_1$, so that the point $(x, z) \in \mathbb{R}^{d+1}$ satisfies $\|(x, z) - (c, 0)\|_1 = r$. In other words, the points (x, z') are the points of $B_1^{d+1}(r, (c, 0))$ with nonnegative last coordinate. Conversely, all such points contribute to the integral above. Therefore, since d is a constant, using [Proposition 4.33](#) and writing ν^{d+1} for the Lebesgue measure on $\mathbb{R}^{d+1}$, we have

$$\text{dist}_1^{\text{mono}}(f) \gtrsim \int_{[0,1]^d} |f| \, dx \gtrsim \nu^{d+1}(B_1^{d+1}(r)) \gtrsim r^{d+1} .$$

We wish this last quantity to be at least $\Omega(\varepsilon)$, so (recalling d is a constant) it suffices to set

$$r \asymp \varepsilon^{\frac{1}{d+1}} .$$

We have established that each function f , for this choice of r and any choice of c , is ε -far from monotone as desired. Our family of functions from which f will be drawn will be given by choices of c such that the balls $B_1^d(r, c)$ are disjoint, so that each query may only rule out one such choice (because queries outside $B_1^d(r, c)$ take value 0). How many disjoint balls $B_1^d(r, c)$ can we fit inside $[0, 1]^d$? It suffices to divide $[0, 1]^d$ into a grid of d -dimensional cells of side $2r$, each of which can contain one ball. The number of such cells is at least (up to a constant factor)

$$(1/r)^d \gtrsim (1/\varepsilon)^{\frac{d}{d+1}} .$$

Hence, to distinguish some f drawn uniformly at random from this family from the constant-0 function with constant probability, any deterministic algorithm must have query complexity at least $\Omega\left((1/\varepsilon)^{\frac{d}{d+1}}\right)$. $\square$

Theorem 4.36 (Lower bound for constant n on the hypergrid). *Let $d \in \mathbb{N}$ be a constant. Any L^1 monotonicity tester (with two-sided error and adaptive queries) for $(\ell^1, 1)$ -Lipschitz functions $f : [n]^d \rightarrow \mathbb{R}$ requires at least $\Omega\left(\min\left\{(n/\varepsilon)^{\frac{d}{d+1}}, n^d\right\}\right)$ queries.*

Proof. We proceed similarly to [Theorem 4.35](#), with small changes for the discrete setting. We again construct functions f based on balls $B_1^d(r, c)$ for suitable choices of r and c . For fixed r and c , f takes value 0 outside the ball and, for each $x \in B_1^d(r, c)$,

$$f(x) = -r + \|x - c\|_1,$$

so that f is $(\ell^1, 1)$ -Lipschitz. Again by a line restriction argument, for any monotone function g we have

$$\int_{[n]^d} |f - g| \, dx \gtrsim \int_{[n]^d} |f| \, dx,$$

and thus

$$\text{dist}_1^{\text{mono}}(f) \gtrsim \frac{1}{n^d} \int_{[n]^d} |f| \, dx. \quad (4.4)$$

When $\varepsilon \leq 1/n^d$, this construction boils down to setting $f(x) = -1$ at a single point x , which requires $\Omega(n^d)$ queries to identify. Now, assume $\varepsilon > 1/n^d$.

Again we may identify the integrand of (4.4) with points on half of $B_1^{d+1}(r, (c, 0))$. As long as $r \geq 1$ and since d is a constant, [Proposition 4.33](#) implies that

$$\text{dist}_1^{\text{mono}}(f) \gtrsim \frac{r^{d+1}}{n^d}.$$

Thus to have $\text{dist}_1^{\text{mono}}(f) \geq \varepsilon$, it suffices (since d is a constant) to set

$$r \asymp n^{\frac{d}{d+1}} \varepsilon^{\frac{1}{d+1}},$$

and indeed this gives $r \geq 1$ since $\varepsilon > 1/n^d$. Then, our functions f are given by choices of c placed on the hypergrid $[n]^d$ inside disjoint cells of side $2r$, of which there are at least (up to a constant factor)

$$\left(\frac{n}{r}\right)^d \gtrsim \left(\frac{n}{\varepsilon}\right)^{\frac{d}{d+1}},$$

and thus any deterministic algorithm requires $\Omega\left((n/\varepsilon)^{\frac{d}{d+1}}\right)$ queries to distinguish between f drawn uniformly at random from this distribution and the constant-0 function. $\square$

The construction for the partial derivative tester lower bounds is simpler: we start with a one-dimensional “step” construction which is flat everywhere except for a small region of negative slope, and then copy this function onto every line along a randomly chosen coordinate i . Then a partial derivative tester must correctly guess both i and the negative-slope region to detect such functions. The following two theorems establish the continuous and discrete cases of [Theorem 4.6](#).

Theorem 4.37 (Lower bound for partial derivative testers on the unit cube). *Any partial derivative L^1 monotonicity tester for $(\ell^1, 1)$ -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ (with two-sided error and adaptive queries) requires at least $\Omega(d/\varepsilon)$ queries.*

Proof. Let $\varepsilon \leq 1/6$. For any $z \in [\frac{1}{3}, \frac{2}{3} - \varepsilon]$, let $g_z : [0, 1] \rightarrow \mathbb{R}$ be the function given by

$$g_z(x) = \begin{cases} \varepsilon & \text{if } x < z, \\ \varepsilon - (x - z) & \text{if } z \leq x \leq z + \varepsilon, \\ 0 & \text{if } x > z + \varepsilon. \end{cases}$$

Note that g_z is 1-Lipschitz. Moreover, we claim that $\text{dist}_1^{\text{mono}}(g_z) \gtrsim \varepsilon$. Indeed, for any $x \in [0, 1/3]$, we have that $g_z(x) = \varepsilon$ and $g_z(2/3 + x) = 0$. On the other hand, for any monotone function $h : [0, 1] \rightarrow \mathbb{R}$ we must have $h(x) \leq h(2/3 + x)$. Thus, for any such h we have $|g_z(x) - h(x)| + |g_z(2/3 + x) - h(2/3 + x)| \geq \varepsilon$. Since this holds for all $x \in [0, 1/3]$, we conclude that for any such h we must have $\mathbb{E}[|g_z - h|] \geq \varepsilon/3$, proving the claim.

Now, for any $i \in [d]$ and $z \in [\frac{1}{3}, \frac{2}{3} - \varepsilon]$, let $f_{i,z} : [0, 1]^d \rightarrow \mathbb{R}$ be given by copying g_z onto f along every line in direction i , i.e. setting $f_{i,z}(x) = g_z(x_i)$ for every $x \in [0, 1]^d$. Note that f is $(\ell^1, 1)$ -Lipschitz (since its partial derivatives are 0 along non- i coordinates), and $\text{dist}_1^{\text{mono}}(f) \gtrsim \varepsilon$ (since the lines in direction i partition the domain).

We construct a set of $\Omega(d/\varepsilon)$ functions $f_{i,z}$ as follows. First, i can be any of the coordinates in $[d]$. Then let $z_1, \dots, z_k$ be given by $z_j = \frac{1}{3} + k\varepsilon$ for $k = \Omega(1/\varepsilon)$, such that for each j we have $z_j \in [\frac{1}{3}, \frac{2}{3} - \varepsilon]$ and, moreover, for distinct $j, \ell \in [k]$, the regions where f_{i,z_j} and f_{i,z_ℓ} take non-zero slope are disjoint. It follows that each partial derivative query may only rule out one such $f_{i,z}$, so any partial derivative tester that distinguishes an $f_{i,z}$ chosen uniformly at random from the constant-0 function must make at least $\Omega(d/\varepsilon)$ queries. $\square$

The argument for the hypergrid is similar, except that the construction cannot be made to occupy an arbitrarily small region of the domain when the domain is discrete. We opt to keep the argument simple and give a proof for constant parameter ε .

Theorem 4.38 (Lower bound for edge testers on the hypergrid). *For sufficiently small constant ε , any partial derivative L^1 monotonicity tester for $(\ell^1, 1)$ -Lipschitz functions $f : [n]^d \rightarrow \mathbb{R}$ (with two-sided error and adaptive queries) requires at least $\Omega(dn)$ queries.*

Proof. Let n be a multiple of 3 for simplicity. For each $z \in \{\frac{n}{3} + 1, \dots, \frac{2n}{3}\}$, define $g_z : [n] \rightarrow \mathbb{R}$ by

$$g_z(x) = \begin{cases} 1 & \text{if } x < z, \\ 0 & \text{if } z \leq x. \end{cases}$$

Then, as before g_z is 1-Lipschitz and $\text{dist}_1^{\text{mono}}(g_z) = \Omega(1)$. Then for each $i \in [d]$ and $z \in [\frac{n}{3} + 1, \frac{2n}{3}]$, we let $f_{i,z} : [n]^d \rightarrow \mathbb{R}$ be given by $f_{i,z}(x) = g_z(x_i)$ for each $x \in [n]^d$; it follows that f is $(\ell^1, 1)$ -Lipschitz and $\text{dist}_1^{\text{mono}}(f) = \Omega(1)$. Note that there are $\Omega(dn)$ such functions. Moreover, each partial derivative query may only rule out one such $f_{i,z}$, and therefore any edge tester that distinguishes an $f_{i,z}$ chosen uniformly at random from the constant-0 function must make at least $\Omega(dn)$ queries. $\square$

4.6 Comparison with prior work on L^p testing for bounded-range functions

[BRY14a] initiated the systematic study of L^p -testing and, most relevant to our results, established the first (and, to our knowledge, only previous) results on L^p testing of monotonicity on the hypergrid and on the discrete line. While our models are broadly compatible, a subtle but crucial distinction must be explained.

In [BRY14a], the authors focused their exposition on the case of functions $f : \Omega \rightarrow [0, 1]$, and in this regime, L^1 testing is only be easier than Hamming testing—which they show via a reduction based on Boolean threshold functions. On the other hand, for functions with other ranges, say $f : \Omega \rightarrow [a, b]$, their definition normalizes the notion of distance by a factor of $\frac{1}{b-a}$. In our terminology, letting $r := b - a$ and $g := f/r$, it follows that $\text{dist}_1^{\text{mono}}(g) = \text{dist}_1^{\text{mono}}(f)/r$, so testing f with proximity parameter ε reduces to testing g with proximity parameter ε/r . For Hamming testers with query complexity that scale linearly with $1/\varepsilon$, this amounts to paying a factor of r in the reduction to the Boolean case⁵. This loss is indeed necessary, because by the same reasoning, testing g with proximity parameter ε reduces to testing f with proximity parameter $r\varepsilon$. Therefore the problems of testing f with proximity parameter ε and testing f/r with proximity parameter ε/r have the same query complexity.

In this thesis, we do not normalize the distance metric by r ; we would like to handle functions f that may take large values as the dimension d grows, as long as f satisfies a Lipschitz assumption, and our goal is to beat the query complexity afforded by the reduction to the Boolean case. We derive such a benchmark in Section 4.6.1 by assuming that the input f is Lipschitz, and inferring an upper bound for r based on the Lipschitz constant and the size of the domain. Combining the hypergrid tester of [BRY14a] and

⁵This factor can also be tracked explicitly in the characterization of the L^1 distance to monotonicity of [BRY14a]: it arises in Lemmas 2.1 and 2.2, where an integral from 0 to 1 must be changed to an integral from a to b , so the best threshold function is only guaranteed to be ε/r -far from monotone.

a discretization argument for the unit cube inspired by [BCS20; HY22], we obtain upper bounds for our setting.

With the discussion above in mind, it is instructive to return to Table 4.1. We note that our upper bounds have polynomially smaller dependence on d than the benchmarks, suggesting that our use of the Lipschitz assumption—via our directed Poincaré inequalities—exploits useful structure underlying the monotonicity testing problem (whereas the benchmark testers must work for every function with bounded range, not only the Lipschitz ones). Our lower bounds feature an almost-linear dependence on the hypergrid length n ; intuitively, this dependence is not implied by previous works such as [BRY14a; BRY14b] because those works construct the violations of monotonicity via Boolean functions, whereas our constructions exploit the fact that a Lipschitz function can vastly exceed the $[0, 1]$ range along a given direction, which exacerbates the L^1 distance to monotonicity in the region where that happens.

4.6.1 Upper bounds derived from [BRY14a]

We now show how the L^1 monotonicity testing upper bounds from [BRY14a] imply testers with query complexity $\tilde{O}\left(\frac{d^2 M}{\varepsilon}\right)$ on the unit cube and $\tilde{O}\left(\frac{d^2 n M}{\varepsilon}\right)$ on the hypergrid for (ℓ^1, M) -Lipschitz functions f . We start with the case of the hypergrid, and first state the upper bound of [BRY14a] for functions with arbitrary range of size r , which without loss of generality (by translation invariance) we denote $[0, r]$:

Theorem 4.39 ([BRY14a]). *There exists an L^1 monotonicity tester for functions $f : [n]^d \rightarrow [0, r]$ that uses $O\left(\frac{rd}{\varepsilon} \log \frac{rd}{\varepsilon}\right)$ value queries. The tester is nonadaptive and has one-sided error.*

We thus obtain the following L^1 monotonicity tester for Lipschitz functions on the hypergrid:

Corollary 4.40. *There is an L^1 monotonicity tester for (ℓ^1, M) -Lipschitz functions $f : [n]^d \rightarrow \mathbb{R}$ that uses $O\left(\frac{d^2 n M}{\varepsilon} \log \left(\frac{dnM}{\varepsilon}\right)\right)$ value queries. The tester is nonadaptive and has one-sided error.*

Proof. The algorithm simulates the tester from Theorem 4.39 with parameters $r = Mnd$ and ε , which gives the announced query complexity. As for correctness, note that since f is (ℓ^1, M) -Lipschitz, for any $x, y \in [n]^d$ we have $|f(x) - f(y)| \leq M\|x - y\|_1 \leq Mnd$.

Thus f has range of size at most Mnd , so the reduction to the tester from [Theorem 4.39](#) is correct. $\square$

We outline how this result implies a tester for Lipschitz functions on the unit cube as well, via the domain reduction or downsampling principle of [\[BCS20; HY22\]](#), and similarly to our work on the one-dimensional case in [Section 4.4](#). Even though the tester above has query complexity that depends on the hypergrid length n , the main observation is that, given an (ℓ^1, M) -Lipschitz function f on $[0, 1]^d$, we may discretize it into an $(\ell^1, M/n)$ -Lipschitz function f' on an arbitrarily fine hypergrid $[n]^d$ by choosing n large enough. Then the term $n(M/n) = M$ remains fixed for any choice of n , so the complexity of the tester above applied to f' does not depend on n . Then, by setting n large enough, we may upper bound the error introduced by the discretization.

Corollary 4.41. *There is an L^1 monotonicity tester for (ℓ^1, M) -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ that uses $O\left(\frac{d^2 M}{\varepsilon} \log\left(\frac{dM}{\varepsilon}\right)\right)$ value queries. The tester is nonadaptive and has one-sided error.*

Proof. For sufficiently large m to be chosen below, the algorithm imposes a hypergrid $[n]^d$ uniformly on $[0, 1]^d$. More precisely, we define a function $f' : [n]^d \rightarrow \mathbb{R}$ via $f'(x) = f(x/n)$ for all $x \in [n]^d$. Note that f' is $(\ell^1, M/n)$ -Lipschitz. Then, the algorithm simulates the tester from [Corollary 4.40](#) on f' with proximity parameter $\Theta(\varepsilon)$, and returns its result. Note that the query complexity is as desired, so it remains to show correctness. If f is monotone so is f' , in which case the algorithm accepts. Now assume that $\text{dist}_1^{\text{mono}}(f) > \varepsilon$, and we need to show that $\text{dist}_1^{\text{mono}}(f') \gtrsim \varepsilon$.

Let $g' : [n]^d \rightarrow \mathbb{R}$ be any monotone function. We obtain a monotone function $g : [0, 1]^d \rightarrow \mathbb{R}$ as follows: for each $x \in [0, 1]^d$, let $\bar{x}$ be obtained by rounding up each coordinate x_i to a positive integer multiple of $1/n$. Then $x' := n\bar{x} \in [n]^d$, and we set $g(x) := g'(x')$. It follows that g is monotone, and by the triangle inequality,

$$\mathbb{E} [|f - g|] \leq \mathbb{E} [|f' - g'|] + \max_{x \in [0, 1]^d} |f(x) - f(\bar{x})| \leq \mathbb{E} [|f' - g'|] + \frac{Md}{n}.$$

In the first inequality, we separately accounted for the cost of turning each value of f into the value at its corresponding “rounded up” point (accounted for by the second summand), and then the cost of turning each equal-sized, constant-valued cell into the value of g on that cell, and these values agree with f' and g' (accounted for by the first summand). In the second inequality, we used the fact that any point x satisfies $\|x - \bar{x}\|_1 \leq \frac{1}{n} \cdot d$, along with the Lipschitz assumption on f .

Therefore, by setting $n > \frac{10Md}{\varepsilon}$, we obtain $\frac{Md}{n} < \frac{\varepsilon}{10}$, and since the inequality above holds for every monotone function g' , we conclude that $\text{dist}_1^{\text{mono}}(f') \geq \text{dist}_1^{\text{mono}}(f) - \varepsilon/10 > 9\varepsilon/10$, as desired. $\square$

Remark 4.42. One may wonder whether the reductions above could yield more efficient testers if combined with Hamming testers for Boolean functions on the hypergrid with better dependence on d (via the reduction from L^1 testing to Hamming testing of [BRY14a], which is behind Theorem 4.39), since e.g. the tester of [BCS23a] has query complexity $\tilde{O}(d^{1/2+o(1)}/\varepsilon^2)$. However, it seems like this cannot be the case, i.e. Corollary 4.40 has the best query complexity of any reduction that upper bounds the size of the range of f by Mnd . The reason is as follows: [KMS18] showed that any nonadaptive, one-sided pair tester for the Boolean cube with query complexity $O(d^\alpha/\varepsilon^\beta)$ must satisfy $\alpha + \frac{\beta}{2} \geq \frac{3}{2}$, so hypergrid testers must also satisfy this as well as $\beta \geq 1$, assuming query complexity independent of n . Then, given a hypergrid tester for Boolean functions with query complexity $\tilde{O}(d^\alpha/\varepsilon^\beta)$, our reduction via the inferred range size $r = Mnd$ gives an L^1 tester with asymptotic query complexity at least $\frac{d^\alpha}{(\varepsilon/r)^\beta} = \frac{d^{\alpha+\beta/2}d^{\beta/2}(nM)^\beta}{\varepsilon^\beta} \geq \frac{d^2nM}{\varepsilon}$.

4.7 Discussion and open questions

In this chapter, we started the systematic study of directed Poincaré inequalities over continuous and discrete domains, and showed that the connection between directed isoperimetry and monotonicity testing extends to the analytic setting by giving an L^1 monotonicity tester for Lipschitz functions over $[0, 1]^d$ via the corresponding directed Poincaré inequality. We also showed that the one-dimensional case of the (discrete or continuous) line enjoys a reduction to the Hamming setting; at the same time, the case of the discrete line $[n]$ shows a strong separation between Hamming testing (which has query complexity $\Theta(\log n)$) and L^1 testing of M -Lipschitz functions (which has query complexity $\tilde{\Theta}(\sqrt{nM}/\varepsilon)$). The one-dimensional setting also establishes (perhaps surprisingly) that our lower bounds of $\Omega\left((M/\varepsilon)^{\frac{d}{d+1}}\right)$ and $\Omega\left((nM/\varepsilon)^{\frac{d}{d+1}}\right)$ for the unit cube and hypergrid, respectively, cannot be improved for general constant dimension d .

Comparing our tester for the solid cube $[0, 1]^d$, which makes $O(dM/\varepsilon)$ queries, with the situation for monotonicity testing of Boolean functions, where $\tilde{O}(\sqrt{d})$ and $d^{1/2+o(1)}$ query complexity bounds are achievable for the Boolean cube and hypergrid, respectively [KMS18; BCS23a], the most immediate question is: is there an L^1 tester for monotonicity of Lipschitz functions with $o(d)$ query complexity? We will begin to tackle this question in the next chapter.

A conjectured (L^1, ℓ^2) -inequality. Classical Poincaré inequalities are usually of the ℓ^2 form, which seems natural e.g. due to basis independence. On the other hand, in the directed setting, the weaker ℓ^1 inequalities (as in [GGLRS00] and this chapter) have more straightforward proofs than ℓ^2 counterparts such as the inequality of [KMS18]. A perhaps related observation is that monotonicity is *not* a basis-independent concept, since it is defined in terms of the standard basis. In light of the parallels between the continuous and discrete settings, we conjecture that the solid unit cube enjoys a directed (L^1, ℓ^2) -Poincaré inequality, mirroring the situation for the Boolean cube [KMS18; PRW22; CCRSW25].

Conjecture 4.43. *For every Lipschitz function $f : [0, 1]^d \rightarrow \mathbb{R}$, it holds that*

$$\text{dist}_1^{\text{mono}}(f) \lesssim \mathbb{E} [\|\nabla^- f\|_2] .$$

While we originally made this conjecture in [Fer23] in hopes of establishing an $o(d)$ tester for the analytic setting, in the rest of this thesis we will actually obtain such a tester—indeed, an $\tilde{O}(\sqrt{d})$ tester—from an (L^2, ℓ^2) -inequality instead. Such an inequality enjoys the more standard ℓ^2 geometry as described above, and appears to be easier to obtain compared to Conjecture 4.43 because L^2 spaces enjoy more useful structure (for example, they are Hilbert spaces) than L^1 spaces. Nevertheless, we believe that a proof or disproof of Conjecture 4.43 will further illuminate the nature of directed isoperimetry in the analytic setting, and its connection to monotonicity.

Chapter 5

Testing with $\tilde{O}(\sqrt{d})$ queries, conditional on (L^2, ℓ^2) -Poincaré

In this chapter, we show that if we can prove a directed (L^2, ℓ^2) -Poincaré inequality for Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$, then we can obtain a monotonicity tester using only $\tilde{O}(\sqrt{d})$ queries, improving upon the $O(d)$ -query tester from the previous chapter. Specifically, we assume for now the following result, which will be proved in [Chapter 7](#).

Theorem 5.1 (Directed Poincaré inequality). *There exists a universal constant $C > 0$ such that, for all Lipschitz $f : [0, 1]^d \rightarrow \mathbb{R}$,*

$$\text{dist}_2^{\text{mono}}(f)^2 \leq C \mathbb{E} [\|\nabla^- f\|_2^2] .$$

Conditional on this theorem, we show

Theorem 5.2. *There exists a nonadaptive, directional derivative L^2 monotonicity tester for M -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ with query complexity $\tilde{O}(\sqrt{d}M^2/\varepsilon^2)$ and one-sided error.*

We make two remarks about this result.

First, as observed in [\[BRY14a\]](#), for $1 \leq p \leq q$ and any (say) Lipschitz f and g , Jensen's inequality gives $\|f - g\|_{L^p([0,1]^d)} \leq \|f - g\|_{L^q([0,1]^d)}$. Thus any L^q monotonicity tester is also an L^p tester, so that the $\tilde{O}(\sqrt{d}M^2/\varepsilon^2)$ upper bound from [Theorem 5.2](#) also holds for L^p testing with $p \in [1, 2]$. At the same time, as remarked in [Chapter 2](#), if

f is M -Lipschitz (with respect to the standard ℓ^2 geometry, which is the assumption in [Theorem 5.2](#)), then f is also (ℓ^1, M) -Lipschitz. Thus the testers from both [Theorem 4.3](#) from the previous chapter and [Theorem 5.2](#) above are L^1 monotonicity testers for M -Lipschitz functions; and in this setting, ignoring logarithmic factors, [Theorem 5.2](#) is better than [Theorem 4.3](#) when $\frac{\varepsilon}{M} > \frac{1}{\sqrt{d}}$ and vice-versa. Curiously, something analogous is true of the testers of [\[KMS18\]](#) and [\[GGLRS00\]](#) for the Boolean setting, which make $\tilde{O}(\sqrt{d}/\varepsilon^2)$ and $O(d/\varepsilon)$ queries respectively.

Second, the $\sqrt{d}$ dependence in the query complexity is optimal for *derivative-pair testers*, which are randomized nonadaptive testers that, at each step, either sample points $x \preceq y$ and use value queries to reject if $f(x) > f(y)$, or sample point x and direction $v \succeq 0$ and use a directional derivative query to reject if $v \cdot \nabla f(x) < 0$. The proof is not difficult, and is given in [Section 5.2](#).

Idea of the algorithm. Our tester may be seen as the natural continuous analogue of the path tester of [\[KMS18\]](#) for the Boolean cube, which samples points $x \preceq y \in \{0, 1\}^d$ connected by a path of length 2^k , for k sampled uniformly from $[\log d]$, and rejects if $f(x) > f(y)$. In our case, a path is replaced by a *direction* $v \in \{0, 1\}^d$, and the condition $f(x) > f(y)$ is replaced by the condition $v \cdot \nabla f(x) < 0$, for a uniformly random point $x \in [0, 1]^d$, via a directional derivative query. Such a tester accepts any monotone function, so the challenge is to ensure that it detects the case when $\text{dist}_2^{\text{mono}}(f) \geq \varepsilon$.

In this case, hiding constant factors, [Theorem 5.1](#) gives that $\mathbb{E}[\|\nabla^- f\|_2^2] \gtrsim \varepsilon^2$. We would like the distribution of v to be such that, for any x , $\mathbb{P}_v[v \cdot \nabla f(x) < 0] \gtrsim \frac{\|\nabla^- f(x)\|_2^2}{\sqrt{d}\|\nabla f(x)\|_2^2}$. Note that $\|\nabla f(x)\|_2^2 \leq M^2$ because f is M -Lipschitz, so if we had such a distribution for v , the tester would reject with probability at least $\mathbb{E}_x \left[\frac{\|\nabla^- f(x)\|_2^2}{\sqrt{d}M^2} \right] \gtrsim \frac{\varepsilon^2}{\sqrt{d}M^2}$, implying that $O(\sqrt{d}M^2/\varepsilon^2)$ queries suffice.

A natural choice would be to sample $v \in \{0, 1\}^d$ so that $v_i = 1$ corresponds to taking edge i in the path tester. We make this concrete as follows: sample p uniformly from $\{1, \frac{1}{2}, \dots, \frac{1}{2^{\lceil \log_2(4d) \rceil}}\}$, and sample $v \in \{0, 1\}^d$ by letting each v_i be 1 independently with probability p . At a high level, [Lemma 5.4](#) below shows that this distribution satisfies the condition above (up to a logarithmic factor) by considering, for each *threshold* τ , the smallest value of p such that, informally, after issuing all its queries the algorithm should expect to sample at least one vector v whose support intersects with at least one entry i from $\nabla f(x)$ satisfying $(\nabla f(x))_i < -\tau$ (this idea allows us to “forget” about all the other negative entries of $\nabla f(x)$, and focus on a simpler “good event”). By ranging over all possible τ , we show that there exists τ and corresponding p such that, with good

Algorithm 2 L^2 monotonicity tester for Lipschitz functions using directional derivative queries

Input: Directional derivative oracle access to M -Lipschitz function $f : [0, 1]^d \rightarrow \mathbb{R}$.

Output: Accept if f is monotone, reject if $\text{dist}_2^{\text{mono}}(f) \geq \varepsilon$.

procedure DIRECTIONALDERIVATIVETESTER(f, d, M, ε)

repeat $\Theta\left(\frac{\sqrt{d}M^2}{\varepsilon^2} \log d\right)$ **times**

 Sample $x \in [0, 1]^d$ uniformly at random.

 Sample $v \in \{0, 1\}^d$ from distribution $\mathcal{D}$ given by [Definition 5.3](#).

Reject if $\nabla f(x) \cdot v < 0$.

end repeat

Accept.

probability, the contribution of the positive entries of $\nabla f(x)$ to $v \cdot \nabla f(x)$ is smaller than τ , so that $v \cdot \nabla f(x) < 0$.

5.1 Algorithm and upper bound

We first define the distribution $\mathcal{D}$ from which the tester, which is given in [Algorithm 2](#), samples each direction v for its directional derivative queries.

Definition 5.3 (Distribution of direction vector). For each $p \in [0, 1]$, we define distribution $\mathcal{D}_p$ over $\{0, 1\}^d$ as follows. To sample $\mathbf{v} \sim \mathcal{D}_p$,

1. Sample $\mathbf{x}_i \sim \text{Ber}(p)$ independently for each $i \in [d]$, where $\text{Ber}(p)$ is the Bernoulli distribution;
2. Produce $\mathbf{v} = \sum_{i=1}^d \mathbf{x}_i \mathbf{e}_i$, where $\mathbf{e}_i$ denotes the i -th standard basis vector.

Then, we define the distribution $\mathcal{D}$ over $\{0, 1\}^d$ as follows. Let $P := \left\{1, \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^{\lceil \log_2(4d) \rceil}}\right\}$. To sample $\mathbf{v} \sim \mathcal{D}$, we first sample $\mathbf{p}$ uniformly at random from P , and then sample $\mathbf{v}$ from $\mathcal{D}_{\mathbf{p}}$.

Our proof of [Theorem 5.2](#) relies on the following lemma, which quantifies the ability of the tester to detect negative entries in the gradient $\nabla f(x)$ via directional derivative queries $v \cdot \nabla f(x)$. This setup may be seen, informally, as a signed version of the classic

group testing problem [Dor43]. We first state the lemma and defer its proof until after the proof of the main theorem.

Lemma 5.4 (Detecting negative entries with subset sums). *There exist universal constants $c, \delta > 0$ such that the distribution $\mathcal{D}$ from Definition 5.3 has the following property: for any nonzero $u \in \mathbb{R}^d$, we have*

$$\mathbb{P}_{\mathbf{v} \sim \mathcal{D}} \left[u \cdot \mathbf{v} < -\frac{\delta}{d} \|u^-\|_2 \right] \geq c \cdot \frac{\|u^-\|_2^2}{\sqrt{d} \log(d) \cdot \|u\|_2^2}.$$

Theorem 5.5 (Refinement of Theorem 5.2). *Algorithm 2 is a nonadaptive, directional derivative L^2 monotonicity tester for M -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ with query complexity $O\left(\frac{\sqrt{d}M^2}{\varepsilon^2} \log d\right)$ and one-sided error.*

Proof. First note that, since f is Lipschitz and hence differentiable almost everywhere in $(0, 1)^d$ by Rademacher's theorem, with probability 1 the algorithm only samples points x at which f is differentiable. Moreover, we have $\|\nabla f\|_2 \leq M$ almost everywhere since f is M -Lipschitz.

The algorithm clearly accepts any monotone function; indeed if f is monotone, then $\nabla f \succeq 0$ and, since $\mathcal{D}$ is supported on $\{0, 1\}^d$, we have $\nabla f \cdot \mathbf{v} \geq 0$ with probability 1. Now, suppose $\text{dist}_2^{\text{mono}}(f) \geq \varepsilon$. Combining Lemma 5.4 and Theorem 5.1, we obtain that the probability that any single iteration of the tester rejects is

$$\begin{aligned} \mathbb{P}[\text{Iteration rejects}] &= \mathbb{P}_{\substack{\mathbf{x} \in [0, 1]^d \\ \mathbf{v} \sim \mathcal{D}}} [\nabla f(\mathbf{x}) \cdot \mathbf{v} < 0] \\ &\geq \mathbb{P}_{\substack{\mathbf{x} \in [0, 1]^d \\ \mathbf{v} \sim \mathcal{D}}} \left[\nabla f(\mathbf{x}) \cdot \mathbf{v} < -\frac{\delta}{d} \|\nabla^- f(\mathbf{x})\|_2 \right] \\ &= \int_{[0, 1]^d} \mathbb{P}_{\mathbf{v} \sim \mathcal{D}} \left[\nabla f(x) \cdot \mathbf{v} < -\frac{\delta}{d} \|\nabla^- f(x)\|_2 \right] dx \\ &\geq \int_{[0, 1]^d} c \cdot \frac{\|\nabla^- f(x)\|_2^2}{\sqrt{d} \log(d) M^2} dx && (\text{Lemma 5.4, } \|\nabla f\|_2 \leq M) \\ &= \frac{c}{M^2 \sqrt{d} \log d} \int_{[0, 1]^d} \|\nabla^- f(x)\|_2^2 dx \\ &\geq \frac{c}{M^2 \sqrt{d} \log d} \cdot \frac{1}{C} \text{dist}_2^{\text{mono}}(f)^2 && (\text{Theorem 5.1}) \\ &\geq \frac{c}{C} \cdot \frac{\varepsilon^2}{M^2 \sqrt{d} \log d}, \end{aligned}$$

where δ, c are the constants from [Lemma 5.4](#) and C is the constant from [Theorem 5.1](#). Thus $\Theta\left(\frac{\sqrt{d}M^2}{\varepsilon^2} \log d\right)$ iterations suffice to reject with probability at least $2/3$. $\square$

Remark 5.6. A slight modification of the proof of [Theorem 5.5](#) also shows that, if we replace the condition $\nabla f(x) \cdot v < 0$ in [Algorithm 2](#) with the more demanding condition $\nabla f(x) \cdot v < -\frac{K\varepsilon}{d}$ for some universal constant $K > 0$, then we still obtain a tester with the same guarantees; in particular, [Algorithm 2](#) does not rely on arbitrary precision. Indeed, [Theorem 5.1](#) gives that $\int \|\nabla^- f(x)\|_2^2 dx \geq \frac{1}{C}\varepsilon^2$ when f is ε -far from monotone, but the points x for which $\|\nabla^- f(x)\|_2^2 \leq \frac{\varepsilon^2}{2C}$ can only contribute at most $\frac{\varepsilon^2}{2C}$ to the integral. Therefore the points satisfying $\|\nabla^- f(x)\|_2^2 > \frac{\varepsilon^2}{2C}$ must contribute at least $\frac{\varepsilon^2}{2C}$ and, at each such point, [Lemma 5.4](#) below guarantees that $\nabla^- f(x) \cdot v < -\frac{\delta}{d}\|\nabla^- f(x)\|_2 < -\frac{\delta}{d} \cdot \frac{\varepsilon}{\sqrt{2C}}$ with at least the probability given in the lemma. Therefore a similar calculation to that of [Theorem 5.5](#) shows that this modified tester also rejects with sufficient probability.

We now prove [Lemma 5.4](#), restated below for convenience.

Lemma 5.4 (Detecting negative entries with subset sums). *There exist universal constants $c, \delta > 0$ such that the distribution $\mathcal{D}$ from [Definition 5.3](#) has the following property: for any nonzero $u \in \mathbb{R}^d$, we have*

$$\mathbb{P}_{v \sim \mathcal{D}} \left[u \cdot v < -\frac{\delta}{d} \|u^-\|_2 \right] \geq c \cdot \frac{\|u^-\|_2^2}{\sqrt{d} \log(d) \cdot \|u\|_2^2}.$$

Proof. We may assume that u contains at least one strictly negative entry, since otherwise $\|u^-\|_2^2 = 0$ and the claim is trivial. Let $\delta := 1/100$, and define $\delta_d := \delta/d$ for convenience.

Recall the distributions $\mathcal{D}$ and $\mathcal{D}_p$, as well as the set P , from [Definition 5.3](#). Let $t := \frac{C\|u^-\|_2^2}{\sqrt{d} \cdot \|u\|_2^2}$, where we let $C := 1/40$. Note that $0 < t < 1$. Letting $c_1 := 1/10$, we will be done if we can show that there exists $p \in P$ such that

$$\mathbb{P}_{v \sim \mathcal{D}_p} [u \cdot v < -\delta_d \|u^-\|_2] \stackrel{?}{\geq} c_1 t,$$

Suppose for a contradiction that this is not the case, i.e. that for every $p \in P$,

$$\mathbb{P}_{v \sim \mathcal{D}_p} [u \cdot v < -\delta_d \|u^-\|_2] < c_1 t.$$

For convenience of notation, let $a := u^-$ and $b := u^+$. Then for every $p \in P$, letting $z, w \sim \mathcal{D}_p$ independently, we conclude that $u \cdot v$ is distributed identically to $b \cdot w - a \cdot z$,

and hence

$$\mathbb{P}_{\mathbf{z}, \mathbf{w} \sim \mathcal{D}_p} [a \cdot \mathbf{z} > b \cdot \mathbf{w} + \delta_d \|a\|_2] < c_1 t.$$

Fix any $\tau \in (2\delta_d \|a\|_2, \|a\|_\infty)$, where the interval is nonempty since $\|a\|_2 \leq \|a\|_1 \leq d\|a\|_\infty$, so that $2\delta_d \|a\|_2 = \frac{2\|a\|_2}{100d} < \|a\|_\infty$. Define $a_\tau \in \mathbb{R}^d$ as the vector obtained from a by only preserving entries that are strictly larger than τ , and zeroing out entries that are at most τ . Since the dot product $a \cdot \mathbf{z}$ only gets smaller if we omit some of its summands, we conclude that for every $p \in P$,

$$\mathbb{P}_{\mathbf{z}, \mathbf{w} \sim \mathcal{D}_p} [a_\tau \cdot \mathbf{z} > b \cdot \mathbf{w} + \delta_d \|a\|_2] < c_1 t.$$

Let E denote the event that $\text{supp}(\mathbf{z}) \cap \text{supp}(a_\tau) \neq \emptyset$, so that $a_\tau \cdot \mathbf{z} > \tau$ when E occurs and $a_\tau \cdot \mathbf{z} = 0$ otherwise. Fix the smallest $p = p(\tau) \in P$ satisfying $p \geq \frac{t}{\|a_\tau\|_0}$, which must exist because $\|a_\tau\|_0 \geq 1$ by the choice of range for τ and $t \leq 1$ as observed above, and therefore $\frac{t}{\|a_\tau\|_0} \leq 1 \in P$. Then

$$\mathbb{P}_{\mathbf{z} \sim \mathcal{D}_{p(\tau)}} [E] = 1 - (1 - p(\tau))^{\|a_\tau\|_0} \geq 1 - e^{-p(\tau)\|a_\tau\|_0} \geq 1 - e^{-t} \geq \frac{t}{2},$$

the last inequality since $e^{-x} \leq 1 - x/2$ for (say) $0 \leq x \leq 1$. Therefore

$$\begin{aligned} c_1 t &> \mathbb{P}_{\mathbf{z}, \mathbf{w} \sim \mathcal{D}_{p(\tau)}} [a_\tau \cdot \mathbf{z} > b \cdot \mathbf{w} + \delta_d \|a\|_2] \\ &= \mathbb{P}_{\mathbf{z} \sim \mathcal{D}_{p(\tau)}} [E] \mathbb{P}_{\mathbf{z}, \mathbf{w} \sim \mathcal{D}_{p(\tau)}} [a_\tau \cdot \mathbf{z} > b \cdot \mathbf{w} + \delta_d \|a\|_2 \mid E] \\ &\quad + \mathbb{P}_{\mathbf{z} \sim \mathcal{D}_{p(\tau)}} [\neg E] \mathbb{P}_{\mathbf{z}, \mathbf{w} \sim \mathcal{D}_{p(\tau)}} [a_\tau \cdot \mathbf{z} > b \cdot \mathbf{w} + \delta_d \|a\|_2 \mid \neg E] \\ &\geq \frac{t}{2} \mathbb{P}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} [\tau > b \cdot \mathbf{w} + \delta_d \|a\|_2]. \end{aligned}$$

We conclude that

$$\mathbb{P}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} [b \cdot \mathbf{w} < \tau - \delta_d \|a\|_2] < 2c_1.$$

We now claim that $p(\tau) \leq \frac{2t}{\|a_\tau\|_0}$. Indeed if this were not the case, then the choice of $p(\tau)$ would imply that $p_{\min} := \min P$ satisfies $p_{\min} > \frac{2t}{\|a_\tau\|_0}$ with $p(\tau) = p_{\min} \leq \frac{1}{4d}$ and hence, since $\tau > 2\delta_d \|a\|_2$,

$$\begin{aligned} \mathbb{P}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} [b \cdot \mathbf{w} < \tau - \delta_d \|a\|_2] &\geq \mathbb{P}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} [b \cdot \mathbf{w} = 0] \geq \mathbb{P}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} [\mathbf{w} = 0] = (1 - p_{\min})^d \\ &\geq 1 - d \cdot p_{\min} \geq \frac{3}{4}, \end{aligned}$$

a contradiction. Thus $p(\tau) \leq \frac{2t}{\|a_\tau\|_0}$. Now, the definition of $\mathcal{D}_{p(\tau)}$, linearity of expectation and the Cauchy-Schwarz inequality yield

$$\mathbb{E}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} [b \cdot \mathbf{w}] = \sum_{i=1}^d \mathbb{P}[\mathbf{w}_i = 1] b_i = p(\tau) \cdot \|b\|_1 \leq p(\tau) \sqrt{d} \|b\|_2,$$

so by Markov's inequality,

$$\mathbb{P}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} \left[b \cdot \mathbf{w} \geq 10p(\tau) \sqrt{d} \|b\|_2 \right] \leq \frac{1}{10}.$$

Now, the union bound implies that

$$\mathbb{P}_{\mathbf{w} \sim \mathcal{D}_{p(\tau)}} \left[\tau - \delta_d \|a\|_2 \leq b \cdot \mathbf{w} < 10p(\tau) \sqrt{d} \|b\|_2 \right] \geq 1 - 2c_1 - \frac{1}{10} > 0,$$

so there exists $\mathbf{w}$ satisfying $\tau - \delta_d \|a\|_2 \leq b \cdot \mathbf{w} < 10p(\tau) \sqrt{d} \|b\|_2$, and hence $\tau - \delta_d \|a\|_2 < 10p(\tau) \sqrt{d} \|b\|_2$. Now, using the fact that $p(\tau) \leq \frac{2t}{\|a_\tau\|_0}$, recalling that $t = \frac{C\|u^-\|_2^2}{\sqrt{d} \cdot \|u\|_2^2}$, observing that $\|u\|_2^2 = \|u^-\|_2^2 + \|u^+\|_2^2$ while $b = u^+$ by definition, and using the inequality $2xy \leq x^2 + y^2$, we obtain

$$\tau - \delta_d \|a\|_2 < \frac{20\sqrt{d} \|b\|_2 t}{\|a_\tau\|_0} = \frac{20C\sqrt{d} \|u^+\|_2 \|u^-\|_2^2}{\|a_\tau\|_0 \cdot \sqrt{d} \cdot \|u\|_2^2} \leq \frac{10C \cdot \|u^-\|_2}{\|a_\tau\|_0}.$$

In summary, recalling that $a = u^-$ by definition, for all $\tau \in (2\delta_d \|a\|_2, \|a\|_\infty)$ we have

$$\frac{\tau}{2} \|a_\tau\|_0 \leq (\tau - \delta_d \|a\|_2) \|a_\tau\|_0 < 10C \|a\|_2,$$

so $\tau \|a_\tau\|_0 < 20C \|a\|_2$ for all $\tau \in (2\delta_d \|a\|_2, \|a\|_\infty)$. On the other hand, if $\tau \in (0, 2\delta_d \|a\|_2]$ then

$$\tau \|a_\tau\|_0 \leq 2\delta_d \|a\|_2 \cdot d = 2\delta \|a\|_2 = \frac{1}{50} \|a\|_2 < \frac{1}{2} \|a\|_2 = 20C \|a\|_2,$$

so in fact $\tau \|a_\tau\|_0 < 20C \|a\|_2$ for all $\tau \in (0, \|a\|_\infty)$. Integrating over τ and using Tonelli's theorem, we conclude that

$$\begin{aligned} \|a\|_\infty \cdot 20C \|a\|_2 &= \int_{(0, \|a\|_\infty)} 20C \|a\|_2 \, d\tau > \int_{(0, \|a\|_\infty)} \tau \|a_\tau\|_0 \, d\tau = \int_{(0, \|a\|_\infty)} \sum_{i=1}^d (\tau \cdot \mathbf{1}[a_i > \tau]) \, d\tau \\ &= \sum_{i=1}^d \int_{(0, \|a\|_\infty)} (\tau \cdot \mathbf{1}[a_i > \tau]) \, d\tau = \sum_{i=1}^d \int_{(0, a_i)} \tau \, d\tau = \sum_{i=1}^d \frac{a_i^2}{2} = \frac{\|a\|_2^2}{2}. \end{aligned}$$

Hence we obtain that $\|a\|_2 < 40C \cdot \|a\|_\infty = \|a\|_\infty$, a contradiction as desired. $\square$

On simulating directional derivative queries using value queries. We remark that the directional derivative queries, while convenient for the analysis and in our opinion conceptually clean, can be replaced with value queries under reasonable smoothness assumptions. First, in [Remark 5.6](#) we observed that the rejection condition $v \cdot \nabla f(x) < 0$ may in fact be replaced with the more robust condition $v \cdot \nabla f(x) < -\Theta(\varepsilon/d)$. Now, suppose the input function f is promised to be twice-differentiable with second derivatives bounded by any constant β . We may then replace any directional derivative query $v \cdot \nabla f(x)$ with the value queries $f(x)$ and $f(y)$, where $y = x + \alpha v$ for sufficiently small $\alpha > 0$ as a function of β, ε and d , and reject if $f(y) < f(x)$. This is possible because, if $v \cdot \nabla f(x) < -\Theta(\varepsilon/d)$ and for such small α , the directional derivative $v \cdot \nabla f(z)$ remains negative on all points z in the line segment between x and y , and hence $f(y) < f(x)$. Conceptually, we view this as confirmation that directional derivative queries are not unreasonably powerful, but rather a useful abstraction for the analytic setting.

5.2 Lower bound

Definition 5.7. A *derivative-pair tester* for monotonicity of Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ is described by a distribution $\mathcal{D}$ over *pair tests* and *directional derivative tests*, where

1. a pair test is an operation that performs a pair of value queries $f(x), f(y)$ for $x \preceq y$, and rejects if and only if $f(x) > f(y)$; and
2. a directional derivative test is an operation that performs a directional derivative query at point x and direction $v \succeq 0$, and rejects if and only if $\nabla f \cdot v < 0$.

The tester independently samples m tests from $\mathcal{D}$, rejects if any of these tests rejects, and accepts otherwise. By definition, every derivative-pair tester is nonadaptive and has one-sided error.

Theorem 5.8. *Every L^1 derivative-pair tester for monotonicity of M -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ must have query complexity $\Omega\left(\frac{\sqrt{d}M}{\varepsilon}\right)$.*

Proof. It suffices to give a distribution over $O(M)$ -Lipschitz functions $\mathbf{f}$ that are $\Omega(\varepsilon)$ -far from monotone in L^1 distance and such that any fixed pair test or directional derivative test has only an $O\left(\frac{\varepsilon}{\sqrt{d}M}\right)$ probability of rejecting a random $\mathbf{f}$. Moreover, to obtain the asymptotic lower bound we may as well assume that (say) $\varepsilon \leq 1 \leq M$.

For each $i \in [d]$, define the function $f_i : [0, 1]^d \rightarrow \mathbb{R}$ by

$$f_i(x) := -\varepsilon x_i + \sum_{j \in [d] \setminus \{i\}} \frac{M}{\sqrt{d}} x_j.$$

Then, define $\mathbf{f} := f_{\mathbf{i}}$ where $\mathbf{i}$ is sampled uniformly at random from $[d]$. Note that each f_i is linear and hence Lipschitz with Lipschitz constant

$$|\nabla f_i(x)| = \sqrt{\varepsilon^2 + (d-1) \left(\frac{M}{\sqrt{d}} \right)^2} \leq \sqrt{2}M = O(M).$$

We claim that $\text{dist}_1^{\text{mono}}(f_i) = \Omega(\varepsilon)$ for each $i \in [d]$; by symmetry it suffices to consider the case $i = d$. Any restriction $f_d(x_{-d}, \cdot)$ of f_d to an axis-aligned line in direction d (where $x_{-d} := (x_1, \dots, x_{d-1})$) is a linear function with slope $-\varepsilon$, so it is $\Omega(\varepsilon)$ -far from monotone in L^1 distance (in fact, $\varepsilon/4$ -far, and the constant average-valued function achieves this). But by Tonelli's theorem, the distance from f_d to any monotone function g is the average of the distance over each such axis-aligned line:

$$\int_{[0,1]^d} |f_d - g| dx = \int_{[0,1]^{d-1}} dx_{-d} \int_{[0,1]} |f_d(x_{-d}, x_d) - g(x_{-d}, x_d)| dx_d \geq \int_{[0,1]^{d-1}} \Omega(\varepsilon) dx_{-d} = \Omega(\varepsilon).$$

We now show that any pair test or directional derivative test rejects a random $\mathbf{f}$ only with probability $O\left(\frac{\varepsilon}{\sqrt{d}M}\right)$. Note that, since each f_i is a linear function, any pair test on points $x \preceq y$ may be simulated by a directional derivative test on point x and direction $v = y - x \succeq 0$. Therefore it suffices to consider directional derivative tests. Moreover, since the gradient of each f_i is constant over the points x , only the direction v is relevant. For simplicity, we will say that direction v *rejects* function f_i if a directional derivative test with direction v rejects f_i .

Fix any direction $v \succeq 0$. Suppose there exists $j \in [d]$ such that

$$\varepsilon v_j \leq \frac{M}{\sqrt{d}} \sum_{k \in [d] \setminus \{j\}} v_k.$$

We then claim that the vector v' given by $v'_j := 0$ and $v'_k := v_k$ for $k \in [d] \setminus \{j\}$ rejects every f_i that v rejects. Indeed, suppose v rejects f_i . There are two cases. If $i = j$, then

$$0 > \nabla f_i(x) \cdot v = -\varepsilon v_j + \sum_{k \in [d] \setminus \{j\}} \frac{M}{\sqrt{d}} v_k,$$

a contradiction, so this case cannot happen. On the other hand, if $i \neq j$, then

$$\nabla f(x) \cdot v' = -\varepsilon v'_i + \sum_{k \in [d] \setminus \{i\}} \frac{M}{\sqrt{d}} v'_k \leq -\varepsilon v_i + \sum_{k \in [d] \setminus \{i\}} \frac{M}{\sqrt{d}} v_k = \nabla f(x) \cdot v < 0,$$

the inequality since $v'_i = v_i$ and $v' \preceq v$. Hence v' also rejects f_i , as claimed. Thus v' rejects a random $\mathbf{f}$ with no less probability than v . Therefore we may assume without loss of generality that, for every $j \in [d]$, either $v_j = 0$ or

$$\varepsilon v_j > \frac{M}{\sqrt{d}} \sum_{k \in [d] \setminus \{j\}} v_k.$$

Let $j^* \in [d]$ be such that v_{j^*} is a minimum nonzero entry of v , and let $n := \|v\|_0$ be the number of nonzero entries of v . Applying the inequality above to $j = j^*$ yields

$$\varepsilon v_{j^*} > \frac{M}{\sqrt{d}} \sum_{k \in [d] \setminus \{j^*\}} v_k \geq \frac{M}{\sqrt{d}} \sum_{k \in [d] \setminus \{j^*\}} \mathbb{1}[v_k > 0] v_{j^*} = \frac{M}{\sqrt{d}} \cdot n v_{j^*},$$

and hence $n < \frac{\sqrt{d}\varepsilon}{M}$. Finally, note that v does not reject f_i if $v_i = 0$, so we conclude that

$$\mathbb{P}_{\mathbf{f}}[v \text{ rejects } \mathbf{f}] = \mathbb{P}_{\mathbf{i}}[v \text{ rejects } f_i] \leq \mathbb{P}_{\mathbf{i}}[v_i > 0] = \frac{n}{d} \leq \frac{\sqrt{d}\varepsilon/M}{d} = \frac{\varepsilon}{\sqrt{d}M},$$

as desired. This concludes the proof. $\square$

Remark 5.9. Since every L^p tester with $p \geq 1$ is also an L^1 tester (see the remark after [Theorem 5.2](#)), the lower bound above holds for L^p testers for all $p \geq 1$.

Interlude

Chapter 6

Directed isoperimetry on graphs via the directed heat flow

In this chapter, we start working towards a proof of a directed (L^2, ℓ^2) -Poincaré inequality for the solid cube $[0, 1]^d$ ([Theorem 5.1](#)) by first showing such an inequality in the discrete setting of functions $f : \{0, 1\}^d \rightarrow \mathbb{R}$ over the Boolean cube. By taking this intermediate step, we untangle two aspects of our eventual proof of [Theorem 5.1](#):

1. A dynamical argument: instead of reasoning about $\text{dist}_2^{\text{mono}}(f)$ as a “static” comparison between the object f and some monotone function f^* , we instead inspect a *time-continuous* process which smoothly takes initial state f to equilibrium state f^* . This idea can already be implemented in the case of functions $f : \{0, 1\}^d \rightarrow \mathbb{R}$.
2. The challenges of continuous space: when we implement this idea in the analytic setting of functions $f : [0, 1]^d \rightarrow \mathbb{R}$, we will need to handle many complications of real- and functional-analytic flavour, such as ensuring regularity and integrability of solutions to partial differential equations. In contrast, the discrete study relies on the more well-behaved theory of finite-dimensional vector spaces, and thus makes for a more appropriate starting point.

It turns out that the most natural way to show an (L^2, ℓ^2) inequality for the Boolean cube is to study the more general setting of *directed graphs* instead. In that setting, classical Poincaré inequalities are intimately connected to the field of *spectral graph theory*, and our argument will proceed by revisiting and extending a spectral theory for directed graphs initiated by [\[Yos16\]](#) in the context of network analysis, via a dynamical system framework introduced by work on the *Paulsen problem* from operator theory [\[KLLR18\]](#).

First, let us briefly recall some elements of spectral graph theory and their connection to Poincaré inequalities. While our proof is mostly self-contained, we refer the reader to [Han14; LP17; Lau25] for a thorough introduction to these ideas.

Let $G = (V, E)$ be an unweighted, undirected graph, with $V = [n]$. One of the central objects in spectral graph theory is the *Laplacian operator* of G , which is the $n \times n$ matrix L given as follows. Let A be the adjacency matrix of G , and let D be the diagonal matrix such that D_i is the degree of vertex i for each $1 \leq i \leq n$. Then the Laplacian operator is $L := D - A$. This matrix is symmetric and real-valued, and therefore enjoys a spectral decomposition; we write $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ for its eigenvalues.

Spectral graph theory shows that the spectrum of L is remarkably effective in capturing the *connectivity properties* of G , roughly speaking. For example, the smallest eigenvalue λ_1 is always equal to zero, and the second eigenvalue λ_2 is nonzero if and only if G is connected. Graph partitioning algorithms work by putting the vertices of G into clusters according to an ordering defined by the eigenvector corresponding to the second eigenvalue λ_2 . This eigenvalue is also related to properties of G such as vertex and edge expansion, which are combinatorial measures of connectivity of G , via so-called *Cheeger inequalities*. Graphs with large λ_2 are *expander graphs*, which are graphs that behave in many ways like random graphs and have far-reaching connections in theoretical computer science, spanning areas such as error-correcting codes, pseudorandomness, probabilistically checkable proofs, and Markov chain theory. Moreover, the action Lf of the Laplacian operator also induces a dynamical process called the *heat flow*, which will be a crucial element of our proof.

The relevance of the Laplacian operator to our goals is revealed by the quadratic form on L : for each $f \in \mathbb{R}^n$, we have

$$f^\top Lf = \sum_{xy \in E} (f(x) - f(y))^2. \quad (6.1)$$

Note that the right-hand side above, which sums up the variation of f along each edge of G , is an unnormalized version of the right-hand side $\mathbb{E} [\|\nabla f\|_2^2]$ of a classical (L^2, ℓ^2) -Poincaré inequality when G is the graph of the hypercube, with an edge between $x, y \in \{0, 1\}^d$ if and only if $\|x - y\|_1 = 1$. Indeed, the Laplacian operator is intimately connected to the Poincaré inequality on G , and this connection provides the starting point for our proof in the next section.

The rest of this chapter is organized as follows. In [Section 6.1](#), we study a spectral theory for directed graphs and prove a general directed Poincaré inequality for any weighted directed graph G , mediated by a quantity called the *dynamical spectral gap* of G . In [Section 6.2](#), we compute the dynamical spectral gap of the hypercube graph H_n , which we

use to derive a directed (L^2, ℓ^2) -Poincaré inequality for functions $f : \{0, 1\}^d \rightarrow \mathbb{R}$. Finally, in [Section 6.3](#) we show a family of *reverse* Poincaré inequalities, showing in particular that the directed (L^2, ℓ^2) -Poincaré inequality is tight up to a factor of d , and thus elucidating the quality of the local-to-global relationship embodied by the directed Poincaré inequality.

6.1 Spectral theory and heat flow for directed graphs

We start by summarizing existing work on spectral theory for directed graphs, and previewing how we will employ and extend this theory to give a dynamical proof of a directed Poincaré inequality for any weighted directed graph G .

Prior work on spectral theory for directed graphs. There has been extensive prior work developing spectral graph theory beyond the classical setting of undirected graphs, toward capturing directed graphs and hypergraphs. Early work of [\[Fil91; Chu05\]](#) associated a certain Hermitian matrix with each directed graph, and showed a Cheeger-type inequality based on the eigenvalues of that matrix, and many subsequent works have built upon that foundation; we refer to the recent thesis [\[Tun25\]](#) for a thorough review, and here we mention two recent lines of work that are closest to our setting. One line of works [\[LTW23; LTW24; Tun25\]](#) has developed a theory of *reweighted eigenvalues* capturing expansion properties of directed graphs and hypergraphs, proved Cheeger inequalities for these settings, and devised efficient algorithms for graph partitioning. Another line of works [\[Yos16; Yos19; FSY21; IMTY22\]](#) has pursued similar goals by analyzing a *nonlinear Laplacian* operator and the heat equation associated with it.

While our interest in a spectral theory for directed graphs is related to these previous works (and indeed we will build upon the approach of [\[Yos16\]](#)), our focus is slightly different. In a nutshell, while prior works have focused on spectral characterizations of good expansion of a directed graph G as captured by directed versions of Cheeger inequalities (for edge conductance or vertex expansion) and mixing time of random walks, our focus will be on the quality of G as the “substrate” for a dynamical process; we will consider G a good directed spectral expander if it affords fast convergence for that process. In particular, our perspective allows for *directed acyclic graphs* to be considered good expanders, which is a stark departure from prior perspectives – as we briefly explain next.

Indeed, a central focus of prior works has been to establish Cheeger inequalities of the type

$$\bar{\lambda}_2 \lesssim \bar{\phi}(G) \lesssim \sqrt{\bar{\lambda}_2}, \quad (6.2)$$

where $\vec{\lambda}_2$ denotes a relevant second eigenvalue related to the directed weighted graph G , and the edge conductance $\vec{\phi}(G)$ of G is

$$\vec{\phi}(G) := \min_{\emptyset \neq S \subsetneq V} \frac{\min \{w(\delta^+(S)), w(\delta^+(V \setminus S))\}}{\min \{\text{vol}_w(S), \text{vol}_w(V \setminus S)\}},$$

where $\delta^+(S)$ denotes the outgoing edge boundary of S and $\text{vol}_w(S)$ denotes the total weighted degree of all vertices in S . Now, if G is not strongly connected, then in general there exists a set S with positive volume but no outgoing edges, which makes $\vec{\phi}(G)$ and thus $\vec{\lambda}_2$ zero. In particular, this is the case for the directed hypercube graph which we are interested in, so if we hope to show a non-trivial directed Poincaré inequality via a spectral argument, such a quantity $\vec{\lambda}_2$ will not do.

Our approach. The type of spectral theory for directed graphs we study in this section was first developed by [Yos16] in the context of network analysis. In that work, [Yos16] defined a nonlinear Laplacian operator acting on real-valued functions defined on the vertices of a directed graph, showed that this operator induces a dynamical process that is a directed analogue of the classical heat flow on graphs, proved that this operator has non-trivial eigenvalues, and established a Cheeger inequality like (6.2) for this setting. While [Yos16] focused on the implications of directed spectral theory for graph partitioning and related problems in network analysis, we focus on the dynamical properties of the heat flow on directed graphs – namely its convergence to a *monotone* limit, and its connection to the directed Poincaré inequality.

Let us briefly motivate the main ideas in our argument by elaborating on the relevance of spectral graph theory to Poincaré inequalities, as previewed in the introduction to this chapter. Given a graph $G = (V, E)$, the following four concepts play a central role:

1. The Laplacian operator¹ $\mathcal{L}$, which acts on function f by outputting another function $\mathcal{L}f$.
2. The Dirichlet energy functional $\mathcal{E}$, which associates with each f an energy $\mathcal{E}(f) \geq 0$ measuring the “local variance” of f along edges of G . (This is essentially the quadratic form shown in (6.1), appropriately normalized by $1/|V|$ and possibly the edge weights of G .)

¹We now write $\mathcal{L}$ as opposed to L because, going forward, we will see $\mathcal{L}$ as an operator acting on a normalized inner product space, as opposed to a matrix acting on unnormalized Euclidean space. This will be made formal in Section 6.1.1.

3. The heat flow semigroup $\tilde{P}_t$, which captures a dynamical process which starts at some initial state f and has its rate of change governed by the Laplacian: $\frac{d}{dt}\tilde{P}_t f = \mathcal{L}\tilde{P}_t f$. The heat flow informally “sends mass” along each edge of G from the vertex with higher f -value to the vertex with lower f -value, causing the system to converge to an equilibrium state.
4. The Poincaré inequality, which states that $\text{Var}[f] \leq \frac{1}{\lambda} \mathcal{E}(f)$. The best constant λ is called the *spectral gap* of G .

The appearance of the Poincaré inequality above hints at the relevance of this theory to our goal of proving a directed Poincaré inequality, and we mentioned above that our strategy toward this goal will be to define a directed version of the spectral gap. As a motivation for this strategy, we observe that the spectral gap ties together essentially all of the elements of the list above; indeed, as summarized in [Han14, Theorem 2.18], the following are equivalent given a constant $c \geq 0$:

1. Poincaré inequality: $\text{Var}[f] \leq c\mathcal{E}(f)$ for all f .
2. Variance decay: $\text{Var}[\tilde{P}_t f] \leq e^{-2t/c} \text{Var}[f]$ for all f, t .
3. Energy decay: $\mathcal{E}(\tilde{P}_t f) \leq e^{-2t/c} \mathcal{E}(f)$ for all f, t .

To prove a directed Poincaré inequality, we replace the Laplacian operator $\mathcal{L}$ with an operator $\mathcal{L}^-$ which, intuitively, only “sends mass” from vertex u to vertex v if (u, v) is a directed edge and $f(u) > f(v)$, i.e. f violates monotonicity along edge (u, v) ; the dynamical system induced by $\mathcal{L}^-$ is the *directed heat flow* on G , which was studied by [Yos16]. The directed heat flow is precisely the *gradient system* for the directed energy functional $\mathcal{E}^-(f)$, which is the directed version of $\mathcal{E}(f)$ that only counts those edges along which f violates monotonicity. Thus, this system intuitively “corrects” the local violations of monotonicity as quickly as possible, and indeed it converges to a monotone function as $t \rightarrow \infty$.

This directed theory cannot fully analogize the classical situation above; for example, variance decay fails to hold, because non-constant monotone functions are (non-unique!) stationary solutions. Instead, we will *define* the *dynamical spectral gap* of G as the best constant characterizing the (directed) energy decay, as in [Item 3](#) above, and then show that (the directed version of) [Item 3](#) implies (the directed version of) [Item 1](#).

Organization. The rest of this section is organized as follows. [Sections 6.1.1 to 6.1.3](#) present the directed versions of the Laplacian operator, the Dirichlet energy functional, and the heat flow, respectively. These subsections are largely an alternative exposition of the ideas covered in [\[Yos16\]](#), but with a different emphasis tailored to our goals². Then, in [Section 6.1.4](#) we define the dynamical spectral gap of a directed weighted graph, and show via an energy decay argument that every directed weighted graph admits a directed Poincaré inequality mediated by the dynamical spectral gap.

Notation. Given a discrete set V , we denote by $L^2(V)$ the Hilbert space obtained by endowing the set of $V \rightarrow \mathbb{R}$ functions with the inner product $\langle f, g \rangle := \mathbb{E}_{x \in V} [f(x)g(x)]$, where the expectation is taken with respect to the uniform distribution over V . This inner product induces the norm $\|f\| = \sqrt{\langle f, f \rangle} = \sqrt{\mathbb{E}[f^2]}$.

For each $u \in V$, we write $e_u \in L^2(V)$ for the standard basis vector given by $e_u(v) := \mathbb{1}[u = v]$.

6.1.1 The directed Laplacian

Let $G = (V, w)$ be a directed weighted graph, where $w : V \times V \rightarrow [0, +\infty)$ is function specifying the weights of edges in G . By convention, we say that $(u, v) \in V \times V$ is an edge in G when $w(u, v) > 0$. We say G is *undirected* if w is a symmetric function.

Definition 6.1 (Directed Laplacian [\[Yos16\]](#)). The *directed Laplacian operator* of G is the operator $\mathcal{L}^- = \mathcal{L}_G^- : L^2(V) \rightarrow L^2(V)$ given by

$$\mathcal{L}^- f := \frac{1}{2} \sum_{u, v \in V} w(u, v) (f(u) - f(v))^+ (e_v - e_u) \quad (6.3)$$

for each $f \in L^2(V)$.

Given $f \in L^2(V)$, we say an edge (u, v) is *f-monotone* if $f(u) \leq f(v)$, and we say it is *f-antimonotone* if $f(u) > f(v)$. We say f is *monotone* if every edge (u, v) of G is *f-monotone*. If we think of f as the distribution of “mass” over the vertices V and of $\mathcal{L}^- \mathbf{f}$ as the rate of change of $\mathbf{f} = \mathbf{f}(t)$ over time t , then [Definition 6.1](#) posits that mass flows along the *f-antimonotone* edges, from the heavier vertex to the lighter one. When G is

²In particular, [\[Yos16\]](#) defined both normalized and unnormalized versions of their Laplacian operator, and focused on the normalized one. We study a single definition for weighted graphs.

undirected, this process is the standard heat flow on G , and indeed [Definition 6.1](#) recovers the standard graph Laplacian in this case:

Observation 6.2. *If G is undirected, then $\mathcal{L}_G^-$ is (half of) the standard (unnormalized) Laplacian operator $\mathcal{L}_G$ of G . Indeed, we can see the action of $\mathcal{L}_G^-$ on $f \in L^2(V)$ as follows: 1) remove the f -monotone edges (u, v) from G ; 2) view the resulting graph G' as undirected; and 3) apply the standard Laplacian operator $\mathcal{L}_{G'}$ to f .*

Remark 6.3. In spectral graph theory one typically defines the Laplacian operator of an undirected graph as $-\mathcal{L}_G$ in our notation, i.e. by replacing $e_v - e_u$ with $e_u - e_v$ in [Definition 6.1](#) (which yields the matrix L from the introduction to this chapter). Our notation follows instead the tradition from probability theory (see e.g. [\[BGL14; Han14\]](#)), which has the advantages 1) that $\mathcal{L}^-$ itself, rather than $-\mathcal{L}^-$, will be the generator of the heat semigroup – our main object of interest; and 2) of consistency with the analytic setting, where we have the Laplacian operator Δ for smooth functions in Euclidean space.

We note that unlike the standard Laplacian operator, the operator $\mathcal{L}^-$ is nonlinear and not self-adjoint in general. Instead, one may think of $\mathcal{L}^-$ as a “piecewise linear” operator on $L^2(V)$, which is in particular Lipschitz continuous.

Lemma 6.4. *The operator $\mathcal{L}^- : L^2(V) \rightarrow L^2(V)$ is Lipschitz continuous.*

Proof. Since all norms are equivalent in a finite-dimensional space, it suffices to let $f, g \in L^2(V)$ differ on a single point $z \in V$ and show that, for all $u \in V$, $|(\mathcal{L}^- f)(u) - (\mathcal{L}^- g)(u)| \leq M|f(z) - g(z)|$ for some constant $M > 0$, which may depend on the graph G but not on f or g .

By [Definition 6.1](#), we have

$$(\mathcal{L}^- f)(u) = \frac{1}{2} \sum_{v \in V} [-w(u, v) (f(u) - f(v))^+ + w(v, u) (f(v) - f(u))^+]$$

and

$$(\mathcal{L}^- g)(u) = \frac{1}{2} \sum_{v \in V} [-w(u, v) (g(u) - g(v))^+ + w(v, u) (g(v) - g(u))^+].$$

By the triangle inequality and the assumption that f and g differ only on coordinate z ,

$$\begin{aligned}
|(\mathcal{L}^- f)(u) - (\mathcal{L}^- g)(u)| &\leq \frac{1}{2} \sum_{v, v' \in V: z \in \{v, v'\}} w(v, v') \left| (f(v) - f(v'))^+ - (g(v) - g(v'))^+ \right| \\
&\leq \frac{1}{2} \sum_{v, v' \in V: z \in \{v, v'\}} w(v, v') |(f(v) - f(v')) - (g(v) - g(v'))| \\
&\leq \frac{1}{2} |f(z) - g(z)| \sum_{v, v' \in V} w(v, v'). \quad \square
\end{aligned}$$

6.1.2 The energy functional

As in the case of the standard Laplacian operator, the directed Laplacian naturally induces an *energy functional* (or Dirichlet form). The following definition corresponds to the Rayleigh quotient defined in [Yos16].

Definition 6.5 (Energy functional). The *directed Dirichlet energy functional* $\mathcal{E}^- : L^2(V) \rightarrow \mathbb{R}$ is given by

$$\mathcal{E}^-(f) := -\langle f, \mathcal{L}^- f \rangle.$$

The directed Dirichlet energy measures the local violations of monotonicity along edges of G , and it is indeed always non-negative, as shown in the following proposition (which is similar to Lemma 4.3 of [Yos16] for the normalized nonlinear Laplacian).

Proposition 6.6 (Energy functional measures local violations). *For each $f \in L^2(V)$, it holds that*

$$\mathcal{E}^-(f) = \frac{1}{2} \mathbb{E}_{u \in V} \left[\sum_{v \in V} w(u, v) ((f(u) - f(v))^+)^2 \right].$$

Proof. We have

$$\begin{aligned}
\mathcal{E}^-(f) &= -\langle f, \mathcal{L}^- f \rangle = -\left\langle f, \frac{1}{2} \sum_{u', v \in V} w(u', v) (f(u') - f(v))^+ (e_v - e_{u'}) \right\rangle \\
&= -\frac{1}{2} \mathbb{E}_{u \in V} \left[f(u) \left(\sum_{u' \in V} w(u', u) (f(u') - f(u))^+ - \sum_{v \in V} w(u, v) (f(u) - f(v))^+ \right) \right] \\
&= -\frac{1}{2} \mathbb{E}_{u \in V} \left[f(u) \sum_{v \in V} \left(w(v, u) (f(v) - f(u))^+ - w(u, v) (f(u) - f(v))^+ \right) \right] \\
&= \frac{1}{2|V|} \sum_{u, v \in V} w(u, v) f(u) (f(u) - f(v))^+ - \frac{1}{2|V|} \sum_{u, v \in V} w(v, u) f(u) (f(v) - f(u))^+ \\
&= \frac{1}{2} \mathbb{E}_{u \in V} \left[\sum_{v \in V} w(u, v) ((f(u) - f(v))^+)^2 \right]. \quad \square
\end{aligned}$$

Observe that since $\mathcal{L}^-$ is “piecewise linear”, the energy functional $\mathcal{E}^-$ is a “piecewise quadratic” functional on $L^2(V)$. Furthermore, at each point $f \in L^2(V)$ the Laplacian $\mathcal{L}^- f$ points at the direction opposite to the gradient of $\mathcal{E}^-$. To formalize this viewpoint, we first need to introduce some standard definitions to clarify what we mean by the gradient of a functional on $L^2(V)$, such as $\mathcal{E}^-$. We recall the following standard definition.

Definition 6.7 (Fréchet derivative and gradient). We say that $F : L^2(V) \rightarrow \mathbb{R}$ is *Fréchet differentiable* at $f \in L^2(V)$ if there exists a bounded linear operator $A : L^2(V) \rightarrow \mathbb{R}$ such that

$$\lim_{\|h\|_2 \rightarrow 0} \frac{|F(f+h) - F(f) - Ah|}{\|h\|_2} = 0.$$

In this case, we call $DF(f) := A$ the *Fréchet derivative* of F at f . Moreover, by the Riesz representation theorem there exists a unique vector $v \in L^2(V)$ satisfying

$$Ah = \langle h, v \rangle \quad (6.4)$$

for all $h \in L^2(V)$. We call $\nabla F(f) := v$ the *gradient* of F at f . Finally, we say that F is (Fréchet) C^1 if it is Fréchet differentiable at every $f \in L^2(V)$ and the map $f \mapsto DF(f)$ (from $L^2(V)$ to the space of bounded linear operators $L^2(V) \rightarrow \mathbb{R}$) is continuous.

The following proposition follows from standard real analysis.

Proposition 6.8. Suppose $F : L^2(V) \rightarrow \mathbb{R}$ has continuous partial derivatives with respect to the standard basis $\{e_u\}_{u \in V}$, that is, for each $u \in V$ the $L^2(V) \rightarrow \mathbb{R}$ function

$$\partial_u F(f) = \lim_{t \rightarrow 0} \frac{F(f + te_u) - F(f)}{t}$$

is continuous. Then F is (Fréchet) C^1 and its gradient $\nabla F : L^2(V) \rightarrow L^2(V)$ is given by

$$(\nabla F(f))(u) = |V| \cdot \partial_u F(f)$$

for each $f \in L^2(V)$ and $u \in V$.

We highlight that since the inner product on $L^2(V)$ used in (6.4) differs from the standard inner product on the finite dimensional Euclidean space $\mathbb{R}^V$ by a factor of $|V|$, the gradient of F also differs from the partial derivatives of F by a factor of $|V|$.

We are now equipped to show that $\mathcal{E}^-$ is a convex C^1 functional, and that the Laplacian $\mathcal{L}^-$ is (half of) the negative gradient of $\mathcal{E}^-$.

Lemma 6.9. The functional $\mathcal{E}^- : L^2(V) \rightarrow [0, +\infty)$ is C^1 and convex.

Proof. We define a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$g(x, y) := \frac{1}{2} ((x - y)^+)^2. \quad (6.5)$$

Class C^1 . By Proposition 6.6, it suffices to show that for each $u, v \in V$, the $L^2(V) \rightarrow \mathbb{R}$ function $f \mapsto ((f(u) - f(v))^+)^2$ is C^1 . By Proposition 6.8, it suffices to show that this function has continuous partial derivatives, which is equivalent to showing that g has continuous partial derivatives. This is indeed the case, with $\partial_1 g(x, y) = (x - y)^+$ and $\partial_2 g(x, y) = -(x - y)^+$.

Convexity. By Proposition 6.6 and recalling that the weights $w(u, v)$ are non-negative, it again suffices to show that the function g is convex. Indeed, let $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ and $\lambda \in [0, 1]$. Then, using the convexity of the functions $z \mapsto z^+$ and $z \mapsto z^2$ along with the fact that the latter function is increasing for $z \geq 0$, we have

$$\begin{aligned} g(\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2)) &= \frac{1}{2} [((\lambda x_1 + (1 - \lambda)x_2) - (\lambda y_1 + (1 - \lambda)y_2))^+]^2 \\ &= \frac{1}{2} [(\lambda(x_1 - y_1) + (1 - \lambda)(x_2 - y_2))^+]^2 \leq \frac{1}{2} [\lambda(x_1 - y_1)^+ + (1 - \lambda)(x_2 - y_2)^+]^2 \\ &\leq \lambda g(x_1, y_1) + (1 - \lambda)g(x_2, y_2). \end{aligned} \quad \square$$

Lemma 6.10. *For any $f \in L^2(V)$, we have $\mathcal{L}^- f = -\frac{1}{2} \nabla \mathcal{E}^-(f)$.*

Proof. Recall that the function g in (6.5) has partial derivatives $\partial_1 g(x, y) = (x - y)^+$ and $\partial_2 g(x, y) = -(x - y)^+$. Now, by Proposition 6.6, for each $u \in V$, the partial derivative $\partial_u \mathcal{E}^-(f)$ (in the notation of Proposition 6.8) is

$$\begin{aligned} \partial_u \mathcal{E}^-(f) &= \frac{1}{|V|} \sum_{v \in V} [w(u, v) \partial_1 g(f(u), f(v)) + w(v, u) \partial_2 g(f(v), f(u))] \\ &= \frac{1}{|V|} \sum_{v \in V} [w(u, v) (f(u) - f(v))^+ - w(v, u) (f(v) - f(u))^+] . \end{aligned}$$

Thus, by Proposition 6.8, the gradient $\nabla \mathcal{E}^-(f)$ (which exists by Lemma 6.9) is given by

$$(\nabla \mathcal{E}^-(f))(u) = |V| \cdot \partial_u \mathcal{E}^-(f) = \sum_{v \in V} [w(u, v) (f(u) - f(v))^+ - w(v, u) (f(v) - f(u))^+]$$

for each $u \in V$. On the other hand, by Definition 6.1,

$$(\mathcal{L}^- f)(u) = \frac{1}{2} \sum_{v \in V} [-w(u, v) (f(u) - f(v))^+ + w(v, u) (f(v) - f(u))^+]$$

for each $u \in V$, so $(\mathcal{L}^- f)(u) = -\frac{1}{2} (\nabla \mathcal{E}^-(f))(u)$ as desired. $\square$

6.1.3 Directed heat flow

As previewed in the Section 6.1.1, the directed Laplacian operator can be thought of as the rate of mass transfer along f -antimonotone edges of G in a dynamical process. Let us make this notion precise.

Given any $f \in L^2(V)$, we define the *directed heat flow* on G with *initial state* f as the dynamical system given by the initial value problem (IVP)

$$\mathbf{f}'(t) = \mathcal{L}^- \mathbf{f}(t) \quad \text{for all } t \geq 0, \quad \mathbf{f}(0) = f. \quad (6.6)$$

Since the operator $\mathcal{L}^-$ is Lipschitz by Lemma 6.4, a standard existence and uniqueness theorem for ordinary differential equations (ODEs) implies that this IVP has a unique solution $\mathbf{f} : [0, +\infty) \rightarrow L^2(V)$; see e.g. [Tes12, Corollary 2.6]. This can also be shown using the theory of maximal monotone operators, as done by [IMTY22] for the heat flow

on hypergraphs, and by [Yos19] in a study that generalizes both the directed graph setting of [Yos16] and the hypergraph setting of [IMTY22].

Moreover, the directed heat flow enjoys the following semigroup structure. Define the operator family $(P_t)_{t \geq 0}$, with $P_t : L^2(V) \rightarrow L^2(V)$ for each $t \geq 0$, as follows: for each $f \in L^2(V)$, let $\mathbf{f} : [0, +\infty) \rightarrow L^2(V)$ be the solution to the IVP (6.6), and let

$$P_t f := \mathbf{f}(t).$$

Then it immediately follows that P_t satisfies the properties of a semigroup, namely

1. P_0 is the identity operator.
2. $P_s P_t = P_{s+t}$ for all $s, t \geq 0$.
3. $\lim_{t \rightarrow 0} P_t f = f$ for all $f \in L^2(V)$ (this follows from the differentiability of the solution $\mathbf{f}(t)$).

We call P_t the *directed heat semigroup* operator. Note that P_t is a nonlinear operator.

Observation 6.11 (Monotone functions are stationary solutions). *If $f \in L^2(V)$ is monotone, then $\mathcal{L}^- f = 0$ and hence $P_t f = f$ for all $t \geq 0$.*

Since monotone functions are stationary solutions to the directed heat flow (Observation 6.11), while any non-monotone f has non-zero Laplacian $\mathcal{L}^- f$ (see Proposition 6.6), it is natural to expect that $P_t f$ always converges to a monotone function as $t \rightarrow \infty$. This is indeed the case, because the directed heat flow is a *gradient system* for the *convex* energy functional $\mathcal{E}^-$.

Proposition 6.12 (Directed heat flow is gradient system). *For all $f \in L^2(V)$ and $t \geq 0$, we have*

$$\frac{d}{dt} P_t f = -\frac{1}{2} \nabla \mathcal{E}^-(P_t f).$$

Proof. By the definition of P_t we have $\frac{d}{dt} P_t f = \mathcal{L}^- P_t f$. The claim then follows from Lemma 6.10. $\square$

Corollary 6.13 (Convergence to monotone equilibrium). *For every $f \in L^2(V)$, there exists a (unique) monotone $f^* \in L^2(V)$ such that $P_t f \rightarrow f^*$ as $t \rightarrow \infty$.*

Proof. It is a standard fact that since $\mathcal{E}^-$ is a convex differentiable function which attains its minimum, its gradient system (which is given by P_t) converges to a minimizer f^* ; see e.g. [AC84, Chapter 3, Theorem 2]. By Proposition 6.6, such a minimizer is monotone. $\square$

In light of Corollary 6.13, we may define the following limit operator.

Definition 6.14 (Monotone equilibrium). We define the operator $P_\infty : L^2(V) \rightarrow L^2(V)$ by $P_\infty f := \lim_{t \rightarrow \infty} P_t f$ for each $f \in L^2(V)$, and call $P_\infty f$ the *monotone equilibrium* of f .

Remark 6.15. Since $P_\infty f$ is monotone, it is constant in each strongly connected component of G .

6.1.4 Dynamical spectral gap

In this subsection, we associate with the directed Laplacian operator $\mathcal{L}^- = \mathcal{L}_G^-$ a quantity $\lambda^- = \lambda^-(G)$, the *dynamical spectral gap* of G , as a natural directed generalization of the classical (undirected) case. In particular, $\lambda^-(G)$ characterizes the rate of energy decay in the directed heat flow as $P_t f$ converges to its monotone equilibrium, and this also implies a directed Poincaré inequality linking the distance between the initial and equilibrium states to the energy of the initial state (i.e. its violations of monotonicity).

In the classical theory, the spectral gap $\lambda(G)$ associated with the classical Laplacian operator $\mathcal{L}_G$ is

$$\lambda(G) = \inf_{f \neq 0: \mathbb{E}[f]=0} \frac{-\langle f, \mathcal{L}_G f \rangle}{\|f\|_2^2}, \quad (6.7)$$

which is the best constant for which the Poincaré inequality

$$\text{Var}[f] \leq \frac{1}{\lambda(G)} \mathcal{E}(f)$$

holds, where the classical Dirichlet energy is $\mathcal{E}(f) = -\langle f, \mathcal{L}_G f \rangle$. As mentioned at the beginning of this section, the spectral gap also characterizes the rate of exponential decay of the variance toward its minimum value under the heat flow. That is, $\lambda(G)$ is the best constant for which

$$\frac{d}{dt} \text{Var}[\tilde{P}_t f] \leq -2\lambda(G) \text{Var}[\tilde{P}_t f] \quad (6.8)$$

holds for all $f \in L^2(V)$ and $t \geq 0$, where $\tilde{P}_t$ is the undirected heat flow. It also similarly characterizes the exponential decay of the Dirichlet energy, i.e. $\lambda(G)$ is the best constant

for which

$$\frac{d}{dt}\mathcal{E}(\tilde{P}_t f) \leq -2\lambda(G)\mathcal{E}(\tilde{P}_t f) \quad (6.9)$$

holds for all $f \in L^2(V)$ and $t \geq 0$. We refer the reader to [Han14] for a comprehensive reference.

We wish to define a natural notion of spectral gap for the directed Laplacian $\mathcal{L}^-$, towards the goal of obtaining a directed Poincaré inequality. One attempt is to take the directed analogue of the ratio in (6.7), namely

$$\frac{\mathcal{E}^-(f)}{\|f\|_2^2} = \frac{-\langle f, \mathcal{L}^- f \rangle}{\|f\|_2^2}. \quad (6.10)$$

(This would essentially be the unnormalized version of the spectral gap defined by [Yos16] for the normalized nonlinear Laplacian.) However, this quantity faces the issue that any monotone function f has $\mathcal{E}^-(f) = 0$, which makes the ratio zero.³ Even if we restricted our attention to non-monotone f , in general $\|f\|_2^2$ could still be made arbitrarily large while keeping $\mathcal{E}^-(f) > 0$ bounded, by putting large values of $f(u)$ that do not introduce violations of monotonicity. Therefore (6.10) does not seem like an informative quantity for our purposes.

Another way to see the conceptual issue is that, unlike in the classical case, in the directed setting we cannot hope to have variance decay as in (6.8), since the monotone equilibrium $P_\infty f$ may have strictly positive variance. However, as our next attempt will reveal, we *can* establish the decay of Dirichlet energy as in (6.9), which will suffice to give a directed Poincaré inequality for G .

We start by observing that, in the classical case, the ratio

$$\inf_{f:\mathcal{E}(f)>0} \frac{\|\mathcal{L} f\|_2^2}{\mathcal{E}(f)} \quad (6.11)$$

equals the smallest *non-zero* eigenvalue of the classical Laplacian $-\mathcal{L}_G$, so for a *connected* graph it also equals the spectral gap $\lambda(G)$. Now, its directed analogue

$$\inf_{f:\mathcal{E}^-(f)>0} \frac{\|\mathcal{L}^- f\|_2^2}{\mathcal{E}^-(f)} \quad (6.12)$$

seems like a reasonable candidate – intuitively, note that not only does the constraint $\mathcal{E}^-(f) > 0$ rule out monotone functions f , but also, f -monotone edges (u, v) with large

³While for strongly connected directed graphs only constant functions are monotone, which renders the objection void, this is not the case for general directed graphs.

$f(v) - f(u)$ contribute nothing to the denominator, suggesting that (6.12) does not have the same shortcoming as (6.10)⁴. Another way to see the intuition behind (6.12) is that, while its classical counterpart (6.11) only differs from (6.7) by ignoring the zero eigenvalues coming from the single pathological case of disconnected graphs, the directed (6.12) ignores many more uninformative situations and captures the connectivity of G in the appropriate, directed sense.

Therefore, we define

Definition 6.16. The *dynamical spectral gap* of G is the quantity $\lambda^-(G) \in [0, +\infty]$ given by

$$\lambda^-(G) := \inf \left\{ \frac{\|\mathcal{L}^- f\|_2^2}{\mathcal{E}^-(f)} \mid f \in L^2(V), \mathcal{E}^-(f) > 0 \right\}.$$

The next proposition shows that every graph has a non-zero dynamical spectral gap.

Proposition 6.17. *The dynamical spectral gap $\lambda^-(G)$ is strictly positive, and it is a real number unless G consists of isolated vertices only.*

Proof. If G contains at least one edge $e = (u, v)$, then any function $f \in L^2(V)$ with $f(u) = 1$ and $f(v) = 0$ has $\mathcal{E}^-(f) > 0$, so $\lambda^-(G) < +\infty$.

We now show that $\lambda^-(G) > 0$. The idea is that, for each function $f \in L^2(V)$, the action of $\mathcal{L}^-$ on f is the same as the action of the standard Laplacian on a subgraph of G (made undirected) that preserves only the f -antimonotone edges (as noted in Observation 6.2). The number of such subgraphs is at most $|V|!$, i.e. the number of permutations induced by f , so $\lambda^-(G)$ is lower bounded by the minimum non-zero eigenvalue out of finitely many graphs.

Formally, let $n := |V|$, fix any $f \in L^2(V)$ satisfying $\mathcal{E}^-(f) > 0$, and let $\pi : V \rightarrow [n]$ be a bijection such that $f_{\pi^{-1}(1)} \leq f_{\pi^{-1}(2)} \leq \dots \leq f_{\pi^{-1}(n)}$. Let $G_\pi = (V, w_\pi)$ be the undirected, weighted graph defined as follows: for each $u, v \in V$,

$$w_\pi(u, v) := \begin{cases} \frac{1}{2}w(u, v) & \text{if } \pi(u) > \pi(v) \\ \frac{1}{2}w(v, u) & \text{otherwise.} \end{cases}$$

⁴For example, let G consist of two disjoint edges (u, v) and (u', v') , and consider the sequence $(f_i)_{i \in \mathbb{N}}$ with $f_i(u) = 0$, $f_i(v) = i$, $f_i(u') = 1$, and $f_i(v') = 0$. Then $\mathcal{E}^-(f_i)/\|f_i\|_2^2 \rightarrow 0$ as $i \rightarrow \infty$, while $\|\mathcal{L}^- f_i\|_2^2/\mathcal{E}^-(f_i)$ remains constant.

Let $\mathcal{L}_\pi := \mathcal{L}_{G_\pi}$ denote the standard Laplacian operator for G_π . We claim that $\mathcal{L}^- f = \mathcal{L}_\pi f$. First, by definition of the standard Laplacian, for any $z \in V$ we have

$$(\mathcal{L}_\pi f)(z) = \sum_{v \in V} w_\pi(z, v) (f(v) - f(z)) .$$

On the other hand, [Definition 6.1](#) and the definition of w_π yield

$$\begin{aligned} (\mathcal{L}^- f)(z) &= \frac{1}{2} \sum_{u, v \in V} w(u, v) (f(u) - f(v))^+ (e_v(z) - e_u(z)) \\ &= \frac{1}{2} \sum_{u: \pi(u) > \pi(z)} w(u, z) (f(u) - f(z)) - \frac{1}{2} \sum_{v: \pi(z) > \pi(v)} w(z, v) (f(z) - f(v)) \\ &= \sum_{v: \pi(v) > \pi(z)} w_\pi(v, z) (f(v) - f(z)) + \sum_{v: \pi(z) > \pi(v)} w_\pi(v, z) (f(v) - f(z)) \\ &= \sum_{v \in V} w_\pi(v, z) (f(v) - f(z)) = (\mathcal{L}_\pi f)(z) , \end{aligned}$$

so indeed $\mathcal{L}^- f = \mathcal{L}_\pi f$. Let λ_π denote the smallest non-zero eigenvalue of $-\mathcal{L}_\pi$, which exists because G_π contains at least one edge since f is not monotone. Then

$$\frac{\|\mathcal{L}^- f\|_2^2}{\mathcal{E}^-(f)} = \frac{\|\mathcal{L}_\pi f\|_2^2}{-\langle f, \mathcal{L}^- f \rangle} = \frac{\|\mathcal{L}_\pi f\|_2^2}{\langle f, -\mathcal{L}_\pi f \rangle} \geq \lambda_\pi .$$

Let $\lambda^* := \min_\pi \lambda_\pi$, where π ranges over all bijections from V to $[n]$ for which $-\mathcal{L}_\pi$ has a non-zero eigenvalue. Then λ^* is strictly positive, and for each $g \in L^2(V)$ with $\mathcal{E}^-(g) > 0$, we have

$$\frac{\|\mathcal{L}^- g\|_2^2}{\mathcal{E}^-(g)} \geq \lambda^* .$$

Thus $\lambda^-(G) > 0$ as claimed. □

Next, we observe that $\lambda^-(G)$ characterizes the rate of exponential decay of $\mathcal{E}^-(P_t f)$. The key fact is that, in a gradient system, the rate of energy decay is governed by the rate of change of the state.

Proposition 6.18. *Let $f \in L^2(V)$. Then the function $t \mapsto \mathcal{E}^-(P_t f)$ is differentiable and, for all $t \geq 0$,*

$$\frac{d}{dt} \mathcal{E}^-(P_t f) = -2 \left\| \frac{d}{dt} P_t f \right\|_2^2 .$$

Proof. By the chain rule (for the Fréchet derivative) and [Proposition 6.12](#),

$$\frac{d}{dt}\mathcal{E}^-(P_t f) = \left\langle \nabla \mathcal{E}^-(P_t f), \frac{d}{dt}P_t f \right\rangle = \left\langle -2 \frac{d}{dt}P_t f, \frac{d}{dt}P_t f \right\rangle = -2 \left\| \frac{d}{dt}P_t f \right\|_2^2. \quad \square$$

Corollary 6.19. *Let $f \in L^2(V)$. Then the function $t \mapsto \mathcal{E}^-(P_t f)$ satisfies, for all $t \geq 0$,*

$$\frac{d}{dt}\mathcal{E}^-(P_t f) \leq -2\lambda^-(G) \cdot \mathcal{E}^-(P_t f). \quad (6.13)$$

Furthermore, $\lambda^-(G)$ is the largest constant for which (6.13) holds for all f, t .

Proof. By [Proposition 6.18](#),

$$\frac{d}{dt}\mathcal{E}^-(P_t f) = -2 \left\| \frac{d}{dt}P_t f \right\|_2^2 = -2 \|\mathcal{L}^- P_t f\|_2^2.$$

By [Definition 6.16](#), $\lambda^-(G) \leq \|\mathcal{L}^- P_t f\|_2^2 / \mathcal{E}^-(P_t f)$, whence (6.13) follows. Conversely, for every $\varepsilon > 0$, by definition of infimum we can find $g \in L^2(V)$ such that $\|\mathcal{L}^- g\|_2^2 / \mathcal{E}^-(g) < \lambda^-(G) + \varepsilon$, in which case we have (recalling that $P_0 g = g$)

$$\frac{d}{dt}\mathcal{E}^-(P_0 g) = -2 \|\mathcal{L}^- P_0 g\|_2^2 > -2(\lambda^-(G) + \varepsilon) \mathcal{E}^-(P_0 g),$$

so the constant $\lambda^-(G)$ is indeed tight. $\square$

We can now show that the dynamical spectral gap mediates an upper bound on the distance from the initial state f to its monotone equilibrium $P_\infty f$ in terms of the directed Dirichlet energy of f , i.e. a directed Poincaré inequality.

Theorem 6.20 (Directed Poincaré inequality). *For all $f \in L^2(V)$, it holds that*

$$\|f - P_\infty f\|_2^2 \leq \frac{1}{\lambda^-(G)} \mathcal{E}^-(f).$$

Proof. Since $P_t f \rightarrow P_\infty f$, it suffices to show that $\|f - P_t f\|_{L^2(V)}^2$ satisfies the claimed upper bound for all $t \geq 0$. By the triangle inequality and [Proposition 6.18](#),

$$\|f - P_t f\|_2 = \left\| \int_0^t \frac{d}{dt}P_t f \, dt \right\|_2 \leq \int_0^t \left\| \frac{d}{dt}P_t f \right\|_2 \, dt = \int_0^t \sqrt{-\frac{1}{2} \cdot \frac{d}{dt}\mathcal{E}^-(P_t f)} \, dt.$$

We now appeal to the exponential rate of decay of $\mathcal{E}^-(P_t f)$ (Corollary 6.19) via a straightforward calculus lemma stated below (Lemma 6.21) to conclude that

$$\|f - P_t f\|_2 \leq \sqrt{\frac{1}{2}} \cdot \frac{2}{\sqrt{2\lambda^-(G)}} \sqrt{\mathcal{E}^-(P_0 f)} = \frac{1}{\sqrt{\lambda^-(G)}} \sqrt{\mathcal{E}^-(f)}. \quad \square$$

We used the following calculus lemma, which appears implicitly in [KLR21, Lemma 4.17].

Lemma 6.21. *Let $t > 0$ and suppose that $F : [0, t] \rightarrow [0, +\infty)$ is differentiable and satisfies $F'(s) \leq -KF(s)$ for all $s \in [0, t]$, for some $K > 0$. Then it holds that*

$$\int_0^t \sqrt{-F'(s)} \, ds \leq \frac{2}{\sqrt{K}} \sqrt{F(0)}.$$

Proof. We first observe that we may assume that $F(s) > 0$ for all $s \in (0, t)$. Indeed, if this is not the case, then let $s^* := \inf\{s \in (0, t) : F(s) = 0\}$. By continuity of F , we obtain that $F(s^*) = 0$; then, since $F \geq 0$ and $F' \leq 0$, we conclude that $F(s) = F'(s) = 0$ for all $s \in [s^*, t]$. Thus it suffices to prove the claim for the interval $[0, s^*]$, and indeed $F(s) > 0$ for all $s \in (0, s^*)$. Thus assume without loss of generality that $F(s) > 0$ for all $s \in (0, t)$.

By the chain rule, the function $s \mapsto \sqrt{F(s)}$ is differentiable with $\frac{d}{ds} \sqrt{F(s)} = \frac{F'(s)}{2\sqrt{F(s)}} \leq 0$ for all $s \in (0, t)$, and thus

$$\begin{aligned} \sqrt{-F'(s)} &= -\frac{2\sqrt{F(s)}}{\sqrt{-F'(s)}} \frac{F'(s)}{2\sqrt{F(s)}} = \frac{2\sqrt{F(s)}}{\sqrt{-F'(s)}} \left(-\frac{d}{ds} \sqrt{F(s)} \right) \\ &\leq \frac{2\sqrt{F(s)}}{\sqrt{KF(s)}} \left(-\frac{d}{ds} \sqrt{F(s)} \right) = -\frac{2}{\sqrt{K}} \frac{d}{ds} \sqrt{F(s)}. \end{aligned}$$

Therefore

$$\int_0^t \sqrt{-F'(s)} \, ds \leq -\frac{2}{\sqrt{K}} \int_0^t \left(\frac{d}{ds} \sqrt{F(s)} \right) ds = \frac{2}{\sqrt{K}} \left(\sqrt{F(0)} - \sqrt{F(t)} \right) \leq \frac{2}{\sqrt{K}} \sqrt{F(0)}. \quad \square$$

6.2 The dynamical spectral gap of the directed hypercube

Let H_d denote the unweighted directed hypercube in dimension d , i.e. $H_d = (\{0, 1\}^d, w)$ where the weight function w is as follows: for each $x, y \in \{0, 1\}^d$, $w(x, y) = 1$ if $\|x - y\|_1 = 1$

with $x \preceq y$, and $w(x, y) = 0$ otherwise. For simplicity of notation, in this section we also let $V := \{0, 1\}^d$.

This section studies the spectral gap of H_d endowed with directed Laplacian operator $\mathcal{L}^- = \mathcal{L}_{H_d}^-$ and associated directed Dirichlet energy functional $\mathcal{E}^-$. Note that the directed Dirichlet energy specialized to H_d satisfies, by [Proposition 6.6](#),

$$\mathcal{E}^-(f) = \frac{1}{2} \mathbb{E} \left[\sum_{u, v \in V} w(u, v) ((f(u) - f(v))^+)^2 \right] = \frac{1}{2} \mathbb{E} [\|\nabla^- f\|_2^2], \quad (6.14)$$

so a directed Poincaré inequality as given by [Theorem 6.20](#) for H_d is indeed a directed (L^2, ℓ^2) -Poincaré inequality in the notation of [Chapter 2](#).

We show the following.

Theorem 6.22 (Dynamical spectral gap of the directed hypercube). *H_d satisfies $\lambda^-(H_d) = 1$.*

Before proving the theorem, we first observe that $\mathcal{L}^-$ and $\mathcal{E}^-$ enjoy a useful coordinate-wise decomposition: we write

$$\mathcal{L}^- = \sum_{i=1}^d \mathcal{L}^{(i)},$$

where each $\mathcal{L}^{(i)} : L^2(V) \rightarrow L^2(V)$ is given by

$$(\mathcal{L}^{(i)} f)(x) := \frac{1}{2} (f(x^{\oplus i}) - f(x)) \mathbf{1} [f(x^{i \rightarrow 0}) > f(x^{i \rightarrow 1})]$$

for each $f \in L^2(V)$ and $x \in \{0, 1\}^d$, where $x^{\oplus i}$ denotes the vertex obtained by flipping the i -th bit of x , and $x^{i \rightarrow b}$ denotes the vertex obtained by setting the i -th bit of x to b . It is straightforward to check that this decomposition agrees with [Definition 6.1](#). Similarly, from [Proposition 6.6](#) we also obtain the decomposition

$$\mathcal{E}^- = \sum_{i=1}^d \mathcal{E}^{(i)},$$

where each $\mathcal{E}^{(i)} : L^2(V) \rightarrow \mathbb{R}$ is given by

$$\mathcal{E}^{(i)}(f) := \frac{1}{4} \cdot \mathbb{E}_{x \in \{0, 1\}^d} \left[((f(x^{i \rightarrow 1}) - f(x^{i \rightarrow 0}))^-)^2 \right]$$

for each $f \in L^2(V)$. (The extra factor of $1/2$ compared to [Proposition 6.6](#) appears because the summation above counts each edge of H_d twice.)

We now prove [Theorem 6.22](#). The upper bound is easy and attained by anti-dictator functions, so the main point is to show the lower bound $\lambda^-(H_d) \geq 1$. To this end, we fix any function f , consider the fraction in [Definition 6.16](#), and expand the operators $\mathcal{L}^-$ and $\mathcal{E}^-$ according to the coordinate-wise decompositions above. The main conceptual ingredient is the positive correlation $\langle \mathcal{L}^{(i)} f, \mathcal{L}^{(j)} f \rangle \geq 0$, which is established by an argument reminiscent of the analysis of the edge tester for monotonicity of functions on the hypercube [\[GGLRS00\]](#)⁵.

Proof of Theorem 6.22. Upper bound. Note that for the anti-dictator function $g(x) = -x_1$, the coordinate-wise decompositions give that $(\mathcal{L}^- g)(x) = \pm \frac{1}{2}$ for every $x \in \{0, 1\}^d$, while the g -antimonotone edges (which contribute to $\mathcal{E}^-$) are precisely those along the first coordinate. Hence

$$\frac{\|\mathcal{L}^- g\|_2^2}{\mathcal{E}^-(g)} = \frac{\mathbb{E}[(\pm 1/2)^2]}{\frac{1}{4}\mathbb{E}[(-1)^2]} = 1,$$

so $\lambda^-(H_d) \leq 1$.

Lower bound. Let $f \in L^2(V)$. It suffices to show that $\|\mathcal{L}^- f\|_2^2 \geq \mathcal{E}^-(f)$. Note that

$$\|\mathcal{L}^- f\|_2^2 = \left\langle \sum_{i=1}^d \mathcal{L}^{(i)} f, \sum_{i=1}^d \mathcal{L}^{(i)} f \right\rangle = \sum_{i=1}^d \|\mathcal{L}^{(i)} f\|_2^2 + 2 \sum_{i < j} \langle \mathcal{L}^{(i)} f, \mathcal{L}^{(j)} f \rangle,$$

and recall that $\mathcal{E}^-(f) = \sum_{i=1}^d \mathcal{E}^{(i)}(f)$. For each $i \in [d]$, we have

$$\begin{aligned} \|\mathcal{L}^{(i)} f\|_2^2 &= \mathbb{E}_{x \in \{0,1\}^d} \left[((\mathcal{L}^{(i)} f)(x))^2 \right] = \mathbb{E}_{x \in \{0,1\}^d} \left[\left(\frac{1}{2} (f(x^{\oplus i}) - f(x)) \mathbb{1}[f(x^{i \rightarrow 0}) > f(x^{i \rightarrow 1})] \right)^2 \right] \\ &= \frac{1}{4} \mathbb{E}_{x \in \{0,1\}^d} \left[((f(x^{i \rightarrow 1}) - f(x^{i \rightarrow 0}))^-)^2 \right] = \mathcal{E}^{(i)}(f). \end{aligned}$$

Therefore, we will be done if we can show that, for each $i \neq j$, it holds that $\langle \mathcal{L}^{(i)} f, \mathcal{L}^{(j)} f \rangle \geq 0$:

⁵[\[GGLRS00\]](#) define a *switch operator* S_i which fixes all violations of monotonicity of a Boolean function f along direction i , by switching the values of f along violating edges. A key lemma in that paper shows that the application of S_i can only make the number of violations of monotonicity along a different direction j smaller – informally, the work along direction i “helps” toward the work along direction j . In our L^2 setting, this is captured by the positive correlation $\langle \mathcal{L}^{(i)} f, \mathcal{L}^{(j)} f \rangle \geq 0$ between the contributions of directions i and j to the action of $\mathcal{L}$.

Claim 6.23. *For every $i, j \in [d]$ with $i \neq j$, we have*

$$\langle \mathcal{L}^{(i)} f, \mathcal{L}^{(j)} f \rangle \geq 0.$$

We prove the claim below, but first we use it to conclude the proof of the theorem. Putting the above together, we obtain

$$\|\mathcal{L}^- f\|_2^2 = \sum_{i=1}^d \|\mathcal{L}^{(i)} f\|_2^2 + 2 \sum_{i < j} \langle \mathcal{L}^{(i)}, \mathcal{L}^{(j)} \rangle \geq \sum_{i=1}^d \mathcal{E}^{(i)}(f) = \mathcal{E}^-(f).$$

Since f was arbitrary, we conclude that $\lambda^-(H_d) \geq 1$. $\square$

Proof of Claim 6.23. Suppose without loss of generality that $i = 1$ and $j = 2$. We first observe that it suffices to consider each “square” obtained by fixing all but the first two coordinates, since the inner product decomposes along these squares. Concretely, for each $y \in \{0, 1\}^{d-2}$, let $g_y : \{0, 1\}^2 \rightarrow \mathbb{R}$ be given by $g_y(x) := f(x, y)$ for each $x \in \{0, 1\}^2$, where we write $f(x, y)$ for the value of f at the input obtained by concatenating x and y . Then

$$\begin{aligned} \langle \mathcal{L}^{(1)} f, \mathcal{L}^{(2)} f \rangle &= \frac{1}{2^d} \sum_{z \in \{0, 1\}^d} ((\mathcal{L}^{(1)} f)(z)) ((\mathcal{L}^{(2)} f)(z)) \\ &= \frac{1}{2^d} \sum_{x \in \{0, 1\}^2} \sum_{y \in \{0, 1\}^{d-2}} \left[\frac{1}{2} (f(x^{\oplus 1}, y) - f(x, y)) \mathbf{1} [f(x^{1 \rightarrow 0}, y) > f(x^{1 \rightarrow 1}, y)] \right. \\ &\quad \left. \cdot \frac{1}{2} (f(x^{\oplus 2}, y) - f(x, y)) \mathbf{1} [f(x^{2 \rightarrow 0}, y) > f(x^{2 \rightarrow 1}, y)] \right] \\ &= \frac{1}{2^d} \sum_{y \in \{0, 1\}^{d-2}} \sum_{x \in \{0, 1\}^2} \mathcal{L}^{(1)} g_y(x) \cdot \mathcal{L}^{(2)} g_y(x). \end{aligned}$$

We will show that for any $g : \{0, 1\}^2 \rightarrow \mathbb{R}$, the sum $\sum_{x \in \{0, 1\}^2} \mathcal{L}^{(1)} g(x) \cdot \mathcal{L}^{(2)} g(x)$ is nonnegative, which will complete the proof. Define

$$\begin{aligned} c &:= g(0, 1), & d &:= g(1, 1), \\ a &:= g(0, 0), & b &:= g(1, 0). \end{aligned}$$

Then we have

$$\begin{aligned}
& \sum_{x \in \{0,1\}^2} \mathcal{L}^{(1)} g(x) \cdot \mathcal{L}^{(2)} g(x) \\
&= \sum_{x \in \{0,1\}^2} (g(x^{\oplus 1}) - g(x)) \mathbf{1} [g(x^{1 \rightarrow 0}) > g(x^{1 \rightarrow 1})] (g(x^{\oplus 2}) - g(x)) \mathbf{1} [g(x^{2 \rightarrow 0}) > g(x^{2 \rightarrow 1})] \\
&= (b-a) \mathbf{1} [a > b] (c-a) \mathbf{1} [a > c] + (a-b) \mathbf{1} [a > b] (d-b) \mathbf{1} [b > d] \\
&\quad + (d-c) \mathbf{1} [c > d] (a-c) \mathbf{1} [a > c] + (c-d) \mathbf{1} [c > d] (b-d) \mathbf{1} [b > d] \\
&= (a-b)^+ (a-c)^+ - (a-b)^+ (b-d)^+ - (c-d)^+ (a-c)^+ + (c-d)^+ (b-d)^+ \\
&= [(a-b)^+ - (c-d)^+] [(a-c)^+ - (b-d)^+] \geq 0,
\end{aligned}$$

where the last inequality is proved in [Lemma 6.24](#) below. $\square$

Lemma 6.24. *For any $a, b, c, d \in \mathbb{R}$, we have*

$$[(a-b)^+ - (c-d)^+] [(a-c)^+ - (b-d)^+] \geq 0.$$

Proof. Let $X := (a-b)^+ - (c-d)^+$ and $Y := (a-c)^+ - (b-d)^+$, so that our goal is to show that $XY \geq 0$, or equivalently that $X > 0 \implies Y \geq 0$ and $Y > 0 \implies X \geq 0$. By switching the roles of b and c , it suffices to prove the first implication, i.e. that $X > 0 \implies Y \geq 0$.

Suppose $X > 0$. Then $a-b > 0$, since otherwise we would have $(a-b)^+ = 0$ and hence $X \leq 0$. Therefore $a-b = (a-b)^+ > (c-d)^+ \geq c-d$. Moreover, if $(b-d)^+ = 0$ then $Y \geq 0$ and we are done, so we may assume that $b-d = (b-d)^+ > 0$. We conclude that

$$Y = (a-c)^+ - (b-d)^+ \geq (a-c) - (b-d) = (a-b) - (c-d) \geq 0,$$

where the first inequality holds since $(a-c)^+ \geq a-c$ while $(b-d)^+ = b-d$, and the second inequality holds since $a-b \geq c-d$. $\square$

Combining [Theorems 6.20](#) and [6.22](#) along with [\(6.14\)](#), we obtain the desired directed (L^2, ℓ^2) -Poincaré inequality for the hypercube.

Theorem 6.25 (Directed (L^2, ℓ^2) -Poincaré inequality for the hypercube). *For all $f : \{0,1\}^d \rightarrow \mathbb{R}$, it holds that*

$$\text{dist}_2^{\text{mono}}(f)^2 \leq \|f - P_\infty f\|_2^2 \leq \frac{1}{\lambda^-(H_d)} \mathcal{E}^-(f) = \mathcal{E}^-(f) = \frac{1}{2} \mathbb{E} [\|\nabla^- f\|_2^2].$$

6.3 Reverse directed Poincaré inequalities

We give a reverse directed Poincaré inequality which establishes the tightness of [Theorem 6.25](#) up to a factor of d . In fact, for a general directed graph G , the reverse directed Poincaré inequality we show is mediated by the maximum total weighted degree of G .

Let $d_w(G)$ denote the maximum total weighted degree of any vertex in G :

$$d_w(G) := \max_{u \in V} \left\{ \sum_{v \in V} (w(u, v) + w(v, u)) \right\}.$$

It will also be natural to generalize the reverse inequality to all (L^p, ℓ^p) types for $p \in [1, 2]$. To that end, define

$$\mathcal{E}_p^-(f) := \frac{1}{p} \cdot \mathbb{E} \left[\sum_{u \in V} w(u, v) ((f(u) - f(v))^+)^p \right], \quad (6.15)$$

so that $\mathcal{E}_2^-(f) = \mathcal{E}^-(f)$ by [Proposition 6.6](#), and moreover when $G = H_d$ we have

$$\mathcal{E}_p^-(f) = \frac{1}{p} \cdot \mathbb{E} [\|\nabla^- f\|_p^p].$$

We then show the following.

Theorem 6.26 (Reverse directed Poincaré inequality). *For all $f : V \rightarrow \mathbb{R}$, all monotone $g : V \rightarrow \mathbb{R}$, and all $p \in [1, 2]$, we have*

$$\|f - g\|_{L^p(V)}^p \geq \frac{1}{d_w(G)} \mathcal{E}_p^-(f).$$

Proof. Let $\Delta \in L^2(V)$ be given by $\Delta(u) := |f(u) - g(u)|$ for each $u \in V$, so that $\|f - g\|_{L^p(V)}^p = \|\Delta\|_{L^p(V)}^p$. For each f -antimonotone edge (u, v) of G , we have $f(u) > f(v)$ while $g(u) \leq g(v)$, which implies via the triangle inequality that

$$\Delta(u) + \Delta(v) \geq f(u) - f(v). \quad (6.16)$$

Using (6.15) and summing (6.16) up over all f -antimonotone edges, we conclude that

$$\begin{aligned}
\mathcal{E}_p^-(f) &= \frac{1}{p} \mathbb{E}_{u \in V} \left[\sum_{v \in V} w(u, v) ((f(u) - f(v))^+)^p \right] \\
&= \frac{1}{p} \mathbb{E}_{u \in V} \left[\sum_{v \in V: f(u) > f(v)} w(u, v) (f(u) - f(v))^p \right] \\
&\leq \frac{1}{p} \mathbb{E}_{u \in V} \left[\sum_{v \in V: f(u) > f(v)} w(u, v) (\Delta(u) + \Delta(v))^p \right] \\
&\leq \mathbb{E}_{u \in V} \left[\sum_{v \in V: f(u) > f(v)} w(u, v) (\Delta(u)^p + \Delta(v)^p) \right] \\
&\leq \mathbb{E}_{u \in V} \left[\Delta(u)^p \left(\sum_{v \in V} (w(u, v) + w(v, u)) \right) \right] \\
&\leq d_w(G) \cdot \|\Delta\|_{L^p(V)}^p,
\end{aligned}$$

where the second inequality used the fact that $(x + y)^p \leq p(x^p + y^p)$ for $x, y \geq 0$ and $p \in [1, 2]$. $\square$

The directed unweighted hypercube H_d has $d_w(H_d) = d$. Hence the theorem above implies

Corollary 6.27. *Fix graph H_d , the directed unweighted hypercube on vertex set $\{0, 1\}^d$. Then for each $f : \{0, 1\}^d \rightarrow \mathbb{R}$ and $p \in [1, 2]$, we have*

$$\text{dist}_p^{\text{mono}}(f)^p \geq \frac{1}{d} \mathcal{E}_p^-(f) = \frac{1}{p \cdot d} \mathbb{E} [\|\nabla^- f\|_p^p].$$

Part II: Continuous setting via PDEs and optimal transport

Chapter 7

An L^2 directed Poincaré inequality via the directed heat equation

The main result of this chapter is a directed (L^2, ℓ^2) -Poincaré inequality for sufficiently regular functions $f : [0, 1]^d \rightarrow \mathbb{R}$ (in particular, Lipschitz continuity suffices; see [Section 7.2.1](#)).

Theorem 7.1 (Directed Poincaré inequality). *There exists a universal constant $C > 0$ such that, for all $f \in H^1((0, 1)^d)$,*

$$\text{dist}_2^{\text{mono}}(f)^2 \leq C \mathbb{E} [\|\nabla^- f\|_2^2] . \quad (7.1)$$

As in the previous chapter, we will prove [Theorem 7.1](#) via a dynamical argument. The two areas of mathematics underpinning our proof are *partial differential equations* [[Eva10](#); [Bre11](#)] and the theory of *optimal transport* [[Vil09](#); [San15](#)]. Our guiding vision is to identify a robust approach to the *classical* Poincaré inequality such that, by “toggling” a single aspect of the approach, we can transform a proof of the classical statement into a proof of its directed counterpart. It turns out that the *heat equation* is the crucial ingredient to implement this vision, as we now outline.

7.1 Proof ideas

7.1.1 Starting point: the heat equation.

The first step is to identify the right classical starting point. For example, the Poincaré inequality can be proved using Fourier analysis, but since the directed problem is highly

nonlinear (due to the ∇^- operator), this approach does not seem suitable. Instead, we take a physically-motivated approach. Let $u = u(t, x)$ where we think of the first variable as time, with each $f = u(t) = u(t, \cdot)$ a function over space. A remarkable property (see e.g. [BGL14, Chapter 4]) of the Poincaré inequality

$$\text{Var}[f] \leq C \mathbb{E} [\|\nabla f\|_2^2]$$

(where we have written $\text{Var}[f]$ in the place of $\text{dist}_2^{\text{const}}(f)^2$, which is the same thing) is that it is equivalent to exponential decay of variance in the heat equation

$$\partial_t u = \Delta u, \tag{7.2}$$

where Δ denotes the Laplacian operator, given appropriate “no-flux” boundary conditions which we do not expand on at the moment. To see this equivalence, assume for simplicity the mean zero condition $\int u(0) dx = \int u(0, x) dx = 0$ (which is then preserved over time), compute $\partial_t \text{Var}[u(t)]$, differentiate under the integral, apply (7.2) and integrate by parts to obtain

$$\partial_t \text{Var}[u(t)] = \partial_t \int u(t)^2 dx = 2 \int u(t) \partial_t u(t) dx = 2 \int u(t) \Delta u(t) dx = -2 \int \nabla u(t) \cdot \nabla u(t) dx.$$

This means that $\partial_t \text{Var}[u(t)] \leq -C \text{Var}[u(t)]$ if and only if $\text{Var}[u(t)] \leq \frac{2}{C} \int \|\nabla u(t)\|_2^2 dx$, i.e. exponential decay of the variance is equivalent to the Poincaré inequality.

We observe that solutions to the heat equation converge to a constant equilibrium (taking the average value of u), and the associated Poincaré inequality is a bound on the distance to a constant function. Accordingly, it seems intuitive that if we replace (7.2) with a PDE that converges to a *monotone* function, we might learn something about the distance to a monotone function instead.

7.1.2 The directed heat equation

The preceding observations suggest that we should study directed analogues of (7.2). However, in the multidimensional setting, the nonlinearities that seem inherent to any such directed analogue prove challenging to analyze. Hence, let us focus on the one-dimensional case for now. In one dimension, the heat equation is

$$\partial_t u = \partial_x \partial_x u.$$

The surface reading of this PDE is that, if we focus on the value of u at a single point x , the PDE tells us how this value changes over time. Another useful perspective is to think

of $u(t, x)$ as the density of “particles” at each point, and ask about how these particles are “moving” over time; this idea can be made formal via the so-called *continuity equation*. Under this view, it turns out that the inner expression $\partial_x u$ represents (up to a sign change) the *flux* (or momentum) field, i.e. the total rate of particle movement at each point, and by taking the derivative of this field, i.e. $\partial_x \partial_x u$, we determine whether, on the balance, there are more “incoming” or “outgoing” particles—quantitatively, the rate of change $\partial_t u$.

With this perspective, a natural candidate PDE—which we call the *directed heat equation*—is

$$\partial_t u = \partial_x \partial_x^- u, \quad (7.3)$$

where $\partial_x^- u := \min\{0, \partial_x u\}$. The idea is that the new flux $\partial_x^- u$ is always nonpositive, which, up to a required sign change, means that particles are only allowed to “move to the right”. Observe also that the RHS of (7.3) is zero for any monotone function, so monotone functions are stationary solutions, while for decreasing functions, this PDE behaves exactly as the heat equation. Thus, intuitively, this PDE seems to take a function $u(0)$ and move it toward a monotone limit over time.

We explained above that the Poincaré inequality is intimately connected to convergence of the heat equation, and we would like to show a similar property in the directed case. One option would be to study the rate of decay of the distance to monotonicity, $\text{dist}_2^{\text{mono}}$. Although this is possible, it does not seem to lead to a proof strategy; also note that, while in the case of the variance we also have tools such as Fourier analysis to reason about the rate of decay directly, here we only have the PDE to work with. Fortunately, there is another relevant quantity which decays exponentially under the heat equation, namely the *Dirichlet energy*

$$\mathcal{E}(f) = \frac{1}{2} \int (\partial_x f)^2 dx.$$

Indeed, another formal computation shows that by differentiating under the integral, switching the partial derivatives, applying (7.2) and integrating by parts, we get

$$\begin{aligned} \partial_t \mathcal{E}(u(t)) &= \frac{1}{2} \partial_t \int (\partial_x u(t))^2 dx = \int (\partial_x u(t)) (\partial_t \partial_x u(t)) dx = \int (\partial_x u(t)) (\partial_x \partial_t u(t)) dx \\ &= \int (\partial_x u(t)) (\partial_x \partial_x \partial_x u(t)) dx = - \int (\partial_x \partial_x u(t))^2 dx \leq -\frac{1}{C} \int (\partial_x u(t))^2 dx \\ &= -\frac{2}{C} \mathcal{E}(u(t)), \end{aligned}$$

where the inequality is an application of a version of the Poincaré inequality for functions that are zero on the boundary, which will be the case for $\partial_x u(t)$ by the no-flux boundary conditions.

Now, it seems reasonable to propose the following directed analogue of the Dirichlet energy:

$$\mathcal{E}^-(f) = \frac{1}{2} \int (\partial_x^- f)^2 dx.$$

If $\mathcal{E}$ measures the “locally non-constant” activity of a function, $\mathcal{E}^-$ measures the “locally non-monotone” activity. Using $\mathcal{E}^-$ to recast the directed heat equation in the language of gradient flows and maximal monotone operators [Bre73], we can show that

1. this PDE has a solution (Proposition 7.20 and Corollary 7.23);
2. the directed Dirichlet energy decays exponentially in time (Proposition 7.25);
3. the solution converges to a monotone equilibrium as $t \rightarrow \infty$ (Proposition 7.68); and
4. the solution, including up to the limit above, satisfies several other essential analytic properties, such as nonexpansiveness (Proposition 7.72) and order preservation (Corollary 7.71).

7.1.3 Transport-energy inequality in one dimension

It is tempting, now, to use these results to conclude the one-dimensional directed Poincaré inequality. The problem is that this seems to lead to a dead end, because we do not know how to *tensorize* the one-dimensional inequality into a multidimensional one¹.

Instead, we keep leaning on what the PDEs naturally tell us. It turns out that there is a different notion of distance, the *Wasserstein* distance, which is much more closely connected to the theory of evolution equations and better suited to the dynamical approach we are undertaking. Informally, the squared Wasserstein distance $W_2^2(\varrho_0, \varrho_1)$ between probability measures ϱ_0 and ϱ_1 is the minimum total cost of a “transport plan” (coupling) moving particles from ϱ_0 to ϱ_1 , where the cost of moving a particle from point x to point y is $|x - y|^2$. The connection to PDEs is via the *Benamou-Brenier formula*, which informally says that

$$W_2^2(\varrho_0, \varrho_1) = \min \left\{ \int_0^1 \|v_t\|_{L^2(\varrho_t)}^2 dt : v_t \text{ is a velocity field taking } \varrho_0 \text{ to } \varrho_1 \text{ from time 0 to 1} \right\}.$$

¹In the classical case, the *law of total variance* does allow such tensorization [BGL14, Chapter 4.3], but this aspect of the problem does not seem to be robust to passing to the directed setting. Also note that, if we only desired a one-dimensional directed Poincaré inequality, then we may obtain a much shorter proof by following the argument from Chapter 4.

Without going into details, it follows that if we can upper bound the directed Dirichlet energy $\mathcal{E}^-(u(t))$, then we can upper bound the total (weighted) magnitude of the velocity field, i.e. $\|v_t\|_{L^2(\varrho_t)}^2$ (recall that the momentum of particles in the directed heat equation is essentially $\partial_x^- u$, so this connection is not arbitrary), and therefore upper bound the Wasserstein distance between the initial state ϱ_0 —which informally corresponds to the function $f = u(0)$ —and the final state ϱ_1 —which informally corresponds to the monotone equilibrium $f^* = \lim_{t \rightarrow \infty} u(t)$.

This strategy yields, at least for bounded functions, the following result. Let $\mathcal{U}$ be a class of “reasonable” initial states which we do not define for now, and let the operator P_∞ map each initial state to its monotone equilibrium according to the directed heat equation. Then

Theorem 7.2 (Transport-energy inequality in one dimension). *There exists a constant $C > 0$ such that the following holds. Let $u \in \mathcal{U}$ be positive, bounded away from zero, and satisfy $\int_{(0,1)} u \, dx = 1$. Define the measures $d\mu := u \, dx$ and $d\mu_\infty := (P_\infty u) \, dx$. Then*

$$W_2^2(\mu, \mu_\infty) \leq \frac{C}{\inf u} \mathcal{E}^-(u).$$

7.1.4 Tensorizing the transport-energy inequality

The advantage of making this detour through the Wasserstein distance is that, modulo the required technical work, tensorization becomes relatively straightforward. Indeed, it is well-known in the theory of optimal transport (c.f. [Vil09, Remark 6.6]) that the W_2 distance scales well with the dimension and reflects useful geometric content. Here the main idea is that, since the cost function for the W_2 distance is the squared Euclidean distance $|x - y|^2$ between points $x, y \in [0, 1]^d$, one way to extend the one-dimensional result above to, say, the unit square $[0, 1]^2$ is to

1. apply the one-dimensional result to each row, making the row restrictions monotone while paying cost $W_2^2(\varrho_0, \varrho_1) = a^2$; and then
2. apply the one-dimensional result to each column, making the column restrictions monotone while paying cost $W_2^2(\varrho_1, \varrho_2) = b^2$.

Then, by combining the transport of particles along rows and columns into a single transport plan, we obtain via the Pythagorean theorem a plan with cost $a^2 + b^2$, so this quantity upper bounds the overall squared distance $W_2^2(\varrho_0, \varrho_2)$. The same idea extends to higher dimensions, and we prove:

Lemma 7.3 (Pythagorean composition of transport plans). *Let $I, J \subseteq [d]$ be nonempty disjoint sets. Let $\mu, \varrho, \nu \in P(\mathbb{R}^d)$ be supported in bounded sets, and let $\gamma^+ \in \Pi_I(\mu \rightarrow \varrho)$ and $\gamma^- \in \Pi_J(\varrho \rightarrow \nu)$. Then there exists $\gamma \in \Pi_{I \cup J}(\mu \rightarrow \nu)$ satisfying $C_2(\gamma)^2 = C_2(\gamma^+)^2 + C_2(\gamma^-)^2$.*

In this statement, $C_2(\gamma)^2$ is the cost of transport plan γ , and $\Pi_I(\mu \rightarrow \nu)$ denotes the set of transport plans between probability measures μ to ν , under two additional restrictions: 1) particles can only move along “ I -aligned” lines, e.g. along rows in the example; and 2) particles can only move “up” in the partial order on $\mathbb{R}^d$, i.e. a particle can move from x to y only if $x \preceq y$ —we call such a transport plan *directed*. The second restriction is important because, when we finally recover a Poincaré inequality from a transport-energy inequality, the information that particles only moved up in the partial order will be reflected in the appearance of the directed gradient ∇^- .

Now, informally, [Theorem 7.2](#) tells us that each step of the above strategy (e.g. moving particles along rows) incurs cost bounded by the directed Dirichlet energy, i.e. the integral $\int (\partial_i^- f)^2 dx$ of the partial derivatives along the (say) rows. Repeating for every $i \in [d]$ via [Lemma 7.3](#), we obtain the directed, multidimensional transport-energy inequality:

Theorem 7.4 (Transport-energy inequality). *There exists a universal constant $C > 0$ such that the following holds. Let $a \in (0, 1)$, and let $f \in \text{Lip}$ satisfy $1 - a \leq f \leq 1 + a$ and $\int_{[0,1]^d} f dx = 1$. Define the probability measures $d\mu := f dx$ and $d\mu^* := f^* dx$ on $[0, 1]^d$. Then*

$$W_2^2(\mu \rightarrow \mu^*) \leq \frac{C(1+a)^2}{(1-a)^3} \int_{[0,1]^d} |\nabla^- f|^2 dx.$$

Moreover, by the order preservation property of our solution to our PDE, each operation along direction i preserves the monotonicity along the directions $j < i$, so the final function f^* in the statement above, which is the result after all d steps, is indeed monotone. The notation $W_2^2(\mu \rightarrow \mu^*)$ indicates that this bound holds even for directed transport plans, as explained above.

7.1.5 Recovering a Poincaré inequality via optimal transport duality

The final step is to recover, from the multidimensional transport-energy inequality, our desired directed Poincaré inequality. Fortunately, there is an intimate connection between

transport and Poincaré inequalities (as well as Sobolev, logarithmic Sobolev, and other isoperimetric and concentration inequalities), and this theory is well established in the classical case (c.f. [Vil09, Chapters 21 and 22]). Closest to the present approach, [Liu20] proved, in the classical setting, the *equivalence* between transport-energy, Poincaré, and other related inequalities. Our task, therefore, is to obtain at least an implication in the directed setting.

The main ingredient toward this goal is the notion of *Kantorovich duality* for the Wasserstein distance. In the classical setting, the weak Kantorovich duality says that, given two probability measures μ and ν , if we can find “certificate functions” ϕ and ψ satisfying

$$\phi(y) - \psi(x) \leq |x - y|^2 \quad \text{for all points } x \text{ and } y, \quad (7.4)$$

then

$$W_2^2(\mu, \nu) \geq \int \phi(y) \, d\nu(y) - \int \psi(x) \, d\mu(x). \quad (7.5)$$

Villani recounts the following analogy by Caffarelli: suppose you operate a consortium of bakeries and cafes in a city, and that $\mu(dx)$ is the supply of bread by a bakery at point x while $\nu(dy)$ is the demand by a cafe at point y . Then if transporting a unit of bread from x to y costs $|x - y|^2$, the minimum cost for transporting all the bread is by definition $W_2^2(\mu, \nu)$. Suppose a transportation company offers to take over the transportation job by buying bread from the bakeries at price $\psi(x)$ and selling it to cafes at price $\phi(y)$. If (7.4) holds then, for each x and y , hiring the transportation company is no more expensive than handling the transportation yourself. Therefore the cost $W_2^2(\mu, \nu)$ is at least as large as how much the bakery-cafe consortium pays the company to fulfill the supply and demand, which is the RHS of (7.5).

In the directed setting, where we can only transport mass from x to y if $x \preceq y$, we expect the transport to be in general more expensive, so “more” certificates should be valid. Indeed, it is not difficult to obtain the following natural directed version of weak duality (Lemma 7.126): if

$$\phi(y) - \psi(x) \leq |x - y|^2 \quad \text{for all points } x \preceq y,$$

then

$$W_2^2(\mu \rightarrow \nu) \geq \int \phi(y) \, d\nu(y) - \int \psi(x) \, d\mu(x).$$

The final main ingredient is an operator which, given a candidate function ϕ , produces the best possible ψ for duality. This is accomplished by a directed analogue of the so-called

Hamilton-Jacobi operator, defined as follows: for each function h on the unit cube,

$$(\vec{H}_t h)(x) := \begin{cases} h(x) & \text{if } t = 0 \\ \sup_{y \succeq x} \{h(y) - \frac{1}{2t}|x - y|^2\} & \text{otherwise.} \end{cases}$$

Then, for each h and setting $t = 1$, this operator yields the following ([Proposition 7.129](#)):

$$\frac{1}{2}W_2^2(\mu \rightarrow \nu) \geq \int h \, d\nu - \int (\vec{H}_1 h) \, d\mu. \quad (7.6)$$

Note that, as $t \rightarrow 0^+$, $(\vec{H}_t h)(x)$ intuitively seeks the direction $y - x \succeq 0$ of steepest ascent of h . This suggests a connection to the directed gradient, and indeed in [Proposition 7.131](#) we show that

$$\limsup_{t \rightarrow 0^+} \frac{\vec{H}_t h(x) - h(x)}{t} \leq \frac{|\nabla^+ h(x)|^2}{2}, \quad (7.7)$$

where $\nabla^+ h := \max\{0, \nabla h\}$.

At this point, instead of trying to reproduce the calculations from [Section 7.4.2](#), let us give some intuition for how the pieces above recover a Poincaré inequality. By homogeneity, it suffices to consider mean-zero functions h . The key idea is to fix a small $t > 0$ and let h play two roles at the same time: 1) as the building block for measures $d\mu(x) = (1 + th) \, dx$ and $d\mu^*(x) = (1 + th^*) \, dx$, where h^* is the monotone function obtained via the coordinate-wise application of the directed heat equation as in the previous section; and 2) in the test function $-th$ for duality via (7.6). That is, after building measures μ and μ^* from h and h^* (as perturbations of the uniform measure, scaled by the parameter t), we *also* employ h to build certificate functions $\phi = -th$ and $\psi = \vec{H}_1(-th)$. This yields, via (7.6), a bound on the Wasserstein distance to monotonicity of (the measure μ produced by) h which is itself written in terms of h .

From the fact that h plays both of these roles, and h (respectively, $\vec{H}_1 h$) appears integrated against ν (respectively, μ) in (7.6), that inequality ends up producing the “interaction term” $\int (h^2 - hh^*) \, dx$, which can be appropriately bounded by $\int (h - h^*)^2 \, dx$; and since h^* is monotone, the inequality $\int (h - h^*) \, dx \geq \text{dist}_2^{\text{mono}}(h)^2$ explains why we should expect the distance to monotonicity to appear. Moreover, after the appropriate calculations, the term involving $\vec{H}_1$ in (7.6) gives rise to the expression in (7.7), so letting $t \rightarrow 0^+$ makes the directed gradient of h appear. Thus $W_2^2(\mu, \mu^*)$ can be lower bounded by an expression involving $\text{dist}_2^{\text{mono}}(h)^2$ and $\int |\nabla^- h|^2 \, dx$. On the other hand, [Theorem 7.4](#) gives that $W_2^2(\mu, \mu^*)$ is upper bounded by an expression involving $\int |\nabla^- h|^2 \, dx$. Therefore, chaining the inequalities, we have precisely the terms required to put (7.1) together.

Organization. The rest of this chapter is organized as follows. In [Section 7.2](#), we study the directed heat equation, showing that this PDE is well-posed and satisfies the properties listed in [Section 7.1.2](#). In [Section 7.3](#), we introduce optimal transport and establish the transport-energy inequality ([Theorem 7.4](#)). Finally, in [Section 7.4](#) we recover the directed Poincaré inequality ([Theorem 7.1](#)) from the transport-energy inequality.

7.2 Directed heat semigroup

In this section, we make concrete the ideas presented in [Section 7.1.2](#) of the proof overview. Recall that we wish to analyze the *directed heat equation*, which we informally specified in [\(7.3\)](#) as

$$\partial_t u = \partial_x \partial_x^- u. \quad (7.8)$$

We require a more rigorous definition before we can study this PDE, since e.g. the quantity $\partial_x^- u = \min\{0, \partial_x u\}$ is not in general differentiable even if u is smooth. Indeed, while the classical heat equation has a smoothing effect, we expect any solution to [\(7.8\)](#) (in the appropriate sense) to have non-differentiable points, so a more technical treatment of this nonlinear PDE is unavoidable.

The first step is to replace the operator $\partial^- u$ with something more amenable to analysis. In [Definition 7.11](#) we define a *canonical representation* $u = u\uparrow + u\downarrow$ such that $u\downarrow$ is nonincreasing and differentiable (more precisely, it is in the Sobolev space H^1), while $u\uparrow$ is nondecreasing and need not be differentiable (it only needs to be in the space L^2 of square-integrable functions). While a priori such a representation need not be unique, we show in [Proposition 7.10](#) that there is a unique representation (after fixing the shift by a constant) minimizing the energy $\int (\partial_x u\downarrow)^2 dx$; intuitively, this means that we pack all the “decreasing behaviour” of u into $u\downarrow$, and all the “increasing behaviour” of u into $u\uparrow$. Note that $u\uparrow$ may even contain jump discontinuities, which our theory must withstand, so this decomposition captures extremely relevant properties of our problem.

Then, we are able to restate our PDE more rigorously in [Definitions 7.14](#) and [7.18](#). Essentially, the “Neumann problem”

$$\begin{cases} z = \partial_x \partial_x u\downarrow & \text{in } (0, 1) \\ \partial_x u\downarrow = 0 & \text{on } \{0, 1\} \end{cases}$$

given in [\(7.9\)](#) (where z will take the role of $\partial_t u$ in [Definition 7.18](#)) replaces $\partial_x^- u$ with $\partial_x u\downarrow$ in [\(7.8\)](#), and then places a “Neumann boundary condition” $\partial_x u\downarrow = 0$ which, as is standard in

PDEs, means that there will be no mass flux at the boundaries of the domain. For technical reasons that are common to most study of PDEs, we need to give in (7.10) a notion of “weak solution”, which is essentially a solution which may not be regular enough to satisfy the specification of the PDE (i.e. it may not be twice-differentiable), but which behaves (via integration by parts) just like a “strong solution” for many analytical purposes, and can sometimes be “promoted”: a weak solution that is regular enough is usually also a strong solution (in fact, our problem automatically enjoys such a promotion thanks to “elliptic regularity”, see Lemma 7.24).

To study our PDE, we bring in the theory of gradient flows and maximal monotone operators [Bre73] by recasting the PDE as the problem of finding the direction of steepest descent² (in the space of L^2 functions) of the directed Dirichlet energy $\mathcal{E}^-(u)$, which our formalism allows us to define in terms of $u\downarrow$ in Definition 7.12. The main results in Section 7.2.2 are Proposition 7.20 on the existence of a unique solution to the gradient flow problem – which allows us to define a semigroup operator P_t , the *directed heat semigroup*, describing the evolution of this solution – and Corollary 7.23, which shows that this solution is also a solution to the Neumann problem (i.e. our PDE of interest).

To conclude the basic study of our PDE, Section 7.2.3 shows, via the theory of gradient flows, that the directed Dirichlet energy $\mathcal{E}^-(P_t u)$ decays exponentially in time (Proposition 7.25), which allows us to conclude that $P_t u$ converges to some limit as $t \rightarrow \infty$ (Proposition 7.27).

Sections 7.2.4 to 7.2.6 are concerned with showing that regular initial states u yield solutions $P_t u$ that are also regular for all times $t > 0$, where “regular” can mean Lipschitz or Sobolev class H^1 . The main result in Section 7.2.4 is Lemma 7.28, a technical lemma which says that for solutions to our PDE, if $\partial_x u\downarrow < 0$ in an interval (a, b) (which we wish to call “ u is decreasing”), it must be the case that $u\uparrow$ is constant in (a, b) (legitimizing that wish). This result is far from being vacuous or tautological – it implies that in regions where our PDE is “active” (because $\partial_x u\downarrow$ is nonzero), the other component $u\uparrow$ of u is “inactive” (constant), which means that solutions to our PDE avoid a host of pathological scenarios and allows crucial technical arguments to go through. Then, using technical arguments and tools from the theory of maximal monotone operators, Section 7.2.5 builds toward Corollary 7.41 – H^1 regularity is preserved over time – and Section 7.2.6 builds toward Corollary 7.48 – Lipschitz regularity is preserved over time.

In Section 7.2.7, we turn our attention to nonexpansiveness and order preservation properties. The main result, Proposition 7.57, is a quantitative order preservation result which implies that if initial states u, v satisfy $u \leq v$, then $P_t u \leq P_t v$ for all times $t > 0$

²This is analogous to our analysis of the heat flow on directed graphs as a gradient system in Chapter 6.

(Corollary 7.59). We conclude Section 7.2.7 with some useful basic observations about the semigroup P_t ; in particular, Proposition 7.66 shows that P_t behaves nicely under certain affine transformations, namely $P_t(\alpha u + \beta) = \alpha P_t u + \beta$ when $\alpha > 0$ and $\beta \in \mathbb{R}$.

Finally, Section 7.2.8 studies the convergence of solutions $P_t u$ as $t \rightarrow \infty$. The main result is Proposition 7.68, which shows that $P_t u$ converges to a *monotone* function, which we define as $P_\infty u$ in Definition 7.69. The remainder of Section 7.2.8 shows that the main properties of P_t shown in the previous subsections – preservation of regularity, nonexpansiveness, behaviour under affine transformations – are also satisfied by P_∞ .

7.2.1 Preliminaries for PDE

We briefly outline some of the main concepts required to study our PDE, and refer the reader to e.g. [Eva10; Bre11] for detailed expositions.

Let $J = (a, b)$ be an open interval. The absolutely continuous (AC) functions $f : \bar{J} \rightarrow \mathbb{R}$ are precisely those for which there exists a function $g \in L^1(J)$ such that, for all $x \in \bar{J}$, $f(x) = f(a) + \int_{(a,x)} g \, dx$. We call g a *weak derivative* of f , and write $\partial_x f$ for any weak derivative of f . The weak derivative is almost everywhere (a.e.) uniquely determined, and moreover, f is classically differentiable a.e. and its classical derivative agrees with $\partial_x f$ almost everywhere.

Let $\Omega := J^d$. Let $k \in \mathbb{Z}_{\geq 0}$ and $p \in [1, +\infty]$. The Sobolev space $W^{k,p}(\Omega)$ is the space of functions $f : \Omega \rightarrow \mathbb{R}$ which have weak derivatives up to order k in $L^p(\Omega)$. The definition of weak derivative in the multidimensional case is more involved than in the one-dimensional case, but the details are not relevant here. In one dimension, we may recursively define $W^{0,p}(J) := L^p(J)$ and, for $k \geq 1$, $W^{k,p}(J)$ as the set of functions $f \in W^{k-1,p}(J)$ such that $\partial_x f \in W^{k-1,p}(J)$.

We identify a.e. equal functions into equivalence classes in $W^{k,p}(\Omega)$, and we write “ $f = g$ a.e.” and “ $f = g$ in $W^{k,p}(\Omega)$ ” interchangeably. Often, we are interested in a continuous representative of f , which is unique when it exists, and by abuse of notation write f for its continuous representative as well. With this interpretation, a standard fact is that $W^{1,\infty}(\Omega)$ is precisely the class of Lipschitz functions on Ω . If $1 \leq p \leq q \leq +\infty$, then $W^{k,q}(\Omega) \subset W^{k,p}(\Omega)$.

The case $p = 2$ is special because then $H^k(\Omega) := W^{k,2}(\Omega)$ is a Hilbert space. In one dimension, the inner product in $H^k(J)$ is

$$\langle f, g \rangle_{H^k(J)} := \sum_{i=0}^k \langle D^i f, D^i g \rangle_{L^2(J)} ,$$

where D^i above denotes the i -th weak derivative. When the space of integration is clear, we drop the subscript from the inner product notation. The inner product above also induces the norm

$$\|f\|_{H^k(J)} = \sqrt{\sum_{i=0}^k \|D^i f\|_{L^2(J)}^2}.$$

For a sequence $(f_n)_{n \in \mathbb{N}} \subset H^k(\Omega)$ and $f \in H^k(\Omega)$, the notation “ $f_n \rightarrow f$ in $H^k(\Omega)$ ” means convergence in norm, i.e. $\|f_n - f\|_{H^k(\Omega)} \rightarrow 0$ as $n \rightarrow \infty$. We write “ $f_n \rightharpoonup f$ weakly in $H^k(\Omega)$ ” to denote weak convergence in this space, which means that $\varphi(f_n) \rightarrow \varphi(f)$ as $n \rightarrow \infty$ for every φ in the dual space of $H^k(\Omega)$. The dual space of $L^2(\Omega)$ is $L^2(\Omega)$ itself with the $L^2(\Omega)$ inner product action $\varphi(f) = \langle \varphi, f \rangle$. The dual space of $H^1(\Omega)$ is larger, but contains $L^2(\Omega)$ with the same action.

It is often useful to approximate a function by a sequence of smooth functions, and for that we will use *mollification*. The following comes from [Eva10, Appendix C]. The *standard mollifier* $\eta \in C^\infty(\mathbb{R}^d)$ is given by

$$\eta(x) := \begin{cases} C \exp\left(\frac{1}{|x|^2-1}\right) & \text{if } |x| < 1 \\ 0 & \text{if } |x| \geq 1, \end{cases}$$

for constant $C > 0$ chosen so that $\int_{\mathbb{R}^d} \eta \, dx = 1$. Then for each $\varepsilon > 0$, we let

$$\eta_\varepsilon(x) := \frac{1}{\varepsilon^d} \eta(x/\varepsilon),$$

which is a $C^\infty(\mathbb{R}^d)$ function satisfying $\int_{\mathbb{R}^d} \eta_\varepsilon \, dx = 1$ and $\eta_\varepsilon(x) = 0$ for $x \notin B(0, \varepsilon)$, where $B(0, \varepsilon)$ is the open ball of radius ε centered at 0. We abuse language and also call η_ε a standard mollifier.

Let $U \subset \mathbb{R}^d$ be an open set and let $U_\varepsilon := \{x \in U : \text{dist}(x, \partial U) > \varepsilon\}$, where ∂U is the boundary of U . For locally integrable $f : U \rightarrow \mathbb{R}$ (meaning that f is integrable on every compact $K \subset U$), we define the *mollification* $f^\varepsilon : U_\varepsilon \rightarrow \mathbb{R}$ of f by

$$f^\varepsilon(x) := (\eta_\varepsilon * f)(x) = \int_U \eta_\varepsilon(x - y) f(y) \, dy = \int_{B(0, \varepsilon)} \eta(y) f(x - y) \, dy.$$

We then have the following properties (see Theorem 7 of [Eva10, Appendix C]):

1. $f^\varepsilon \in C^\infty(U_\varepsilon)$.

2. $f^\varepsilon \rightarrow f$ a.e. as $\varepsilon \rightarrow 0$.
3. If $f \in C(U)$, then $f^\varepsilon \rightarrow f$ uniformly on compact subsets of U .
4. If $1 \leq p < \infty$ and $f \in L^p_{\text{loc}}(U)$, then $f^\varepsilon \rightarrow f$ in $L^p_{\text{loc}}(U)$,

where $L^p_{\text{loc}}(U)$ is the space of measurable functions whose every restriction to compact $K \subset U$ is in $L^p(K)$, and convergence in $L^p_{\text{loc}}(U)$ means convergence in $L^p(K)$ for every compact $K \subset U$.

We also use the following abstract spaces. Let X be a real Banach space (for us, typically $L^2(J)$), let $1 \leq p \leq +\infty$ and let $T \in (0, +\infty]$. Then the *Bochner space* $L^p(0, T; X)$ is the space of Bochner measurable functions (whose precise definition is not important here) $\mathbf{u} : [0, T] \rightarrow X$ whose norm $\|\mathbf{u}\|_{L^p(0, T; X)}$ is finite, where

$$\|\mathbf{u}\|_{L^p(0, T; X)} := \begin{cases} \left(\int_{(0, T)} \|\mathbf{u}(t)\|_X^p dt \right)^{1/p} & 1 \leq p < +\infty \\ \text{ess sup}_{t \in (0, T)} \|\mathbf{u}(t)\|_X & p = +\infty. \end{cases}$$

As usual, $C([0, T]; X)$ denotes the set of continuous functions $\mathbf{u} : [0, T] \rightarrow X$. When X possesses the so-called Radon-Nikodym property, which in particular is the case for $X = L^2(J)$, and for $T < +\infty$, the characterization of absolutely continuous functions as those possessing an integrable weak derivative introduced earlier extends to functions $\mathbf{u} : [0, T] \rightarrow X$ (see [DU77, pp. 217–219]).

Above and hereafter, we use boldface symbols, e.g. $\mathbf{u}$, to denote Banach-valued functions whose domain we think of as the time in an evolution equation.

Hereafter, we write I for the unit interval $(0, 1)$ unless otherwise specified, which will always be clearly indicated.

7.2.2 Neumann and gradient flow problems

The next definition captures the “natural” class of functions we need to handle in our PDE study; very informally, those whose “decreasing regions are differentiable”.

Definition 7.5 (Down- H^1 functions). We define the set $\mathcal{U}$ of *down- H^1 functions* as

$$\mathcal{U} := \{u \in L^2(I) : \mathcal{R}(u) \neq \emptyset\},$$

where $\mathcal{R}(u)$ is the set of admissible *representations* for u as follows:

$$\mathcal{R}(u) := \left\{ (u^{(1)}, u^{(2)}) \in L^2(I) \times H^1(I) : \right. \\ \left. \begin{aligned} &u^{(1)} \text{ is nondecreasing, } u^{(2)} \text{ is nonincreasing,} \\ &\int_I u^{(2)} \, dx = 0, \text{ and } u = u^{(1)} + u^{(2)} \text{ a.e.} \end{aligned} \right\}.$$

Remark 7.6. The properties “nondecreasing” and “nonincreasing” in the definition above technically do not apply directly to members of $L^2(I)$ and $H^1(I)$, which are equivalence classes of a.e. equal functions. Rather, we mean that $u^{(1)}$ is a member of $L^2(I)$ which has a representative function $I \rightarrow \mathbb{R}$ that is nondecreasing, and likewise for $u^{(2)}$. More generally, when f is an object of any space that identifies a.e. equal functions, we abuse language and write “ f is nonincreasing” (resp. nondecreasing) to mean that f has a nonincreasing (resp. nondecreasing) representative.

Remark 7.7. It is easy to check that $H^1(I) \subset \mathcal{U}$, since for any $u \in H^1(I)$ we may separate the positive and negative parts of $\partial_x u$ into $u^{(1)}$ and $u^{(2)}$, respectively, and shift $u^{(1)}$ and $u^{(2)}$ by appropriate constants so that they satisfy the conditions of [Definition 7.5](#). Then, since $H^1(I)$ is dense in $L^2(I)$, so is $\mathcal{U}$.

Definition 7.8 (Directed Dirichlet energy). For each $u \in \mathcal{U}$ and $(u^{(1)}, u^{(2)}) \in \mathcal{R}(u)$, define

$$\mathcal{D}^-(u^{(1)}, u^{(2)}) := \frac{1}{2} \int_I (\partial_x u^{(2)})^2 \, dx.$$

Definition 7.9 (Optimal representations). For each $u \in \mathcal{U}$, define $\mathcal{R}^*(u) \subseteq \mathcal{R}(u)$ as the set of representations minimizing $\mathcal{D}^-$, i.e.

$$\mathcal{R}^*(u) := \left\{ (u^{(1)}, u^{(2)}) \in \mathcal{R}(u) : \mathcal{D}^-(u^{(1)}, u^{(2)}) = \inf_{\mathcal{R}(u)} \mathcal{D}^- \right\}.$$

Proposition 7.10 (Existence and uniqueness of optimal representation). *For all $u \in \mathcal{U}$, $\mathcal{R}^*(u)$ contains exactly one element.*

Proof. See [Appendix B](#). □

Owing to [Proposition 7.10](#), we may define for each $u \in \mathcal{U}$ a canonical representation and directed Dirichlet energy (by a slight abuse of notation):

Definition 7.11 (Canonical representation). For each $u \in \mathcal{U}$, the *canonical representation* of u is the unique element of $\mathcal{R}^*(u)$, denoted $(u\uparrow, u\downarrow)$. We define the *directed Dirichlet energy* of u by $\mathcal{D}^-(u) := \mathcal{D}^-(u\uparrow, u\downarrow)$.

The following definition, which extends the directed Dirichlet energy functional to all of $L^2(I)$ by assigning $+\infty$ to functions outside of $\mathcal{U}$, will enable us to find and study a solution to our PDE using tools from the theory of maximal monotone operators and gradient flows. Our main reference for this theory is [Gre73].

Definition 7.12 (Energy functional). Define the functional $\mathcal{E}^- : L^2(I) \rightarrow [0, +\infty]$ by

$$\mathcal{E}^-(u) := \begin{cases} \mathcal{D}^-(u) & \text{if } u \in \mathcal{U} \\ +\infty & \text{otherwise.} \end{cases}$$

Proposition 7.13. *The functional $\mathcal{E}^- : L^2(I) \rightarrow [0, +\infty]$ is convex, proper and lower semicontinuous.*

Proof. See [Appendix B](#). □

Definition 7.14 (Static Neumann problem). Let $u \in \mathcal{U}$ and $z \in L^2(I)$. We say that u, z form a *weak solution* to the *static Neumann problem*

$$\begin{cases} z = \partial_x \partial_x u \downarrow & \text{in } I \\ \partial_x u \downarrow = 0 & \text{on } \{0, 1\} \end{cases} \quad (7.9)$$

if for all $\phi \in H^1(I)$ we have

$$\int_I z \phi \, dx = - \int_I (\partial_x u \downarrow) (\partial_x \phi) \, dx. \quad (7.10)$$

Remark 7.15. The equation $z = \partial_x \partial_x u \downarrow$ in (7.9) is also called the Poisson equation, while $\partial_x u \downarrow = 0$ on $\{0, 1\}$ is called the (homogeneous) Neumann boundary condition.

Definition 7.16 (Nice evolution function). Let $\mathbf{u} \in C([0, +\infty); L^2(I))$ and $\mathbf{u}' : (0, +\infty) \rightarrow L^2(I)$. We say that $\mathbf{u}$ is a *nice evolution function* (with weak derivative $\mathbf{u}'$) if 1) $\mathbf{u}(t) = \mathbf{u}(0) + \int_{(0,t)} \mathbf{u}'(s) \, ds$ for all $t > 0$; 2) $\mathbf{u}'_{(\delta, +\infty)} \in L^\infty(\delta, +\infty; L^2(I))$ for all $\delta > 0$, where $\mathbf{u}'_J$ denotes the restriction of $\mathbf{u}'$ to domain $J \subseteq (0, +\infty)$; and 3) $\mathbf{u}'_{(0, \delta)} \in L^2(0, \delta; L^2(I))$ for all $\delta > 0$.

Remark 7.17. [Definition 7.16](#) captures functions that are Lipschitz away from zero, and absolutely continuous with square-integrable weak derivative near zero.

Definition 7.18 (Neumann evolution problem). Let $u_0 \in \mathcal{U}$ and let $\mathbf{u} \in C([0, +\infty); L^2(I))$ be a nice evolution function. We say $\mathbf{u}$ is a *weak solution* to the *Neumann evolution problem* with initial state (or initial data) u_0 if 1) $\mathbf{u}(0) = u_0$; 2) $\mathbf{u}(t) \in \mathcal{U}$ for all $t > 0$; and 3) $\mathbf{u}(t), \mathbf{u}'(t)$ form a weak solution to the static Neumann problem for a.e. $t > 0$.

Before introducing the gradient flow problem, we need some notation. Let $\varphi : L^2(I) \rightarrow [0, +\infty]$ be a convex, proper, lower semicontinuous function. We write $D(\varphi) := \{u \in L^2(I) : \varphi(u) < +\infty\}$ for the *domain* of φ . For any $u \in L^2(I)$, we define the *subdifferential* of φ at u as

$$\partial\varphi(u) := \{z \in L^2(I) : \forall v \in L^2(I) . \varphi(v) \geq \varphi(u) + \langle z, v - u \rangle\},$$

and we write $D(\partial\varphi) := \{u \in L^2(I) : \partial\varphi(u) \neq \emptyset\}$ for the *domain* of $\partial\varphi$. Using the fact that φ is proper, it is easy to check that $D(\partial\varphi) \subseteq D(\varphi)$. It is standard that $\partial\varphi$ is a *maximal monotone operator*; we will not use (the definition of) this property explicitly, but rather rely on the theory of such operators as presented in [\[Bre73\]](#).

We can now define the gradient flow problem:

Definition 7.19 (Gradient flow problem). Let $u_0 \in \mathcal{U}$ and let $\mathbf{u}$ be a nice evolution function. We say $\mathbf{u}$ is a *solution* to the *gradient flow problem* with initial state (or initial data) u_0 if 1) $\mathbf{u}(0) = u_0$; 2) $\mathbf{u}(t) \in D(\partial\mathcal{E}^-)$ for all $t > 0$; and 3) $\mathbf{u}'(t) \in -\partial\mathcal{E}^-(\mathbf{u}(t))$ for a.e. $t > 0$.

Note that in both [Definitions 7.18](#) and [7.19](#), the pointwise condition at $t = 0$ makes sense by the requirement that $\mathbf{u}$ be continuous.

The theory of maximal monotone operators and gradient flows immediately yields that the gradient flow problem has a unique solution, as follows:

Proposition 7.20. *Let $u_0 \in \mathcal{U}$. Then there exists a unique solution $\mathbf{u} \in C([0, +\infty); L^2(I))$ to the gradient flow problem. Moreover, for all $t > 0$ we have*

$$\mathcal{E}^-(u_0) - \mathcal{E}^-(\mathbf{u}(t)) = \int_{(0,t)} \|\mathbf{u}'(s)\|_{L^2(I)}^2 ds.$$

For $u_0, v_0 \in \mathcal{U}$, the corresponding solutions $\mathbf{u}, \mathbf{v}$ have $\|\mathbf{u}(t) - \mathbf{v}(t)\|_{L^2(I)} \leq \|u_0 - v_0\|_{L^2(I)}$ for all $t > 0$.

Proof. This is a direct application of [Bre73, Theorems 3.1 and 3.2 and Proposition 3.1]. $\square$

Directed heat semigroup. As described in [Bre73], the nonexpansive property of solutions $\mathbf{u}$ allows us to define an operator $P_t : L^2(I) \rightarrow L^2(I)$ such that $P_t u = \mathbf{u}(t)$ for each initial state $u \in \mathcal{U}$ with corresponding solution $\mathbf{u}$, and with $P_t u$ defined by continuous extension when $u \in L^2(I) \setminus \mathcal{U}$ (recall that $\mathcal{U}$ is dense in $L^2(I)$). Then for every $u \in L^2(I)$ and $t > 0$ it holds that $P_t u \in D(\partial\mathcal{E}^-)$, and $(P_t)_{t \geq 0}$ forms a nonexpansive semigroup in $L^2(I)$, i.e.

1. $P_{t+s} = P_t P_s$ for all $t, s \in \mathbb{R}_{\geq 0}$, with P_0 the identity;
2. $P_t u \rightarrow u$ in $L^2(I)$ as $t \downarrow 0$, for all $u \in L^2(I)$; and
3. $\|P_t u - P_t v\|_{L^2(I)} \leq \|u - v\|_{L^2(I)}$ for all $u, v \in L^2(I)$ and $t \in \mathbb{R}_{\geq 0}$.

The last property also implies that, for each $t \geq 0$, $P_t : L^2(I) \rightarrow L^2(I)$ is continuous. We say that P_t is the semigroup *generated* by $-\partial\mathcal{E}^-$.

It turns out that the solution to the gradient flow problem is also a weak solution to the Neumann evolution problem, as the following results show.

Proposition 7.21. *Let $u \in D(\partial\mathcal{E}^-)$ and $z \in -\partial\mathcal{E}^-(u)$. Then u, z form a weak solution to the static Neumann problem.*

Proof. Note that we have $u \in D(\partial\mathcal{E}^-) \subseteq D(\mathcal{E}^-) = \mathcal{U}$. Let $\phi \in H^1(I)$; we will prove that (7.10) holds. Let $\alpha \neq 0$, let $\psi_\alpha := \alpha\phi$ and let $v_\alpha := u + \psi_\alpha$. Since $-z \in \partial\mathcal{E}^-(u)$, we have

$$\mathcal{E}^-(v_\alpha) \geq \mathcal{E}^-(u) + \langle -z, v_\alpha - u \rangle = \mathcal{E}^-(u) + \langle -z, \psi_\alpha \rangle. \quad (7.11)$$

Now, the fact that $u \in \mathcal{U}$ implies that $\mathcal{E}^-(u) = \mathcal{D}^-(u)$. We now claim that $v_\alpha \in \mathcal{U}$ as well, which will imply that $\mathcal{E}^-(v_\alpha) = \mathcal{D}^-(v_\alpha)$. In fact, we establish the following:

Claim 7.22. *We have $v_\alpha \in \mathcal{U}$. Moreover, $\mathcal{D}^-(v_\alpha) \leq \frac{1}{2} \int_I (\partial_x u \downarrow + \partial_x \psi_\alpha)^2 dx$.*

Proof. We define a nondecreasing function $v^{(1)} \in L^2(I)$ and a nonincreasing function $v^{(2)} \in H^1(I)$ as follows: for each $x \in I$,

$$\begin{aligned} v^{(1)}(x) &:= u \uparrow(x) + \psi_\alpha(0) + \int_{(0,x)} (\partial_x \psi_\alpha(y) - |\partial_x u \downarrow(y)|)^+ dy \quad \text{and} \\ v^{(2)}(x) &:= u \downarrow(x) + \int_{(0,x)} (\partial_x \psi_\alpha(y) \wedge |\partial_x u \downarrow(y)|) dy. \end{aligned}$$

It is clear that $v^{(1)}, v^{(2)} \in L^2(I)$ with $v^{(1)}$ nondecreasing. Since $u \downarrow \in H^1(I)$ by definition of $\mathcal{R}(u)$, while $\psi_\alpha \in H^1(I)$ because $\phi \in H^1(I)$, we also obtain that $v^{(2)} \in H^1(I)$. To see that $v^{(2)}$ is nonincreasing, note that for a.e. $x \in I$ we have

$$\partial_x v^{(2)}(x) = \partial_x u \downarrow(x) + (\partial_x \psi_\alpha(x) \wedge |\partial_x u \downarrow(x)|) \leq 0.$$

Observe also that $v^{(1)} + v^{(2)} = u + \psi_\alpha = v_\alpha$. Finally, by translating $v^{(1)}$ and $v^{(2)}$ by a constant if necessary, we can ensure that $\int_I v^{(2)} dx = 0$ without invalidating the other properties. Thus $v_\alpha \in \mathcal{U}$.

To show the second part of the claim, we note that for a.e. $x \in I$,

$$\left| \underbrace{\partial_x v^{(2)}(x)}_{\leq 0} \right| = \left| \underbrace{\partial_x u \downarrow(x)}_{\leq 0} + \underbrace{(\partial_x \psi_\alpha(x) \wedge |\partial_x u \downarrow(x)|)}_{\leq |\partial_x u \downarrow(x)|} \right| \leq \left| \partial_x u \downarrow(x) + \partial_x \psi_\alpha(x) \right|,$$

where the inequality follows from inspecting the cases $\partial_x \psi_\alpha(x) \leq |\partial_x u \downarrow(x)|$ and $\partial_x \psi_\alpha(x) > |\partial_x u \downarrow(x)|$. We conclude that

$$\mathcal{D}^-(v_\alpha) \leq \mathcal{D}^-(v^{(1)}, v^{(2)}) = \frac{1}{2} \int_I (\partial_x v^{(2)})^2 dx \leq \frac{1}{2} \int_I (\partial_x u \downarrow(x) + \partial_x \psi_\alpha(x))^2 dx. \quad \square$$

Now, putting together (7.11) and the claim, we get

$$\frac{1}{2} \int_I (\partial_x u \downarrow)^2 dx - \int_I z \psi_\alpha dx \leq \mathcal{D}^-(v_\alpha) \leq \frac{1}{2} \int_I (\partial_x u \downarrow + \partial_x \psi_\alpha)^2 dx.$$

Simplifying and substituting $\psi_\alpha = \alpha \phi$, we obtain

$$\alpha \int_I z \phi dx \geq -\frac{1}{2} \alpha^2 \int_I (\partial_x \phi)^2 dx - \alpha \int_I (\partial_x u \downarrow)(\partial_x \phi) dx.$$

Taking the limits $\alpha \rightarrow 0^+$ and $\alpha \rightarrow 0^-$, we conclude that

$$\int_I z \phi dx = - \int_I (\partial_x u \downarrow)(\partial_x \phi) dx,$$

which is (7.10) as needed. $\square$

Corollary 7.23. *Let $u_0 \in \mathcal{U}$ and suppose $\mathbf{u}$ is a solution to the gradient flow problem. Then $\mathbf{u}$ is also a weak solution to the Neumann evolution problem.*

Proof. The condition $\mathbf{u}(0) = u_0$ holds by definition of solution to the gradient flow problem, and for all $t > 0$ we have $\mathbf{u}(t) \in D(\partial \mathcal{E}^-) \subseteq D(\mathcal{E}^-) = \mathcal{U}$. Finally, for a.e. $t > 0$ we have $\mathbf{u}'(t) \in -\partial \mathcal{E}^-(\mathbf{u}(t))$, which by Proposition 7.21 implies that $\mathbf{u}(t), \mathbf{u}'(t)$ form a solution to the static Neumann problem. Hence $\mathbf{u}$ is a weak solution to the Neumann evolution problem. $\square$

7.2.3 Exponential decay of directed Dirichlet energy

The following elliptic regularity result is standard, and essentially shows that weak solutions to the static Neumann problem are in fact strong solutions whose regularity is two degrees higher than that of z ; in particular, even if we only have $z \in L^2(I)$, we gain one degree of regularity by obtaining $u \downarrow \in H^2(I)$ when we only assumed $u \downarrow \in H^1(I)$. Recall that $H^0(I)$ is the same as $L^2(I)$.

Lemma 7.24 (Elliptic regularity; see [Mik78, Chapter IV Section 2, Theorems 3 and 4]). *Let $k \geq 0$ be an integer and let $u \in \mathcal{U}, z \in H^k(I)$ form a weak solution to the static Neumann problem*

$$\begin{cases} z = \partial_x \partial_x u \downarrow & \text{in } I \\ \partial_x u \downarrow = 0 & \text{on } \{0, 1\}. \end{cases}$$

Then $u \downarrow \in H^{k+2}(I)$, and moreover $\partial_x u \downarrow = 0$ on $\{0, 1\}$ and $z = \partial_x \partial_x u \downarrow$ a.e. in I .

We now give our energy decay result.

Proposition 7.25. *There exists a constant $K > 0$ such that the following holds. Let $u \in \mathcal{U}$. Then for all $t > 0$,*

$$\mathcal{E}^-(P_t u) \leq e^{-Kt} \mathcal{E}^-(u).$$

Proof. Let $u \in \mathcal{U}$ and let $\mathbf{u}(t) = P_t u$ be the corresponding solution to the gradient flow problem. Recall that Proposition 7.20 gives, for all $t > 0$,

$$\mathcal{E}^-(\mathbf{u}(t)) = \mathcal{E}^-(u) + \int_{(0,t)} -\|\mathbf{u}'(s)\|_{L^2(I)}^2 ds.$$

It follows that $t \mapsto \mathcal{E}^-(\mathbf{u}(t))$ is absolutely continuous on every interval $[0, T]$ with weak derivative $\partial_t \mathcal{E}^-(\mathbf{u}(t)) = -\|\mathbf{u}'(t)\|_{L^2(I)}^2$. Moreover for a.e. $t > 0$ we have that $\mathbf{u}(t), \mathbf{u}'(t)$ form a weak solution to the static Neumann problem, and thus $\mathbf{u}'(t) = \partial_x \partial_x \mathbf{u}(t) \downarrow$ a.e. in I and $\partial_x \mathbf{u}(t) \downarrow = 0$ on $\{0, 1\}$ by Lemma 7.24. Therefore the Poincaré inequality (for zero-on-the-boundary functions) yields

$$\mathcal{E}^-(\mathbf{u}(t)) = \frac{1}{2} \|\partial_x \mathbf{u}(t) \downarrow\|_{L^2(I)}^2 \leq \frac{1}{2} C \|\partial_x \partial_x \mathbf{u}(t) \downarrow\|_{L^2(I)}^2 = \frac{C}{2} \|\mathbf{u}'(t)\|_{L^2(I)}^2$$

for some constant $C > 0$. It follows that for a.e. $t > 0$, we have

$$\partial_t \mathcal{E}^-(\mathbf{u}(t)) = -\|\mathbf{u}'(t)\|_{L^2(I)}^2 \leq -\frac{2}{C} \mathcal{E}^-(\mathbf{u}(t)),$$

which implies that for all $t > 0$,

$$\mathcal{E}^-(\mathbf{u}(t)) \leq e^{-2t/C} \mathcal{E}^-(u). \quad \square$$

The exponential decay of $t \mapsto \mathcal{E}^-(P_t u)$ allows us to find a Cauchy sequence in $(P_t u)_{t \geq 0}$, and thus establish its strong convergence to some limit in $L^2(I)$. Later on, we will say more about this limit by reasoning about the weak convergence of $P_t u$ (to the same limit).

Lemma 7.26 (Cauchy sequence). *Let $u \in \mathcal{U}$, and let $(u_k)_{k \in \mathbb{N}}$ be given by $u_k := P_k u$. Then $(u_k)_k$ is Cauchy as a sequence in $L^2(I)$. As a consequence, $u_k \rightarrow u^*$ in $L^2(I)$ for some $u^* \in L^2(I)$.*

Proof. The existence of a strong limit from the Cauchy property follows from the fact that $L^2(I)$ is a complete normed space. Let us now establish the Cauchy property. Fix any $n > m$ in $\mathbb{N}$. Then

$$\begin{aligned} \|u_n - u_m\|_{L^2(I)} &= \left\| \int_{(m,n)} \mathbf{u}'(t) dt \right\|_{L^2(I)} && \text{(Absolute continuity)} \\ &\leq \sum_{j=m}^{n-1} \int_{(j,j+1)} \|\mathbf{u}'(t)\|_{L^2(I)} dt && \text{(Triangle inequality)} \\ &\leq \sum_{j=m}^{n-1} \left(\int_{(j,j+1)} \|\mathbf{u}'(t)\|_{L^2(I)}^2 dt \right)^{1/2} && \text{(Jensen's inequality)} \\ &= \sum_{j=m}^{n-1} (\mathcal{E}^-(u_j) - \mathcal{E}^-(u_{j+1}))^{1/2} && \text{(Proposition 7.20)} \\ &\leq \sum_{j=m}^{\infty} \mathcal{E}^-(u_j)^{1/2} && (\mathcal{E}^- \text{ is nonnegative}) \\ &\leq \sum_{j=m}^{\infty} (e^{-Kj} \mathcal{E}^-(u))^{1/2} && \text{(Proposition 7.25)} \\ &= \mathcal{E}^-(u)^{1/2} \frac{e^{-Km/2}}{1 - e^{-K/2}} && \text{(Geometric series)} \\ &= A \mathcal{E}^-(u)^{1/2} e^{-Bm} \end{aligned}$$

for constants $A, B > 0$ that only depend on the constant K from [Proposition 7.25](#). It follows that, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for all $n > m > N$,

$$\|u_n - u_m\|_{L^2(I)} \leq A \mathcal{E}^-(u)^{1/2} e^{-Bm} \leq A \mathcal{E}^-(u)^{1/2} e^{-BN} < \varepsilon,$$

so $(u_k)_k$ is Cauchy. $\square$

Proposition 7.27 (Strong convergence). *Let $u \in \mathcal{U}$. Then $P_t u$ converges in $L^2(I)$ to some $u^* \in L^2(I)$ as $t \rightarrow \infty$.*

Proof. Let u^* be such that $(P_k u)_{k \in \mathbb{N}}$ converges to u^* in $L^2(I)$ as $k \rightarrow \infty$, as given by [Lemma 7.26](#). We claim that $P_t u \rightarrow u^*$. Let $\varepsilon > 0$, and let $N \in \mathbb{N}$ be such that $\|P_k u - u^*\|_{L^2(I)} < \varepsilon$ for all $k \geq N$, as given by the convergence of the sequence $(P_k u)_k$. Then for any $t \geq N$, letting $j := \lfloor t \rfloor$, we have

$$\|P_t u - u^*\|_{L^2(I)} \leq \|P_j u - u^*\|_{L^2(I)} + \|P_t u - P_j u\|_{L^2(I)}$$

by the triangle inequality. We have $\|P_j u - u^*\|_{L^2(I)} < \varepsilon$ by the choice of N , and on the other hand,

$$\begin{aligned} \|P_t u - P_j u\|_{L^2(I)} &= \left\| \int_{(j,t)} \mathbf{u}'(s) \, ds \right\|_{L^2(I)} && \text{(Absolute continuity)} \\ &\leq \int_{(j,t)} \|\mathbf{u}'(s)\|_{L^2(I)} \, ds && \text{(Triangle inequality)} \\ &\leq (t-j)^{1/2} \left(\int_{(j,t)} \|\mathbf{u}'(s)\|_{L^2(I)}^2 \, ds \right)^{1/2} && \text{(Jensen's inequality)} \\ &\leq (\mathcal{E}^-(P_j u) - \mathcal{E}^-(P_t u))^{1/2} && (t-j \in [0, 1), \text{ [Proposition 7.20](#)}) \\ &\leq e^{-Kj/2} \mathcal{E}^-(u)^{1/2} && (\mathcal{E}^- \geq 0, \text{ [Proposition 7.25](#)}). \end{aligned}$$

Thus by letting N be large enough, we can ensure that

$$\|P_t u - P_j u\|_{L^2(I)} \leq e^{-Kj/2} \mathcal{E}^-(u)^{1/2} \leq e^{-KN/2} \mathcal{E}^-(u)^{1/2} < \varepsilon,$$

and hence $\|P_t u - u^*\|_{L^2(I)} < 2\varepsilon$. Thus $P_t u \rightarrow u^*$ in $L^2(I)$. $\square$

7.2.4 Auxiliary results for studying regularity of solutions

The following lemma establishes the intuitive property that, if $\partial_x u \downarrow < 0$ in a certain region, then $u \uparrow$ is constant in that region, i.e. u has no increasing component.

Lemma 7.28 (Functions in $D(\partial \mathcal{E}^-)$ are well-behaved). *Suppose $u \in D(\partial \mathcal{E}^-)$. Let $(a, b) \in I$ be a nonempty interval and suppose that $\partial_x u \downarrow(x) < 0$ for all $x \in (a, b)$. Then $u \uparrow$ is constant in (a, b) .*

Proof. Let $z \in L^2(I)$ be such that $-z \in \partial\mathcal{E}^-(u)$, which is nonempty by hypothesis. By [Proposition 7.21](#) and [Lemma 7.24](#), we conclude that $u\downarrow \in H^2(I)$ (which in particular justifies writing the condition that $\partial_x u\downarrow < 0$ on (a, b) , which we take to mean in terms of the continuous representative of $\partial_x u\downarrow \in H^1(I)$).

By the continuity of $\partial_x u\downarrow$ and the boundary conditions $\partial_x u\downarrow(0) = \partial_x u\downarrow(1) = 0$ (which follows from [Lemma 7.24](#)), we may assume without loss of generality (by extending the interval (a, b) if necessary) that $\partial_x u\downarrow(a) = \partial_x u\downarrow(b) = 0$.

Let

$$Z := \begin{cases} \inf_{[a,b]} u\uparrow & \text{if } a > 0 \\ \frac{1}{b-a} \int_{(a,b)} u\uparrow dx & \text{if } a = 0, \end{cases}$$

which is finite in the first case because the infimum must be no smaller than (say) $u\uparrow(\frac{a}{2})$, and in the second case because $u\uparrow \in L^2(I)$. Next, define $v \in L^2(I)$ by

$$v(x) := \begin{cases} u(x) & \text{if } x \in (0, a) \cup [b, 1) \\ u\downarrow(x) + Z & \text{if } x \in [a, b]. \end{cases}$$

We first observe that $v \in \mathcal{U}$, since $v = v^{(1)} + v^{(2)}$ with $v^{(2)} := u\downarrow$ and $v^{(1)} \in L^2(I)$ given by

$$v^{(1)}(x) := \begin{cases} u\uparrow(x) & \text{if } x \in (0, a) \cup [b, 1) \\ Z & \text{if } x \in [a, b], \end{cases}$$

which is nondecreasing because $u\uparrow$ is and by the definition of Z . Therefore $v \in D(\mathcal{E}^-)$ and

$$\mathcal{E}^-(v) \leq \mathcal{D}^-(v^{(1)}, v^{(2)}) = \frac{1}{2} \int_I (\partial_x v^{(2)})^2 dx = \frac{1}{2} \int_I (\partial_x u\downarrow)^2 dx = \mathcal{D}^-(u\uparrow, u\downarrow) = \mathcal{E}^-(u).$$

On the other hand, we claim that $\mathcal{E}^-(v) \geq \mathcal{E}^-(u)$ as well. Indeed, let $(w^{(1)}, w^{(2)}) \in \mathcal{R}(v)$ be arbitrary. Construct $u^{(1)} \in L^2(I)$ and $u^{(2)} \in H^1(I)$ by $u^{(2)} := w^{(2)}$ and

$$u^{(1)}(x) := \begin{cases} w^{(1)}(x) & \text{if } x \in (0, a) \cup [b, 1) \\ u(x) - u^{(2)}(x) & \text{if } x \in [a, b], \end{cases}$$

That $u = u^{(1)} + u^{(2)}$ is clear by construction. To show that $(u^{(1)}, u^{(2)}) \in \mathcal{R}(u)$, it remains to show that $u^{(1)}$ is nondecreasing. Certainly it is nondecreasing on $(0, a) \cup [b, 1)$ since $w^{(1)}$ is nondecreasing. Note that for $x \in [a, b]$, we have

$$u^{(1)}(x) = u(x) - u^{(2)}(x) - w^{(1)}(x) + w^{(1)}(x) = u(x) - v(x) + w^{(1)}(x) = u\uparrow(x) + w^{(1)}(x) - Z.$$

Thus $u^{(1)}$ is nondecreasing on $[a, b)$, since $u\uparrow$ and $w^{(1)}$ are. For $x \in (0, a)$ and $y \in [a, b)$, which only applies when $a > 0$, the inequality $u^{(1)}(x) \leq u^{(1)}(y)$ follows from the nondecreasing monotonicity of $w^{(1)}$ and the inequality $u\uparrow(y) \geq Z$, which holds by the (first case of the) definition of Z .

Now, we must show that $u^{(1)}(x) \leq u^{(1)}(y)$ if $x \in [a, b)$ and $y \in [b, 1)$. In fact, it suffices to consider $y \in (b, 1)$, because if the inequality holds in all such cases, then $u^{(1)}$ can be made monotone on all of I by possibly changing its value at b , which does not affect the a.e. equality $u = u^{(1)} + u^{(2)}$. Therefore let $x \in [a, b)$ and $y \in (b, 1)$, which in particular only applies when $b < 1$. Recalling that every monotone function has limits from the left and from the right at every point, the key observation is that

$$\begin{aligned} \lim_{s \rightarrow b^+} w^{(1)}(s) - \lim_{s \rightarrow b^-} w^{(1)}(s) &= \left[\lim_{s \rightarrow b^+} v(s) - w^{(2)}(s) \right] - \left[\lim_{s \rightarrow b^-} v(s) - w^{(2)}(s) \right] \\ &= \left[\lim_{s \rightarrow b^+} u\uparrow(s) + u\downarrow(s) \right] - w^{(2)}(b) - \left[\lim_{s \rightarrow b^-} u\downarrow(s) + Z \right] + w^{(2)}(b) \\ &= \lim_{s \rightarrow b^+} u\uparrow(s) - Z, \end{aligned}$$

where we used the fact that $(w^{(1)}, w^{(2)}) \in \mathcal{R}(v)$ in the first equality, the definition of v and continuity of $w^{(2)}$ in the second equality, and the continuity of $u\downarrow$ in the third equality. Now, we have

$$\begin{aligned} u^{(1)}(x) \leq u^{(1)}(y) &\iff u\uparrow(x) + u\downarrow(x) - w^{(2)}(x) + Z - Z \leq w^{(1)}(y) \\ &\iff u\uparrow(x) + w^{(1)}(x) - Z \leq w^{(1)}(y) \\ &\iff u\uparrow(x) + w^{(1)}(x) - Z \leq \lim_{s \rightarrow b^+} w^{(1)}(s) \\ &\iff u\uparrow(x) + w^{(1)}(x) - Z \leq \lim_{s \rightarrow b^-} w^{(1)}(s) + \lim_{s \rightarrow b^+} u\uparrow(s) - Z \\ &\iff u\uparrow(x) \leq \lim_{s \rightarrow b^+} u\uparrow(s) \quad \text{and} \quad w^{(1)}(x) \leq \lim_{s \rightarrow b^-} w^{(1)}(s), \end{aligned}$$

which is true because $u\uparrow$ and $w^{(1)}$ are nondecreasing and $x < b$. This establishes that $u^{(1)}$ is nondecreasing and thus $(u^{(1)}, u^{(2)}) \in \mathcal{R}(u)$. It follows that

$$\mathcal{E}^-(u) \leq \mathcal{D}^-(u^{(1)}, u^{(2)}) = \frac{1}{2} \int_I (\partial_x u^{(2)})^2 dx = \frac{1}{2} \int_I (\partial_x w^{(2)})^2 dx = \mathcal{D}^-(w^{(1)}, w^{(2)}),$$

and thus $\mathcal{E}^-(u) \leq \mathcal{E}^-(v)$ as claimed. Thus $\mathcal{E}^-(u) = \mathcal{E}^-(v)$, which in particular implies that $\mathcal{E}^-(v) = \mathcal{D}^-(v^{(1)}, v^{(2)})$ and hence $(v\uparrow, v\downarrow) = (v^{(1)}, v^{(2)})$.

By definition of subdifferential, the fact that $-z \in \partial \mathcal{E}^-(u)$ implies that

$$\frac{1}{2} \int_I (\partial_x v \downarrow)^2 dx = \mathcal{E}^-(v) \geq \mathcal{E}^-(u) + \langle -z, v - u \rangle = \frac{1}{2} \int_I (\partial_x u \downarrow)^2 dx - \langle z, v - u \rangle .$$

Since $v = u$ on $I \setminus [a, b]$ and $v \downarrow = u \downarrow$, we conclude that

$$0 \leq \langle z, v - u \rangle = \langle z, v - u \rangle_{L^2(a,b)} = \langle z, Z - u \uparrow \rangle_{L^2(a,b)} .$$

Now, since $z = \partial_x \partial_x u \downarrow$ a.e. and $\partial_x u \downarrow(a) = \partial_x u \downarrow(b) = 0$, we have

$$\langle z, Z \rangle_{L^2(a,b)} = Z \langle \partial_x \partial_x u \downarrow, 1 \rangle_{L^2(a,b)} = Z(\partial_x u \downarrow(b) - \partial_x u \downarrow(a)) = 0 ,$$

and hence

$$\langle z, u \uparrow \rangle_{L^2(a,b)} \leq 0 .$$

Now, suppose for a contradiction that $u \uparrow$ is not constant in (a, b) . Then since $u \uparrow$ is nondecreasing, there must exist a', b' with $a < a' < b' < b$ and $u \uparrow(a') < u \uparrow(b')$, i.e. $\delta := u \uparrow(b') - u \uparrow(a') > 0$. Let $\alpha := -\sup_{(a', b')} \partial_x u \downarrow$, and note that $\alpha > 0$ by the extreme value theorem together with the continuity of $\partial_x u \downarrow$ and the fact that $\partial_x u \downarrow < 0$ in $[a', b']$. Then [Lemma 7.29](#) below, applied to $f = u \uparrow$, $g = \partial_x u \downarrow$ and $a', b' \in J = (a, b)$, implies that $\langle z, u \uparrow \rangle_{L^2(a,b)} = \langle \partial_x \partial_x u \downarrow, u \uparrow \rangle_{L^2(a,b)} \geq \alpha \delta > 0$, which is the desired contradiction. $\square$

We relied on the following technical lemma, which intuitively says that if f is nondecreasing on an interval and $g \leq 0$ attains 0 at the boundaries of that interval, then f and $\partial_x g$ are positively correlated.

Lemma 7.29. *Let $J \subset \mathbb{R}$ be a finite, nonempty open interval. Let $f \in L^2(J)$ be nondecreasing and let $g \in H^1(J)$ be such that $g < 0$ in J and $g = 0$ on ∂J , i.e. the endpoints of J . Let $[a, b] \subset J$, let $\delta := f(b) - f(a)$ and let $\alpha := -\sup_{(a,b)} g$. Then $\langle f, \partial_x g \rangle_{L^2(J)} \geq \alpha \delta$.*

Proof. See [Appendix B](#). $\square$

The following auxiliary result confirms the natural expectation that, if u has a first derivative and some degree of regularity, then its components $u \uparrow$ and $u \downarrow$ enjoy the same regularity and, moreover, can be obtained by taking the positive and negative parts of the derivative $\partial_x u$, respectively.

Proposition 7.30. *Let $1 \leq p \leq \infty$, and let $u \in \mathcal{U} \cap W^{1,p}(I)$. Then $u \uparrow, u \downarrow \in W^{1,p}(I)$ and, moreover, we have $\partial_x u \uparrow = \partial_x^+ u$ and $\partial_x u \downarrow = \partial_x^- u$ a.e. in I .*

Proof. Let $u^{(1)}, u^{(2)} : I \rightarrow \mathbb{R}$ be given by

$$u^{(1)}(y) := u(0) + C + \int_{(0,y)} \partial_x^+ u(x) \, dx \quad \text{and} \quad u^{(2)}(y) := -C + \int_{(0,y)} \partial_x^- u(x) \, dx$$

for each $y \in I$, where $C \in \mathbb{R}$ is implicitly defined so as to satisfy

$$\int_I u^{(2)} \, dx = 0.$$

Then $u^{(1)}$ is nondecreasing, $u^{(2)}$ is nonincreasing, both are absolutely continuous with $\partial_x u^{(1)} = \partial_x^+ u$ and $\partial_x u^{(2)} = \partial_x^- u$ a.e. in I , and

$$(u^{(1)} + u^{(2)})(y) = u(0) + \int_{(0,y)} (\partial_x^+ u(x) + \partial_x^- u(x)) \, dx = u(x)$$

for all $y \in I$ by the absolute continuity of u . It is also clear that $u^{(1)}, u^{(2)} \in L^\infty(I) \subset L^p(I)$, since they are pointwise upper bounded in magnitude by $|u(0)| + |C| + \|\partial_x u\|_{L^1(I)} < +\infty$. Moreover, since $u \in W^{1,p}(I)$, we have $\partial_x u \in L^p(I)$ and hence $\partial_x^+ u, \partial_x^- u \in L^p(I)$, yielding that $u^{(1)}, u^{(2)} \in W^{1,p}(I)$.

The proof will be concluded if we show that $(u^\uparrow, u^\downarrow) = (u^{(1)}, u^{(2)})$. However, at this point we cannot even state that $(u^{(1)}, u^{(2)}) \in \mathcal{R}(u)$ because we have not established that $u^{(2)} \in H^1(I) = W^{1,2}(I)$ (unless $p \geq 2$, of course). However, if we can show that any $(v^{(1)}, v^{(2)}) \in \mathcal{R}(u)$ satisfies

$$\int_I (\partial_x v^{(2)})^2 \, dx \stackrel{?}{\geq} \int_I (\partial_x u^{(2)})^2 \, dx, \quad (7.12)$$

then using the assumption that $u \in \mathcal{U}$ and the definition of $u^\uparrow, u^\downarrow$, we will conclude that indeed $(u^\uparrow, u^\downarrow) = (u^{(1)}, u^{(2)})$, as needed.

Let $(v^{(1)}, v^{(2)}) \in \mathcal{R}(u)$. We claim that $|\partial_x v^{(2)}(x)| \geq |\partial_x u^{(2)}(x)|$ for a.e. $x \in I$, which will imply (7.12). Recall that $v^{(2)} \in H^1(I) \subset W^{1,1}(I)$. Since $u = v^{(1)} + v^{(2)}$ a.e. and $u \in W^{1,p}(I) \subset W^{1,1}(I)$, we conclude that $v^{(1)} \in W^{1,1}(I)$. Hence $u, u^{(1)}, u^{(2)}, v^{(1)}, v^{(2)}$ are all absolutely continuous. The fundamental theorem of calculus for the Lebesgue integral implies that, almost everywhere in I , these functions are all differentiable, and their classical and weak derivatives agree with $\partial_x u = \partial_x u^{(1)} + \partial_x u^{(2)} = \partial_x v^{(1)} + \partial_x v^{(2)}$. On any such point x , the monotonicity of $u^{(1)}, u^{(2)}, v^{(1)}, v^{(2)}$ implies that $\partial_x u^{(1)}(x), \partial_x v^{(1)}(x) \geq 0$ and $\partial_x u^{(2)}(x), \partial_x v^{(2)}(x) \leq 0$. Therefore

$$\begin{aligned} |\partial_x v^{(2)}(x)| &= -\partial_x v^{(2)}(x) = -\partial_x u(x) + \partial_x v^{(1)}(x) = -\partial_x u^{(2)}(x) - \partial_x u^{(1)}(x) + \partial_x v^{(1)}(x) \\ &= |\partial_x u^{(2)}(x)| - \partial_x u^{(1)}(x) + \partial_x v^{(1)}(x). \end{aligned}$$

Now, if $\partial_x u(x) \geq 0$, then $\partial_x u^{(2)}(x) = 0$ by definition, so $|\partial_x v^{(2)}(x)| \geq |\partial_x u^{(2)}(x)|$ holds trivially. Otherwise, we conversely have $\partial_x u^{(1)}(x) = 0$ while $\partial_x v^{(1)} \geq 0$, and hence, by the above,

$$|\partial_x v^{(2)}(x)| \geq |\partial_x u^{(2)}(x)|,$$

which concludes the proof. $\square$

7.2.5 Preservation of H^1 regularity

We wish to show that if the initial state u is in $H^1(I)$, then $P_t u$ remains in $H^1(I)$ for all times $t > 0$. To that end, define $\varphi : L^2(I) \rightarrow [0, +\infty]$ by

$$\varphi(u) := \begin{cases} \|\partial_x u\|_{L^2(I)}^2 & \text{if } u \in H^1(I) \\ +\infty & \text{otherwise.} \end{cases}$$

The theory of maximal monotone operators gives us a recipe to establish that $t \mapsto \varphi(P_t u)$ is nonincreasing. The key ingredients are [Claim 7.31](#) and [Lemma 7.33](#).

Claim 7.31. *The functional φ is convex, proper and lower semicontinuous.*

Proof. Convexity and properness are straightforward; it remains to verify lower semicontinuity. Since $L^2(I)$ is a metric space, it suffices to check sequential lower semicontinuity. Let $(u_n)_{n \in \mathbb{N}}$ be a sequence in $L^2(I)$ such that $u_n \rightarrow u$ in $L^2(I)$. We need to show that

$$\varphi(u) \stackrel{?}{\leq} \liminf_{n \rightarrow \infty} \varphi(u_n).$$

The only relevant case is when the RHS above is finite, so suppose there exists a subsequence $(u_{n_k})_{k \in \mathbb{N}}$ such that $\lim_{k \rightarrow \infty} \varphi(u_{n_k}) = A < +\infty$. By extracting a subsequence if necessary, we may assume that $\varphi(u_{n_k}) < +\infty$, and thus $u_{n_k} \in H^1(I)$, for every k .

We claim that $(u_{n_k})_k$ is bounded in $H^1(I)$. Indeed suppose this is not the case. Then since $\|u_{n_k}\|_{H^1(I)}^2 = \|u_{n_k}\|_{L^2(I)}^2 + \|\partial_x u_{n_k}\|_{L^2(I)}^2$ and $\|\partial_x u_{n_k}\|_{L^2(I)}^2$ remains bounded due to the fact that $\varphi(u_{n_k}) \rightarrow A$, we conclude that $\|u_{n_k}\|_{L^2(I)}$ gets arbitrarily large as $k \rightarrow \infty$. But this contradicts the fact that $u_{n_k} \rightarrow u$ in $L^2(I)$, so the claim holds.

It follows that we may extract a weakly convergent subsequence $(u_{n_{k_\ell}})_{\ell \in \mathbb{N}}$, and by uniqueness of weak limits, we obtain that $u \in H^1(I)$ and $u_{n_{k_\ell}} \rightharpoonup u$ weakly in $H^1(I)$. By weak lower semicontinuity the norm in $H^1(I)$,

$$\|u\|_{L^2(I)}^2 + \|\partial_x u\|_{L^2(I)}^2 = \|u\|_{H^1(I)}^2 \leq \liminf_{\ell \rightarrow \infty} \|u_{n_{k_\ell}}\|_{H^1(I)}^2 = \liminf_{\ell \rightarrow \infty} \|u_{n_{k_\ell}}\|_{L^2(I)}^2 + \|\partial_x u_{n_{k_\ell}}\|_{L^2(I)}^2.$$

Since $u_{n_{k_\ell}} \rightarrow u$ in $L^2(I)$, we have $\|u_{n_{k_\ell}}\|_{L^2(I)}^2 \rightarrow \|u\|_{L^2(I)}^2$ and hence

$$\varphi(u) = \|\partial_x u\|_{L^2(I)}^2 \leq -\|u\|_{L^2(I)}^2 + \liminf_{\ell \rightarrow \infty} \|u_{n_{k_\ell}}\|_{L^2(I)}^2 + \|\partial_x u_{n_{k_\ell}}\|_{L^2(I)}^2 = \liminf_{\ell \rightarrow \infty} \varphi(u_{n_{k_\ell}}). \quad \square$$

We will need the following standard fact.

Fact 7.32 (See e.g. [EG15, Theorem 4.4]). *Let $1 \leq p < \infty$ and let $f \in W^{1,p}(I)$. Then $\partial_x f = 0$ a.e. on $\{f = 0\}$.*

Lemma 7.33. *Let $u \in D(\partial\mathcal{E}^-)$ and let $z \in -\partial\mathcal{E}^-(u)$. Then for all $\lambda > 0$, $\varphi(u - \lambda z) \geq \varphi(u)$.*

Proof. By Proposition 7.21, z is a weak solution to the static Neumann problem, and by Lemma 7.24 we have $u \downarrow \in H^2(I)$ with $\partial_x u \downarrow = 0$ on $\{0, 1\}$ and $z = \partial_x \partial_x u \downarrow$ in $L^2(I)$.

Note that the result holds trivially if $u - \lambda z \notin H^1(I)$, in which case $\varphi(u - \lambda z) = +\infty$. Therefore assume that $u - \lambda z \in H^1(I)$. We consider four cases: $u \in H^1(I)$; $u \in W^{1,1}(I) \setminus H^1(I)$, i.e. u is AC but not in $H^1(I)$; u is continuous but not AC; and u is not continuous.³

Case 1. Suppose $u \in H^1(I)$. Note that in this case the assumption that $u - \lambda z \in H^1(I)$ implies that $z \in H^1(I)$ as well, so in particular $\partial_x u, \partial_x z \in L^2(I)$. Additionally, Lemma 7.24 also yields $u \downarrow \in H^3(I)$. We have

$$\varphi(u - \lambda z) = \int_I (\partial_x(u - \lambda z))^2 dx = \varphi(u) + \lambda^2 \|\partial_x z\|_{L^2(I)}^2 - 2\lambda \int_I (\partial_x u)(\partial_x z) dx.$$

Therefore it suffices to show that

$$\int_I (\partial_x u)(\partial_x z) dx \stackrel{?}{\leq} 0. \quad (7.13)$$

Recalling that $u = u \uparrow + u \downarrow$ with $u \uparrow, u \downarrow \in H^1(I) = W^{1,2}(I)$ by Proposition 7.30, we have

$$\int_I (\partial_x u)(\partial_x z) dx = \int_I (\partial_x u \uparrow)(\partial_x z) dx + \int_I (\partial_x u \downarrow)(\partial_x z) dx,$$

³As usual, phrases such as “ u is continuous” should be understood as “the object $u \in L^2(I)$ has a continuous representative”, and in particular the condition “ u is continuous but not AC” makes sense because the continuous representative, if it exists, is unique.

with the second term in the RHS satisfying

$$\int_I (\partial_x u \downarrow)(\partial_x z) \, dx = (\partial_x u \downarrow)z|_0^1 - \int_I (\partial_x \partial_x u \downarrow)z \, dx = -\|\partial_x \partial_x u \downarrow\|_{L^2(I)}^2 \leq 0.$$

By [Proposition 7.30](#), $\partial_x u \uparrow = \partial_x^+ u$ and $\partial_x u \downarrow = \partial_x^- u$ in $L^2(I)$. Since $z = \partial_x \partial_x u \downarrow$ in $L^2(I)$, we get

$$\int_I (\partial_x u \uparrow)(\partial_x z) \, dx = \int_I (\partial_x^+ u)(\partial_x \partial_x \partial_x u \downarrow) \, dx.$$

We claim that the quantity above is zero. Indeed, fixing any representative of $\partial_x u \in L^2(I)$, let $S := \{x \in I : \partial_x u > 0\}$. First, we have $(\partial_x^+ u)(\partial_x \partial_x \partial_x^- u) = \partial_x^+ u = 0$ on $I \setminus S$. Second, we have $\partial_x u \downarrow = \partial_x^- u = 0$ on S , so applying [Fact 7.32](#) twice (recall that $u \downarrow \in H^3(I)$) gives that $\partial_x \partial_x \partial_x u \downarrow = 0$ a.e. in S , thus establishing the claim. Thus [\(7.13\)](#) indeed holds, which concludes the proof in Case 1.

Case 2. Suppose $u \in W^{1,1}(I) \setminus H^1(I)$. We will derive a contradiction, showing that this case cannot happen. Note that, since $u - \lambda z \in H^1(I)$ by assumption, we conclude that $z \in W^{1,1}(I) \setminus H^1(I)$, and in particular u, z are AC with $\partial_x u, \partial_x z \in L^1(I)$. As in the previous case, we have

$$\varphi(u - \lambda z) = \int_I (\partial_x(u - \lambda z))^2 \, dx \tag{7.14}$$

$$= \int_I (\partial_x u - \lambda \partial_x z)^2 \, dx \tag{7.15}$$

$$= \int_I ((\partial_x u)^2 - 2\lambda(\partial_x u)(\partial_x z) + \lambda^2(\partial_x z)^2) \, dx. \tag{7.16}$$

We claim that the function $(\partial_x u)(\partial_x z) \in L^1(I)$. First, by [Proposition 7.30](#) we have $u \uparrow \in W^{1,1}(I)$ (while $u \downarrow \in H^1(I)$ since $u \downarrow \in H^2(I)$), as well as $\partial_x u \uparrow = \partial_x^+ u$ and $\partial_x u \downarrow = \partial_x^- u$ a.e. in I . Hence

$$\int_I |(\partial_x u)(\partial_x z)| \, dx = \int_I |(\partial_x u \uparrow)(\partial_x z) + (\partial_x u \downarrow)(\partial_x z)| \, dx \leq \int_I \left[|(\partial_x u \uparrow)(\partial_x z)| + |(\partial_x u \downarrow)(\partial_x z)| \right] \, dx.$$

We claim that $(\partial_x u \uparrow)(\partial_x z), (\partial_x u \downarrow)(\partial_x z) \in L^1(I)$. First, we again have that $(\partial_x u \uparrow)(\partial_x z) = (\partial_x u \uparrow)(\partial_x \partial_x \partial_x u \downarrow) = 0$ a.e. as in the previous case, where in particular we are allowed to apply [Fact 7.32](#) twice to $\partial_x u \downarrow$ because $\partial_x u \downarrow \in W^{2,1}(I)$ by virtue of the fact that $\partial_x \partial_x u \downarrow =$

$z \in W^{1,1}(I)$ in the current case. Hence $(\partial_x u \uparrow)(\partial_x z) \in L^1(I)$. Second, note that $\partial_x u \downarrow$ is AC and hence bounded, while $\partial_x z \in L^1(I)$ since z is AC. Therefore

$$\int_I |(\partial_x u \downarrow)(\partial_x z)| dx \leq \|\partial_x u \downarrow\|_{L^\infty(I)} \|\partial_x z\|_{L^1(I)} < +\infty,$$

and hence $(\partial_x u \downarrow)(\partial_x z) \in L^1(I)$. Hence $|(\partial_x u \uparrow)(\partial_x z)| + |(\partial_x u \downarrow)(\partial_x z)| \in L^1(I)$ and

$$\int_I |(\partial_x u)(\partial_x z)| dx \leq \int_I |(\partial_x u \uparrow)(\partial_x z)| dx + \int_I |(\partial_x u \downarrow)(\partial_x z)| dx < +\infty,$$

so $(\partial_x u)(\partial_x z) \in L^1(I)$ as claimed. Now, since $\varphi(u - \lambda z) < +\infty$ by assumption, (7.16) shows that $(\partial_x u)^2 - 2\lambda(\partial_x u)(\partial_x z) + \lambda^2(\partial_x z)^2 \in L^1(I)$, while we have just established that $2\lambda(\partial_x u)(\partial_x z) \in L^1(I)$. We conclude that $(\partial_x u)^2 + \lambda^2(\partial_x z)^2 \in L^1(I)$, i.e.

$$\int_I ((\partial_x u)^2 + \lambda^2(\partial_x z)^2) dx < +\infty.$$

On the other hand, the fact that $u, z \in W^{1,1}(I) \setminus H^1(I)$ implies that

$$\int_I (\partial_x u)^2 dx = +\infty \quad \text{and} \quad \int_I (\partial_x z)^2 dx = +\infty,$$

which is the desired contradiction. This concludes the proof in Case 2.

Case 3. Suppose u is continuous but not AC (and hence the same is true of $u \uparrow$). Then by the definition of absolute continuity, there exists $\varepsilon > 0$ such that, for all $\delta > 0$, there exists a set of pairwise disjoint intervals $((a_i, b_i))_{i \in [k]}$ in I such that $\sum_{i=1}^k (b_i - a_i) < \delta$ and $\sum_{i=1}^k |u(a_i) - u(b_i)| > \varepsilon$.

We claim that, moreover, the sequences $((a_i, b_i))_{i \in [k]}$ above can always be taken to satisfy $u(a_i) < u(b_i)$ for every $i \in [k]$. Indeed, let $\varepsilon > 0$ be as in the paragraph above, let $\delta > 0$, and let $((a_i, b_i))_{i \in [k]}$ be the corresponding sequence. Let $S := \{i \in [k] : u(a_i) \geq u(b_i)\}$. Then for each $i \in S$, we have

$$|u(a_i) - u(b_i)| = u(a_i) - u(b_i) = \underbrace{[u \downarrow(a_i) - u \downarrow(b_i)]}_{\geq 0} + \underbrace{[u \uparrow(a_i) - u \uparrow(b_i)]}_{\leq 0} \leq |u \downarrow(a_i) - u \downarrow(b_i)|.$$

Now, since $u \downarrow$ is AC, let $\delta > 0$ be small enough so that

$$\sum_{i \in S} |u \downarrow(a_i) - u \downarrow(b_i)| \leq \frac{\varepsilon}{2}.$$

It follows that

$$\varepsilon < \sum_{i=1}^k |u(a_i) - u(b_i)| \leq \sum_{i \in S} |u \downarrow(a_i) - u \downarrow(b_i)| + \sum_{i \in [k] \setminus S} |u(a_i) - u(b_i)| \leq \frac{\varepsilon}{2} + \sum_{i \in [k] \setminus S} |u(a_i) - u(b_i)|,$$

and hence

$$\sum_{i \in [k] \setminus S} |u(a_i) - u(b_i)| > \frac{\varepsilon}{2},$$

and of course $u(a_i) < u(b_i)$ for each $i \in [k] \setminus S$ and $\sum_{i \in [k] \setminus S} (b_i - a_i) < \delta$. This establishes the claim.

Let $v := u - \lambda z$ for convenience, and fix any sequence $((a_i, b_i))_{i \in [k]}$ of pairwise disjoint intervals satisfying $u(a_i) < u(b_i)$ for each $i \in [k]$. Using [Lemma 7.35](#) below, we map each interval (a_i, b_i) into an interval $(a'_i, b'_i) \subseteq (a_i, b_i)$ such that

$$|v(a'_i) - v(b'_i)| \geq \frac{|u(a_i) - u(b_i)|}{2}.$$

This implies that v is not AC and hence $\varphi(v) = +\infty$, thus concluding the proof in this case.

Case 4. Suppose u is not continuous. By [Lemma 7.36](#) below, $u - \lambda z$ is not continuous, which implies that $u - \lambda z \notin H^1(I)$ and $\varphi(u - \lambda z) = +\infty$, concluding the proof. $\square$

We now give [Lemmas 7.35](#) and [7.36](#), which we used above. The following fact is an immediate application of the Sobolev embedding theorem (see e.g. [\[Eva10; Maz11\]](#)).

Fact 7.34. *Let $f \in H^2(I)$. Then $f \in C^{1,1/2}(\bar{I})$. In particular, f is continuously differentiable.*

Lemma 7.35. *Let $u \in D(\partial \mathcal{E}^-)$, suppose u is continuous, and let $z \in -\partial \mathcal{E}^-(u)$. Let $\lambda > 0$ and let $v := u - \lambda z$. Let $0 < a < b < 1$ and suppose $u(a) < u(b)$. Then for all $\varepsilon > 0$, there exist a', b' with $a \leq a' < b' \leq b$ such that*

$$v(b') - v(a') \geq |u(a) - u(b)| - \varepsilon.$$

Proof. Recall that, by [Proposition 7.21](#), u, z form a weak solution to the static Neumann problem, and by [Lemma 7.24](#) we have $u \downarrow \in H^2(I)$ with $\partial_x u \downarrow = 0$ on $\{0, 1\}$ and $z = \partial_x \partial_x u \downarrow$ in $L^2(I)$. In particular, $u \downarrow$ is continuously differentiable by [Fact 7.34](#).

We first observe that it cannot be the case that $\partial_x u \downarrow(x) < 0$ for all $x \in (a, b)$, since otherwise [Lemma 7.28](#) would imply that $u \uparrow$ is constant in (a, b) , which by continuity would imply that $u \uparrow(a) = u \uparrow(b)$ and hence $u(a) \geq u(b)$, a contradiction.

We first construct $a' \in (a, b)$. In particular, we wish a' to satisfy

$$u(a') \stackrel{?}{\leq} u(a) + \frac{\varepsilon}{2} \quad \text{and} \quad z(a') \stackrel{?}{\geq} 0. \quad (7.17)$$

Let $x^* \in [a, b)$ be given by

$$x^* := \inf\{x \in (a, b) : \partial_x u \downarrow(x) = 0\},$$

which is well-defined by the observation above and the fact that $\partial_x u \downarrow \leq 0$ in all of I (since $u \downarrow$ is nonincreasing and continuously differentiable). By the continuity of $\partial_x u \downarrow$, we have that $\partial_x u \downarrow(x^*) = 0$. Since $\partial_x u \downarrow(a) \leq 0$, it must be the case that either $a = x^*$ or

$$\mathcal{L}\{x \in (a, x^*) : \partial_x \partial_x u \downarrow(x) \geq 0\} > 0. \quad (7.18)$$

We consider each case separately. First, suppose $a = x^*$. We claim that for all $\delta > 0$, there exists $x \in (a, a + \delta)$ such that $z(x) \geq 0$. Suppose for a contradiction that this is not the case, and fix $\delta > 0$ such that for all $x \in (a, a + \delta)$, $z(x) < 0$. Then since $z = \partial_x \partial_x u \downarrow$ almost everywhere, we conclude that $\partial_x \partial_x u \downarrow < 0$ a.e. in $(a, a + \delta)$. Hence $\partial_x u \downarrow(x) < 0$ for all $x \in (a, a + \delta)$, contradicting the assumption that $a = x^*$ given the definition of x^* . Thus the claim holds. Now, using the continuity of u , choose $\delta > 0$ small enough and choose $a' \in (a, a + \delta)$ so that $z(a') \geq 0$ and moreover $u(a') \leq u(a) + \varepsilon/2$. This choice of a' satisfies (7.17).

Second, suppose $a < x^*$ and (7.18) holds. Since $z = \partial_x \partial_x u \downarrow$ almost everywhere, choose $a' \in (a, x^*)$ such that $z(a') \geq 0$, which is possible by (7.18). Now, since $\partial_x u \downarrow < 0$ for all $x \in (a, a') \subset (a, x^*)$ by the choice of x^* , [Lemma 7.28](#) implies that $u \uparrow$ is constant in (a, a') . By the continuity of $u \uparrow$, we conclude that $u \uparrow(a) = u \uparrow(a')$ and hence, since $u \downarrow$ is nonincreasing, we have

$$u(a') = u \uparrow(a') + u \downarrow(a') \leq u \uparrow(a) + u \downarrow(a) = u(a),$$

and again (7.17) is satisfied. This concludes the choice of a' .

Now, we may assume without loss of generality that $\varepsilon < |u(a) - u(b)|/2$. Therefore our choice of a' yields an interval (a', b) which, using (7.17) and recalling that $u(a) < u(b)$, satisfies $u(a') < u(b)$. Therefore, repeating a symmetric version of the argument above yields a choice of $b' \in (a', b)$ satisfying

$$u(b') \geq u(b) - \frac{\varepsilon}{2} \quad \text{and} \quad z(b') \leq 0. \quad (7.19)$$

Combining (7.17) and (7.19), we conclude that

$$v(b') - v(a') = \left[u(b') - u(a') \right] - \underbrace{\lambda \left[z(b') - z(a') \right]}_{\leq 0} \geq u(b) - u(a) - \varepsilon = |u(a) - u(b)| - \varepsilon. \quad \square$$

Lemma 7.36. *Let $u \in D(\partial\mathcal{E}^-)$, suppose u is not continuous, and let $z \in -\partial\mathcal{E}^-(u)$. Let $\lambda > 0$. Then $u - \lambda z$ is not continuous.*

Proof. Recall that, by Proposition 7.21, u, z form a weak solution to the static Neumann problem, and by Lemma 7.24 we have $u\downarrow \in H^2(I)$ with $\partial_x u\downarrow = 0$ on $\{0, 1\}$ and $z = \partial_x \partial_x u\downarrow$ in $L^2(I)$. In particular, $u\downarrow$ is continuously differentiable by Fact 7.34.

Since $u\downarrow \in H^2(I)$ is continuous while u is not continuous, we conclude that $u\uparrow$ is not continuous and, since it is monotone, it contains only jump discontinuities. Let $x_0 \in I$ be a point of jump discontinuity of $u\uparrow$, i.e. a point such that $L_- < L_+$ where

$$L_- := \lim_{x \rightarrow x_0^-} u\uparrow(x) \quad \text{and} \quad L_+ := \lim_{x \rightarrow x_0^+} u\uparrow(x).$$

We first claim that $\partial_x u\downarrow(x_0) = 0$. Indeed, it is clear that $\partial_x u\downarrow \leq 0$ since $u\downarrow$ is nonincreasing and continuously differentiable. If we had $\partial_x u\downarrow(x_0) < 0$, then by continuity $\partial_x u\downarrow$ would be strictly negative in a neighbourhood of x_0 , in which case Lemma 7.28 would imply that $u\uparrow$ is constant in a neighbourhood of x_0 , contradicting the fact that $L_- < L_+$. Hence the claim holds.

We now claim that, for all $\delta > 0$, there exists $x \in (x_0, x_0 + \delta)$ such that $z(x) \leq 0$. Suppose for a contradiction that there exists $\delta > 0$ such that for all $x \in (x_0, x_0 + \delta)$, $z(x) > 0$. Then, since $z = \partial_x \partial_x u\downarrow$ a.e. and $\partial_x u\downarrow(x_0) = 0$, we conclude that $\partial_x u\downarrow > 0$ in $(x_0, x_0 + \delta)$, contradicting the fact that $\partial_x u\downarrow \leq 0$. Hence the claim holds. By the same reasoning, we conclude that for all $\delta > 0$ there exists $x \in (x_0 - \delta, x_0)$ such that $z(x) \geq 0$.

Thus, for all $\delta > 0$ there exist points $x_- \in (x_0 - \delta, x_0)$, $x_+ \in (x_0, x_0 + \delta)$ such that $z(x_+) - z(x_-) \leq 0$ and hence

$$\begin{aligned} (u - \lambda z)(x_+) - (u - \lambda z)(x_-) &= \left[u\uparrow(x_+) - u\uparrow(x_-) \right] + \left[u\downarrow(x_+) - u\downarrow(x_-) \right] - \lambda \left[z(x_+) - z(x_-) \right] \\ &\geq \left[u\uparrow(x_+) - u\uparrow(x_-) \right] + \left[u\downarrow(x_+) - u\downarrow(x_-) \right]. \end{aligned}$$

Since $u\uparrow \rightarrow L_+$ as $x \downarrow x_0$ and $u\uparrow \rightarrow L_-$ as $x \uparrow x_0$ and $u\uparrow$ is nondecreasing, we have $u\uparrow(x_+) - u\uparrow(x_-) \geq L_+ - L_-$. By the continuity of $u\downarrow$, we can let $\delta > 0$ be small enough so

that $u\downarrow(x_+) - u\downarrow(x_-) \geq -\frac{L_+ - L_-}{2}$. We conclude that, for all sufficiently small $\delta > 0$, there exist points $x_- \in (x_0 - \delta, x_0)$, $x_+ \in (x_0, x_0 + \delta)$ such that

$$(u - \lambda z)(x_+) - (u - \lambda z)(x_-) \geq \frac{L_+ - L_-}{2},$$

and thus $u - \lambda z$ is not continuous. $\square$

We are ready to obtain our preservation of regularity result. We will use the following tool from the theory of maximal monotone operators.

Lemma 7.37 (Specialization of [Bre73, Theorem 4.4]). *Let H be a Hilbert space and let $\tau : H \rightarrow [0, +\infty]$ be a convex, proper, and lower semicontinuous functional such that $\tau(\text{Proj}_{\overline{D(A)}}x) \leq \tau(x)$ for all $x \in H$. Let $A : H \rightarrow 2^H$ be a maximal monotone operator and let S_t be the semigroup generated by $-A$. Then the following are equivalent:*

1. $\tau((I + \lambda A)^{-1}x) \leq \tau(x)$ for all $x \in H$ and $\lambda > 0$; and
2. $\tau(S_t x) \leq \tau(x)$ for all $x \in \overline{D(A)}$ and $t \geq 0$.

In the statement above, $(I + \lambda A)^{-1} : H \rightarrow D(A)$ is the *resolvent* of A (also denoted by J_λ , see e.g. [Eva10, p. 563]).

We will need the following fact and observation about the density of our domains of interest in $L^2(I)$.

Fact 7.38 (See e.g. [Bar76, Chapter II, Corollary 2.1]). *The set $D(\partial\mathcal{E}^-)$ is a dense subset of $D(\mathcal{E}^-) = \mathcal{U}$.*

Observation 7.39. *Since $\mathcal{U}$ contains $H^1(I)$, which is dense in $L^2(I)$, Fact 7.38 implies that*

$$\overline{D(\partial\mathcal{E}^-)} = \overline{D(\mathcal{E}^-)} = \overline{\mathcal{U}} = L^2(I).$$

Proposition 7.40 (φ -monotonicity of solutions). *Let $u_0 \in \mathcal{U}$ and let $\mathbf{u} \in C([0, +\infty); L^2(I))$ be the solution to the gradient flow problem with initial data u_0 . Then for all $0 \leq t_1 \leq t_2 < +\infty$, we have $\varphi(\mathbf{u}(t_1)) \geq \varphi(\mathbf{u}(t_2))$.*

Proof. This is a direct consequence of Lemma 7.37. Indeed, letting $H := L^2(I)$ and $A := \partial\mathcal{E}^-$ (which is maximal monotone as remarked earlier), we first observe that indeed

$\varphi(\text{Proj}_{\overline{D(A)}} f) \leq \varphi(f)$ for all $f \in L^2(I)$, since by [Observation 7.39](#) we have $\text{Proj}_{\overline{D(A)}} f = f$. Thus it suffices to show that for all $f \in L^2(I)$ and all $\lambda > 0$,

$$\varphi((I + \lambda A)^{-1} f) \stackrel{?}{\leq} \varphi(f).$$

But this is equivalent to showing that, for all $u \in D(A)$, $-z \in A(u)$ and $\lambda > 0$,

$$\varphi(u) \stackrel{?}{\leq} \varphi(u - \lambda z),$$

which is precisely [Lemma 7.33](#). $\square$

Corollary 7.41 (Preservation of H^1 regularity). *Suppose $u_0 \in H^1(I)$, and let $\mathbf{u} \in C([0, +\infty); L^2(I))$ be the solution to the gradient flow problem with initial data u_0 . Then $\mathbf{u}(t) \in H^1(I)$ for all $t > 0$.*

Proof. This is an immediate consequence of [Proposition 7.40](#) and the definition of φ . $\square$

7.2.6 Preservation of Lipschitz regularity

It will also be useful to control the Lipschitz regularity of solutions. At a high level, we follow a similar strategy as in [Section 7.2.5](#). Recall that $W^{1,\infty}(I)$ is equivalent to the space of Lipschitz real-valued functions on I , up to identification of almost everywhere equal functions. Define $\psi : L^2(I) \rightarrow [0, +\infty]$ by

$$\psi(u) := \begin{cases} \|\partial_x u\|_{L^\infty(I)} & \text{if } u \in W^{1,\infty}(I) \\ +\infty & \text{otherwise.} \end{cases}$$

Fact 7.42. *Let $u \in L^2(I)$ and let $M \in \mathbb{R}_{\geq 0}$. Then $\psi(u) \leq M$ if and only if $u = f$ a. e. for some M -Lipschitz function $f : I \rightarrow \mathbb{R}$. Moreover, in this case f is the (unique) continuous representative of u and $\psi(u)$ is the Lipschitz constant of f .*

Claim 7.43. *The functional ψ is convex, proper and lower semicontinuous.*

Proof. Convexity and properness are straightforward; it remains to verify lower semicontinuity. Since $L^2(I)$ is a metric space, it suffices to check sequential lower semicontinuity. Let $(u_n)_{n \in \mathbb{N}}$ be a sequence in $L^2(I)$ such that $u_n \rightarrow u$ in $L^2(I)$. We need to show that

$$\psi(u) \stackrel{?}{\leq} \liminf_{n \rightarrow \infty} \psi(u_n).$$

The only relevant case is when the RHS above is finite, so suppose there exists a subsequence $(u_{n_k})_{k \in \mathbb{N}}$ such that $\lim_{k \rightarrow \infty} \psi(u_{n_k}) = M < +\infty$. By extracting a subsequence if necessary, we may assume that $\psi(u_{n_k}) < +\infty$, and thus $u_{n_k} \in W^{1,\infty}(I)$, for every k . For simplicity and using [Fact 7.42](#), fix for each n_k the continuous representative of u_{n_k} , which we denote by the same name. Then each u_{n_k} is M_k -Lipschitz with $M_k \rightarrow M$.

Since $u_{n_k} \rightarrow u$ in $L^2(I)$, it is standard that we may extract a subsequence that converges to u pointwise almost everywhere. Denote this further subsequence again by $(u_{n_k})_k$, and let $N \subset I$ be a measure zero set such that $u_{n_k} \rightarrow u$ pointwise in $I \setminus N$. Then for each $x \neq y$ in $I \setminus N$, we have

$$|u(x) - u(y)| = \lim_{k \rightarrow \infty} |u_{n_k}(x) - u_{n_k}(y)| \leq \lim_{k \rightarrow \infty} M_k |x - y| = M |x - y|.$$

By [Lemma 7.44](#) below, u is a.e. equal to an M -Lipschitz function, so $\psi(u) \leq M$ by [Fact 7.42](#), as needed. $\square$

We used the following simple lemma.

Lemma 7.44 (From “almost Lipschitz” to Lipschitz). *Let $M \in \mathbb{R}_{\geq 0}$ and let $N \subset I$ be a measure zero set. Suppose $f : I \rightarrow \mathbb{R}$ satisfies $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in I \setminus N$. Then there exists a M -Lipschitz function $g : I \rightarrow \mathbb{R}$ such that $f = g$ in $I \setminus N$.*

Proof. See [Appendix B](#). $\square$

Definition 7.45. For any $w : I \rightarrow \mathbb{R}$ and distinct $x, y \in I$, let

$$\text{slope}_w(x, y) := \frac{w(y) - w(x)}{y - x}.$$

Lemma 7.46. *Let $u \in D(\partial \mathcal{E}^-)$ and let $z \in -\partial \mathcal{E}^-(u)$. Then for all $\lambda > 0$, $\psi(u - \lambda z) \geq \psi(u)$.*

Proof. We follow the proof outline from [Lemma 7.33](#), but some of the technical details are different. As in that proof, we have $u \downarrow \in H^2(I)$ with $\partial_x u \downarrow = 0$ on $\{0, 1\}$ and $z = \partial_x \partial_x u \downarrow$ in $L^2(I)$.

We may assume that $u - \lambda z \in W^{1,\infty}(I)$, since otherwise $\psi(u - \lambda z) = +\infty$ and there is nothing to prove. In particular, this establishes that $u - \lambda z$ is continuous.

When u is not continuous, [Lemma 7.36](#) yields that $u - \lambda z$ is not continuous, a contradiction. Therefore we may assume that u is continuous. Since both u and $u - \lambda z$ are

continuous, we conclude that z is also continuous. Let $v := u - \lambda z$. By [Fact 7.42](#), it suffices to show that for every $(a, b) \subset I$ and all $\varepsilon > 0$, there exists $(a', b') \subset I$ such that

$$|\text{slope}_v(a', b')| \stackrel{?}{\geq} |\text{slope}_u(a, b)| - \varepsilon. \quad (7.20)$$

If $u(a) < u(b)$, then the interval $(a', b') \subseteq (a, b)$ given by [Lemma 7.35](#) with parameter $\varepsilon(b - a)$ satisfies (7.20); and if $u(a) = u(b)$, then any interval will do. Therefore suppose $u(a) > u(b)$, so in particular $\text{slope}_u(a, b) < 0$.

Note that a sufficient condition for (7.20) is that $|\text{slope}_u(a', b')| \geq |\text{slope}_u(a, b)| - \varepsilon$ with $u(a') > u(b')$, $z(a') \leq 0$ and $z(b') \geq 0$, since in that case,

$$\begin{aligned} |\text{slope}_v(a', b')| &= \frac{|(u - \lambda z)(b') - (u - \lambda z)(a')|}{b' - a'} = \left| \underbrace{u(b') - u(a')}_{<0} - \lambda \underbrace{(z(b') - z(a'))}_{\geq 0} \right| \left(\frac{1}{b' - a'} \right) \\ &\geq \frac{|u(b') - u(a')|}{b' - a'} = |\text{slope}_u(a', b')| \geq |\text{slope}_u(a, b)| - \varepsilon, \end{aligned}$$

which is (7.20). We now find a' and b' satisfying the aforementioned conditions.

For each point $x \in I$, say x is *left-favourable* if every neighbourhood $(x - \delta, x + \delta)$ contains a point x' such that $z(x') \leq 0$. Similarly, say x is *right-favourable* if every neighbourhood $(x - \delta, x + \delta)$ contains a point x' such that $z(x') \geq 0$.

We claim that there exist points $a^* < b^*$ in I such that a^* is left-favourable, b^* is right-favourable, and $|\text{slope}_u(a^*, b^*)| \geq |\text{slope}_u(a, b)|$ with $u(a^*) > u(b^*)$. Let us first show that this claim yields the desired points a' and b' , and then proceed to prove the claim. Suppose we have a^* and b^* as claimed. Then by the continuity of u , we may fix sufficiently small $\delta > 0$ and find points $a' \in (a^* - \delta, a^* + \delta)$ and $b' \in (b^* - \delta, b^* + \delta)$ such that $a' < b'$, $z(a') \leq 0$, $z(b') \geq 0$, $u(a') > u(b')$, and, for $\alpha := \frac{\varepsilon}{2} \cdot \frac{b^* - a^*}{u(a^*) - u(b^*)}$ and $\beta := \frac{\varepsilon(b^* - a^*)}{2}$,

$$\begin{aligned} |\text{slope}_u(a', b')| &= \frac{u(a') - u(b')}{b' - a'} \geq \frac{u(a^*) - u(b^*) - \beta}{(b^* - a^*)(1 + \alpha)} \geq \frac{u(a^*) - u(b^*)}{b^* - a^*} (1 - \alpha) - \frac{\varepsilon/2}{1 + \alpha} \\ &\geq \frac{u(a^*) - u(b^*)}{b^* - a^*} - \varepsilon = |\text{slope}_u(a^*, b^*)| - \varepsilon \geq |\text{slope}_u(a, b)| - \varepsilon, \end{aligned}$$

as desired, where we used the inequality $\frac{1}{1 + \alpha} \geq 1 - \alpha$. We now establish the existence of a^* and b^* .

We first find a^* . If a is left-favourable, then choose $a^* = a$; note that, of course, we have $u(a^*) > u(b)$ and $|\text{slope}_u(a^*, b)| \geq |\text{slope}_u(a, b)|$. Otherwise, a is not left-favourable, which

gives some $\delta > 0$ such that $z(x) > 0$ for every $x \in (a - \delta, a + \delta)$, and hence $\partial_x \partial_x u \downarrow > 0$ a.e. in $(a - \delta, a + \delta)$. We now consider two cases. Recall that $\text{slope}_u(a, b) < 0$.

First, suppose $\partial_x u \downarrow(a) \geq \text{slope}_u(a, b)$. Note that it cannot be the case that $\partial_x u \downarrow(x) > \text{slope}_u(a, b)$ for all $x \in (a, b)$, since otherwise we would have

$$u(b) = u \uparrow(b) + u \downarrow(b) \geq u \uparrow(a) + u \downarrow(a) + \int_{(a,b)} \partial_x u \downarrow \, dx > u(a) + (b - a) \text{slope}_u(a, b) = u(b),$$

a contradiction. Therefore we may let

$$a^* := \inf \{x \in (a, b) : \partial_x u \downarrow(x) \leq \text{slope}_u(a, b)\}.$$

By the continuity of $\partial_x u \downarrow$ (recall that $u \downarrow \in H^2(I)$ is continuously differentiable), we conclude that $\partial_x u \downarrow(a^*) = \text{slope}_u(a, b)$. We also observe that we must have $a^* > a$ since, as noted above, we have $\partial_x \partial_x u \downarrow(x) > 0$ a.e. in some neighbourhood $(a - \delta, a + \delta)$, which implies that $\partial_x u \downarrow(x) > \partial_x u \downarrow(a) \geq \text{slope}_u(a, b)$ for all $x \in (a, a + \delta)$. We claim that a^* is left-favourable. Indeed, otherwise there would be some $\delta > 0$ such that $z(x) > 0$ for all $x \in (a^* - \delta, a^* + \delta)$, and since $z = \partial_x \partial_x u \downarrow$ a.e. we would conclude that $\partial_x u \downarrow(x) < \partial_x u \downarrow(a^*) = \text{slope}_u(a, b)$ for $x \in (a^* - \delta, a^*)$, contradicting the choice of a^* . Hence we have found a left-favourable a^* in this case; we claim that a^* also satisfies $u(a^*) > u(b)$ and $|\text{slope}_u(a^*, b)| \geq |\text{slope}_u(a, b)|$. The first inequality holds since

$$\begin{aligned} u(a^*) - u(b) &= u \downarrow(a^*) + u \uparrow(a^*) - [u(a) + (b - a) \text{slope}_u(a, b)] \\ &\geq u \downarrow(a) + \int_{(a,a^*)} \partial_x u \downarrow \, dx + u \uparrow(a) - u \uparrow(a) - u \downarrow(a) - (b - a) \text{slope}_u(a, b) \\ &= \int_{(a,a^*)} \left(\underbrace{\partial_x u \downarrow(x)}_{> \text{slope}_u(a, b)} - \text{slope}_u(a, b) \right) dx - (b - a^*) \underbrace{\text{slope}_u(a, b)}_{< 0} \\ &> 0, \end{aligned}$$

and the second inequality holds since

$$\begin{aligned}
& (b - a^*)(b - a) \left(|\text{slope}_u(a^*, b)| - |\text{slope}_u(a, b)| \right) \\
&= (b - a^*)(b - a) \left(\frac{u(a^*) - u(b)}{b - a^*} - \frac{u(a) - u(b)}{b - a} \right) \\
&= \left(u(a^*) - u(b) \right) (b - a) - \left(u(a) - u(b) \right) (b - a^*) \\
&= \left(u\uparrow(a^*) + u\downarrow(a) + \int_{(a, a^*)} \underbrace{\partial_x u\downarrow(x)}_{> \text{slope}_u(a, b)} dx - u(b) \right) (b - a) - \left(u(a) - u(b) \right) (b - a^*) \\
&> \left(u\uparrow(a) + u\downarrow(a) + (a^* - a) \text{slope}_u(a, b) - u(b) \right) (b - a) - \left(u(a) - u(b) \right) (b - a^*) \\
&= \left(u(a) + (a^* - a) \frac{u(b) - u(a)}{b - a} - u(b) \right) (b - a) - \left(u(a) - u(b) \right) (b - a^*) \\
&= \left(u(a) - u(b) \right) \left((b - a) - (b - a^*) - (a^* - a) \right) \\
&= 0.
\end{aligned}$$

Therefore in the first case we have found a left-favourable $a^* \in (a, b)$ such that $u(a^*) > u(b)$ and $|\text{slope}_u(a^*, b)| \geq |\text{slope}_u(a, b)|$.

Second, suppose $\partial_x u\downarrow(a) < \text{slope}_u(a, b)$. We proceed similarly, but with points to the left of a instead. Namely, it cannot be the case that $\partial_x u\downarrow(x) < \text{slope}_u(a, b)$ for all $x \in (0, a)$, since by continuity this would imply that $\partial_x u\downarrow(0) \leq \text{slope}_u(a, b) < 0$, a contradiction. Hence we may define

$$a^* := \sup \{x \in (0, a) : \partial_x u\downarrow(x) \geq \text{slope}_u(a, b)\}.$$

By continuity, we conclude that $\partial_x u\downarrow(a^*) = \text{slope}_u(a, b)$, which also implies that $a^* < a$. We claim that a^* is left-favourable. Indeed, if it was not, then for some $\delta > 0$ we would have $z(x) > 0$ for all $x \in (a^* - \delta, a^* + \delta)$, and since $z = \partial_x \partial_x u\downarrow$ a.e. we would conclude that $\partial_x u\downarrow(x) > \partial_x u\downarrow(a^*) = \text{slope}_u(a, b)$ for all $x \in (a^*, a^* + \delta)$, contradicting the choice of a^* . We now claim that a^* also satisfies $u(a^*) > u(b)$ and $|\text{slope}_u(a^*, b)| \geq |\text{slope}_u(a, b)|$.

To prove the first inequality, we first observe that $\partial_x u\downarrow(x) < \text{slope}_u(a, b) < 0$ for all $x \in (a^*, a)$, which by [Lemma 7.28](#) implies that $u\uparrow$ is constant in (a^*, a) . Since $u\uparrow$ is continuous (because u and $u\downarrow$ are), we conclude that $u\uparrow(a^*) = u\uparrow(a)$. Therefore we have

$$u(a^*) - u(a) = u\downarrow(a^*) - u\downarrow(a) \geq 0$$

since $u\downarrow$ is nonincreasing, and hence $u(a^*) \geq u(a) > u(b)$. As for the second inequality, we have

$$\begin{aligned}
& (b - a^*)(b - a) \left(|\text{slope}_u(a^*, b)| - |\text{slope}_u(a, b)| \right) \\
&= (b - a^*)(b - a) \left(\frac{u(a^*) - u(b)}{b - a^*} - \frac{u(a) - u(b)}{b - a} \right) \\
&= \left(u(a^*) - u(b) \right) (b - a) - \left(u(a) - u(b) \right) (b - a^*) \\
&= \left(u\uparrow(a) + u\downarrow(a) - \int_{(a^*, a)} \underbrace{\partial_x u\downarrow(x)}_{< \text{slope}_u(a, b)} dx - u(b) \right) (b - a) - \left(u(a) - u(b) \right) (b - a^*) \\
&> \left(u(a) - (a - a^*) \frac{u(b) - u(a)}{b - a} - u(b) \right) (b - a) - \left(u(a) - u(b) \right) (b - a^*) \\
&= \left(u(a) - u(b) \right) \left((b - a) - (b - a^*) + (a - a^*) \right) \\
&= 0.
\end{aligned}$$

Therefore in any case we have found a left-favourable $a^* \in (0, b)$ such that $u(a^*) > u(b)$ and $|\text{slope}_u(a^*, b)| \geq |\text{slope}_u(a, b)|$.

Repeating an analogous argument for the right endpoint of the interval (a^*, b) , we find a right-favourable $b^* \in (a^*, 1)$ such that $u(a^*) > u(b^*)$ and $|\text{slope}_u(a^*, b^*)| \geq |\text{slope}_u(a^*, b)|$, which concludes the proof as explained above. $\square$

Proposition 7.47 (ψ -monotonicity of solutions). *Let $u_0 \in \mathcal{U}$ and let $\mathbf{u} \in C([0, +\infty); L^2(I))$ be the solution to the gradient flow problem with initial data u_0 . Then for all $0 \leq t_1 \leq t_2 < +\infty$, we have $\psi(\mathbf{u}(t_1)) \geq \psi(\mathbf{u}(t_2))$.*

Proof. As in the proof of [Proposition 7.40](#), this follows from [Lemmas 7.37](#) and [7.46](#). $\square$

Corollary 7.48 (Preservation of Lipschitz regularity). *Suppose $u_0 \in W^{1,\infty}(I)$, and let $\mathbf{u} \in C([0, +\infty); L^2(I))$ be the solution to the gradient flow problem with initial data u_0 . Then $\mathbf{u}(t) \in W^{1,\infty}(I)$ for all $t > 0$.*

Proof. This is an immediate consequence of [Proposition 7.47](#) and the definition of ψ . $\square$

7.2.7 Nonexpansiveness and order preservation

Our goal in this section is to establish that the semigroup P_t is nonexpansive and order preserving. These properties will help us show that applying P_t to line restrictions in the multidimensional setting behaves as expected by “making progress” toward monotonicity with each application.

In the context of PDEs, one desirable way to show that a property is preserved through time is to differentiate in time, and then pass the derivative inside the integral to exploit the definition of the PDE. However, we need to establish some technical results before we can justify such calculations.

We start with the following lemma, slightly adapted from [CT80], which reveals a close connection between order preservation and nonexpansiveness (here, in the supremum norm) of operators. Hence our strategy will be to establish order preservation, and conclude nonexpansiveness.

Lemma 7.49. *Let (Ω, Σ, μ) be a finite measure space, and let $1 \leq p \leq +\infty$. Let $L^\infty(\Omega) \subset C \subset L^p(\Omega)$ have the property that for all $f \in C$ and $r \in \mathbb{R}$, $f + r \in C$. Let $T : C \rightarrow L^p(\Omega)$ be continuous as a map from $L^p(\Omega)$ to $L^p(\Omega)$ (the continuity requirement may be dropped if $p = \infty$) satisfying*

$$T(f + r) = T(f) + r \text{ a.e.} \quad (7.21)$$

for every $f \in C$ and $r \in \mathbb{R}$. Then the following are equivalent:

- (a) *For all $f, g \in C$, if $f \leq g$ a.e. then $Tf \leq Tg$ a.e. .*
- (b) *For all $f, g \in C$, $(Tf - Tg)^+ \leq \text{ess sup}(f - g)^+ \text{ a.e. .}$*
- (c) *For all $f, g \in C$, $|Tf - Tg| \leq \text{ess sup}|f - g| \text{ a.e. .}$*

Proof. We show (a) $\implies$ (b) $\implies$ (c) $\implies$ (a). First, assume (a) holds, and let $f, g \in C$. Suppose $r := \text{ess sup}(f - g)^+ < +\infty$, since otherwise there is nothing to prove. Note that $g + r \geq f$ and $g + r \geq g$ both a.e. . Thus by (a) and (7.21), we have

$$Tg + (Tf - Tg)^+ = Tf \vee Tg \leq T(g + r) = Tg + r \text{ a.e. ,}$$

so $(Tf - Tg)^+ \leq r$ a.e. as needed. Next, suppose (b) holds, and let $f, g \in C$. Let $r_1 := \text{ess sup}(f - g)^+$ and $r_2 := \text{ess sup}(g - f)^+$; note that $\text{ess sup}|f - g| = \max\{r_1, r_2\}$, so

we may assume that $r_1, r_2 < +\infty$, since otherwise there is nothing to show. Then indeed, using (b),

$$|Tf - Tg| = \max \{ (Tf - Tg)^+, (Tg - Tf)^+ \} \leq \max \{ r_1, r_2 \} = \text{ess sup} |f - g| \text{ a.e. },$$

as needed. Finally, suppose (c) holds, and let $f, g \in C$ satisfy $f \leq g$ a.e. . First, suppose $f, g \in L^\infty(\Omega)$. Let $r := \text{ess sup}(g - f)$, so that $0 \leq r < +\infty$ by assumption. Then, by (c) and (7.21),

$$\begin{aligned} \text{ess sup} \{ Tf - Tg + r \} &= \text{ess sup} \{ T(f + r) - Tg \} \leq \text{ess sup} |T(f + r) - Tg| \\ &\leq \text{ess sup} |f - g + r| \leq r, \end{aligned}$$

so $Tf \leq Tg$ a.e. as needed.

Now, for general $f, g \in C$ with $f \leq g$ a.e. , we may approximate them by sequences $(f_n)_n, (g_n)_n$ in $L^\infty(\Omega) \subset C$ such that $f_n \rightarrow f$ and $g_n \rightarrow g$ in $L^p(\Omega)$ and moreover $f_n \leq g_n$ a.e. for each n ; indeed, letting $f_n := \max\{-n, \min\{n, f\}\}$ and likewise for g_n , it is immediate that $f_n, g_n \in L^\infty(\Omega)$ with $f_n \leq g_n$ a.e. , and the convergences $f_n \rightarrow f$ and $g_n \rightarrow g$ in $L^p(\Omega)$ follow from the dominated convergence theorem: since $|f - f_n|^p \rightarrow 0$ pointwise and $|f - f_n| \leq |f|$ pointwise for each n , we have $\lim_{n \rightarrow \infty} \int_\Omega |f_n - f|^p d\mu = 0$.

The previous case yields $Tf_n \leq Tg_n$ a.e. for each n , and we claim that the continuity of T implies that $Tf \leq Tg$ a.e. as well. Indeed, we have $Tf_n \rightarrow Tf$ and $Tg_n \rightarrow Tg$ in $L^p(\Omega)$ and thus

$$\begin{aligned} \|(Tf - Tg)^+ - (Tf_n - Tg_n)^+\|_{L^p(\Omega)} &\leq \|(Tf - Tg) - (Tf_n - Tg_n)\|_{L^p(\Omega)} \\ &\leq \|Tf - Tf_n\|_{L^p(\Omega)} + \|Tg - Tg_n\|_{L^p(\Omega)} \rightarrow 0, \end{aligned}$$

which means that $(Tf_n - Tg_n)^+ \rightarrow (Tf - Tg)^+$ in $L^p(\Omega)$. But since $(Tf_n - Tg_n)^+ = 0$ a.e. and hence in $L^p(\Omega)$ for each n , we conclude that $(Tf - Tg)^+ = 0$ a.e. , i.e. $Tf \leq Tg$ a.e. as needed. $\square$

The following lemma is a specific formulation of the well-known Leibniz rule for differentiating under the integral sign. It is a specialization of the version stated in [Che], and can be proved by a standard argument using the Fubini-Tonelli theorem.

Lemma 7.50 (Differentiating under the integral sign). *Let (Ω, Σ, μ) be a measure space and let $[a, b] \subset \mathbb{R}$. Let $f : [a, b] \times \Omega \rightarrow \mathbb{R}$ be a jointly measurable function satisfying the following:*

1. $f(t, \cdot) \in L^1(\Omega)$ for a. e. $t \in (a, b)$.
2. $f(\cdot, x)$ is AC for a. e. $x \in \Omega$. (Its a. e. defined partial derivative is denoted by $\partial_t f$ as usual.)
3. It holds that

$$\int_{(a,b)} \int_{\Omega} |\partial_t f(t, x)| \, d\mu \, dt < +\infty.$$

Then the function $t \mapsto \int_{\Omega} f(t, x) \, d\mu$ is AC and

$$\partial_t \int_{\Omega} f(t, x) \, d\mu = \int_{\Omega} \partial_t f(t, x) \, d\mu \quad \text{for a. e. } t \in (a, b).$$

The following fact is a standard consequence of the definition of absolute continuity.

Fact 7.51. *Let $J \subset \mathbb{R}$ be a compact interval. If $f, g : J \rightarrow \mathbb{R}$ are AC, then $f \vee g$ is AC.*

We will also use the following standard fact about the weak derivative of the positive part of a function.

Fact 7.52 (See e.g. [EG15, Theorem 4.4]). *Let $J \subset \mathbb{R}$ be a compact interval. Let $1 \leq p < \infty$ and let $f \in W^{1,p}(J)$. Then $f^+ \in W^{1,p}(J)$ and $\partial_x(f^+) = \chi_{\{f>0\}} \partial_x f$ a. e. in J .*

We also use the following standard formulation of a chain rule for Sobolev functions. This version follows e.g. from [EG15, Theorem 4.4], which is stated for globally Lipschitz functions F , by using the fact that the image of f on J is bounded, so that the local Lipschitz condition suffices (e.g. extend F linearly outside the image of f to obtain a $C^1(\mathbb{R})$, globally Lipschitz function $\tilde{F}$).

Fact 7.53. *Let $J \subset \mathbb{R}$ be a compact interval. Let $1 \leq p < \infty$, let $f \in W^{1,p}(J)$, and let $F \in C^1(\mathbb{R})$ be locally Lipschitz. Then $F \circ f \in W^{1,p}(J)$ and $\partial_x(F \circ f)(x) = F'(f(x)) \partial_x f(x)$ for a. e. $x \in J$.*

The following lemma essentially says that, given an element of a Bochner space, we can get a handle on a concrete jointly measurable function “representing” that element in a precise sense. This makes concrete the intuitive expectation that a solution $\mathbf{u}$ to our PDEs, which maps each point in time to an element of $L^2(I)$, should also give us a specific value at each point in time and space “ $\mathbf{u}(t, x)$ ”.

Lemma 7.54. [[DS58](#), Theorem 17, p. 198] Let (S, Σ_S, μ) and (T, Σ_T, λ) be measure spaces which are either both finite or both positive and σ -finite, and let (R, Σ_R, ρ) be their product. Let $1 \leq p \leq \infty$ and let F be a μ -integrable function on S to $L^p(T, \Sigma_T, \lambda, \mathcal{X})$ where $\mathcal{X}$ is a real or complex Banach space. Then there is a ρ -measurable function f on R to $\mathcal{X}$, which is uniquely determined except for a set of ρ -measure zero, such that $f(s, \cdot) = F(s)$ for μ -almost all s in S . Moreover $f(\cdot, t)$ is μ -integrable on S for λ -almost all t and the integral $\int_S f(s, t) \mu(ds)$, as a function of t is equal to the element $\int_S F(s) \mu(ds)$ of $L^p(T, \Sigma_T, \lambda, \mathcal{X})$.

Corollary 7.55. Let $[a, b] \subset \mathbb{R}$ be a compact interval endowed with the Lebesgue measure, and let $F \in L^1(a, b; L^2(I))$. Then there exists a jointly measurable function $f^* : [a, b] \times I \rightarrow \mathbb{R}$ satisfying

1. $f^*(s, \cdot) = F(s)$ in $L^2(I)$ for a.e. $s \in (a, b)$.
2. $f^*(\cdot, x) \in L^1(a, b)$ for all $x \in I$.
3. For all $s \in [a, b]$, the functions $x \mapsto \int_{(a,s)} f^*(r, x) dr$ and $\int_{(a,s)} F(r) dr$ are equal in $L^2(I)$.

Proof. Let condition $\tilde{2}$ denote condition 2 with “all $x \in I$ ” replaced by “a.e. $x \in I$ ”. We first apply [Lemma 7.54](#) with $S = [a, b]$ and $T = I$, both endowed with the Lebesgue measure, to obtain a function f satisfying conditions 1 and $\tilde{2}$, as well condition 3 for $s = b$.

We now show that condition 3 also holds for other values of $s \in [a, b]$ using the a.e. uniqueness given by [Lemma 7.54](#). For any $s \in [a, b]$, we may apply that lemma with $\tilde{S} = [a, s]$ instead to obtain a function $\tilde{f}$ satisfying $x \mapsto \int_{(a,s)} \tilde{f}(r, x) dr = \int_{(a,s)} F(r) dr$ in $L^2(I)$. But since both $g = \tilde{f}$ and $g = f$ satisfy that $g(r, \cdot) = F(r)$ in $L^2(I)$ for a.e. $r \in (a, s)$, [Lemma 7.54](#) implies that $\tilde{f} = f$ except for a subset of $[a, s] \times I$ of joint measure zero. We conclude that, for a.e. $x \in I$, we have $\tilde{f}(\cdot, x) = f(\cdot, x)$ a.e. in (a, s) and hence $\int_{(a,s)} f(r, x) dr = \int_{(a,s)} \tilde{f}(r, x) dr$. It follows that $x \mapsto \int_{(a,s)} f(r, x) dr = x \mapsto \int_{(a,s)} \tilde{f}(r, x) dr = \int_{(a,s)} F(r)$ in $L^2(I)$, as claimed.

The final step is to construct a jointly measurable function $f^* : [a, b] \times I \rightarrow \mathbb{R}$ satisfying condition 2 rather than just $\tilde{2}$, while preserving conditions 1 and 3. Let $N \subset I$ be a measure zero set such that $f(\cdot, x) \in L^1(a, b)$ for all $x \in I \setminus N$. Define f^* by

$$f^*(s, x) := \begin{cases} f(s, x) & \text{if } x \in I \setminus N \\ 0 & \text{otherwise.} \end{cases}$$

We note that f^* is jointly measurable; indeed, this follows from the facts that $f = f^*$ on $[a, b] \times (I \setminus N)$ and that the set $[a, b] \times (I \setminus N)$ is jointly measurable. By construction, f^* satisfies condition 2, and it is clear that f^* also satisfies conditions 1 and 3 since f does and N is a null set. $\square$

Lemma 7.56. *Let $0 \leq a < b < +\infty$, and let $\mathbf{u} \in C([a, b]; L^2(I))$ be the restriction to domain $[a, b]$ of any solution to the gradient flow problem, with $\mathbf{u}' : [a, b] \rightarrow L^2(I)$ its weak derivative restricted in the same way. Then there exists a jointly measurable function $\tilde{\mathbf{u}}' : [a, b] \times I \rightarrow \mathbb{R}$ satisfying*

1. $\tilde{\mathbf{u}}'(t, \cdot) = \mathbf{u}'(t)$ in $L^2(I)$ for a.e. $t \in (a, b)$.
2. $\tilde{\mathbf{u}}'(\cdot, x) \in L^1(a, b)$ for all $x \in I$.
3. For all $t \in [a, b]$, the functions $x \mapsto \int_{(a,t)} \tilde{\mathbf{u}}'(s, x) \, ds$ and $\int_{(a,t)} \mathbf{u}'(s) \, ds$ are equal in $L^2(I)$.

Moreover, fixing any representative of $\mathbf{u}(a) \in L^2(I)$, the function $\tilde{\mathbf{u}} : [a, b] \times I \rightarrow \mathbb{R}$ given by

$$\tilde{\mathbf{u}}(t, x) := \mathbf{u}(a)(x) + \int_{(a,t)} \tilde{\mathbf{u}}'(s, x) \, ds$$

is jointly measurable and satisfies

4. For each $t \in [a, b]$, $\tilde{\mathbf{u}}(t, \cdot) = \mathbf{u}(t)$ in $L^2(I)$.
5. For each $x \in I$, $\tilde{\mathbf{u}}(\cdot, x)$ is absolutely continuous with weak derivative $\partial_t \tilde{\mathbf{u}}(t, x) = \tilde{\mathbf{u}}'(t, x)$.

Proof. Apply [Corollary 7.55](#) to $\mathbf{u}'$, which is integrable since it is the weak derivative of $\mathbf{u}$, to obtain jointly measurable $\tilde{\mathbf{u}}' : [a, b] \times I \rightarrow \mathbb{R}$ satisfying properties 1–3. We now verify that $\tilde{\mathbf{u}}$ satisfies properties 4 and 5. Note that these two properties then imply that $\tilde{\mathbf{u}}$ is a Carathéodory function and hence jointly measurable (see e.g. [\[AB06, Lemma 4.51\]](#)).

For each $t \in [a, b]$, the definition of $\mathbf{u}$ (in particular its absolute continuity) implies that

$$\mathbf{u}(t) - \mathbf{u}(a) = \left[\mathbf{u}(0) + \int_{(0,t)} \mathbf{u}'(s) \, ds \right] - \left[\mathbf{u}(0) + \int_{(0,a)} \mathbf{u}'(s) \, ds \right] = \int_{(a,t)} \mathbf{u}'(s) \, ds$$

in $L^2(I)$. Property 3 implies that, for a.e. $x \in I$,

$$\mathbf{u}(t)(x) = \mathbf{u}(a)(x) + \int_{(a,t)} \tilde{\mathbf{u}}'(s, x) \, ds = \tilde{\mathbf{u}}(t, x),$$

which is property 4. Finally, property 5 is an immediate consequence of the definition of $\tilde{\mathbf{u}}$. $\square$

We are now prepared to differentiate in time in order to establish that P_t is order preserving.

Proposition 7.57. *Let $u_0, v_0 \in H^1(I)$, and let $\mathbf{u}, \mathbf{v}$ be the solutions to the gradient flow problem with initial data u_0, v_0 respectively. Then the function $\Delta : [0, +\infty) \rightarrow [0, +\infty)$ given by*

$$\Delta(t) := \frac{1}{2} \int_I [(\mathbf{u}(t) - \mathbf{v}(t))^+]^2 \, dx$$

is nonincreasing.

Proof. Let $0 \leq a < b < +\infty$, so that it suffices to show that $\Delta(a) \geq \Delta(b)$. Apply [Lemma 7.56](#) to $\mathbf{u}$ and $\mathbf{v}$ to obtain functions $\tilde{\mathbf{u}}', \tilde{\mathbf{u}}, \tilde{\mathbf{v}}',$ and $\tilde{\mathbf{v}}$ with the properties stated in that lemma. Define the function $f : [a, b] \times I \rightarrow \mathbb{R}$ by

$$f(t, x) := \frac{1}{2} [(\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x))^+]^2.$$

Note that f is jointly measurable, since both $\tilde{\mathbf{u}}$ and $\tilde{\mathbf{v}}$ are. We also obtain that, for each $t \in [a, b]$,

$$\Delta(t) = \int_I f(t, x) \, dx.$$

We now verify that f satisfies the conditions of [Lemma 7.50](#). Let us verify the first condition. Since $\tilde{\mathbf{u}}(t, \cdot), \tilde{\mathbf{v}}(t, \cdot) \in L^2(I)$, it follows that $f(t, \cdot) \in L^1(I)$, as desired.

We now verify the second condition. We already have that, for all $x \in I$, $\tilde{\mathbf{u}}(\cdot, x)$ and $\tilde{\mathbf{v}}(\cdot, x)$ are AC. [Fact 7.52](#) implies that, for all $x \in I$, the function $t \mapsto (\tilde{\mathbf{u}}(\cdot, x) - \tilde{\mathbf{v}}(\cdot, x))^+$ is AC. But the function $y \mapsto y^2$ is locally Lipschitz, so [Fact 7.53](#) implies that $f(\cdot, x)$ is AC for each x , so the second condition is satisfied. Moreover, using [Facts 7.52](#) and [7.53](#), its weak derivative is

$$\begin{aligned} \partial_t f(t, x) &= [(\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x))^+] \partial_t [(\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x))^+] \\ &= [(\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x))^+] \chi_{\{\tilde{\mathbf{u}}(t, x) > \tilde{\mathbf{v}}(t, x)\}} [\partial_t \tilde{\mathbf{u}}(t, x) - \partial_t \tilde{\mathbf{v}}(t, x)] \\ &= \chi_{\{\tilde{\mathbf{u}}(t, x) > \tilde{\mathbf{v}}(t, x)\}} [\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x)] [\tilde{\mathbf{u}}'(t, x) - \tilde{\mathbf{v}}'(t, x)]. \end{aligned}$$

We claim that $\int_{(a,b)} \int_I |\partial_t f(t, x)| \, dx \, dt < +\infty$. Indeed, we have

$$\begin{aligned}
& \int_{(a,b)} \int_I |\partial_t f(t, x)| \, dx \, dt \\
&= \int_{(a,b)} \int_I \left| \chi_{\{\tilde{\mathbf{u}}(t,x) > \tilde{\mathbf{v}}(t,x)\}} [\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x)] [\tilde{\mathbf{u}}'(t, x) - \tilde{\mathbf{v}}'(t, x)] \right| \, dx \, dt \\
&\leq \frac{1}{2} \int_{(a,b)} \int_I (\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x))^2 \, dx \, dt + \frac{1}{2} \int_{(a,b)} \int_I (\tilde{\mathbf{u}}'(t, x) - \tilde{\mathbf{v}}'(t, x))^2 \, dx \, dt \\
&\leq \int_{(a,b)} \int_I \tilde{\mathbf{u}}(t, x)^2 \, dx \, dt + \int_{(a,b)} \int_I \tilde{\mathbf{v}}(t, x)^2 \, dx \, dt + \int_{(a,b)} \int_I \tilde{\mathbf{u}}'(t, x)^2 \, dx \, dt + \int_{(a,b)} \int_I \tilde{\mathbf{v}}'(t, x)^2 \, dx \, dt.
\end{aligned}$$

We claim that each of the four terms above is finite. First, using Tonelli's theorem, the definition of $\tilde{\mathbf{u}}$ and Jensen's inequality, and letting $L := b - a$, we have

$$\begin{aligned}
\int_{(a,b)} \int_I \tilde{\mathbf{u}}(t, x)^2 \, dx \, dt &= \int_I \int_{(a,b)} \left(\mathbf{u}(a)(x) + \int_{(a,t)} \tilde{\mathbf{u}}'(s, x) \, ds \right)^2 \, dt \, dx \\
&\leq 2 \int_I \int_{(a,b)} \mathbf{u}(a)(x)^2 \, dt \, dx + 2L \int_I \int_{(a,b)} \int_{(a,t)} \tilde{\mathbf{u}}'(s, x)^2 \, ds \, dt \, dx \\
&\leq 2 \int_{(a,b)} \int_I \mathbf{u}(a)(x)^2 \, dx \, dt + 2L \int_{(a,b)} \int_{(a,b)} \int_I \tilde{\mathbf{u}}'(s, x)^2 \, dx \, ds \, dt \\
&= 2L \|\mathbf{u}(a)\|_{L^2(I)}^2 + 2L^2 \int_{(a,b)} \int_I \tilde{\mathbf{u}}'(s, x)^2 \, dx \, ds,
\end{aligned}$$

where in the last line we write $\|\mathbf{u}(a)\|_{L^2(I)}^2$, which is finite, since $\mathbf{u}(a) \in L^2(I)$. Hence, since the argument for the $\mathbf{v}$ terms proceeds identically, all we need to show is that

$$\int_{(a,b)} \int_I \tilde{\mathbf{u}}'(t, x)^2 \, dx \, dt \stackrel{?}{<} +\infty.$$

Since $\tilde{\mathbf{u}}'(t, \cdot) = \mathbf{u}'(t)$ in $L^2(I)$ for a.e. $t \in (a, b)$, this is equivalent to showing that

$$\int_{(a,b)} \|\mathbf{u}'(t)\|_{L^2(I)}^2 \, dt \stackrel{?}{<} +\infty. \tag{7.22}$$

Recall that, by [Definition 7.16](#), $\mathbf{u}' \in L^2(0, b; L^2(I))$, so in particular $\mathbf{u}' \in L^2(a, b; L^2(I))$, which by definition implies (7.22). Hence the claim holds and the third condition of [Lemma 7.50](#) is satisfied.

Therefore, [Lemma 7.50](#) implies that $\Delta(\cdot) = \int_I f(\cdot, x) dx$ is AC and, for a.e. $t \in (a, b)$,

$$\begin{aligned}
\partial_t \Delta(t) &= \partial_t \int_I f(t, x) dx = \int_I \partial_t f(t, x) dx \\
&= \int_I \chi_{\{\tilde{u}(t, x) > \tilde{v}(t, x)\}} [\tilde{\mathbf{u}}(t, x) - \tilde{\mathbf{v}}(t, x)] [\tilde{\mathbf{u}}'(t, x) - \tilde{\mathbf{v}}'(t, x)] dx && \text{(Shown above)} \\
&= \int_I \chi_{\{\mathbf{u}(t) > \mathbf{v}(t)\}} [\mathbf{u}(t) - \mathbf{v}(t)] [\mathbf{u}'(t) - \mathbf{v}'(t)] dx && \text{(By [Lemma 7.56](#))} \\
&= \int_I [\mathbf{u}(t) - \mathbf{v}(t)]^+ \partial_x \partial_x [\mathbf{u}(t) \downarrow - \mathbf{v}(t) \downarrow] dx && \text{(By [Lemma 7.24](#)).}
\end{aligned}$$

By [Corollary 7.41](#), $\mathbf{u}(t), \mathbf{v}(t) \in H^1(I)$ and hence are AC, and by [Fact 7.52](#), $[\mathbf{u}(t) - \mathbf{v}(t)]^+$ is AC. Therefore we can integrate by parts and, using [Fact 7.52](#) and the boundary condition $\partial_x \mathbf{u}(t) \downarrow = \partial_x \mathbf{v}(t) \downarrow = 0$ on $\{0, 1\}$ from [Lemma 7.24](#), we obtain

$$\begin{aligned}
\partial_t \Delta(t) &= - \int_I \chi_{\{\mathbf{u}(t) > \mathbf{v}(t)\}} [\partial_x (\mathbf{u}(t) - \mathbf{v}(t))] [\partial_x (\mathbf{u}(t) \downarrow - \mathbf{v}(t) \downarrow)] \\
&= - \int_I \chi_{\{\mathbf{u}(t) > \mathbf{v}(t)\}} [\partial_x \mathbf{u}(t) - \partial_x \mathbf{v}(t)] [\partial_x^- \mathbf{u}(t) - \partial_x^- \mathbf{v}(t)] \quad \text{(By [Proposition 7.30](#)).}
\end{aligned}$$

But for any numbers $\alpha, \beta \in \mathbb{R}$, it is the case that $\alpha \geq \beta \implies (\alpha \wedge 0) \geq (\beta \wedge 0)$ and $\alpha \leq \beta \implies (\alpha \wedge 0) \leq (\beta \wedge 0)$. Hence the integrand above is pointwise nonnegative, and we conclude that $\partial_t \Delta(t) \leq 0$ for a.e. $t \in (a, b)$. Hence $\Delta(a) \geq \Delta(b)$, as needed. $\square$

Corollary 7.58 (“Directed nonexpansiveness” of P_t). *Let $u, v \in L^2(I)$. Then for all $t > 0$,*

$$\int_I [(P_t u - P_t v)^+]^2 dx \leq \int_I [(u - v)^+]^2 dx.$$

Proof. This is an immediate consequence of [Proposition 7.57](#) when $u, v \in H^1(I)$, and the general case follows by approximating u and v by $H^1(I)$ functions and using the continuity of the map $w \mapsto P_t w$ from $L^2(I)$ to $L^2(I)$. $\square$

Corollary 7.59 (P_t is order preserving). *Let $u, v \in L^2(I)$, and suppose $u \leq v$ a.e. . Then $P_t u \leq P_t v$ a.e. for all $t > 0$.*

Proof. This is an immediate consequence of [Corollary 7.58](#). $\square$

We also note below that the semigroup P_t has no effect on monotone functions.

Lemma 7.60. *Every nondecreasing $u \in L^2(I)$ is a stationary point of P_t , i. e. $P_t u = u$ for all t .*

Proof. By definition, $u = u\uparrow + u\downarrow$ with $u\uparrow = u$ and $u\downarrow = 0$, so $\mathcal{E}^-(u) = 0$. Thus for every $v \in L^2(I)$,

$$\mathcal{E}^-(v) \geq 0 = \mathcal{E}^-(u) + \langle 0, v - u \rangle ,$$

so $0 \in \partial\mathcal{E}^-(u)$. Thus $\mathbf{u}(t) = u$ is the solution to the gradient flow problem with initial state u . $\square$

Observation 7.61. *Corollary 7.59 in particular implies that if $u \geq a$ a. e. for some $a \in \mathbb{R}$, then $P_t u \geq P_t a = a$ a. e. for all $t > 0$, the equality by Lemma 7.60.*

The following results show that P_t is degree one positively homogeneous and additive when one argument is a constant function.

Lemma 7.62. *Let $u \in \mathcal{U}$, $\beta \in \mathbb{R}$, and $v := u + \beta$. Then $v \in \mathcal{U}$ with $v\uparrow = u\uparrow + \beta$ and $v\downarrow = u\downarrow$, and thus $\mathcal{E}^-(v) = \mathcal{E}^-(u)$.*

Proof. This is a straightforward consequence of the definition of $v\uparrow, v\downarrow$. $\square$

Lemma 7.63. *Let $u \in \mathcal{U}$, $\alpha > 0$, and $v := \alpha u$. Then $v \in \mathcal{U}$ with $v\uparrow = \alpha u\uparrow$ and $v\downarrow = \alpha u\downarrow$, and thus $\mathcal{E}^-(v) = \alpha^2 \mathcal{E}^-(u)$.*

Proof. This is a straightforward consequence of the definitions of $v\uparrow, v\downarrow$ and $\mathcal{E}^-$. $\square$

Lemma 7.64. *Let $u \in D(\partial\mathcal{E}^-)$, $\beta \in \mathbb{R}$, and $v := u + \beta$. Then $\partial\mathcal{E}^-(u) = \partial\mathcal{E}^-(v)$.*

Proof. By symmetry, it suffices to prove that $\partial\mathcal{E}^-(u) \subseteq \partial\mathcal{E}^-(v)$. Let $z \in \partial\mathcal{E}^-(u)$. Then for all $w \in \mathcal{U}$, using Lemma 7.62 twice and the definition of subdifferential,

$$\mathcal{E}^-(w) = \mathcal{E}^-(w - \beta) \geq \mathcal{E}^-(u) + \langle z, w - \beta - u \rangle = \mathcal{E}^-(v) + \langle z, w - v \rangle ,$$

and hence $z \in \partial\mathcal{E}^-(v)$ as desired. $\square$

Lemma 7.65. *Let $u \in D(\partial\mathcal{E}^-)$, $\alpha > 0$, and $v := \alpha u$. Then $\partial\mathcal{E}^-(v) = \alpha \partial\mathcal{E}^-(u)$.*

Proof. By symmetry, it suffices to prove that $\alpha\partial\mathcal{E}^-(u) \subseteq \partial\mathcal{E}^-(v)$. Let $z \in \partial\mathcal{E}^-(u)$; we claim that $\alpha z \in \partial\mathcal{E}^-(v)$. Indeed for all $w \in \mathcal{U}$, using [Lemma 7.63](#) twice and the definition of subdifferential,

$$\mathcal{E}^-(w) = \alpha^2 \mathcal{E}^-\left(\frac{1}{\alpha}w\right) \geq \alpha^2 \left(\mathcal{E}^-(u) + \langle z, \frac{1}{\alpha}w - u \rangle\right) = \mathcal{E}^-(v) + \langle \alpha z, w - v \rangle,$$

and hence $\alpha z \in \partial\mathcal{E}^-(v)$ as desired. $\square$

Proposition 7.66 (Effect of certain affine transformations on P_t). *Let $u \in L^2(I)$, $\alpha > 0$, and $\beta \in \mathbb{R}$. Then $P_t(\alpha u + \beta) = \alpha P_t u + \beta$.*

Proof. We suffices to prove the statement for $u \in \mathcal{U}$; the general case then follows by approximating $\mathcal{U}$ by $H^1(I)$ functions and the continuity of P_t from $L^2(I)$ to $L^2(I)$. For $v := u + \beta$, we observe that $\mathbf{v}(t) := \mathbf{u}(t) + \beta$ with $\mathbf{v}'(t) := \mathbf{u}'(t)$ is a solution to the gradient flow problem with initial state v , since for each $t > 0$ for which $-\mathbf{u}'(t) \in \partial\mathcal{E}^-(\mathbf{u}(t))$, [Lemma 7.64](#) implies that $-\mathbf{v}'(t) \in \partial\mathcal{E}^-(\mathbf{v}(t))$.

Similarly, let $w := \alpha u$. Then $\mathbf{w}(t) := \alpha \mathbf{u}(t)$ with $\mathbf{w}'(t) := \alpha \mathbf{u}'(t)$ is the solution to the gradient flow problem with initial state w , since for each $t > 0$ for which $-\mathbf{u}'(t) \in \partial\mathcal{E}^-(\mathbf{u}(t))$, [Lemma 7.65](#) implies that $-\mathbf{w}'(t) \in \partial\mathcal{E}^-(\mathbf{w}(t))$. $\square$

7.2.8 Convergence to monotone equilibrium

Since the directed Dirichlet energy $\mathcal{E}^-(P_t u)$ decays over time and $P_t u$ converges to some limit as $t \rightarrow \infty$ (by [Proposition 7.27](#)), we expect this limit to be a monotone function. Let us establish this fact and other properties of that limit.

Lemma 7.67. *Let $u \in \mathcal{U}$ satisfy $\mathcal{E}^-(u) = 0$. Then u is nondecreasing.*

Proof. Since $\mathcal{E}^-(u) = 0$, we have $\partial_x u \downarrow = 0$ a.e. in I . Hence $u \downarrow$ is a constant function, while $u \uparrow$ is nondecreasing by definition. $\square$

Proposition 7.68. *Let $u \in L^2(I)$. Then there exists a nondecreasing $u^* \in \mathcal{U}$, unique as an element of $L^2(I)$, such that $P_t u \rightarrow u^*$ in $L^2(I)$ as $t \rightarrow \infty$.*

Proof. Since $\mathcal{E}^-$ achieves its minimum (namely 0, on e.g. constant functions) and $P_t u \in D(\partial\mathcal{E}^-)$ for all $t > 0$, [[AC84](#), Theorem 2, p. 160] implies that there exists a minimizer u^* of $\mathcal{E}^-$ such that $P_t u \rightharpoonup u^*$ weakly in $L^2(I)$. This means that $u^* \in \mathcal{U}$ with $\mathcal{E}^-(u^*) = 0$, so by [Lemma 7.67](#) u^* is nondecreasing. By [Proposition 7.27](#) (which we may apply because,

fixing any $t_0 > 0$, we have $P_{t_0}u \in D(\partial\mathcal{E}^-) \subseteq D(\mathcal{E}^-) = \mathcal{U}$, P_tu also converges strongly in $L^2(I)$, and it is standard that the weak and strong limits agree and that this limit is unique. $\square$

Therefore the following definition is justified.

Definition 7.69 (Monotone equilibrium). Let $P_\infty : L^2(I) \rightarrow \mathcal{U}$ be the operator mapping each $u \in L^2(I)$ to the unique (as an element of $L^2(I)$) nondecreasing $u^* \in \mathcal{U}$ such that $P_tu \rightarrow u^*$ in $L^2(I)$ as $t \rightarrow \infty$. We call $P_\infty u$ the *monotone equilibrium* of u .

We now pass to the limit P_∞ some useful properties of P_t .

Proposition 7.70 (“Directed nonexpansiveness” of P_∞). *Let $u, v \in L^2(I)$. Then*

$$\int_I [(P_\infty u - P_\infty v)^+]^2 dx \leq \int_I [(u - v)^+]^2 dx.$$

Proof. This follows by a standard limit argument as follows. Let $u, v \in L^2(I)$. By [Corollary 7.58](#), $\|(P_t u - P_t v)^+\|_{L^2(I)} \leq \|(u - v)^+\|_{L^2(I)}$ for all $t > 0$. By definition of P_∞ , we have $P_t u \rightarrow P_\infty u$ and $P_t v \rightarrow P_\infty v$ in $L^2(I)$ as $t \rightarrow \infty$. Using the triangle inequality, we obtain

$$\begin{aligned} \|(P_t u - P_t v)^+ - (P_\infty u - P_\infty v)^+\|_{L^2(I)} &\leq \|(P_t u - P_t v) - (P_\infty u - P_\infty v)\|_{L^2(I)} \\ &\leq \|P_t u - P_\infty u\|_{L^2(I)} + \|P_t v - P_\infty v\|_{L^2(I)} \rightarrow 0, \end{aligned}$$

so $(P_t u - P_t v)^+ \rightarrow (P_\infty u - P_\infty v)^+$ in $L^2(I)$ as $t \rightarrow \infty$. Since $\|(P_t u - P_t v)^+\|_{L^2(I)} \leq \|(u - v)^+\|_{L^2(I)}$ for all $t > 0$, we conclude that $\|(P_\infty u - P_\infty v)^+\|_{L^2(I)} \leq \|(u - v)^+\|_{L^2(I)}$, which gives the conclusion. $\square$

Corollary 7.71 (P_∞ is order preserving). *Let $u, v \in L^2(I)$, and suppose $u \leq v$ a. e. . Then $P_\infty u \leq P_\infty v$ a. e. .*

Proof. This is a direct consequence of [Proposition 7.70](#). $\square$

Proposition 7.72. P_∞ is nonexpansive, and therefore continuous, as an $L^2(I) \rightarrow L^2(I)$ map.

Proof. Let $u, v \in L^2(I)$. In the proof of [Proposition 7.70](#), we showed that $\|(P_t u - P_t v) - (P_\infty u - P_\infty v)\|_{L^2(I)} \rightarrow 0$, which implies that $\|P_t u - P_t v\|_{L^2(I)} \rightarrow \|P_\infty u - P_\infty v\|_{L^2(I)}$. But $\|P_t u - P_t v\|_{L^2(I)} \leq \|u - v\|_{L^2(I)}$ for all $t > 0$ since $(P_t)_{t \geq 0}$ is a nonexpansive semigroup, so the conclusion follows. $\square$

Lemma 7.73. *For every nondecreasing $u \in L^2(I)$, we have $P_\infty u = u$.*

Proof. This follows from [Lemma 7.60](#) along with the convergence $P_t u \rightarrow P_\infty u$ in $L^2(I)$. $\square$

Strong convergence to the monotone equilibrium, together with the preservation of regularity results from [Section 7.2.5](#), allows us to obtain regularity of the monotone equilibrium as well:

Proposition 7.74 (H^1 regularity of the monotone equilibrium). *Let $u \in H^1(I)$. Then $P_\infty u \in H^1(I)$ with $\varphi(P_\infty u) \leq \varphi(u)$.*

Proof. Recall the functional φ from [Section 7.2.5](#). Since $u \in H^1(I)$, $\varphi(u) < +\infty$. By [Proposition 7.40](#), we conclude that $\varphi(P_t u) \leq \varphi(u) < +\infty$ for all $t > 0$. Since φ is lower semicontinuous by [Claim 7.31](#) and $P_t u \rightarrow P_\infty u$ in $L^2(I)$ by [Proposition 7.27](#), we conclude that $\varphi(P_\infty u) \leq \varphi(u) < +\infty$. Hence $P_\infty u \in H^1(I)$. $\square$

Proposition 7.75 (Lipschitz regularity of the monotone equilibrium). *Let $u \in W^{1,\infty}(I)$. Then $P_\infty u \in W^{1,\infty}(I)$ with $\psi(P_\infty u) \leq \psi(u)$.*

Proof. Recall the functional ψ from [Section 7.2.6](#). Since $u \in W^{1,\infty}(I)$, $\psi(u) < +\infty$. By [Proposition 7.47](#), we conclude that $\psi(P_t u) \leq \psi(u) < +\infty$ for all $t > 0$. Since ψ is lower semicontinuous by [Claim 7.43](#) and $P_t u \rightarrow P_\infty u$ in $L^2(I)$ by [Proposition 7.27](#), we conclude that $\psi(P_\infty u) \leq \psi(u) < +\infty$. Hence $P_\infty u \in W^{1,\infty}(I)$. $\square$

We observe that P_∞ also behaves nicely under the appropriate class of affine transformations.

Proposition 7.76 (Effect of certain affine transformations on P_∞). *Let $u \in L^2(I)$, $\alpha > 0$, and $\beta \in \mathbb{R}$. Then $P_\infty(\alpha u + \beta) = \alpha P_\infty u + \beta$.*

Proof. Since $P_t v \rightarrow P_\infty v$ as $t \rightarrow \infty$ for each $v \in L^2(I)$, applying [Proposition 7.66](#) yields

$$P_\infty(\alpha u + \beta) = \lim_{t \rightarrow \infty} P_t(\alpha u + \beta) = \lim_{t \rightarrow \infty} [\alpha P_t u + \beta] = \alpha \left[\lim_{t \rightarrow \infty} P_t u \right] + \beta = \alpha P_\infty u + \beta. \quad \square$$

We can finally conclude, via [Lemma 7.49](#), that P_∞ is nonexpansive in the L^∞ norm.

Proposition 7.77 (P_∞ is nonexpansive in L^∞ norm). *Let $u, v \in L^2(I)$. Then $|P_\infty u - P_\infty v| \leq \text{ess sup}|u - v|$ a. e. in I .*

Proof. We verify the conditions of [Lemma 7.49](#) with (Ω, Σ, μ) the set I endowed with the Lebesgue measure, $p = 2$, $C = L^2(I)$, and $T = P_\infty$. It is clear that for all $f \in L^2(I)$ and $r \in \mathbb{R}$, $f + r \in L^2(I)$. By [Proposition 7.72](#), P_∞ is a continuous $L^2(I) \rightarrow L^2(I)$ map. Moreover, for all $f \in L^2(I)$ and $r \in \mathbb{R}$, the condition $P_\infty(f + r) = P_\infty f + r$ holds by [Proposition 7.76](#). Finally, for all $f, g \in L^2(I)$ with $f \leq g$ a.e., we have that $P_\infty f \leq P_\infty g$ a.e. by [Corollary 7.71](#). The conclusion follows. $\square$

7.3 Directed transport-energy inequality

In this section, we establish a connection between the PDE studied above and the Wasserstein distance between the initial state u and its monotone equilibrium $P_\infty u$, via the dynamical approach embodied by the *Benamou-Brenier formula*, and then *tensorize* this result from one dimension to $[0, 1]^d$. For optimal transport concepts, we follow the presentation and formalism of [\[San15\]](#), and also refer to [\[AGS05; Vil09\]](#).

This section is organized as follows. After preliminary considerations in [Section 7.3.1](#), [Section 7.3.2](#) shows that the PDE results from [Section 7.2](#) imply a result about optimal transport (Wasserstein) distance, namely a one-dimensional *transport-energy inequality*, thereby implementing the ideas described in [Section 7.1.3](#) of the proof overview. The idea is to show that our solution to the directed heat equation

$$\partial_t u = \partial_x \partial_x^- u$$

satisfies the so-called *continuity equation*

$$\partial_t \varrho_t + \partial_x(\varrho_t v_t) = 0,$$

which prescribes that “particles” (distributed according to measure ϱ_t) move according to the velocity field v_t ; in our setting, we have $d\varrho_t = u dx$ and $v_t = -\frac{\partial_x u \downarrow}{u}$ ([Proposition 7.85](#)), so that indeed $\partial_x(\varrho_t v_t) = -\partial_x \partial_x u \downarrow = -\partial_t \varrho_t$. For functions u bounded close to 1, we can informally disregard the denominator in v_t and imagine, for simplicity, that $v_t \approx -\partial_x u \downarrow$. Then thanks to the exponential decay of

$$\mathcal{E}^-(u) = \frac{1}{2} \int (\partial_x u \downarrow)^2 dx$$

from the previous section, we conclude that the right-hand side of the Benamou-Brenier formula

$$W_2^2(\mu, \nu) = \min \left\{ \int_0^1 \|v_t\|_{L^2(\varrho_t)}^2 dt : \partial_t \varrho_t + \partial_x(\varrho_t v_t) = 0, \varrho_0 = \mu, \varrho_1 = \nu \right\}$$

from [Proposition 7.83](#) is roughly upper bounded by $\mathcal{E}^-(u)$, where (informally) the measures $d\mu = u dx$ and $d\nu = (P_\infty u) dx$ correspond to the initial state and monotone equilibrium of the directed heat equation, as probability measures⁴. Therefore we conclude in [Theorem 7.2](#) the transport-energy inequality in one dimension,

$$W_2^2(\mu, \mu_\infty) \lesssim \mathcal{E}^-(u).$$

Finally, the continuity equation also lets us conclude natural and useful properties of solutions of the directed heat equation, such as mass conservation ([Corollaries 7.86](#) and [7.87](#)) and the property that solutions can only ever lose mass in any given prefix of the interval $[0, 1]$ ([Corollary 7.91](#)).

The next goal is to establish the *tensorized* (multidimensional) transport-energy inequality, thereby implementing the ideas described in [Section 7.1.4](#) of the proof overview. In [Section 7.3.3](#), we develop the definitions and technical tools required to show our “Pythagorean composition” result ([Lemma 7.3](#)), which yields a recipe for proving multidimensional transport-energy inequalities (with respect to the “directed Wasserstein distance”) by induction ([Lemma 7.110](#)).

Finally, [Section 7.3.4](#) ties together the tensorization recipe described above with the one-dimensional results specific to our PDE from the previous sections. We show that, by taking an initial state $f : [0, 1]^d \rightarrow \mathbb{R}$ with corresponding measure $d\mu = f dx$, applying the directed heat semigroup one coordinate at a time, and putting together the one-dimensional transport-energy inequalities at each of these steps via properties of our PDE such as order preservation and nonexpansiveness (which enable the induction to go through), we indeed obtain a *monotone* final state $f^* : [0, 1]^d \rightarrow \mathbb{R}$ with corresponding measure $d\mu^* = f^* dx$ for which the multidimensional transport-energy inequality applies ([Theorem 7.4](#)):

$$W_2^2(\mu \rightarrow \mu^*) \lesssim \int_{[0,1]^d} |\nabla^- f|^2 dx.$$

7.3.1 Preliminaries for optimal transport

We start by introducing notation and definitions relevant to the theory of optimal transport.

⁴More formally, we decompose the evolution from time $t = 0$ to time $t \rightarrow \infty$ into intervals of unit time length, but by the exponential energy decay, this summation is dominated by the initial term $\mathcal{E}^-(u)$.

Projections. Say $X \times Y$ is a product space and $z = (x, y)$ is an element in this space. Then the projection operator onto the first coordinate, denoted interchangeably by π_1 or π_x depending on the context, is given by $\pi_1(z) = \pi_x(z) = x$. Similarly, $\pi_2(z) = \pi_y(z) = y$. We extend this definition in the natural way to projections from larger product spaces onto smaller product spaces. For example, given the product space $X_1 \times \cdots \times X_d$ and index set $I \subset [d]$, $I = \{i_1, \dots, i_n\}$, the operator π_I projects any element in this space down to an element of $X_{i_1} \times \cdots \times X_{i_n}$. We also use the shorthand $\pi_{-I} := \pi_{[d] \setminus I}$.

Now, let $\Omega \subset \mathbb{R}^d$ be a Borel set and let $I \subseteq [d]$. For a point $x \in \Omega$, we write the projections $x_I := \pi_I(x)$ and $x_{-I} := \pi_{-I}(x)$, and we also write $x = (x_I, x_{-I})$. Then, for the set Ω , we write the projections $\Omega_I := \{x_I : x \in \Omega\}$ and $\Omega_{-I} := \{x_{-I} : x \in \Omega\}$. For small sets I or $[d] \setminus I$, we also use shorthand notation such as $x_i := x_{\{i\}}$ and $\Omega_{-i-j} := \Omega_{-\{i,j\}}$.

Pushforward measure. Given measurable spaces (X, Σ_X) and (Y, Σ_Y) , measure μ on X , and measurable map $T : X \rightarrow Y$, the *pushforward measure* $T_{\#}\mu$ on Y is the measure satisfying

$$\begin{aligned} (T_{\#}\mu)(B) &= \mu(T^{-1}(B)) && \text{for all } B \in \Sigma_Y, \text{ or equivalently,} \\ \int_Y \phi(y) d(T_{\#}\mu)(y) &= \int_X \phi(T(x)) d\mu(x) && \text{for all measurable } \phi : Y \rightarrow \mathbb{R}. \end{aligned}$$

For a measure γ on product space $X \times Y$, we say that $(\pi_1)_{\#}\gamma$ and $(\pi_2)_{\#}\gamma$, which are measures on X and Y respectively, are the first and second *marginals* of γ , respectively.

If (Ω_3, Σ_3) is another measurable space and $S : Y \rightarrow Z$ a measurable map, then it holds that $(S \circ T)_{\#}\mu = S_{\#}(T_{\#}\mu)$.

Transport plans. For two probability spaces $(\Omega_1, \Sigma_1, \mu_1)$ and $(\Omega_2, \Sigma_2, \mu_2)$, we write $\Pi(\mu_1, \mu_2)$ for the set of *couplings*, or *transport plans*, between μ_1 and μ_2 , namely probability measures γ on the product space $\Omega_1 \times \Omega_2$ whose first and second marginals are μ_1 and μ_2 , respectively.

Space of probability measures. For any Borel set $\Omega \subset \mathbb{R}^d$, let $P(\Omega)$ denote the space of all (Borel) probability measures on Ω . We endow $P(\Omega)$ with the weak topology, which is the topology of weak convergence with respect to bounded continuous functionals. Namely, we say μ_n converges weakly to μ in $P(\Omega)$, and write $\mu_n \rightharpoonup \mu$, if $\int_{\Omega} \phi d\mu_n \rightarrow \int_{\Omega} \phi d\mu$ for all bounded continuous $\phi : \Omega \rightarrow \mathbb{R}$.

Wasserstein distances. Let $\Omega \subset \mathbb{R}^d$ be a *bounded* Borel set and let $p \in [1, +\infty)$. Let $\mu, \nu \in P(\Omega)$. Given transport plan $\gamma \in \Pi(\mu, \nu)$, we define the *cost* $C_p(\gamma)$ by

$$C_p(\gamma) := \left(\int_{\Omega \times \Omega} |x - y|^p d\gamma(x, y) \right)^{1/p}.$$

We then define the *p-Wasserstein* distance between μ and ν by

$$W_p(\mu, \nu) := \inf_{\gamma \in \Pi(\mu, \nu)} C_p(\gamma),$$

and we also often refer to the quantity $W_p^p(\mu, \nu) := W_p(\mu, \nu)^p$. It is standard that $W_p(\cdot, \cdot)$ is a distance metric on $P(\Omega)$.

Fact 7.78 (Wasserstein distance metrizes weak convergence; see e.g. [San15, Theorems 5.10 and 5.11]). *Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set, and let $p \in [1, \infty)$. Then for a sequence $(\mu_n)_n$ in $P(\Omega)$ and $\mu \in P(\Omega)$, we have $\mu_n \rightharpoonup \mu$ if and only if $W_p(\mu_n, \mu) \rightarrow 0$.*

Corollary 7.79. *Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set, and let $p \in [1, \infty)$. Then for sequences $(\mu_n)_n$ and $(\nu_n)_n$ in $P(\Omega)$ with $\mu_n \rightharpoonup \mu$ and $\nu_n \rightharpoonup \nu$ for $\mu, \nu \in P(\Omega)$, we have $W_p(\mu_n, \nu_n) \rightarrow W_p(\mu, \nu)$. In other words, the *p-Wasserstein* distance is continuous in (the weak topology on) $P(\Omega)$.*

Proof. By the triangle inequality, we have

$$W_p(\mu, \nu) - W_p(\mu, \mu_n) - W_p(\nu_n, \nu) \leq W_p(\mu_n, \nu_n) \leq W_p(\mu_n, \mu) + W_p(\mu, \nu) + W_p(\nu, \nu_n).$$

By Fact 7.78, the LHS and RHS converge to $W_p(\mu, \nu)$ as $n \rightarrow \infty$, so the conclusion follows. $\square$

7.3.2 Optimal transport via Benamou-Brenier

Let $\{\varrho_t : t \in [0, T]\}$ be a family of measures on $\bar{I}$ and $\{v_t : t \in [0, T]\}$ be a family of *velocity fields* such that $v_t \in L^1(\varrho_t)$ for each t . We start by defining what it means for the family (ϱ_t, v_t) to solve the *continuity equation*

$$\partial_t \varrho_t + \partial_x(\varrho_t v_t) = 0.$$

Definition 7.80 (Weak solution; see [San15, Section 4.1.2]). Let (ϱ_t, v_t) be a family of measure/velocity field pairs indexed by $t \in [0, T]$ such that $v_t \in L^1(\varrho)$ for each t . We say that (ϱ_t, v_t) is a *weak solution to the continuity equation* if, for every test function $\psi \in C^1(\bar{I})$, the function $t \mapsto \int_{\bar{I}} \psi \, d\varrho_t$ is absolutely continuous in t and, for a.e. t , we have

$$\partial_t \int_{\bar{I}} \psi \, d\varrho_t = \int_{\bar{I}} (\partial_x \psi) v_t \, d\varrho_t. \quad (7.23)$$

In this case, we call ϱ_0 and ϱ_T the *initial* and *final states* of the solution, respectively (this makes sense because the above implies that $t \mapsto \varrho_t$ is continuous for the weak convergence of measures).

Remark 7.81. We will only work with absolutely continuous measures $d\varrho_t = u \, dx$, $u \in L^2(I)$.

Remark 7.82. Applying (7.23) with a constant test function ψ shows that every solution to the continuity equation is *mass conserving*. In particular, Proposition 7.85 below implies that this is true of the semigroup P_t ; see Corollary 7.86.

The continuity equation is connected to Wasserstein distance via the *Benamou-Brenier formula*:

Proposition 7.83 (Benamou-Brenier formula; see e.g. [San15, Theorem 5.28]). *Let μ, ν be probability measures on $\bar{I}$. Then*

$$W_2^2(\mu, \nu) = \min \left\{ \int_0^1 \|v_t\|_{L^2(\varrho_t)}^2 \, dt : \partial_t \varrho_t + \partial_x(\varrho_t v_t) = 0, \varrho_0 = \mu, \varrho_1 = \nu \right\},$$

where the constraint above means that $(\varrho_t, v_t)_{t \in [0,1]}$ is a weak solution to the continuity equation with initial state μ and final state ν .

We now show our solution to the directed heat equation yields a solution to the continuity equation. This allows us to conclude several nice properties of this solution, such as mass conservation, as well as apply the Benamou-Brenier formula to obtain a *transport-energy inequality*.

Condition 7.84. $u \in \mathcal{U}$ is a.e. positive and bounded away from zero, and satisfies $\int_I u \, dx = 1$.

Proposition 7.85. *Let $u \in \mathcal{U}$ satisfy [Condition 7.84](#). Let $\mathbf{u}(t)$ be the solution to the gradient flow problem with initial state u , and $\mathbf{u}'(t)$ its weak derivative for each $t \geq 0$. Let $T > 0$ and define the measure/velocity field family $(\varrho_t, v_t)_{t \in [0, T]}$ by*

$$d\varrho_t := \mathbf{u}(t) dx \quad \text{and} \quad v_t := -\frac{\partial_x \mathbf{u}(t) \downarrow}{\mathbf{u}(t)}$$

for each $t \in [0, T]$. Then (ϱ_t, v_t) is a weak solution to the continuity equation with initial state $\mathbf{u}(0) dx$ and final state $\mathbf{u}(T) dx$.

Proof. We first note that, by [Observation 7.61](#), each $\mathbf{u}(t)$ is a.e. positive and bounded away from zero, uniformly in t . In particular, this justifies the denominator in the definition of v_t . Furthermore, recall that $\mathbf{u}(t) \in \mathcal{U}$ for all $t \geq 0$ by [Corollary 7.23](#) and [Definition 7.18](#), so the numerator in the definition of v_t is defined up to sets of measure zero for each t . (In fact, by [Definition 7.19](#), [Proposition 7.21](#), and [Lemma 7.24](#) we have $\mathbf{u}(t) \downarrow \in H^2(I)$ for a.e. $t > 0$, in which case the numerator is even pointwise well-defined.)

We now verify that (ϱ_t, v_t) satisfies [Definition 7.80](#). The initial and final states are as claimed by construction. Let $\psi \in C^1(\bar{I})$. Let $\tilde{\mathbf{u}}', \tilde{\mathbf{u}} : [0, T] \times I \rightarrow \mathbb{R}$ be obtained by applying [Lemma 7.56](#) to $\mathbf{u}$ with time domain $[0, T]$. Define the (jointly measurable) function $f : [0, T] \times I \rightarrow \mathbb{R}$ by

$$f(t, x) := \psi(x) \tilde{\mathbf{u}}(t, x).$$

We claim that f satisfies the properties of [Lemma 7.50](#). It is clear that $f(t, \cdot) \in L^2(I) \subset L^1(I)$ for each $t \in [0, T]$ since $\mathbf{u}(t, \cdot) \in L^2(I)$ while ψ is bounded, so the first property is satisfied. Also, for each $x \in I$ we have that $\tilde{\mathbf{u}}(\cdot, x)$ is AC and hence so is $f(\cdot, x)$, so the second property is satisfied. Finally, by properties 1 and 5 of [Lemma 7.56](#) and the Cauchy-Schwarz inequality,

$$\int_{(0, T)} \int_I |\partial_t f(t, x)| dx dt = \int_{(0, T)} \int_I |\psi(x) \tilde{\mathbf{u}}'(t, x)| dx dt \leq \|\psi\|_{L^2(I)} \int_{(0, T)} \|\mathbf{u}'(t)\|_{L^2(I)} dt < +\infty,$$

the last inequality since $\psi \in C^1(\bar{I})$ while $\mathbf{u}' \in L^2(0, T; L^2(I))$ by [Definition 7.16](#). Hence the third property is satisfied and [Lemma 7.50](#) applies. Thus the function mapping each $t \in [0, T]$ to

$$\int_{\bar{I}} \psi d\varrho_t = \int_I \psi \mathbf{u}(t) dx = \int_I \psi(x) \tilde{\mathbf{u}}(t, x) dx = \int_I f(t, x) dx$$

is absolutely continuous and, for a.e. $t \in (0, T)$,

$$\begin{aligned}
\partial_t \int_I \psi \, d\varrho_t &= \int_I \partial_t f(t, x) \, dx = \int_I \psi(x) \tilde{\mathbf{u}}'(t, x) \, dx && \text{(Property 5 of Lemma 7.56)} \\
&= \int_I \psi \mathbf{u}'(t) \, dx && \text{(Property 1 of Lemma 7.56)} \\
&= \int_I \psi \partial_x \partial_x \mathbf{u}(t) \downarrow \, dx && \text{(Proposition 7.21 and Lemma 7.24)} \\
&= - \int_I (\partial_x \psi) (\partial_x \mathbf{u}(t) \downarrow) \, dx && \text{(Integration by parts, Lemma 7.24)} \\
&= \int_I (\partial_x \psi) \left(- \frac{\partial_x \mathbf{u}(t) \downarrow}{\mathbf{u}(t)} \right) \mathbf{u}(t) \, dx = \int_I (\partial_x \psi) v_t \, d\varrho_t. \quad \square
\end{aligned}$$

Corollary 7.86 (P_t is mass conserving). *Let $u \in L^2(I)$. Then for all $t > 0$, $\int_I P_t u \, dx = \int_I u \, dx$.*

Proof. If u satisfies Condition 7.84, then this follows from Proposition 7.85 by taking any constant test function ψ in Definition 7.80.

If $u \in H^1(I)$, then it is bounded, so let $\alpha > 0$, $\beta \in \mathbb{R}$ be such that $v := \alpha u + \beta \in H^1(I)$ satisfies Condition 7.84. Then $u = \frac{1}{\alpha} v - \frac{\beta}{\alpha}$ and, by Proposition 7.66 and the above,

$$\begin{aligned}
\int_I P_t u \, dx &= \int_I \left(\frac{1}{\alpha} P_t v - \frac{\beta}{\alpha} \right) \, dx = -\frac{\beta}{\alpha} + \frac{1}{\alpha} \int_I P_t v \, dx \\
&= -\frac{\beta}{\alpha} + \frac{1}{\alpha} \int_I v \, dx = \int_I \left(\frac{1}{\alpha} v - \frac{\beta}{\alpha} \right) \, dx = \int_I u \, dx.
\end{aligned}$$

Finally, let $u \in \mathcal{U}$ be arbitrary. Since $H^1(I)$ is dense in $L^2(I)$, let $(u_n)_n$ be a sequence in $H^1(I)$ such that $u_n \rightarrow u$ in $L^2(I)$. By the continuity of P_t from $L^2(I)$ to $L^2(I)$, and using the above,

$$\int_I P_t u \, dx = \lim_{n \rightarrow \infty} \int_I P_t u_n \, dx = \lim_{n \rightarrow \infty} \int_I u_n \, dx = \int_I u \, dx. \quad \square$$

By passing to the limit $P_t u \rightarrow P_\infty u$ in $L^2(I)$, we also conclude

Corollary 7.87 (P_∞ is mass conserving). *Let $u \in L^2(I)$. Then $\int_I P_\infty u \, dx = \int_I u \, dx$.*

We now upper bound the Wasserstein distance between $u \in \mathcal{U}$ and its monotone equilibrium by combining the Benamou-Brenier formula with the exponential decay of the directed Dirichlet energy, at least as long as u is positive and bounded away from zero.

Theorem 7.2 (Transport-energy inequality in one dimension). *There exists a constant $C > 0$ such that the following holds. Let $u \in \mathcal{U}$ be positive, bounded away from zero, and satisfy $\int_{(0,1)} u \, dx = 1$. Define the measures $d\mu := u \, dx$ and $d\mu_\infty := (P_\infty u) \, dx$. Then*

$$W_2^2(\mu, \mu_\infty) \leq \frac{C}{\inf u} \mathcal{E}^-(u).$$

Proof. Let $a := \inf u$. By Corollaries 7.59 and 7.86, $P_t u$ satisfies Condition 7.84 with $P_t u \geq a$ a.e. for all $t \geq 0$. For each $t \geq 0$, define the measure $d\varrho_t := (P_t u) \, dx$. Let $\ell \in \mathbb{Z}_{\geq 0}$. Then Proposition 7.85 applies to time interval $[\ell, \ell+1]$ (we refrain from introducing excessive notation for such relabeling of intervals) with initial/final states ϱ_ℓ and $\varrho_{\ell+1}$, respectively, so that Proposition 7.83 gives

$$\begin{aligned} W_2^2(\varrho_\ell, \varrho_{\ell+1}) &\leq \int_\ell^{\ell+1} \int_I \left(-\frac{\partial_x \mathbf{u}(t) \downarrow}{\mathbf{u}(t)} \right)^2 \mathbf{u}(t) \, dx \, dt = \int_\ell^{\ell+1} \int_I \frac{(\partial_x \mathbf{u}(t) \downarrow)^2}{\mathbf{u}(t)} \, dx \, dt \\ &\leq \frac{1}{a} \int_\ell^{\ell+1} \int_I (\partial_x \mathbf{u}(t) \downarrow)^2 \, dx \, dt = \frac{2}{a} \int_\ell^{\ell+1} \mathcal{E}^-(P_t u) \, dt \\ &\leq \frac{2}{a} \mathcal{E}^-(u) \int_\ell^{\ell+1} e^{-Kt} \, dt < \frac{2}{a} \mathcal{E}^-(u) \int_\ell^\infty e^{-Kt} \, dt = \frac{2e^{-K\ell}}{aK} \mathcal{E}^-(u), \end{aligned}$$

where the penultimate inequality and the constant K come from Proposition 7.25. Now, by the triangle inequality (on the distance W_2 , not the squared distance W_2^2), for every $\ell \in \mathbb{Z}_{\geq 1}$ we have

$$W_2(\varrho_0, \varrho_\ell) \leq \sum_{j=0}^{\ell-1} W_2(\varrho_j, \varrho_{j+1}) \leq \left(\frac{2}{aK} \mathcal{E}^-(u) \right)^{1/2} \sum_{j=0}^{\ell-1} e^{-Kj/2} = \left(\frac{2}{aK} \mathcal{E}^-(u) \right)^{1/2} \frac{1}{1 - e^{-K/2}}.$$

Therefore, extracting the appropriate constant C from the terms involving K , we obtain

$$W_2(\varrho_0, \varrho_\ell) \leq \frac{C}{a^{1/2}} (\mathcal{E}^-(u))^{1/2}.$$

Recall that $P_t u \rightharpoonup P_\infty u$ weakly in $L^2(I)$ by Proposition 7.68. Hence, defining the absolutely continuous measure $d\varrho_\infty := (P_\infty u) \, dx$, we conclude that $\varrho_t \rightharpoonup \varrho_\infty$ as $t \rightarrow \infty$. Then by Corollary 7.79,

$$W_2(\varrho_0, \varrho_\infty) = \lim_{\ell \rightarrow \infty} W_2(\varrho_0, \varrho_\ell) \leq \frac{C}{a^{1/2}} (\mathcal{E}^-(u))^{1/2}. \quad \square$$

The definition of the velocity field v_t in Proposition 7.85, along with the fact that $\partial_x \mathbf{u}(t) \downarrow \leq 0$, gives $v_t \geq 0$, which intuitively says that “particles only move to the right”. This suggests that the solution $\mathbf{u}(t)$ can only lose mass in any prefix of the interval $(0, 1)$ over time. The following items make this observation rigorous.

Lemma 7.88. *Let $\delta \in I$ and let $J := (0, \delta)$. Let $u \in L^2(I)$. Then for all $t > 0$,*

$$\int_J (P_t u) dx \leq \int_J u dx.$$

Proof. Suppose u satisfies [Condition 7.84](#); the general case will then follow by the same arguments as in the proof of [Corollary 7.86](#). Let $J' := (-\infty, \delta)$, let $\chi_{J'} : \mathbb{R} \rightarrow [0, 1]$ be the characteristic function for the set J' , and for each $\varepsilon > 0$, let $\chi_{J'}^\varepsilon$ be its mollification by the standard mollifier. Note that each $\chi_{J'}^\varepsilon$ is nonincreasing. Also note that $\chi_{J'} = \chi_J$ pointwise in I . Now, fix $T > 0$. Since the family $(\varrho_t, v_t)_{t \in [0, T]}$ given by [Proposition 7.85](#) solves the continuity equation, applying (7.23) with test function $\chi_{J'}^\varepsilon \in C^\infty(\bar{I})$ gives that $t \mapsto \int_I \chi_{J'}^\varepsilon \mathbf{u}(t) dx$ is AC and, for a.e. $t \in [0, T]$,

$$\partial_t \int_I \chi_{J'}^\varepsilon \mathbf{u}(t) dx = \int_I \underbrace{(\partial_x \chi_{J'}^\varepsilon)}_{\leq 0} \underbrace{v_t}_{\geq 0} \underbrace{\mathbf{u}(t)}_{\geq 0} dx \leq 0.$$

Hence $t \mapsto \int_I \chi_{J'}^\varepsilon \mathbf{u}(t) dx$ is nonincreasing, and obtain

$$\langle \chi_{J'}^\varepsilon, u \rangle_{L^2(I)} = \int_I \chi_{J'}^\varepsilon \mathbf{u}(0) dx \geq \int_I \chi_{J'}^\varepsilon \mathbf{u}(T) dx = \langle \chi_{J'}^\varepsilon, P_T u \rangle_{L^2(I)}.$$

Now, since $\chi_{J'} \in L^2_{\text{loc}}(\mathbb{R})$, we have that $\chi_{J'}^\varepsilon \rightarrow \chi_{J'}$ in $L^2_{\text{loc}}(\mathbb{R})$, so in particular $\chi_{J'}^\varepsilon \rightarrow \chi_{J'} = \chi_J$ in $L^2(I)$. Thus we have

$$\int_J (P_T u) dx = \langle \chi_J, P_T u \rangle_{L^2(I)} = \lim_{\varepsilon \rightarrow 0} \langle \chi_{J'}^\varepsilon, P_T u \rangle_{L^2(I)} \leq \lim_{\varepsilon \rightarrow 0} \langle \chi_{J'}^\varepsilon, u \rangle_{L^2(I)} = \langle \chi_J, u \rangle_{L^2(I)} = \int_J u dx. \quad \square$$

Corollary 7.89. *Let $\delta \in I$ and let $J := (0, \delta)$. Let $u \in L^2(I)$. Then*

$$\int_J (P_\infty u) dx \leq \int_J u dx.$$

Proof. This follows from [Lemma 7.88](#) along with the convergence $P_t u \rightarrow P_\infty u$ in $L^2(I)$. $\square$

Definition 7.90. Let μ and ν be probability measures on $\mathbb{R}$. We say μ *dominates* ν , and write $\mu \succeq \nu$, if for every $x \in \mathbb{R}$ we have $\mu(-\infty, x) \geq \nu(-\infty, x)$.

Corollary 7.91. *Let $u \in \mathcal{U}$ satisfy [Condition 7.84](#). Define the measures $d\mu := u dx$ and $d\mu_\infty := (P_\infty u) dx$. Then $\mu \succeq \mu_\infty$.*

Proof. We may view μ and μ_∞ as absolutely continuous measures on all of $\mathbb{R}$, taking zero outside $[0, 1]$. Now, for each $x \in [0, 1]$, [Corollary 7.89](#) gives

$$\mu(-\infty, x) = \int_{(0, x)} u \, dy \geq \int_{(0, x)} (P_\infty u) \, dy = \mu_\infty(-\infty, x).$$

On the other hand, for $x < 0$ both sides are zero, and for $x > 1$, both sides are 1 by [Corollary 7.87](#). $\square$

7.3.3 Directed optimal transport and Pythagorean composition

We now introduce directed versions of basic optimal transport concepts and theory, with the goal of combining a directed version of weak Kantorovich duality with a multidimensional directed version of the transport-energy inequality from [Theorem 7.2](#) to obtain our desired directed Poincaré inequality via a perturbation argument. In the interest of space, we refrain starting a systematic study of directed optimal transport, but rather limit ourselves to the results we require.

Definition 7.92 (Quasimetric). Let X be a set. An extended real-valued function $d : X \times X \rightarrow [0, +\infty]$, not necessarily symmetric, is called a *quasimetric* on X if it satisfies the following:

1. For all $x, y \in X$, $d(x, y) \geq 0$ with $d(x, y) = 0$ if and only if $x = y$.
2. For all $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$.

Definition 7.93. Let $\Omega \subset \mathbb{R}^d$ be a Borel set, and let μ and ν be probability measures on Ω . We define the set $\Pi(\mu \rightarrow \nu)$ of *directed couplings*, or *directed transport plans*, from μ to ν as

$$\Pi(\mu \rightarrow \nu) := \left\{ \gamma \in \Pi(\mu, \nu) : \int_{\Omega \times \Omega} \chi_{\{x \not\preceq y\}} \, d\gamma(x, y) = 0 \right\},$$

where the integral is well-defined because the set $\{(x, y) \in \mathbb{R}^d \times \mathbb{R}^d : x \not\preceq y\}$ is open and hence Borel measurable. Note that the condition could also be written as $\gamma(\{x \not\preceq y\}) = 0$, or $\gamma(\{x \preceq y\}) = 1$.

Definition 7.94 (Directed Wasserstein distance). Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set and let $p \in [1, \infty)$. Given two probability distributions μ, ν over Ω , we define the *directed p -Wasserstein distance from μ to ν* by

$$W_p(\mu \rightarrow \nu) := \inf_{\gamma \in \Pi(\mu \rightarrow \nu)} C_p(\gamma),$$

and we write $W_p^p(\mu \rightarrow \nu) := W_p(\mu \rightarrow \nu)^p$. Note that we may have $W_p(\mu \rightarrow \nu) = +\infty$, and that $W_p(\cdot \rightarrow \cdot)$ is not symmetric in general.

Observation 7.95. *For all measures μ and ν , we have $\Pi(\mu \rightarrow \nu) \subset \Pi(\mu, \nu)$ and hence $W_p(\mu, \nu) \leq W_p(\mu \rightarrow \nu)$.*

Lemma 7.96 (Gluing lemma; see e.g. [San15, Lemma 5.5]). *Let $\Omega \subset \mathbb{R}^d$ be a Borel set. Let μ, ϱ, ν be probability measures on Ω , and let $\gamma^+ \in \Pi(\mu, \varrho)$ and $\gamma^- \in \Pi(\varrho, \nu)$. Then there exists a probability measure σ on $\Omega \times \Omega \times \Omega$ such that $(\pi_{x,y})_{\#}\sigma = \gamma^+$ and $(\pi_{y,z})_{\#}\sigma = \gamma^-$.*

The proof of that $W_p(\cdot \rightarrow \cdot)$ is a quasimetric follows the presentation of [San15, Lemma 5.4], with a simple additional argument to handle *directed* couplings.

Lemma 7.97 (Composition of directed transport plans). *In Lemma 7.96, if $\gamma^+ \in \Pi(\mu \rightarrow \varrho)$ and $\gamma^- \in \Pi(\varrho \rightarrow \nu)$, then $(\pi_{x,z})_{\#}\sigma \in \Pi(\mu \rightarrow \nu)$.*

Proof. Let $\gamma := (\pi_{x,z})_{\#}\sigma$. First, since $(\pi_x)_{\#}\gamma = (\pi_x \circ \pi_{x,z})_{\#}\sigma = (\pi_x \circ \pi_{x,y})_{\#}\sigma = (\pi_x)_{\#}\gamma^+ = \mu$, and similarly $(\pi_z)_{\#}\gamma = \nu$, we have $\gamma \in \Pi(\mu, \nu)$. Moreover, by definition of pushforward measure we have

$$\begin{aligned} \int_{\Omega \times \Omega} \chi_{\{x \not\leq z\}} d\gamma(x, z) &= \int_{\Omega \times \Omega \times \Omega} \chi_{\{x \not\leq z\}} d\sigma(x, y, z) \leq \int_{\Omega \times \Omega \times \Omega} (\chi_{\{x \not\leq y\}} + \chi_{\{y \not\leq z\}}) d\sigma(x, y, z) \\ &= \int_{\Omega \times \Omega \times \Omega} \chi_{\{x \not\leq y\}} d\sigma(x, y, z) + \int_{\Omega \times \Omega \times \Omega} \chi_{\{y \not\leq z\}} d\sigma(x, y, z) \\ &= \int_{\Omega \times \Omega} \chi_{\{x \not\leq y\}} d\gamma^+(x, y) + \int_{\Omega \times \Omega} \chi_{\{y \not\leq z\}} d\gamma^-(y, z) = 0, \end{aligned}$$

the last equality since γ^+ and γ^- are directed couplings. Hence $\gamma \in \Pi(\mu \rightarrow \nu)$ as claimed. $\square$

Proposition 7.98. *Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set and $p \in [1, \infty)$. Then $W_p(\cdot \rightarrow \cdot)$ is a quasimetric on $P(\Omega)$.*

Proof. It is clear that $W_p(\mu \rightarrow \nu) \geq 0$ always. If $\mu = \nu$ then the identity coupling shows that $W_p(\mu \rightarrow \nu) = 0$, and conversely if $W_p(\mu \rightarrow \nu) = 0$ then, by Observation 7.95, $W_p(\mu, \nu) = 0$ and hence $\mu = \nu$. It remains to show that $W_p(\cdot \rightarrow \cdot)$ satisfies the triangle inequality.

Let μ, ϱ, ν be probability measures on Ω . Let $\gamma^+ \in \Pi(\mu \rightarrow \varrho)$ and $\gamma^- \in \Pi(\varrho \rightarrow \nu)$. Then since we also have $\gamma^+ \in \Pi(\mu, \varrho)$ and $\gamma^- \in \Pi(\varrho, \nu)$, apply Lemma 7.96 to obtain a

probability measure σ on $\Omega \times \Omega \times \Omega$ such that $(\pi_{x,y})_{\#}\sigma = \gamma^+$ and $(\pi_{y,z})_{\#}\sigma = \gamma^-$. Let $\gamma := (\pi_{x,z})_{\#}\sigma$, so that $\gamma \in \Pi(\mu \rightarrow \nu)$ by [Lemma 7.97](#). Then

$$\begin{aligned} W_p(\mu \rightarrow \nu) &\leq \left(\int_{\Omega \times \Omega} |x - z|^p d\gamma(x, z) \right)^{1/p} = \left(\int_{\Omega \times \Omega \times \Omega} |x - z|^p d\sigma(x, y, z) \right)^{1/p} \\ &= \| |x - z| \|_{L^p(\sigma)} \leq \| |x - y| + |y - z| \|_{L^p(\sigma)} \leq \| |x - y| \|_{L^p(\sigma)} + \| |y - z| \|_{L^p(\sigma)} \\ &= \left(\int_{\Omega \times \Omega \times \Omega} |x - y|^p d\sigma(x, y, z) \right)^{1/p} + \left(\int_{\Omega \times \Omega \times \Omega} |y - z|^p d\sigma(x, y, z) \right)^{1/p} \\ &= \left(\int_{\Omega \times \Omega} |x - y|^p d\gamma^+(x, y) \right)^{1/p} + \left(\int_{\Omega \times \Omega} |y - z|^p d\gamma^-(y, z) \right)^{1/p}. \end{aligned}$$

Since $\gamma^+ \in \Pi(\mu \rightarrow \varrho)$, $\gamma^- \in \Pi(\varrho \rightarrow \nu)$ were arbitrary, $W_p(\mu \rightarrow \nu) \leq W_p(\mu \rightarrow \varrho) + W_p(\varrho \rightarrow \nu)$. $\square$

The directed Wasserstein distance arises naturally in the one-dimensional case when one probability measure dominates the other in the sense of [Definition 7.90](#), as we now show.

Proposition 7.99 (Specialization of [\[San15, Theorem 2.9\]](#)). *Let $p \in (1, +\infty)$. Let μ and ν be two absolutely continuous probability measures on $\bar{I}$ with strictly positive densities. Then there exists a unique $\gamma \in \Pi(\mu, \nu)$ attaining $C_p(\gamma) = W_p(\mu, \nu)$, and in fact $\gamma = (\text{id}, T_{\text{mon}})_{\#}\mu$ where $T_{\text{mon}} : \bar{I} \rightarrow \bar{I}$ is a nondecreasing function that is a.e. uniquely determined, and id denotes the identity map.*

Corollary 7.100. *If in [Proposition 7.99](#) we have $\mu \succeq \nu$, then in fact $\gamma \in \Pi(\mu \rightarrow \nu)$. As a consequence, $W_p(\mu \rightarrow \nu) = W_p(\mu, \nu)$.*

Proof. We claim that $T_{\text{mon}}(x) \geq x$ for every $x \in I$. Indeed, suppose $T_{\text{mon}}(x) < x$ for some $x \in I$. Then

$$\nu(0, T_{\text{mon}}(x)) = (T_{\text{mon}\#}\mu)(0, T_{\text{mon}}(x)) = \mu(T_{\text{mon}}^{-1}(0, T_{\text{mon}}(x))) \geq \mu(0, x),$$

the inequality because certainly $(0, x) \subseteq T_{\text{mon}}^{-1}(0, T_{\text{mon}}(x))$, but we do not rule out at this point that T remains constant for a while after x . But since $T_{\text{mon}}(x) < x$ and ν has strictly positive density by assumption, we conclude that

$$\nu(0, x) > \nu(0, T_{\text{mon}}(x)) \geq \mu(0, x),$$

contradicting the assumption that $\mu \succeq \nu$. Hence $T_{\text{mon}}(x) \geq x$ for every $x \in I$ as claimed. We now show that $\gamma \in \Pi(\mu \rightarrow \nu)$. Let $S := \{(x, y) \in \bar{I} \times \bar{I} : x \leq y\}$. Since $\gamma = (\text{id}, T_{\text{mon}})_{\#} \mu$, we have

$$\gamma(S) = \mu((\text{id}, T_{\text{mon}})^{-1}(S)) = \mu(\{x \in \bar{I} : x \leq T_{\text{mon}}(x)\}) = \mu(\bar{I}) = 1,$$

so $\gamma \in \Pi(\mu \rightarrow \nu)$ as claimed. $\square$

We now introduce the formal language for two related operations: constructing a probability measure by its marginal and conditional distributions, and the inverse process of extracting marginal and conditionals from a probability measure, which is also called *disintegration*. We refer the reader to [AGS05, Section 5.3] for an overview of these ideas from a measure-theoretic perspective.

We use the following definition from [AGS05]. For X and Y separable metric spaces and $x \in X \mapsto \mu_x \in P(Y)$ a measure-valued map, we say μ_x is a Borel map if $x \mapsto \mu_x(B)$ is a Borel map for any Borel set $B \subset Y$, or equivalently if this holds for any open set $A \subset Y$. In this case we also have that

$$x \mapsto \int_Y f(x, y) d\mu_x(y)$$

is Borel for any bounded (or nonnegative) Borel function $f : X \times Y \rightarrow \mathbb{R}$.

Definition 7.101 (Construction and disintegration). Let S be a finite set and let $I \subset S$. Let $x_{S \setminus I} \in \mathbb{R}^{S \setminus I} \mapsto \mu_{|x_{S \setminus I}} \in P(\mathbb{R}^I)$ be a Borel map (the *conditionals*). Then for any bounded (or nonnegative) Borel function $f : \mathbb{R}^S \rightarrow \mathbb{R}$, the $\mathbb{R}^{S \setminus I} \rightarrow \mathbb{R}$ map

$$x_{S \setminus I} \mapsto \int_{\mathbb{R}^I} f(x_{S \setminus I}, x_I) d\mu_{|x_{S \setminus I}}(x_I) \quad (7.24)$$

is Borel. Therefore, for any $\mu_{S \setminus I} \in P(\mathbb{R}^{S \setminus I})$ (the *marginal*), we define $\mu \in P(\mathbb{R}^S)$ implicitly by

$$\int_{\mathbb{R}^S} f d\mu = \int_{\mathbb{R}^{S \setminus I}} d\mu_{S \setminus I}(x_{S \setminus I}) \int_{\mathbb{R}^I} f(x_{S \setminus I}, x_I) d\mu_{|x_{S \setminus I}}(x_I), \quad (7.25)$$

and we formally write $\mu = \int_{\mathbb{R}^{S \setminus I}} \mu_{|x_{S \setminus I}} d\mu_{S \setminus I}$.

Conversely, given $\mu \in P(\mathbb{R}^S)$, we let $\mu_{S \setminus I} := (\pi_{S \setminus I})_{\#} \mu \in P(\mathbb{R}^{S \setminus I})$ and write $(\mu_{|x_{S \setminus I}})_{x_{S \setminus I} \in \mathbb{R}^{S \setminus I}} \subset P(\mathbb{R}^I)$ for the $\mu_{S \setminus I}$ -a.e. uniquely determined Borel family of probability measures such that $\mu = \int_{\mathbb{R}^{S \setminus I}} \mu_{|x_{S \setminus I}} d\mu_{S \setminus I}$, and we call this decomposition a *disintegration* of μ .

Examples and simplified notational conventions. Since the index notation in the definition above can be somewhat laborious to parse, let us briefly give two concrete settings we will use below, and introduce simplified notational conventions for those settings. The simplifications are supposed to be mnemonic for what we have already defined for projections.

1. For a probability measure $\mu \in P(\mathbb{R}^d)$ and index $i \in [d]$, we may disintegrate μ into a marginal on index set $[d] \setminus \{i\}$ and conditionals supported along i -th coordinate, and write $\mu = \int_{\mathbb{R}^{[d] \setminus \{i\}}} \mu_{|x_{[d] \setminus \{i\}}} d\mu_{[d] \setminus \{i\}}$. We simplify this notation by writing $\mu = \int_{\mathbb{R}_{-i}^d} \mu_{|x_{-i}} d\mu_{-i}$.
2. For a transport plan $\gamma \in P(\mathbb{R}^d \times \mathbb{R}^d)$, we write the index set as $[2d] = S = S_x \cup S_y$ for $S_x := [d]$ and $S_y := [2d] \setminus [d]$, so we write each element in the support of γ as a tuple (x, y) with $x \in \mathbb{R}^{S_x}$ and $y \in \mathbb{R}^{S_y}$. Then, given index set $I \subset [d]$, we may disintegrate γ into a marginal on index set $S_M := (S_x \setminus I_x) \cup (S_y \setminus I_y)$ for $I_x := I$ and $I_y := \{d + i : i \in I\}$, so that we think of the marginal as determining all but the I -indexed coordinates of the points x and y ; and we view the conditionals as supported along the $I_x \cup I_y$ directions. This disintegration is $\gamma = \int_{\mathbb{R}^{S_M}} \gamma_{|x_{S_M}} d\gamma_{S_M}$. To be clear, here we have $\gamma_{S_M} \in P(\mathbb{R}^{S_M})$ and $\gamma_{|x_{S_M}} \in P(\mathbb{R}^{I_x \cup I_y})$ for each $x_{S_M} \in \mathbb{R}^{S_M}$. We make the notation somewhat more mnemonic by writing $\gamma = \int_{\mathbb{R}_{-I}^d \times \mathbb{R}_{-I}^d} \gamma_{|x_{-I}, y_{-I}} d\gamma_{-I_x - I_y}$. When we are considering a singleton $I = \{i\}$, we further simplify the notation by directly writing i in the place of I in this formula.

For absolutely continuous probability measures on $[0, 1]^d$, the following more familiar characterization of disintegrations will be useful:

Proposition 7.102. *Let μ be an absolutely continuous probability measure whose density is Borel measurable and supported in $[0, 1]^d$. Namely, write $d\mu = u dx$ where $u : [0, 1]^d \rightarrow [0, +\infty)$ is a Borel map. Let $i \in [d]$. Then the disintegration $\mu = \int_{\mathbb{R}_{-i}^d} \mu_{|x_{-i}} d\mu_{-i}$ is as follows: μ_{-i} and each $\mu_{|x_{-i}}$ are absolutely continuous probability measures with Borel densities supported in $[0, 1]_{-i}^d$ and $[0, 1]$ respectively; and writing $d\mu_{-i} = u_{-i} dx_{-i}$, and $d\mu_{|x_{-i}} = u_{|x_{-i}} dx_i$ for each x_{-i} , where the (Borel) density functions are $u_{-i} : [0, 1]_{-i}^d \rightarrow [0, +\infty)$ and $u_{|x_{-i}} : [0, 1] \rightarrow [0, +\infty)$ for each $x_{-i} \in [0, 1]_{-i}^d$, we have (marginal density)*

$$u_{-i}(x_{-i}) = \int_I u(x_{-i}, x_i) dx_i$$

for each $x_{-i} \in [0, 1]_{-i}^d$, and (conditional density)

$$u_{|x_{-i}}(x_i) = \frac{u(x)}{u_{-i}(x_{-i})}$$

for each $x \in [0, 1]^d$ where the denominator is nonzero, or 1 by convention otherwise.

Proof. First, note that it is standard that u_{-i} is a Borel map since it is the integral of the section $u(x_{-i}, \cdot)$ at each point x_{-i} . It is hence immediate that each $u_{|x_{-i}}$ is also Borel. Thus μ_{-i} and all $\mu_{|x_{-i}}$ are indeed absolutely continuous probability measures.

We now check the conditions of [Definition 7.101](#). We claim that the map $x_{-i} \in \mathbb{R}_{-i}^d \mapsto \mu_{|x_{-i}} \in P(\mathbb{R})$ is Borel. Let $B \subset \mathbb{R}$ be a Borel set; we may assume that $B \subset [0, 1]$ without loss of generality, since all measures assign zero outside this interval. Then the map $x_{-i} \in \mathbb{R}_{-i}^d \mapsto \mu_{|x_{-i}}(B)$ takes value zero outside $[0, 1]_{-i}^d$, and for $x_{-i} \in [0, 1]_{-i}^d$, it is

$$x_{-i} \mapsto \frac{1}{u_{-i}(x_{-i})} \int_B u(x_{-i}, x_i) dx_i.$$

This is again, at each point, the integral of a section over the subspace B , so it is standard that this map is Borel. Hence the claim holds.

It remains to verify [\(7.25\)](#). Let $f : \mathbb{R}^d \rightarrow [0, +\infty)$ be a nonnegative Borel function (and the result for bounded f will also follow). By the definition of the measures and Tonelli's theorem, we have

$$\begin{aligned} \int_{\mathbb{R}^{-i}} d\mu_{-i}(x_{-i}) \int_{\mathbb{R}} f(x_{-i}, x_i) du_{|x_{-i}}(x_i) &= \int_{[0, 1]_{-i}^d} u_{-i}(x_{-i}) dx_{-i} \int_I f(x_{-i}, x_i) u_{|x_{-i}}(x_i) dx_i \\ &= \int_{[0, 1]^d} f(x) u_{-i}(x_{-i}) u_{|x_{-i}}(x_i) dx = \int_{[0, 1]^d} f(x) u(x) dx, \end{aligned}$$

where the last equality holds because $u(x) = u_{-i}(x_{-i})u_{|x_{-i}}(x_i)$ whenever $u_{-i}(x_{-i}) > 0$, and if $u_{-i}(x_{-i}) = 0$ then $u(x_{-i}, \cdot) = 0$ almost everywhere, so $\{x \in [0, 1]^d : u(x) > 0 \text{ and } u_{-i}(x_{-i}) = 0\}$ is Borel and has measure zero (by another application of Tonelli's theorem). $\square$

Definition 7.103 (Aligned transport plans). Let $I \subsetneq [d]$ be nonempty and let $\mu, \nu \in P(\mathbb{R}^d)$. Let $\gamma \in \Pi(\mu, \nu)$ and write its disintegration $\gamma = \int_{\mathbb{R}_{-I}^d \times \mathbb{R}_{-I}^d} \gamma_{|x_{-I}, y_{-I}} d\gamma_{-I, x_{-I}, y_{-I}}$. We say γ is *I-aligned* if its marginal $\gamma_{-I, x_{-I}, y_{-I}} \in P(\mathbb{R}_{-I}^d \times \mathbb{R}_{-I}^d)$ satisfies the following:

$$\gamma_{-I, x_{-I}, y_{-I}}(\{(x_{-I}, y_{-I}) \in \mathbb{R}_{-I}^d \times \mathbb{R}_{-I}^d : x_{-I} \neq y_{-I}\}) = 0.$$

Note that this condition is well-defined because the set being measured is open and hence Borel. By convention, we say that every $\gamma \in \Pi(\mu, \nu)$ is $[d]$ -aligned. For any nonempty $I \subseteq [d]$, we denote the set of I -aligned transport plans by $\Pi_I(\mu, \nu)$. When we have a singleton $I = \{i\}$, we write i directly in the place of I in this definition.

Definition 7.104. For nonempty $I \subseteq [d]$ and $\mu, \nu \in P(\mathbb{R}^d)$, write $\Pi_I(\mu \rightarrow \nu) := \Pi(\mu \rightarrow \nu) \cap \Pi_I(\mu, \nu)$.

Lemma 7.105 (Alternative characterization of I -aligned plans). *Let $I \subsetneq [d]$ be nonempty and let $\mu, \nu \in P(\mathbb{R}^d)$. Then $\gamma \in \Pi(\mu, \nu)$ is I -aligned if and only if*

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-I} \neq y_{-I}\}} d\gamma(x, y) = 0.$$

Proof. Writing the disintegration $\gamma = \int_{\mathbb{R}_{-I}^d \times \mathbb{R}_{-I}^d} \gamma_{|x_{-I}, y_{-I}} d\gamma_{-I, x_{-I}, y_{-I}}$, we have

$$\begin{aligned} \gamma_{-I, x_{-I}, y_{-I}}(\{x_{-I} \neq y_{-I}\}) &= \int_{\mathbb{R}_{-I}^d \times \mathbb{R}_{-I}^d} \chi_{\{x_{-I} \neq y_{-I}\}} d\gamma_{-I, x_{-I}, y_{-I}} \underbrace{\int_{\mathbb{R}^I \times \mathbb{R}^I} d\gamma_{|x_{-I}, y_{-I}}(x_I, y_I)}_{=1} \\ &= \int_{\mathbb{R}_{-I}^d \times \mathbb{R}_{-I}^d} d\gamma_{-I, x_{-I}, y_{-I}} \int_{\mathbb{R}^I \times \mathbb{R}^I} \chi_{\{x_{-I} \neq y_{-I}\}} d\gamma_{|x_{-I}, y_{-I}}(x_I, y_I) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-I} \neq y_{-I}\}} d\gamma. \end{aligned} \quad \square$$

While we defined I -aligned transport plans for probability measures on all of $\mathbb{R}^d$, we extend the definition to measures on subsets $\Omega \subset \mathbb{R}^d$ (e.g. on the cube) by canonically extending any such measure to all of $\mathbb{R}^d$ by assigning zero outside of Ω .

Lemma 7.106 (Composition of aligned transport plans). *Let $I, J \subseteq [d]$ be nonempty. Then in [Lemma 7.96](#), if $\gamma^+ \in \Pi_I(\mu, \varrho)$ and $\gamma^- \in \Pi_J(\varrho, \nu)$, then $(\pi_{x,z})_{\#} \sigma \in \Pi_{I \cup J}(\mu, \nu)$.*

Proof. Let $\gamma := (\pi_{x,z})_{\#} \sigma$ and $K := I \cup J$. As in the proof of [Lemma 7.97](#), we do have $\gamma \in \Pi(\mu, \nu)$, so it remains to show that γ is K -aligned (and we may assume that $K \subsetneq [n]$,

otherwise there is nothing to prove). We have

$$\begin{aligned}
& \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-K} \neq z_{-K}\}} d\gamma \\
&= \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-K} \neq z_{-K}\}}(x, z) d\sigma(x, y, z) && \text{(Pushforward)} \\
&\leq \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} (\chi_{\{x_{-K} \neq y_{-K}\}}(x, y) + \chi_{\{y_{-K} \neq z_{-K}\}}(y, z)) d\sigma(x, y, z) \\
&= \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-K} \neq y_{-K}\}} d\sigma + \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} \chi_{\{y_{-K} \neq z_{-K}\}} d\sigma \\
&= \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-K} \neq y_{-K}\}} d\gamma^+ + \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{y_{-K} \neq z_{-K}\}} d\gamma^- && \text{(Pushforward)} \\
&\leq \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-I} \neq y_{-I}\}} d\gamma^+ + \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{y_{-J} \neq z_{-J}\}} d\gamma^- && (I, J \subseteq K) \\
&= 0,
\end{aligned}$$

the last step since γ^+ is I -aligned and γ^- is J -aligned and by [Lemma 7.105](#); and again by the latter, γ is K -aligned. $\square$

The following results let us find aligned transport plans by transporting mass only within i -aligned lines.

Lemma 7.107 (Measurable selection; specialization of [\[Vil09, Corollary 5.22\]](#)). *Let $\Omega \subset \mathbb{R}^d$ be a bounded set. Let $p \in [1, \infty)$. Let A be a measurable space and let $a \mapsto (\mu_a, \nu_a)$ be a measurable function $A \rightarrow P(\mathbb{R}) \times P(\mathbb{R})$ with each μ_a and ν_a having bounded support. Then there is a measurable choice $a \mapsto \gamma_a$ such that, for each $a \in A$, $\gamma_a \in \Pi(\mu_a, \nu_a)$ and $C_p(\gamma_a) = W_p(\mu_a, \nu_a)$.*

Lemma 7.108. *Let $p \in [1, \infty)$, $i \in [d]$, and let $\mu, \nu \in P(\mathbb{R}^d)$ be supported inside some bounded set. Suppose $\mu_{-i} = \nu_{-i}$. Then there exists $\gamma \in \Pi_i(\mu, \nu)$ satisfying*

$$C_p(\gamma)^p = \int_{\mathbb{R}_{-i}^d} W_p^p(\mu_{|x_{-i}}, \nu_{|x_{-i}}) d\mu_{-i}(x_{-i}).$$

Proof. Let $A := \mathbb{R}_{-i}^d$ be equipped with the Borel σ -algebra on $\mathbb{R}^{[d] \setminus \{i\}}$. Then the $A \rightarrow P(\mathbb{R}) \times P(\mathbb{R})$ function $x_{-i} \mapsto (\mu_{|x_{-i}}, \nu_{|x_{-i}})$ is Borel measurable because, by disintegration, both $x_{-i} \mapsto \mu_{|x_{-i}}$ and $x_{-i} \mapsto \nu_{|x_{-i}}$ are Borel measurable. By [Lemma 7.107](#), we obtain a

Borel measurable map $x_{-i} \mapsto \gamma_{x_{-i}}$ such that, for each $x_{-i} \in \mathbb{R}_{-i}^d$, $\gamma_{x_{-i}} \in \Pi(\mu|_{x_{-i}}, \nu|_{x_{-i}})$ and $C_p(\gamma_{x_{-i}})^p = W_p^p(\mu|_{x_{-i}}, \nu|_{x_{-i}})$.

Define $\gamma \in P(\mathbb{R}^d \times \mathbb{R}^d)$ by its disintegration $\gamma = \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} \gamma_{|x_{-i}, y_{-i}} d\gamma_{-i_x - i_y}$ as follows. First, for the marginal $\gamma_{-i_x - i_y} \in P(\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d)$, for each Borel $Z \subset \mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d$ we set

$$\gamma_{-i_x - i_y}(Z) := \int_{\mathbb{R}_{-i}^d} \chi_Z(x_{-i}, x_{-i}) d\mu_{-i}(x_{-i}),$$

where the integral is well-defined because the set $\{x_{-i} \in \mathbb{R}_{-i}^d : (x_{-i}, x_{-i}) \in Z\}$ is Borel by standard arguments. We remark that $\gamma_{-i_x - i_y}$ is indeed a probability measure because it is nonnegative, it assigns zero to the empty set, it is additive over countable disjoint unions, and

$$\gamma_{-i_x - i_y}(\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d) = \int_{\mathbb{R}_{-i}^d} d\mu_{-i} = 1.$$

It follows that for every bounded (or nonnegative) Borel function $f : \mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d \rightarrow \mathbb{R}$, we have

$$\int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} f(x_{-i}, y_{-i}) d\gamma_{-i_x - i_y}(x_{-i}, y_{-i}) = \int_{\mathbb{R}_{-i}^d} f(x_{-i}, x_{-i}) d\mu_{-i}(x_{-i}). \quad (7.26)$$

Second, for the conditionals, we simply set $\gamma_{|x_{-i}, y_{-i}} := \gamma_{x_{-i}}$ for each $(x_{-i}, y_{-i}) \in \mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d$.

We claim that $\gamma \in \Pi_i(\mu, \nu)$. The i -aligned condition holds by construction, so it remains to show that $\gamma \in \Pi(\mu, \nu)$. We first verify that $(\pi_1)_\# \gamma = \mu$. It suffices to verify that these measures agree on each rectangle $X := X_{-i} \times X_i$ of Borel sets $X_{-i} \subset \mathbb{R}_{-i}^d$, $X_i \subset \mathbb{R}$. We have

$$\begin{aligned} & ((\pi_1)_\# \gamma)(X) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_X(x) d\gamma(x, y) && \text{(Pushforward)} \\ &= \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} d\gamma_{-i_x - i_y}(x_{-i}, y_{-i}) \int_{\mathbb{R} \times \mathbb{R}} \chi_X(x_{-i}, x_i) d\gamma_{|x_{-i}, y_{-i}}(x_i, y_i) && \text{(Disintegration)} \\ &= \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} \chi_{X_{-i}}(x_{-i}) d\gamma_{-i_x - i_y}(x_{-i}, y_{-i}) \int_{\mathbb{R} \times \mathbb{R}} \chi_{X_i}(x_i) d\gamma_{x_{-i}}(x_i, y_i) \\ & && \text{(Definition of } X, \gamma_{|x_{-i}, y_{-i}}) \\ &= \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} \chi_{X_{-i}}(x_{-i}) ((\pi_1)_\# \gamma_{x_{-i}})(X_i) d\gamma_{-i_x - i_y}(x_{-i}, y_{-i}) && \text{(Pushforward)} \end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} \chi_{X_{-i}}(x_{-i}) \mu_{|x_{-i}}(X_i) d\gamma_{-i, x_{-i}y}(x_{-i}, y_{-i}) && (\gamma_{x_{-i}} \in \Pi(\mu_{|x_{-i}}, \nu_{|x_{-i}})) \\
&= \int_{\mathbb{R}_{-i}^d} \chi_{X_{-i}}(x_{-i}) \mu_{|x_{-i}}(X_i) d\mu_{-i}(x_{-i}) && (\text{Application of (7.26)}) \\
&= \int_{\mathbb{R}_{-i}^d} \chi_{X_{-i}}(x_{-i}) d\mu_{-i}(x_{-i}) \int_{\mathbb{R}} \chi_{X_i}(x_i) d\mu_{|x_{-i}}(x_i) \\
&= \int_{\mathbb{R}_{-i}^d} d\mu_{-i}(x_{-i}) \int_{\mathbb{R}} \chi_X(x_{-i}, x_i) d\mu_{|x_{-i}}(x_i) && (\text{Definition of } X) \\
&= \int_{\mathbb{R}^d} \chi_X d\mu = \mu(X), && (\text{Disintegration})
\end{aligned}$$

as desired. Thanks to the hypothesis that $\mu_{-i} = \nu_{-i}$, an analogous calculation on the second variable yields that $(\pi_2)_\# \gamma = \nu$ and hence $\gamma \in \Pi(\mu, \nu)$. Thus $\gamma \in \Pi_i(\mu, \nu)$ as claimed.

Finally, we compute the cost of the plan γ . Again using its disintegration and (7.26), we have

$$\begin{aligned}
C_p(\gamma)^p &= \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^p d\gamma(x, y) \\
&= \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} d\gamma_{-i, x_{-i}y}(x_{-i}, y_{-i}) \int_{\mathbb{R} \times \mathbb{R}} |(x_{-i}, x_i) - (y_{-i}, y_i)|^p d\gamma_{|x_{-i}, y_{-i}}(x_i, y_i) \\
&= \int_{\mathbb{R}_{-i}^d} d\mu_{-i}(x_{-i}) \int_{\mathbb{R} \times \mathbb{R}} |(x_{-i}, x_i) - (x_{-i}, y_i)|^p d\gamma_{|x_{-i}, x_{-i}}(x_i, y_i) \\
&= \int_{\mathbb{R}_{-i}^d} d\mu_{-i}(x_{-i}) \int_{\mathbb{R} \times \mathbb{R}} |x_i - y_i|^p d\gamma_{x_{-i}}(x_i, y_i) \\
&= \int_{\mathbb{R}_{-i}^d} d\mu_{-i}(x_{-i}) C_p(\gamma_{x_{-i}})^p = \int_{\mathbb{R}_{-i}^d} W_p^p(\mu_{|x_{-i}}, \nu_{|x_{-i}}) d\mu_{-i}(x_{-i}). \quad \square
\end{aligned}$$

We may combine the above to find *directed* transport plans by transporting mass within axis-aligned lines, as long as the first probability distribution dominates the second in each such line.

Lemma 7.109. *Let $p \in (1, +\infty)$, and let $\mu, \nu \in P([0, 1]^d)$ be absolutely continuous probability measures with strictly positive densities. Let $i \in [d]$ and suppose that 1) $\mu_{-i} = \nu_{-i}$*

and 2) $\mu|_{x_{-i}} \succeq \nu|_{x_{-i}}$ for each $x_{-i} \in [0, 1]_{-i}^d$. Then there exists $\gamma \in \Pi_i(\mu \rightarrow \nu)$ satisfying

$$C_p(\gamma)^p = \int_{[0,1]_{-i}^d} W_p^p(\mu|_{x_{-i}}, \nu|_{x_{-i}}) d\mu_{-i}(x_{-i}). \quad (7.27)$$

Proof. Let $\gamma \in \Pi_i(\mu, \nu)$ be the plan obtained from [Lemma 7.108](#), which satisfies (7.27). After recalling the definition of γ in [Lemma 7.108](#), [Corollary 7.100](#) implies that $\gamma|_{x_{-i}, y_{-i}} \in \Pi(\mu|_{x_{-i}} \rightarrow \nu|_{x_{-i}})$ for each $(x_{-i}, y_{-i}) \in [0, 1]_{-i}^d \times [0, 1]_{-i}^d$. It remains to confirm that $\gamma \in \Pi(\mu \rightarrow \nu)$. Indeed,

$$\begin{aligned} & \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x \not\leq y\}} d\gamma(x, y) \\ &= \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} d\gamma_{-i, y_{-i}}(x_{-i}, y_{-i}) \int_{\mathbb{R} \times \mathbb{R}} \chi_{\{x \not\leq y\}}((x_{-i}, x_i), (y_{-i}, y_i)) d\gamma|_{x_{-i}, y_{-i}}(x_i, y_i) \\ & \hspace{15em} \text{(Disintegration)} \\ &\leq \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} d\gamma_{-i, y_{-i}}(x_{-i}, y_{-i}) \int_{\mathbb{R} \times \mathbb{R}} \chi_{\{x_{-i} \not\leq y_{-i}\}}(x_{-i}, y_{-i}) d\gamma|_{x_{-i}, y_{-i}}(x_i, y_i) \\ & \quad + \underbrace{\int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} d\gamma_{-i, y_{-i}}(x_{-i}, y_{-i}) \int_{\mathbb{R} \times \mathbb{R}} \chi_{\{x_i > y_i\}}(x_i, y_i) d\gamma|_{x_{-i}, y_{-i}}(x_i, y_i)}_{= 0 \text{ since } \gamma|_{x_{-i}, y_{-i}} \text{ is a directed plan}} \\ &= \int_{\mathbb{R}_{-i}^d \times \mathbb{R}_{-i}^d} \chi_{\{x_{-i} \not\leq y_{-i}\}} d\gamma_{-i, y_{-i}} \underbrace{\int_{\mathbb{R} \times \mathbb{R}} d\gamma|_{x_{-i}, y_{-i}}}_{=1} = \int_{\mathbb{R}_{-i}^d} \underbrace{\chi_{\{x_{-i} \not\leq x_{-i}\}}}_{=0} d\mu_{-i}(x_{-i}) \quad (\text{By (7.26)}) \\ &= 0. \quad \square \end{aligned}$$

Now, we show that the directed, aligned 2-Wasserstein distance enjoys a nice “Pythagorean” composition property: composing I -aligned and J -aligned directed transport plans, for I and J disjoint, yields an $I \cup J$ -aligned directed transport plan whose cost is obtained from the costs of the two other plans via the Pythagorean theorem. Arguments of this nature are well-known in the undirected case, e.g. a similar statement appears in [\[Vil09, p. 572\]](#).

Lemma 7.3 (Pythagorean composition of transport plans). *Let $I, J \subseteq [d]$ be nonempty disjoint sets. Let $\mu, \varrho, \nu \in P(\mathbb{R}^d)$ be supported in bounded sets, and let $\gamma^+ \in \Pi_I(\mu \rightarrow \varrho)$ and $\gamma^- \in \Pi_J(\varrho \rightarrow \nu)$. Then there exists $\gamma \in \Pi_{I \cup J}(\mu \rightarrow \nu)$ satisfying $C_2(\gamma)^2 = C_2(\gamma^+)^2 + C_2(\gamma^-)^2$.*

Proof. Using Lemma 7.96, we obtain $\sigma \in P(\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d)$ such that $(\pi_{1,2})_{\#}\sigma = \gamma^+$ and $(\pi_{2,3})_{\#}\sigma = \gamma^-$, and by Lemma 7.97 and Lemma 7.106, we have $\gamma := (\pi_{1,3})_{\#}\sigma \in \Pi_{I \cup J}(\mu \rightarrow \nu)$. To compute the cost of γ , we first claim that σ assigns zero measure to points x, y, z such that $x_{-I} \neq y_{-I}$ or $y_{-J} \neq z_{-J}$. Indeed,

$$\begin{aligned} 0 &\leq \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} (1 - \chi_{\{x_{-I}=y_{-I} \text{ and } y_{-J}=z_{-J}\}}) d\sigma(x, y, z) \\ &\leq \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-I} \neq y_{-I}\}} d\sigma(x, y, z) + \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} \chi_{\{y_{-J} \neq z_{-J}\}} d\sigma(x, y, z) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-I} \neq y_{-I}\}} d\gamma^+(x, y) + \int_{\mathbb{R}^d \times \mathbb{R}^d} \chi_{\{y_{-J} \neq z_{-J}\}} d\gamma^-(y, z) = 0, \end{aligned}$$

the last step by Lemma 7.105 since γ^+ and γ^- are I - and J -aligned, respectively. Hence $\int f d\sigma = \int f \chi_{\{x_{-I}=y_{-I} \text{ and } y_{-J}=z_{-J}\}} d\sigma$ for any bounded or nonnegative Borel function f . Therefore, using the assumption that I and J are disjoint to apply the Pythagorean theorem, we obtain

$$\begin{aligned} C_2(\gamma)^2 &= \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - z|^2 d\gamma(x, z) = \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} |x - z|^2 d\sigma(x, y, z) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-I}=y_{-I} \text{ and } y_{-J}=z_{-J}\}} |x - z|^2 d\sigma(x, y, z) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} \chi_{\{x_{-I}=y_{-I} \text{ and } y_{-J}=z_{-J}\}} (|x - y|^2 + |y - z|^2) d\sigma(x, y, z) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} (|x - y|^2 + |y - z|^2) d\sigma(x, y, z) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 d\sigma(x, y, z) + \int_{\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d} |y - z|^2 d\sigma(x, y, z) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 d\gamma^+(x, y) + \int_{\mathbb{R}^d \times \mathbb{R}^d} |y - z|^2 d\gamma^-(y, z) \\ &= C_2(\gamma^+)^2 + C_2(\gamma^-)^2. \end{aligned} \quad \square$$

Lemma 7.110 (Induction over coordinates). *Let $(\mu^{(i)})_{i \in [d+1]}$ be a family of absolutely continuous probability measures on $[0, 1]^d$ with strictly positive densities. Suppose that, for each $i \in [d]$, we have 1) $\mu^{(i)}_{-i} = \mu^{(i+1)}_{-i}$ and 2) $\mu^{(i)}|_{x_{-i}} \succeq \mu^{(i+1)}|_{x_{-i}}$ for each $x_{-i} \in [0, 1]_{-i}^d$. Then*

$$W_2^2(\mu^{(1)} \rightarrow \mu^{(d+1)}) \leq \sum_{i=1}^d \int_{[0,1]_{-i}^d} W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) d\mu^{(i)}_{-i}(x_{-i}).$$

Proof. For each $k \in [d + 1]$, let $A(k)$ be the following proposition: there exists $\gamma \in \Pi_{[k]}(\mu^{(1)} \rightarrow \mu^{(k+1)})$ satisfying

$$C_2(\gamma)^2 = \sum_{i=1}^k \int_{[0,1]_{-i}^d} W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) d\mu^{(i)}_{-i}(x_{-i}).$$

Note that $A(d)$ implies the result we want to prove, while $A(1)$ holds by [Lemma 7.109](#). Now let $2 \leq k \leq d$. Suppose $A(k-1)$ holds, and thus let $\gamma^+ \in \Pi_{[k-1]}(\mu^{(1)} \rightarrow \mu^{(k)})$ satisfy

$$C_2(\gamma^+)^2 = \sum_{i=1}^{k-1} \int_{[0,1]_{-i}^d} W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) d\mu^{(i)}_{-i}(x_{-i}).$$

Apply [Lemma 7.109](#) with $i = k$ to obtain a plan $\gamma^- \in \Pi_k(\mu^{(k)} \rightarrow \mu^{(k+1)})$ satisfying

$$C_2(\gamma^-)^2 = \int_{[0,1]_{-k}^d} W_2^2(\mu^{(k)}|_{x_{-k}}, \mu^{(k+1)}|_{x_{-k}}) d\mu^{(k)}_{-k}(x_{-k}).$$

Then, use [Lemma 7.3](#) with $I = [k-1]$ and $J = \{k\}$ to obtain $\gamma \in \Pi_{[k]}(\mu^{(1)} \rightarrow \mu^{(k+1)})$ satisfying

$$C_2(\gamma)^2 = C_2(\gamma^+)^2 + C_2(\gamma^-)^2 = \sum_{i=1}^k \int_{[0,1]_{-i}^d} W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) d\mu^{(i)}_{-i}(x_{-i}),$$

which implies that $A(k)$ holds. Thus $A(d)$ follows by induction. $\square$

7.3.4 From one-dimensional PDE to optimal transport in the cube

We now tie most of the foregoing theory together by showing how to inductively transform a function on the unit cube into a monotone function, one coordinate at a time via the directed heat semigroup machinery, keeping track of the cost using the directed optimal transport machinery.

For this part of the proof, it is convenient to restrict our attention to Lipschitz functions. As we will see, the Lipschitz property of $f : [0, 1]^d \rightarrow \mathbb{R}$ is preserved under taking the monotone equilibrium along each axis-aligned direction, which enables an inductive argument over the coordinates.

Denote the set of Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$ by Lip (the dimension d will always be clear from context).

Observation 7.111. A function $f : [0, 1]^d \rightarrow \mathbb{R}$ is Lipschitz if and only if its restriction to every axis-aligned line is uniformly Lipschitz, i.e. there exists some $M > 0$ such that, for every $i \in [d]$ and $x_{-i} \in [0, 1]_{-i}^d$, $f(x_{-i}, \cdot) : [0, 1] \rightarrow \mathbb{R}$ is M -Lipschitz.

Definition 7.112 (i - and I -monotonicity). Let $i \in [d]$. We say $f : [0, 1]^d \rightarrow \mathbb{R}$ is i -monotone if its every restriction along direction i is nondecreasing, i.e. if for every $x_{-i} \in [0, 1]_{-i}^d$, $f(x_{-i}, \cdot)$ is nondecreasing. For $I \subseteq [d]$, we say f is I -monotone if it is i -monotone for every $i \in I$. We write Mon_I for the set of I -monotone functions.

Observation 7.113. $\text{Mon}_{[d]}$ is simply the set of monotone functions on $[0, 1]^d$.

The following useful characterization of i -monotone functions in terms of k -aligned lines follows by definition:

Observation 7.114. Let $i, k \in [d]$ be distinct. Then for all $f : [0, 1]^d \rightarrow \mathbb{R}$, f is i -monotone if and only if the following holds: for every $z_{-i-k} \in [0, 1]_{-i-k}^d$, and for all $x_i \leq y_i$ in $[0, 1]$, $f(z_{-i-k}, x_i, \cdot) \leq f(z_{-i-k}, y_i, \cdot)$.

Definition 7.115 (Monotone equilibrium operator). Let $k \in [d]$. We define the operator $\mathcal{M}_k : \text{Lip} \rightarrow ([0, 1]^d \rightarrow \mathbb{R})$ as follows: for each function $f \in \text{Lip}$, $x_{-k} \in [0, 1]_{-k}^d$ and $x_k \in [0, 1]$, the function $\mathcal{M}_k f : [0, 1]^d \rightarrow \mathbb{R}$ satisfies

$$(\mathcal{M}_k f)(x_{-k}, x_k) := (P_\infty f(x_{-k}, \cdot))(x_k),$$

where the application of P_∞ is well-defined because each $f(x_{-k}, \cdot)$ is Lipschitz and hence in $L^2(I)$.

The following lemma shows that $\mathcal{M}_k$ preserves Lipschitzness, confirming that $\mathcal{M}_k f$ is well-defined pointwise as a real-valued function, as opposed to only defined up to sets of measure zero.

Lemma 7.116. Let $k \in [d]$. Let $f \in \text{Lip}$. Then $\mathcal{M}_k f \in \text{Lip}$.

Proof. Let $M > 0$ be such that every axis-aligned line restriction of f is M -Lipschitz. We claim that every axis-aligned line restriction of $\mathcal{M}_k f$ is M -Lipschitz. We first check the k -aligned lines. For any $x_{-k} \in [0, 1]_{-k}^d$ [Proposition 7.75](#) implies that

$$\psi((\mathcal{M}_k f)(x_{-k}, \cdot)) = \psi(P_\infty f(x_{-k}, \cdot)) \leq \psi(f(x_{-k}, \cdot)) \leq M,$$

the last inequality and the conclusion that $(\mathcal{M}_k f)(x_{-k}, \cdot)$ is M -Lipschitz by [Fact 7.42](#).

It remains to check the other line restrictions. Let $i \in [d] \setminus \{k\}$ and let $z_{-i-k} \in [0, 1]_{-i-k}^d$. Let $x_i \neq y_i \in [0, 1]$. Since the axis-aligned line restrictions of f (in particular, in direction i) are M -Lipschitz, we have

$$\sup_{w_k \in [0, 1]} |f(z_{-i-k}, x_i, w_k) - f(z_{-i-k}, y_i, w_k)| \leq M|x_i - y_i|.$$

Proposition 7.77 implies that

$$\begin{aligned} |P_\infty f(z_{-i-k}, x_i, \cdot) - P_\infty f(z_{-i-k}, y_i, \cdot)| &\leq \operatorname{ess\,sup}_{w_k \in [0, 1]} |f(z_{-i-k}, x_i, w_k) - f(z_{-i-k}, y_i, w_k)| \\ &\leq M|x_i - y_i| \end{aligned} \tag{7.28}$$

a.e. in I , and since $P_\infty f(z_{-i-k}, x_i, \cdot)$ and $P_\infty f(z_{-i-k}, y_i, \cdot)$ are (Lipschitz) continuous as observed above, we conclude that (7.28) holds pointwise. Since (7.28) holds for every $i \in [d] \setminus \{k\}$ and $z_{-i-k} \in [0, 1]_{-i-k}^d$, and since

$$|(\mathcal{M}_k f)(z_{-i-k}, x_i, \cdot) - (\mathcal{M}_k f)(z_{-i-k}, y_i, \cdot)| = |P_\infty f(z_{-i-k}, x_i, \cdot) - P_\infty f(z_{-i-k}, y_i, \cdot)|$$

pointwise, we get that every axis-aligned line restriction of $\mathcal{M}_k f$ is M -Lipschitz, and $\mathcal{M}_k f \in \operatorname{Lip}$. $\square$

Lemma 7.117 ($\mathcal{M}_k$ application makes progress). *Let $k \in [d]$, and let $I \subseteq [d] \setminus \{k\}$. Let $f \in \operatorname{Lip} \cap \operatorname{Mon}_I$. Then $\mathcal{M}_k f \in \operatorname{Lip} \cap \operatorname{Mon}_{I \cup \{k\}}$.*

Proof. The fact that $\mathcal{M}_k f \in \operatorname{Lip}$ is given by [Lemma 7.116](#), and the fact that $\mathcal{M}_k f$ is k -monotone follows from the definition of $\mathcal{M}_k$ and the fact that the monotone equilibrium $P_\infty f(x_{-k}, \cdot)$ is nondecreasing. Now, let $i \in I$, so that f is i -monotone. By [Observation 7.114](#), we have that for every $z_{-i-k} \in [0, 1]_{-i-k}^d$, and for all $x_i \leq y_i$ in $[0, 1]$, $f(z_{-i-k}, x_i, \cdot) \leq f(z_{-i-k}, y_i, \cdot)$. By [Corollary 7.71](#), we obtain that $P_\infty f(z_{-i-k}, x_i, \cdot) \leq P_\infty f(z_{-i-k}, y_i, \cdot)$ a.e. in I , and since these two functions are (Lipschitz) continuous, this inequality holds pointwise. Thus $(\mathcal{M}_k f)(z_{-i-k}, x_i, \cdot) \leq (\mathcal{M}_k f)(z_{-i-k}, y_i, \cdot)$, and again by [Observation 7.114](#), $\mathcal{M}_k f$ is i -monotone. This concludes the proof. $\square$

Lemma 7.118 ($\mathcal{M}_k$ preserves bounds). *Let $f \in \operatorname{Lip}$. Let $a \leq b$ be real numbers and suppose $a \leq f \leq b$. Then $a \leq \mathcal{M}_k f \leq b$.*

Proof. This follows by the definition of $\mathcal{M}_k$ together with [Corollary 7.71](#) and [Lemma 7.73](#). $\square$

Lemma 7.119 ($\mathcal{M}_k$ is nonexpansive in L^2). *Let $f, g \in \text{Lip}$. Then*

$$\int_{[0,1]^d} (\mathcal{M}_k f - \mathcal{M}_k g)^2 dx \leq \int_{[0,1]^d} (f - g)^2 dx.$$

Proof. This follows from [Proposition 7.72](#) and Tonelli's theorem, as follows:

$$\begin{aligned} \int_{[0,1]^d} (\mathcal{M}_k f - \mathcal{M}_k g)^2 dx &= \int_{[0,1]_{-k}^d} dx_{-k} \int_I [(\mathcal{M}_k f)(x_{-k}, x_k) - (\mathcal{M}_k g)(x_{-k}, x_k)]^2 dx_k \\ &= \int_{[0,1]_{-k}^d} dx_{-k} \int_I [(P_\infty f(x_{-k}, \cdot))(x_k) - (P_\infty g(x_{-k}, \cdot))(x_k)]^2 dx_k \\ &\leq \int_{[0,1]_{-k}^d} dx_{-k} \int_I [f(x_{-k}, x_k) - g(x_{-k}, x_k)]^2 dx_k \\ &= \int_{[0,1]^d} (f - g)^2 dx. \end{aligned} \quad \square$$

Similarly, combining [Corollary 7.87](#) with Fubini's theorem yields

Lemma 7.120. *Let $f \in \text{Lip}$. Then*

$$\int_{[0,1]^d} (\mathcal{M}_k f) dx = \int_{[0,1]^d} f dx.$$

The following result is essentially an L^2 version of [\[Fer23, Proposition 3.14\]](#).

Lemma 7.121 (Effect of $\mathcal{M}_k$ on the directed Dirichlet energy). *Let $i, k \in [d]$ be distinct and let $f \in \text{Lip}$. Then*

$$\int_{[0,1]_{-i}^d} \mathcal{E}^-((\mathcal{M}_k f)(x_{-i}, \cdot)) dx_{-i} \leq \int_{[0,1]_{-i}^d} \mathcal{E}^-(f(x_{-i}, \cdot)) dx_{-i}.$$

Proof. For any $g \in \text{Lip}$, $i \in [d]$ and $h \in \mathbb{R} \setminus \{0\}$, define the $Q_{g,i,h} : [0,1]^d \rightarrow \mathbb{R}$ by

$$Q_{g,i,h}(x) := \begin{cases} \frac{g(x_{-i}, x_i+h) - g(x_{-i}, x_i)}{h} & \text{if } x_i + h \in (0, 1) \\ 0 & \text{otherwise.} \end{cases} \quad (7.29)$$

By Rademacher's theorem on the open domain $(0, 1)^d$, g is differentiable almost everywhere. In particular, the function $D_{g,i} : [0, 1]^d \rightarrow \mathbb{R}$ given by

$$D_{g,i}(x) := \begin{cases} \lim_{h \rightarrow 0} Q_{g,i,h}(x) & \text{if the limit exists} \\ 0 & \text{otherwise} \end{cases}$$

gives the partial derivative of g in direction i for a.e. $x \in [0, 1]^d$, and is measurable as the limit of the measurable functions $Q_{g,i,h}$. Moreover, letting $M > 0$ be such that the axis-aligned line restrictions of g are M -Lipschitz, we have $|Q_{g,i,h}| \leq M$ pointwise. Finally, fixing any $x_{-i} \in [0, 1]_{-i}^d$, $g(x_{-i}, \cdot)$ is (Lipschitz and hence) in $H^1(I)$, and the partial derivative $D_{g,i}(x_{-i}, \cdot)$ agrees with any weak derivative $\partial_{x_i} g(x_{-i}, \cdot)$ a.e. in $[0, 1]$. In particular, all of these considerations apply to the functions f and $\mathcal{M}_k f$ by [Lemma 7.116](#). We conclude that, for $g = f$ or $g = \mathcal{M}_k f$ and each $x_{-i} \in [0, 1]_{-i}^d$,

$$\begin{aligned} \mathcal{E}^-(g(x_{-i}, \cdot)) &= \frac{1}{2} \int_I [\partial_{x_i}^- g(x_{-i}, x_i)]^2 dx_i = \frac{1}{2} \int_I [D_{g,i}(x_{-i}, x_i)^-]^2 dx_i \\ &= \frac{1}{2} \int_I \lim_{h \rightarrow 0} [Q_{g,i,h}(x_{-i}, x_i)^-]^2 dx_i, \end{aligned}$$

where for simplicity we write the limit in the last expression with the understanding that it is only defined almost everywhere in $[0, 1]$. Note that the measurability of the last integrand over $[0, 1]^d$ justifies writing the integrals in the statement of the lemma via Tonelli's theorem (as the computation below shows). Define for each $h \in \mathbb{R}$ the set

$$I_h := \begin{cases} (0, 1 - h) & \text{if } h \geq 0 \\ (-h, 1) & \text{otherwise,} \end{cases}$$

so that $Q_{g,i,h}(x)$ is defined by the first case of (7.29) if and only if $x_i \in I_h$. Then, by repeated applications of Tonelli's theorem and the dominated convergence theorem, we obtain

$$\begin{aligned} \int_{[0,1]_{-i}^d} \mathcal{E}^-(\mathcal{M}_k f)(x_{-i}, \cdot) dx_{-i} &= \frac{1}{2} \int_{[0,1]_{-i}^d} dx_{-i} \int_I \lim_{h \rightarrow 0} [Q_{\mathcal{M}_k f,i,h}(x_{-i}, x_i)^-]^2 dx_i \\ &= \frac{1}{2} \int_{[0,1]^d} \lim_{h \rightarrow 0} \underbrace{[Q_{\mathcal{M}_k f,i,h}(x)^-]^2}_{\leq M^2} dx = \frac{1}{2} \lim_{h \rightarrow 0} \int_{[0,1]^d} [Q_{\mathcal{M}_k f,i,h}(x)^-]^2 dx \\ &= \frac{1}{2} \lim_{h \rightarrow 0} \int_{[0,1]_{-i-k}^d} dx_{-i-k} \int_I dx_i \int_I [Q_{\mathcal{M}_k f,i,h}(x_{-i-k}, x_i, x_k)^-]^2 dx_k \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \lim_{h \rightarrow 0} \int_{[0,1]_{-i-k}^d} dx_{-i-k} \int_{I_h} dx_i \\
&\quad \int_I \left[\left(\frac{(\mathcal{M}_k f)(x_{-i-k}, x_i + h, x_k) - (\mathcal{M}_k f)(x_{-i-k}, x_i, x_k)}{h} \right)^- \right]^2 dx_k \\
&= \frac{1}{2} \lim_{h \rightarrow 0} \int_{[0,1]_{-i-k}^d} dx_{-i-k} \int_{I_h} dx_i \\
&\quad \int_I \left[\left(\frac{(P_\infty f)(x_{-i-k}, x_i + h, \cdot)(x_k) - (P_\infty f)(x_{-i-k}, x_i, \cdot)(x_k)}{h} \right)^- \right]^2 dx_k \\
&\leq \frac{1}{2} \lim_{h \rightarrow 0} \int_{[0,1]_{-i-k}^d} dx_{-i-k} \int_{I_h} dx_i \int_I \left[\left(\frac{f(x_{-i-k}, x_i + h, x_k) - f(x_{-i-k}, x_i, x_k)}{h} \right)^- \right]^2 dx_k \\
&= \frac{1}{2} \lim_{h \rightarrow 0} \int_{[0,1]_{-i-k}^d} dx_{-i-k} \int_I dx_i \int_I [Q_{f,i,h}(x_{-i-k}, x_i, x_k)^-]^2 dx_k \\
&= \frac{1}{2} \lim_{h \rightarrow 0} \int_{[0,1]^d} \underbrace{[Q_{f,i,h}(x)^-]^2}_{\leq M^2} dx = \frac{1}{2} \int_{[0,1]^d} \lim_{h \rightarrow 0} [Q_{f,i,h}(x)^-]^2 dx \\
&= \frac{1}{2} \int_{[0,1]_{-i}^d} dx_{-i} \int_I \lim_{h \rightarrow 0} [Q_{f,i,h}(x_{-i}, x_i)^-]^2 dx_i = \int_{[0,1]_{-i}^d} \mathcal{E}^-(f(x_{-i}, \cdot)) dx_{-i},
\end{aligned}$$

the inequality by [Proposition 7.70](#) via the identity $[(a - b)^+]^2 = [(b - a)^-]^2$. $\square$

Definition 7.122 (Coordinate-wise monotone equilibrium). For each $f \in \text{Lip}$, define $f^* \in \text{Lip}$, the *coordinate-wise monotone equilibrium* of f , by

$$f^* := \mathcal{M}_d \mathcal{M}_{d-1} \cdots \mathcal{M}_1 f.$$

Note that indeed f^* is indeed Lipschitz by [Lemma 7.116](#), and it is monotone:

Proposition 7.123. *Let $f \in \text{Lip}$. Then f^* is monotone.*

Proof. This follows by repeatedly applying [Lemma 7.117](#), and recalling that $\text{Mon}_{[d]}$ is the set of monotone functions on $[0, 1]^d$. $\square$

By [Proposition 7.76](#) and the definition of $\mathcal{M}_k$, we also observe that the coordinate-wise monotone equilibrium behaves nicely with respect to certain affine transformations:

Observation 7.124. *Let $f \in \text{Lip}$. Let $\alpha > 0$ and $\beta \in \mathbb{R}$. Then $(\alpha f + \beta)^* = \alpha f^* + \beta$.*

Similarly, [Lemma 7.119](#) implies that taking the coordinate-wise monotone equilibrium is a nonexpansive operation:

Observation 7.125. *Let $f, g \in \text{Lip}$. Then $\|f^* - g^*\|_{L^2((0,1)^d)} \leq \|f - g\|_{L^2((0,1)^d)}$. Since $0^* = 0$ by [Lemma 7.73](#) (where we write 0 for the constant zero function), we conclude in particular that $\|f^*\|_{L^2((0,1)^d)} \leq \|f\|_{L^2((0,1)^d)}$.*

We are now ready to combine the ingredients from the previous sections and establish the transport-energy inequality. Note that this inequality is most effective when the function f is pointwise bounded close to 1.

Theorem 7.4 (Transport-energy inequality). *There exists a universal constant $C > 0$ such that the following holds. Let $a \in (0, 1)$, and let $f \in \text{Lip}$ satisfy $1 - a \leq f \leq 1 + a$ and $\int_{[0,1]^d} f \, dx = 1$. Define the probability measures $d\mu := f \, dx$ and $d\mu^* := f^* \, dx$ on $[0, 1]^d$. Then*

$$W_2^2(\mu \rightarrow \mu^*) \leq \frac{C(1+a)^2}{(1-a)^3} \int_{[0,1]^d} |\nabla^- f|^2 \, dx.$$

Proof. Define the family $(f^{(i)})_{i \in [d+1]} \subset \text{Lip}$ by $f^{(1)} := f$ and $f^{(i+1)} := \mathcal{M}_i f^{(i)}$ for each $i \in [d]$, so that $f^{(d+1)} = f^*$. Note that we have $1 - a \leq f^{(i)} \leq 1 + a$ and $\int_{[0,1]^d} f^{(i)} \, dx = 1$ for each $i \in [d+1]$ by [Lemmas 7.118](#) and [7.120](#). Then, define the family $(\mu^{(i)})_{i \in [d+1]}$ of probability measures on $[0, 1]^d$ by $d\mu^{(i)} := f^{(i)} \, dx$, so that $\mu^{(1)} = \mu$ and $\mu^{(d+1)} = \mu^*$.

We claim $(\mu^{(i)})_{i \in [d+1]}$ satisfies the conditions of [Lemma 7.110](#). First, they have strictly positive densities because $1 - a \leq f^{(i)} \leq 1 + a$ for each $i \in [d+1]$ as observed above. They are also absolutely continuous with Borel measurable density because each $f^{(i)}$ is continuous and hence Borel.

Recall that, by [Proposition 7.102](#), for each $i, j \in [d+1]$ we may characterize the disintegration $\mu^{(j)} = \int_{\mathbb{R}^d} \mu^{(j)}|_{x_{-i}} d\mu^{(j)}_{-i}$ as follows: the marginal and all conditional measures are both absolutely continuous supported in $[0, 1]_{-i}^d$ and $[0, 1]$ respectively, and we have $d\mu^{(j)}_{-i} = f^{(j)}_{-i} \, dx_{-i}$ where

$$f^{(j)}_{-i}(x_{-i}) = \int_I f^{(j)}(x_{-i}, x_i) \, dx_i \quad (7.30)$$

for each $x_{-i} \in [0, 1]_{-i}^d$, and $d\mu^{(j)}|_{x_{-i}} = f^{(j)}|_{x_{-i}} dx_i$ where

$$f^{(j)}|_{x_{-i}}(x_i) = \frac{f^{(j)}(x)}{f^{(j)}_{-i}(x_{-i})} \quad (7.31)$$

for each $x \in [0, 1]^d$ (note that the denominator is nonzero since $f^{(j)} \geq 1 - a$).

Now, fix any $i \in [d]$. Then the condition $\mu^{(i)}_{-i} = \mu^{(i+1)}_{-i}$ is implied by the pointwise condition

$$\int_I f^{(i)}(x_{-i}, x_i) dx_i = \int_I f^{(i+1)}(x_{-i}, x_i) dx_i \quad \forall x_{-i} \in [0, 1]_{-i}^d, \quad (7.32)$$

which in turn holds by the definitions of $f^{(i+1)}$ and $\mathcal{M}_i$, and [Corollary 7.87](#). Finally, fixing $x_{-i} \in [0, 1]_{-i}^d$, we verify the condition $\mu^{(i)}|_{x_{-i}} \succeq \mu^{(i+1)}|_{x_{-i}}$. By [Corollary 7.91](#), it suffices to show that

$$f^{(i+1)}|_{x_{-i}} \stackrel{?}{=} P_\infty f^{(i)}|_{x_{-i}}. \quad (7.33)$$

By (7.30) and (7.31), this is equivalent to

$$\frac{1}{\int_I f^{(i+1)}(x_{-i}, x_i) dx_i} f^{(i+1)}(x_{-i}, \cdot) \stackrel{?}{=} P_\infty \left(\frac{1}{\int_I f^{(i)}(x_{-i}, x_i) dx_i} f^{(i)}(x_{-i}, \cdot) \right).$$

Since $f^{(i+1)}(x_{-i}, \cdot) = P_\infty f^{(i)}(x_{-i}, \cdot)$, and applying [Proposition 7.76](#), the above is equivalent to

$$\frac{1}{\int_I f^{(i+1)}(x_{-i}, x_i) dx_i} P_\infty f^{(i)}(x_{-i}, \cdot) \stackrel{?}{=} \frac{1}{\int_I f^{(i)}(x_{-i}, x_i) dx_i} P_\infty f^{(i)}(x_{-i}, \cdot),$$

which is true again by (7.32), so indeed $\mu^{(i)}|_{x_{-i}} \succeq \mu^{(i+1)}|_{x_{-i}}$. Thus, by [Lemma 7.110](#), we have

$$\begin{aligned} W_2^2(\mu \rightarrow \mu^*) &\leq \sum_{i=1}^d \int_{[0,1]_{-i}^d} W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) d\mu^{(i)}_{-i}(x_{-i}) \\ &= \sum_{i=1}^d \int_{[0,1]_{-i}^d} W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) f^{(i)}_{-i}(x_{-i}) dx_{-i} \\ &\leq (1+a) \sum_{i=1}^d \int_{[0,1]_{-i}^d} W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) dx_{-i}, \end{aligned}$$

the last inequality because $1 - a \leq f^{(i)} \leq 1 + a$ also implies that $1 - a \leq f^{(i)}_{-i} \leq 1 + a$. Now, applying [Theorem 7.2](#) via (7.33), we have

$$W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) \leq \frac{C}{\inf f^{(i)}|_{x_{-i}}} \mathcal{E}^-(f^{(i)}|_{x_{-i}})$$

for each $i \in [d]$ and $x_{-i} \in [0, 1]_{-i}^d$. Note that $f^{(i)}|_{x_{-i}} \geq \frac{1-a}{1+a}$ by the pointwise bounds on $f^{(i)}$ and $f^{(i)}_{-i}$, and that $\mathcal{E}^-(\alpha g) = \alpha^2 \mathcal{E}^-(g)$ for any $\alpha \geq 0$ and (say) Lipschitz g by the definition of $\mathcal{E}^-$. Therefore, applying (7.31), we obtain

$$W_2^2(\mu^{(i)}|_{x_{-i}}, \mu^{(i+1)}|_{x_{-i}}) \leq \frac{C(1+a)}{1-a} \left(\frac{1}{f^{(i)}_{-i}(x_{-i})} \right)^2 \mathcal{E}^-(f^{(i)}(x_{-i}, \cdot)) \leq \frac{C(1+a)}{(1-a)^3} \mathcal{E}^-(f^{(i)}(x_{-i}, \cdot)).$$

Hence we have

$$W_2^2(\mu \rightarrow \mu^*) \leq \frac{C(1+a)^2}{(1-a)^3} \sum_{i=1}^d \int_{[0,1]_{-i}^d} \mathcal{E}^-(f^{(i)}(x_{-i}, \cdot)) \, dx_{-i}.$$

Recall that $f^{(1)} = f$ while $f^{(i)} = \mathcal{M}_{i-1} \cdots \mathcal{M}_1 f$ for each $i = 2, \dots, d+1$. Hence, by inductively applying Lemma 7.121 for each $i \geq 2$, and by Tonelli's theorem and the definition of $\mathcal{E}^-$, we obtain

$$\begin{aligned} W_2^2(\mu \rightarrow \mu^*) &\leq \frac{C(1+a)^2}{(1-a)^3} \sum_{i=1}^d \int_{[0,1]_{-i}^d} \mathcal{E}^-(f(x_{-i}, \cdot)) \, dx_{-i} \\ &= \frac{C(1+a)^2}{2(1-a)^3} \sum_{i=1}^d \int_{[0,1]_{-i}^d} dx_{-i} \int_I (\partial_{x_i}^- f(x_{-i}, x_i))^2 \, dx_i \\ &= \frac{C(1+a)^2}{2(1-a)^3} \int_{[0,1]^d} \sum_{i=1}^d (\partial_i^- f)^2 \, dx \\ &= \frac{C(1+a)^2}{2(1-a)^3} \int_{[0,1]^d} |\nabla^- f|^2 \, dx. \end{aligned} \quad \square$$

7.4 Directed Poincaré from Wasserstein and Kantorovich

In this section, we establish our directed Poincaré inequality (Theorem 7.1) by combining the transport-energy inequality from Theorem 7.4 with a directed version of (weak) Kantorovich duality via a perturbation argument, thereby implementing the ideas from Section 7.1.5 of the proof overview. The general theme of going between transport inequalities and Poincaré inequalities, as well as the spirit of the technical arguments presented here, are well-known in the undirected case; we refer the reader to [Vil09, Chapters 7 and

22] for a comprehensive presentation and primary references, and to [Liu20] for an undirected version of the precise reduction we instantiate here (that is, from a transport-energy inequality to a Poincaré inequality).

For simplicity, we do not attempt to initiate a systematic study of duality in directed optimal transport, or to state results in the broadest possible generality, but rather limit ourselves to what is required for the aforementioned goal. In [Section 7.4.1](#), we introduce the notion of directed weak Kantorovich duality and establish some of its properties. The main results are [Proposition 7.129](#), which gives a *lower bound* on the directed Wasserstein distance between two measures in terms the so-called (directed) *Hamilton-Jacobi operator* $\vec{H}$ (recall that the previous section gave an *upper bound* on this distance via the transport-energy inequality), and [Proposition 7.131](#), which relates the action $\vec{H}_t h(x)$ of this operator back to the directed gradient $\nabla^+ h(x)$ (which, recall, we wish to have in the right-hand side of our Poincaré inequality). Then, in [Section 7.4.2](#), we apply these tools via a careful choice of dual function (a “perturbation argument”) to conclude the directed Poincaré inequality, first for continuously differentiable functions ([Theorem 7.133](#)) and then, via a standard approximation argument, for all H^1 functions ([Theorem 7.1](#)).

7.4.1 Directed weak Kantorovich duality and Hamilton-Jacobi operator

Kantorovich duality plays a fundamental role in the study of Wasserstein distance. The weak part of this duality allows one to lower bound the Wasserstein distance between two distributions by means of “test functions” satisfying a certain constraint. Here, we translate a small part of this rich theory to the directed setting.

Lemma 7.126 (Weak duality). *Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set and let $\mu, \nu \in P(\Omega)$. Suppose $\phi \in L^1(\nu)$ and $\psi \in L^1(\mu)$ satisfy*

$$\phi(y) - \psi(x) \leq |x - y|^2 \quad \forall x \preceq y \text{ in } \Omega.$$

Then

$$W_2^2(\mu \rightarrow \nu) \geq \int_{\Omega} \phi \, d\nu - \int_{\Omega} \psi \, d\mu.$$

Proof. Suppose $\gamma \in \Pi(\mu \rightarrow \nu)$. Then

$$C_2(\gamma)^2 = \int_{\Omega \times \Omega} |x - y|^2 \, d\gamma(x, y)$$

$$\begin{aligned}
&= \int_{\Omega \times \Omega} \chi_{\{x \preceq y\}} |x - y|^2 \, d\gamma(x, y) && \text{(Since } \gamma \in \Pi(\mu \rightarrow \nu)\text{)} \\
&\geq \int_{\Omega \times \Omega} \chi_{\{x \preceq y\}} (\phi(y) - \psi(x)) \, d\gamma(x, y) && \text{(Hypothesis)} \\
&= \int_{\Omega \times \Omega} (\phi(y) - \psi(x)) \, d\gamma(x, y) && \text{(Since } \gamma \in \Pi(\mu \rightarrow \nu)\text{)} \\
&= \int_{\Omega} \phi \, d\nu - \int_{\Omega} \psi \, d\mu && \text{(Since } (\pi_1)_\# \gamma = \mu, (\pi_2)_\# \gamma = \nu\text{).}
\end{aligned}$$

Since this holds for every $\gamma \in \Pi(\mu \rightarrow \nu)$, the result follows. $\square$

This duality results motivates the following definition. Let $C_b(\Omega)$ denote the set of bounded continuous functions $\Omega \rightarrow \mathbb{R}$.

Definition 7.127 (Directed Hamilton-Jacobi operator). Let $\Omega \subset \mathbb{R}^d$. For each $t \geq 0$, define the *directed Hamilton-Jacobi operator* $\vec{H}_t : C_b(\Omega) \rightarrow (\Omega \rightarrow \mathbb{R})$ as follows: for each $h \in C_b(\Omega)$ and $x \in \Omega$, we set

$$(\vec{H}_t h)(x) := \begin{cases} h(x) & \text{if } t = 0 \\ \sup_{y \succeq x} \{h(y) - \frac{1}{2t}|x - y|^2\} & \text{otherwise.} \end{cases} \quad (7.34)$$

Remark 7.128. The definition above is a directed analogue of the so-called (backward) *Hamilton-Jacobi-Hopf-Lax-Oleinik semigroup*. We do not claim that $\vec{H}_t$ forms a semigroup.

Proposition 7.129. Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set, and let $\mu, \nu \in P(\Omega)$. Let $h \in C_b(\Omega)$. Then

$$\frac{1}{2}W_2^2(\mu \rightarrow \nu) \geq \int_{\Omega} h \, d\nu - \int_{\Omega} (\vec{H}_1 h) \, d\mu.$$

Proof. Let $\phi := 2h$ and $\psi := 2\vec{H}_1 h$. Note that $\phi \in L^1(\nu)$ since h is bounded. The fact that h is bounded also implies that $\vec{H}_1 h$ is bounded, as can be verified from its definition, and hence $\psi \in L^1(\mu)$. Finally, for each $x \preceq y$ in Ω we have

$$\begin{aligned}
\phi(y) - \psi(x) &= 2h(y) - 2 \sup_{y' \succeq x} \left\{ h(y') - \frac{1}{2}|x - y'|^2 \right\} \\
&= |x - y|^2 + 2 \underbrace{\left[\left(h(y) - \frac{1}{2}|x - y|^2 \right) - \sup_{y' \succeq x} \left\{ h(y') - \frac{1}{2}|x - y'|^2 \right\} \right]}_{\leq 0} \leq |x - y|^2.
\end{aligned}$$

Thus [Lemma 7.126](#) gives the conclusion. $\square$

The following simple lemma plays a key role in the perturbation argument.

Lemma 7.130. *Let $\Omega \subset \mathbb{R}^d$. Let $t > 0$ and let $h \in C_b(\Omega)$. Then $\vec{H}_1(th) = t\vec{H}_t h$.*

Proof. Indeed, for each $x \in \Omega$ we have

$$(\vec{H}_1(th))(x) = \sup_{y \succeq x} \left\{ th(y) - \frac{1}{2}|x - y|^2 \right\} = t \sup_{y \succeq x} \left\{ h(y) - \frac{1}{2t}|x - y|^2 \right\} = t((\vec{H}_t h)(x)). \quad \square$$

Notation. For $\Omega \subset \mathbb{R}^d$ an open set and $k \in \mathbb{N} \cup \{+\infty\}$, write $C^k(\overline{\Omega})$ for the set of restrictions to $\overline{\Omega}$ of functions in $C^k(\mathbb{R}^d)$.

For $\Omega \subset \mathbb{R}^d$, point $x \in \Omega$, and $r > 0$, let $B_\Omega^\circ(x, r) := B^\circ(x, r) \cap \Omega$, where $B^\circ(x, r) := \{y \in \mathbb{R}^d : 0 < |x - y| < r\}$ is the open ball of radius r centered at x with x itself excluded, and let $B_\Omega^+(x, r) := B_\Omega^\circ(x, r) \cap \{y \in \mathbb{R}^d : y \succeq x\}$.

The following result gives a directed analogue to some of the properties of the Hamilton-Jacobi semigroup, as presented in [Vil09, Theorem 22.16].⁵

Proposition 7.131 (Some properties of the directed Hamilton-Jacobi operator). *Let $\Omega \subset \mathbb{R}^d$ be a bounded open set. Let $h \in C^1(\overline{\Omega})$.*

(a) *Let $C := \sup h - \inf h$, which is well-defined because h is continuous over the compact set $\overline{\Omega}$ and thus bounded. Then for each $t > 0$ and $x \in \Omega$, the supremum in (7.34) may be taken over the set $B_\Omega^+(x, \sqrt{2Ct})$.*

(b) *For each $x \in \Omega$,*

$$\limsup_{t \rightarrow 0^+} \frac{\vec{H}_t h(x) - h(x)}{t} \leq \frac{|\nabla^+ h(x)|^2}{2}.$$

(c) *The quotient*

$$\frac{\vec{H}_t h(x) - h(x)}{t}$$

is nonnegative and bounded, uniformly in $t > 0$ and $x \in \Omega$.

⁵Note that our use of the notation $\nabla^+ h = 0 \vee \nabla h$, $\nabla^- h = 0 \wedge \nabla h$ is unrelated to the use of similar notation in [Vil09], where it denotes a notion of norm of the gradient for functions in more general spaces where the usual derivative may not be defined. In particular, their definition agrees with the norm of the gradient for differentiable functions, while the point is the our definition (the directed gradient) does not.

Proof. Let us first show (a). For any $x \preceq y$ in Ω and $t > 0$, if $|x - y| \geq \sqrt{2Ct}$ then

$$h(y) - \frac{1}{2t}|x - y|^2 \leq h(x) + C - \frac{1}{2t}2Ct = h(x),$$

so y is irrelevant to the supremum in (7.34). Moreover, by the continuity of h , we may drop x itself from the supremum. Thus y is irrelevant to the supremum whenever $y \notin B_\Omega^+(x, \sqrt{2Ct})$, as claimed.

We now show (b) using (a). Let $x \in \Omega$ and $t > 0$. We have

$$\begin{aligned} \frac{\vec{H}_t h(x) - h(x)}{t} &= \frac{\sup_{y \in B_\Omega^+(x, \sqrt{2Ct})} \left\{ h(y) - \frac{1}{2t}|x - y|^2 \right\} - h(x)}{t} \\ &= \sup_{y \in B_\Omega^+(x, \sqrt{2Ct})} \left\{ \frac{h(y) - h(x)}{t} - \frac{1}{2t^2}|x - y|^2 \right\} \\ &\leq \sup_{y \in B_\Omega^+(x, \sqrt{2Ct})} \left\{ \frac{(h(y) - h(x))^+}{t} - \frac{1}{2t^2}|x - y|^2 \right\} \\ &= \sup_{y \in B_\Omega^+(x, \sqrt{2Ct})} \left\{ \frac{(h(y) - h(x))^+}{|x - y|} \frac{|x - y|}{t} - \frac{1}{2} \left(\frac{|x - y|}{t} \right)^2 \right\}. \end{aligned}$$

Using the inequality $\alpha\beta - \frac{1}{2}\beta^2 \leq \frac{1}{2}\alpha^2$, we obtain

$$\frac{\vec{H}_t h(x) - h(x)}{t} \leq \frac{1}{2} \sup_{y \in B_\Omega^+(x, \sqrt{2Ct})} \left(\frac{(h(y) - h(x))^+}{|x - y|} \right)^2. \quad (7.35)$$

Using Lemma 7.132 and the fact that the term inside the supremum is nonnegative, we conclude that

$$\limsup_{t \rightarrow 0^+} \frac{\vec{H}_t h(x) - h(x)}{t} \leq \frac{1}{2} \left(\lim_{t \rightarrow 0^+} \sup_{y \in B_\Omega^+(x, \sqrt{2Ct})} \frac{(h(y) - h(x))^+}{|x - y|} \right)^2 = \frac{|\nabla^+ h(x)|^2}{2},$$

as desired. We may also obtain (c) from (7.35) as follows. First note that $\vec{H}_t h(x) \geq h(x)$ for all $t \geq 0$ and $x \in \Omega$. It is standard that C^1 functions are Lipschitz on compact sets, so h is Lipschitz on $\bar{\Omega}$. Let $M > 0$ be its Lipschitz constant. Then from (7.35) we conclude that, for each $x \in \Omega$ and $t > 0$,

$$0 \leq \frac{\vec{H}_t h(x) - h(x)}{t} \leq \frac{1}{2} \sup_{y \in B_\Omega^+(x, \sqrt{2Ct})} \left(\frac{M|x - y|}{|x - y|} \right)^2 = \frac{M^2}{2},$$

which establishes (c). □

Lemma 7.132. Let $\Omega \subset \mathbb{R}^d$ be an open set. Let $h \in C^1(\overline{\Omega})$. Then for each $x \in \Omega$,

$$|\nabla h(x)| = \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{(h(y) - h(x))^+}{|x - y|}$$

and

$$|\nabla^+ h(x)| = \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^+(x, r)} \frac{(h(y) - h(x))^+}{|x - y|}.$$

Proof. The first part of the statement is known, but let us give a proof for completeness. Let $x \in \Omega$. We have

$$\lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{h(y) - h(x)}{|x - y|} = \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{h(y) - h(x) - \nabla h(x)^\top(y - x) + \nabla h(x)^\top(y - x)}{|x - y|}.$$

By definition of derivative, we have

$$\lim_{y \rightarrow x} \frac{|h(y) - h(x) - \nabla h(x)^\top(y - x)|}{|x - y|} = 0,$$

and hence, by ℓ^2 norm duality,

$$\lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{h(y) - h(x)}{|x - y|} = \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{\nabla h(x)^\top(y - x)}{|x - y|} = \sup_{v \in \mathbb{R}^d: |v|=1} \nabla h(x)^\top v = |\nabla h(x)|.$$

Similarly, we also have

$$\begin{aligned} \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{|h(y) - h(x)|}{|x - y|} &= \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{|h(y) - h(x) - \nabla h(x)^\top(y - x) + \nabla h(x)^\top(y - x)|}{|x - y|} \\ &\leq \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{|h(y) - h(x) - \nabla h(x)^\top(y - x)| + |\nabla h(x)^\top(y - x)|}{|x - y|} \\ &= \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^\circ(x, r)} \frac{|\nabla h(x)^\top(y - x)|}{|x - y|} = \sup_{v \in \mathbb{R}^d: |v|=1} |\nabla h(x)^\top v| = |\nabla h(x)|. \end{aligned}$$

Since $h(y) - h(x) \leq (h(y) - h(x))^+ \leq |h(y) - h(x)|$, the first part of the statement follows.

We now establish the second part. Write the partition $[d] = P \cup N$ for

$$\begin{aligned} P &:= \{i \in [d] : \partial_i h(x) > 0\}, \\ N &:= \{i \in [d] : \partial_i h(x) \leq 0\}. \end{aligned}$$

Assume for a moment⁶ that $P \neq \emptyset$. Define $\Omega^P \subset \mathbb{R}^P$ by

$$\Omega^P := \{y_P : y \in \Omega \text{ satisfies } \text{supp}(y - x) \subseteq P\},$$

and let $h^P \in C^1(\overline{\Omega^P})$ be given by $h^P(y_P) := h(y_P, x_{-P})$ for each $y_P \in \Omega_P$, i.e. h^P is the restriction of h to the space obtained by fixing the N -coordinates of inputs to those of x . Note that $\partial_i h(x) = \partial_i h^P(x_P)$ for each $i \in P$, and hence $|\nabla^+ h(x)| = |\nabla h^P(x_P)|$. Applying the first part of the statement to Ω^P , h^P and x_P gives

$$|\nabla^+ h(x)| = |\nabla h^P(x_P)| = \lim_{r \rightarrow 0^+} \sup_{y_P \in B_{\Omega^P}^\circ(x_P, r)} \frac{(h^P(y_P) - h^P(x_P))^+}{|x_P - y_P|} \quad (7.36)$$

We claim that

$$\lim_{r \rightarrow 0^+} \sup_{y_P \in B_{\Omega^P}^\circ(x_P, r)} \frac{(h^P(y_P) - h^P(x_P))^+}{|x_P - y_P|} \stackrel{?}{=} \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^+(x, r)} \frac{(h(y) - h(x))^+}{|x - y|}, \quad (7.37)$$

with the LHS replaced by 0 if $P = \emptyset$, which will conclude the proof.

For each set $A \subseteq [d]$, let $X_A := \{y \in \Omega : y_A \succeq x_A\}$ and $Y_A := \{y \in \Omega : y_A \preceq x_A\}$. Note that for all $r > 0$, $B_\Omega^+(x, r) = B_\Omega^\circ(x, r) \cap X_P \cap X_N$ and moreover, defining $B'_\Omega(x, r) := B_\Omega^\circ(x, r) \cap X_N \cap Y_N$, when $P \neq \emptyset$ we have

$$\lim_{r \rightarrow 0^+} \sup_{y_P \in B_{\Omega^P}^\circ(x_P, r)} \frac{(h^P(y_P) - h^P(x_P))^+}{|x_P - y_P|} = \lim_{r \rightarrow 0^+} \sup_{y \in B'_\Omega(x, r)} \frac{(h(y) - h(x))^+}{|x - y|} \quad (7.38)$$

by definition of the objects involved.

Note that, when $P \neq \emptyset$, $\min_{i \in P} \partial_i h(x) > 0$ by the definition of P . Since h is continuously differentiable, it follows that there exists $r_0 > 0$ such that, for all $y \in B_\Omega^\circ(x, r_0)$ and every $i \in P$, we have $\partial_i h(y) > 0$. Since Ω is open, we can also let r_0 be small enough so that $B(x, r_0) \subset \Omega$.

Now, suppose $y \in B_\Omega^\circ(x, r_0)$ and $i \in P$ are such that $y_i < x_i$. Then define $z \in \mathbb{R}^d$ by $z_i := y_i + 2(x_i - y_i) > x_i$, and $z_j := y_j$ for $j \in [d] \setminus \{i\}$. Then $|x - z| = |x - y|$, and in particular $z \in B^\circ(x, r_0) \subset \Omega$. Also, since $i \in P$ and z agrees with y on all $j \neq i$, we have that $y \in X_N$ (resp. Y_N) if and only if $z \in X_N$ (resp. Y_N). Moreover, since $\partial_i h(w) > 0$ for all w in the line segment connecting y and z (which is contained in $B_\Omega^\circ(x, r_0) \cup \{x\}$), the fundamental theorem of calculus implies that $h(z) > h(y)$.

⁶We track the edge case $P = \emptyset$ as we go along the proof; the main idea is the same.

Repeating this argument inductively for each index $i \in P$ for which $y_i < x_i$, we conclude that for all $r \in (0, r_0)$ and $y \in B'_\Omega(x, r)$, there exists $z \in B'_\Omega(x, r) \cap X_P$ such that $|x - z| = |x - y|$ and $h(z) > h(y)$. It follows that, when $P \neq \emptyset$,

$$\lim_{r \rightarrow 0^+} \sup_{y \in B'_\Omega(x, r)} \frac{(h(y) - h(x))^+}{|x - y|} = \lim_{r \rightarrow 0^+} \sup_{y \in B'_\Omega(x, r) \cap X_P} \frac{(h(y) - h(x))^+}{|x - y|}. \quad (7.39)$$

Combining (7.37), (7.38) and (7.39), we see that it remains to show that

$$\lim_{r \rightarrow 0^+} \sup_{y \in B'_\Omega(x, r) \cap X_P} \frac{(h(y) - h(x))^+}{|x - y|} \stackrel{?}{=} \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^+(x, r)} \frac{(h(y) - h(x))^+}{|x - y|}, \quad (7.40)$$

again with the LHS replaced by 0 if $P = \emptyset$. Note that for each $r > 0$ it holds by definition that $B'_\Omega(x, r) \cap X_P \subseteq B_\Omega^+(x, r)$, namely $B'_\Omega(x, r) \cap X_P = B_\Omega^+(x, r) \cap Y_N$. Thus we already have

$$\lim_{r \rightarrow 0^+} \sup_{y \in B'_\Omega(x, r) \cap X_P} \frac{(h(y) - h(x))^+}{|x - y|} \leq \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^+(x, r)} \frac{(h(y) - h(x))^+}{|x - y|},$$

which also holds when the LHS is replaced by 0, and it remains to prove the reverse inequality. Let $\varepsilon > 0$. As before, recalling that $\partial_i h(x) \leq 0$ for every $i \in N$ and using the fact that h is continuously differentiable, we can let $r_0 > 0$ be small enough so that for all $y \in B^\circ(x, r_0)$ and every $i \in N$, we have $\partial_i h(y) \leq \varepsilon$. Again since Ω is open, we may also let r_0 be small enough so that $B(x, r_0) \subset \Omega$.

Now, let $r \in (0, r_0)$ and $y \in B_\Omega^+(x, r)$. Define $y^P \in \mathbb{R}^d$ as follows: $y_i^P := y_i$ for each $i \in P$, and $y_j^P := x_j$ for each $j \in N$. Then $\text{supp}(y^P - x) \subseteq P$ by construction, so $y^P \in X_N \cap Y_N$. We also have $y^P \succeq x$ and $|x - y^P| \leq |x - y|$, and hence $y^P \in (B^+(x, r) \cap Y_N) \cup \{x\} = (B'_\Omega(x, r) \cap X_P) \cup \{x\}$.

By a standard multivariate version of the mean value theorem, there exists a point z in the line segment connecting y^P and y such that

$$h(y) - h(y^P) = \nabla h(z)^\top (y - y^P).$$

By construction, we have $\text{supp}(y - y^P) \subseteq N$. Moreover, since $y \in B_\Omega^+(x, r)$ and hence $y \succeq x$, while $y_j^P = x_j$ for each $j \in N$, we have that $y - y^P \succeq \vec{0}$. Finally, it is clear from the

construction of y^P that $|y - y^P| \leq |x - y|$. We conclude that

$$\begin{aligned}
h(y) - h(y^P) &= \nabla h(z)^\top (y - y^P) \\
&\leq \nabla^+ h(z)^\top (y - y^P) && \text{(Since } y - y^P \succeq \vec{0}\text{)} \\
&= \sum_{i \in N} (\partial_i^+ h(z))(y_i - y_i^P) && \text{(Since } \text{supp}(y - y^P) \subseteq N\text{)} \\
&\leq \left(\sum_{i \in N} (\partial_i^+ h(z))^2 \right)^{1/2} |y - y^P| && \text{(Cauchy-Schwarz inequality)} \\
&\leq \left(\sum_{i \in N} \varepsilon^2 \right)^{1/2} |x - y| && \text{(Choice of } r_0 \text{ and last observation)} \\
&\leq \varepsilon \sqrt{d} |x - y|.
\end{aligned}$$

Therefore, when $P \neq \emptyset$ we have

$$\begin{aligned}
\lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^+(x, r)} \frac{(h(y) - h(x))^+}{|x - y|} &\leq \lim_{r \rightarrow 0^+} \sup_{y \in B_\Omega^+(x, r)} \frac{\left(h(y^P) - h(x) + \varepsilon \sqrt{d} |x - y| \right)^+}{|x - y|} \\
&\leq \varepsilon \sqrt{d} + \lim_{r \rightarrow 0^+} \sup_{y \in B'_\Omega(x, r) \cap X_P} \frac{(h(y) - h(x))^+}{|x - y|},
\end{aligned}$$

the second inequality since $|x - y^P| \leq |x - y|$ and since we showed that, for all $r < r_0$ and $y \in B_\Omega^+(x, r)$, we have $y^P \in (B'_\Omega(x, r) \cap X_P) \cup \{x\}$, and moreover $(h(y^P) - h(x))^+ = 0$ when $y^P = x$. Similarly, when $P = \emptyset$ we have $y^P = x$ always, so in this case we obtain the upper bound $\varepsilon \sqrt{d}$, i.e. the last limit superior may indeed be replaced by 0. Since this inequality holds for every $\varepsilon > 0$, (7.40) follows, and this concludes the proof. $\square$

7.4.2 Perturbation argument

We can now obtain the main result, first for $C^1([0, 1]^d)$ functions and then for all of $H^1((0, 1)^d)$ by an approximation argument. The key idea of the perturbation argument is to use the same (mean zero) function h in two different roles: 1) to construct absolutely continuous probability measures from h and its coordinate-wise monotone equilibrium, namely $d\mu = (1 + th)dx$ and $d\mu^* = (1 + th^*)dx$, for small $t > 0$; and 2) in the “test function” $-th$ for weak duality on $W_2^2(\mu \rightarrow \mu^*)$ via the directed Hamilton-Jacobi operator.

Theorem 7.133. *There exists a universal constant $C > 0$ such that the following holds. Define $\Omega := (0, 1)^d$, and let $h \in C^1(\overline{\Omega})$. Then*

$$\text{dist}_2^{\text{mono}}(h)^2 \leq \int_{\Omega} (h - h^*)^2 dx \leq C \int_{\Omega} |\nabla^- h|^2 dx,$$

where h^* is the coordinate-wise monotone equilibrium of h .

Proof. Note that h^* is well-defined because C^1 functions are Lipschitz on compact sets. We first show that we may assume without loss of generality that h is bounded and has mean zero. For any $\alpha > 0$ and $\beta \in \mathbb{R}$, we have that $\alpha g + \beta$ is monotone if and only if g is monotone, so it follows that

$$\text{dist}_2^{\text{mono}}(\alpha h + \beta)^2 = \alpha^2 \text{dist}_2^{\text{mono}}(h)^2.$$

Moreover, [Observation 7.124](#) gives that $(\alpha h + \beta)^* = \alpha h^* + \beta$, and clearly $|\nabla^-(\alpha h + \beta)|^2 = \alpha^2 |\nabla^- h|^2$ pointwise. Thus as long as we show the result for h satisfying $\int_{\Omega} h dx = 0$ and (say) $-0.1 \leq h \leq 0.1$, then we may write any other function in $C^1(\overline{\Omega})$ as $\alpha h + \beta$ for some h satisfying these conditions, and conclude the result. Hence assume that $\int_{\Omega} h dx = 0$ and that $-0.1 \leq h \leq 0.1$ pointwise.

Let $t \in (0, 1)$, and define $f \in C^1(\overline{\Omega})$ by $f := 1 + th$. Note that f is Lipschitz and bounded between 0.9 and 1.1, and it satisfies $\int_{\Omega} f dx = 1$. Define the absolutely continuous probability measures $d\mu := f dx$ and $d\mu^* := f^* dx$. [Theorem 7.4](#) then implies that, for some universal constant $C > 0$,

$$W_2^2(\mu \rightarrow \mu^*) \leq C \int_{\Omega} |\nabla^- f|^2 dx = Ct^2 \int_{\Omega} |\nabla^- h|^2 dx = Ct^2 \int_{\Omega} |\nabla^+(-h)|^2 dx,$$

the first equality by the definition of f and the fact that $t > 0$. On the other hand, noting that $-th \in C^1(\overline{\Omega})$ and hence $-th \in C_b(\Omega)$ as well, we have

$$\begin{aligned} \frac{1}{2} W_2^2(\mu \rightarrow \mu^*) &\geq \int_{\Omega} (-th) d\mu^* - \int_{\Omega} (\vec{H}_1(-th)) d\mu && \text{(Proposition 7.129)} \\ &= -t \int_{\Omega} h(1 + th)^* dx - \int_{\Omega} (\vec{H}_1(-th))(1 + th) dx && \text{(Definition of } \mu, \mu^*). \end{aligned}$$

Recalling that $\int_{\Omega} h dx = 0$ and that $(1 + th)^* = 1 + th^*$ by [Observation 7.124](#), the first term in the last line above is

$$-t \int_{\Omega} h(1 + th)^* dx = -t \int_{\Omega} h(1 + th^*) dx = -t^2 \int_{\Omega} hh^* dx.$$

Recalling that $1 + th$ is bounded between 0.9 and 1.1, the second term is

$$\begin{aligned}
& - \int_{\Omega} (\vec{H}_1(-th))(1 + th) \, dx \\
& = - \int_{\Omega} (t\vec{H}_t(-h))(1 + th) \, dx \quad (\text{Lemma 7.130}) \\
& = -t^2 \int_{\Omega} \left(\frac{\vec{H}_t(-h) - (-h) - h}{t} \right) (1 + th) \, dx \\
& = -t^2 \int_{\Omega} \underbrace{\left(\frac{\vec{H}_t(-h) - (-h)}{t} \right)}_{\geq 0 \text{ by Proposition 7.131(c)}} \underbrace{(1 + th)}_{\leq 2} \, dx + t \int_{\Omega} h(1 + th) \, dx \\
& \geq -2t^2 \int_{\Omega} \frac{\vec{H}_t(-h) - (-h)}{t} \, dx + t \underbrace{\int_{\Omega} h \, dx}_{=0} + t^2 \int_{\Omega} h^2 \, dx \\
& = -2t^2 \int_{\Omega} \frac{\vec{H}_t(-h) - (-h)}{t} \, dx + t^2 \int_{\Omega} h^2 \, dx.
\end{aligned}$$

Putting all of the above together, we conclude that

$$\frac{C}{2} t^2 \int_{\Omega} |\nabla^+(-h)|^2 \, dx \geq \frac{1}{2} W_2^2(\mu \rightarrow \mu^*) \geq -t^2 \int_{\Omega} h h^* \, dx - 2t^2 \int_{\Omega} \frac{\vec{H}_t(-h) - (-h)}{t} \, dx + t^2 \int_{\Omega} h^2 \, dx$$

and hence, since $t > 0$,

$$\int_{\Omega} h^2 \, dx - \int_{\Omega} h h^* \, dx \leq \frac{C}{2} \int_{\Omega} |\nabla^+(-h)|^2 \, dx + 2 \int_{\Omega} \frac{\vec{H}_t(-h) - (-h)}{t} \, dx.$$

Since this holds for all sufficiently small $t > 0$, we may pass the inequality to the limit superior as $t \rightarrow 0^+$. Since $\frac{\vec{H}_t(-h) - (-h)}{t}$ is uniformly bounded by Proposition 7.131(c), we may apply the reverse Fatou lemma and Proposition 7.131(b) to obtain

$$\begin{aligned}
\int_{\Omega} h^2 \, dx - \int_{\Omega} h h^* \, dx & \leq \frac{C}{2} \int_{\Omega} |\nabla^+(-h)|^2 \, dx + 2 \limsup_{t \rightarrow 0^+} \int_{\Omega} \frac{\vec{H}_t(-h) - (-h)}{t} \, dx \\
& \leq \frac{C}{2} \int_{\Omega} |\nabla^+(-h)|^2 \, dx + 2 \int_{\Omega} \limsup_{t \rightarrow 0^+} \frac{(\vec{H}_t(-h))(x) - (-h)(x)}{t} \, dx \\
& \leq \frac{C}{2} \int_{\Omega} |\nabla^+(-h)|^2 \, dx + \int_{\Omega} |\nabla^+(-h)(x)|^2 \, dx = \left(1 + \frac{C}{2}\right) \int_{\Omega} |\nabla^-h|^2 \, dx.
\end{aligned}$$

By [Observation 7.125](#), we have that $\int_{\Omega} (h^*)^2 dx \leq \int_{\Omega} h^2 dx$ and hence, since h^* is monotone,

$$\begin{aligned} \frac{1}{2} \text{dist}_2^{\text{mono}}(h)^2 &\leq \frac{1}{2} \int_{\Omega} (h - h^*)^2 dx = \frac{1}{2} \int_{\Omega} h^2 dx + \frac{1}{2} \int_{\Omega} (h^*)^2 dx - \int_{\Omega} h h^* dx \\ &\leq \int_{\Omega} h^2 dx - \int_{\Omega} h h^* dx \leq \left(1 + \frac{C}{2}\right) \int_{\Omega} |\nabla^- h|^2 dx. \end{aligned} \quad \square$$

The following denseness result is an immediate application of [\[Maz11, Theorem 1, p. 10\]](#).

Fact 7.134. *Define $\Omega := (0, 1)^d$. The space $C^\infty(\overline{\Omega})$ is dense in $H^1(\Omega)$.*

Thus we may promote the main result to all of $H^1(\Omega)$ by an approximation argument, which yields the main result.

Theorem 7.1 (Directed Poincaré inequality). *There exists a universal constant $C > 0$ such that, for all $f \in H^1((0, 1)^d)$,*

$$\text{dist}_2^{\text{mono}}(f)^2 \leq C \mathbb{E} [\|\nabla^- f\|_2^2]. \quad (7.1)$$

Proof. By [Fact 7.134](#), we may find a sequence $(f_n)_{n \in \mathbb{N}} \subset C^\infty(\overline{\Omega})$ such that $f_n \rightarrow f$ in $H^1(\Omega)$. Since each f_n belongs in particular to $C^1(\overline{\Omega})$, [Theorem 7.133](#) gives that, for each $n \in \mathbb{N}$,

$$\|f_n - f_n^*\|_{L^2(\Omega)}^2 \leq C \int_{\Omega} |\nabla^- f_n|^2 dx.$$

Let $\varepsilon > 0$. Since $f_n \rightarrow f$ in $H^1(\Omega)$ implies in particular that $f_n \rightarrow f$ in $L^2(\Omega)$, we have that, for all sufficiently large n , $\|f - f_n\|_{L^2(\Omega)} \leq \varepsilon$ and hence, by the triangle inequality,

$$\text{dist}_2^{\text{mono}}(f) \leq \|f - f_n^*\|_{L^2(\Omega)} \leq \|f - f_n\|_{L^2(\Omega)} + \|f_n - f_n^*\|_{L^2(\Omega)} \leq \varepsilon + \sqrt{C \int_{\Omega} |\nabla^- f_n|^2 dx}. \quad (7.41)$$

Now, since the norm in $H^1(\Omega)$ is given by

$$\|u\|_{H^1(\Omega)}^2 = \|u\|_{L^2(\Omega)}^2 + \sum_{i=1}^d \int_{\Omega} |\partial_i u|^2 dx,$$

the fact that $f_n \rightarrow f$ in $H^1(\Omega)$ implies that $\|f - f_n\|_{H^1(\Omega)} \rightarrow 0$ and hence

$$0 = \lim_{n \rightarrow \infty} \sum_{i=1}^d \int_{\Omega} |\partial_i (f - f_n)|^2 dx = \lim_{n \rightarrow \infty} \sum_{i=1}^d \int_{\Omega} |(\partial_i f) - (\partial_i f_n)|^2 dx.$$

Since $0 \leq |(a \wedge 0) - (b \wedge 0)| \leq |a - b|$ for any $a, b \in \mathbb{R}$, we conclude that

$$\lim_{n \rightarrow \infty} \sum_{i=1}^d \int_{\Omega} |(\partial_i^- f) - (\partial_i^- f_n)|^2 dx = 0. \quad (7.42)$$

Observing that $\partial_i^- f, \partial_i^- f_n \in L^2(\Omega)$ for each $i \in [d]$ and $n \in \mathbb{N}$ (because $\partial_i f, \partial_i f_n \in L^2(\Omega)$ by the definition of $H^1(\Omega)$, and if $u \in L^2(\Omega)$ then $u \wedge 0 \in L^2(\Omega)$), (7.42) implies that $\partial_i^- f_n \rightarrow \partial_i^- f$ in $L^2(\Omega)$ for each $i \in [d]$, and hence

$$\int_{\Omega} |\nabla^- f|^2 dx = \sum_{i=1}^d \int_{\Omega} |\partial_i^- f|^2 dx = \lim_{n \rightarrow \infty} \sum_{i=1}^d \int_{\Omega} |\partial_i^- f_n|^2 dx = \lim_{n \rightarrow \infty} \int_{\Omega} |\nabla^- f_n|^2 dx.$$

Thus, for all sufficiently large n we have

$$\int_{\Omega} |\nabla^- f_n|^2 dx \leq \varepsilon + \int_{\Omega} |\nabla^- f|^2 dx.$$

Combining with (7.41) we obtain, for all sufficiently large n ,

$$\begin{aligned} \text{dist}_2^{\text{mono}}(f)^2 &\leq \left(\varepsilon + \sqrt{C \int_{\Omega} |\nabla^- f_n|^2 dx} \right)^2 \leq 2\varepsilon^2 + 2C \int_{\Omega} |\nabla^- f_n|^2 dx \\ &\leq 2\varepsilon^2 + 2C \left(\varepsilon + \int_{\Omega} |\nabla^- f|^2 dx \right) = 2\varepsilon^2 + 2C\varepsilon + 2C \int_{\Omega} |\nabla^- f|^2 dx. \end{aligned}$$

Since this holds for all $\varepsilon > 0$, we conclude that

$$\text{dist}_2^{\text{mono}}(f)^2 \leq 2C \int_{\Omega} |\nabla^- f|^2 dx.$$

□

Chapter 8

Conclusion

In this thesis, we initiated the study of property testing in the analytic setting, where the input is a continuous real-valued function and the goal is to test with respect to the L^p distance. This research direction builds upon the current state of the art in property testing in two ways: it extends the applicability of that paradigm to settings that are inherently continuous in nature, and it reveals new conceptual and technical facets of long-studied problems, hence contributing to a more comprehensive understanding of the mathematical underpinnings of approximate decision making.

We took the first steps toward a theory of analytic property testing by focusing on the fundamental problem of monotonicity testing, and in particular its connection to the phenomenon of directed isoperimetry. We gave a systematic account of directed isoperimetry in continuous and discrete domains as captured by (L^p, ℓ^q) -Poincaré inequalities, and showed that these inequalities imply efficient monotonicity testers for real-valued functions $f : [0, 1]^d \rightarrow \mathbb{R}$ defined on the solid unit cube. While our initial study in [Chapter 4](#), where we obtained an $O(d)$ query tester, leveraged the intuition from the classic edge tester [\[GGLRS00\]](#) by defining and analyzing the appropriate continuous analogues of the discrete arguments (namely via monotone rearrangements), our improved $\tilde{O}(\sqrt{d})$ query tester relies on a new perspective on directed isoperimetry: its relationship with the convergence properties of the directed heat equation. We showed that the connection between isoperimetry and the heat equation, which dates back to the work of Poincaré in mathematical physics [\[Poi90\]](#), enjoys a natural directed analogue whose convergence properties we studied using the theories of nonlinear partial differential equations and optimal transport.

Naturally, there is no lack of open questions and exciting avenues for future research, as we now outline. We divide our discussion in two parts: questions about monotonicity

testing, and proposed directions for analytic property testing.

8.1 Monotonicity testing

Within monotonicity testing, the first open question that emerges from this thesis is that of lower bounds for L^p monotonicity testing of Lipschitz functions. This thesis gave two types of lower bounds:

1. Strong lower bounds for specific classes of testers. In [Chapters 4 and 5](#), we showed that the query complexity upper bounds given in [Theorems 1.2 and 1.5](#) are essentially optimal for the type of tester given by those theorems, i.e. partial derivative testers and directional derivative testers. In particular, the $\tilde{O}(\sqrt{d})$ bound from [Theorem 1.5](#) cannot be improved by any *pair tester*, i.e. an algorithm that repeatedly samples points $x \preceq y$ and rejects if $f(x) > f(y)$, or any tester using the directional derivative analogue of such a pair test. In the discrete regimes of the Boolean cube and hypergrid, the best known monotonicity testers are indeed pair testers [[KMS18](#); [BCS23a](#)].
2. Lower bounds for constant dimension d . In [Chapter 4](#), we gave lower bounds showing that, for each constant d , at least $\Omega\left((M/\varepsilon)^{\frac{d}{d+1}}\right)$ queries are required to test monotonicity of M -Lipschitz functions on $[0, 1]^d$, and we gave similar results for hypergrid domains. These bounds show that the dependence of our upper bounds on M/ε is close to correct. Interestingly, the exponent $\frac{d}{d+1}$ in the lower bound cannot be improved in general, because in [Chapter 4](#) we also gave a nearly-matching upper bound for the special case of the line ([Theorem 4.4](#)).

The immediate open problem, then, is to give general lower bounds in this model with good dependence on the dimension d . We believe that the $\sqrt{d}$ dependence from [Theorem 1.5](#) is optimal.

Problem 8.1. *Give a $d^{\Omega(1)}$ lower bound for L^p monotonicity testing of 1-Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$. As an easier problem, can we show such a bound for nonadaptive testers?*

We may also ask about *tolerant* testing [[PRR06](#)]. In tolerant testing, we require the tester to accept not only when the input f is monotone, but also when it is *close* to a

monotone function. A closely related formulation refers to *distance approximation*: given an input function f , and approximation parameters $\alpha > 0$ (corresponding to additive error) and $\nu > 1$ (corresponding to multiplicative error), a (ν, α) -approximation algorithm for monotonicity (with respect to some fixed distance metric dist) is an algorithm whose output $\widehat{\varepsilon}$ satisfies, with probability at least $2/3$,

$$\text{dist}^{\text{mono}}(f) - \alpha \leq \widehat{\varepsilon} \leq \nu \cdot \text{dist}^{\text{mono}}(f).$$

Early works on monotonicity testing gave or implied $(O(d), \alpha)$ -approximation algorithms with $\text{poly}(d, 1/\alpha)$ query complexity for Boolean functions on hypergrid domains, as well as similar bounds with a dependence on $\log n$ for real-valued functions on the hypergrid domains [DGLRRS99; GGLRS00; FR10]. In more recent progress, [PRW22] extended the work of [KMS18] to obtain an $(\widetilde{O}(\sqrt{d}), \alpha)$ -approximation algorithm with $\text{poly}(d, 1/\alpha)$ query complexity, and they showed that the $\widetilde{O}(\sqrt{d})$ multiplicative factor is essentially optimal. In the L^1 setting, [BRY14a] gave distance approximation algorithms with a purely additive error guarantee for the cases of the discrete line and the 2-dimensional grid.

For the special case of Boolean functions $f : \{0, 1\}^d \rightarrow \{0, 1\}$, a simple variant of the edge tester of [GGLRS00] suffices to obtain an $(O(d), \alpha)$ -approximation: instead of rejecting whenever a violating edge is observed, compute $\widehat{\varepsilon}$ by estimating the fraction of violating edges in the hypercube. This works thanks to a reverse directed Poincaré inequality which, together with Theorem 1.1, implies that the (relative Hamming) distance $\text{dist}_H^{\text{mono}}(f)$ of f to monotonicity and its number $V(f)$ of violating edges are related by the inequalities $\text{dist}_H^{\text{mono}}(f) \cdot 2^d \leq V(f) \leq \text{dist}_H^{\text{mono}}(f) \cdot d \cdot 2^d$. Hence a constant-factor approximation to $V(f)$ yields an $O(d)$ -approximation to $\text{dist}_H^{\text{mono}}(f)$.

Indeed, since we also showed in Section 6.3 reverse (L^1, ℓ^1) -Poincaré inequalities for the hypergrid $[n]^d$ up to a factor of d (which is the maximum degree of the directed hypergrid graph), while our (L^1, ℓ^1) -Poincaré inequality from Theorem 4.2 takes a factor of n , a similar argument gives an $(O(nd), \alpha)$ -approximation algorithm for the L^1 distance to monotonicity of (ℓ^1, M) -Lipschitz functions $f : [n]^d \rightarrow \mathbb{R}$ using $\text{poly}(n, d, M, 1/\alpha)$ queries. However, we do not know how to implement this idea in the continuous case. Hence, we leave the following open question:

Question 8.2. *Is there a $\text{poly}(d, 1/\alpha)$ -query $(O(d), \alpha)$ -approximation algorithm for the L^p distance to monotonicity of 1-Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$? How about $(o(d), \alpha)$ -approximation?*

Finally, we may ask about the power of adaptivity for testing monotonicity in the analytic setting. Indeed, whether adaptivity helps with testing monotonicity of Boolean

functions is a notable open question in monotonicity testing – the best known tester of [KMS18] is nonadaptive, and indeed achieves optimal query complexity $\tilde{O}(\sqrt{d}/\varepsilon^2)$ for non-adaptive testers, while the best lower bound for adaptive testers is $\tilde{\Omega}(d^{1/3})$ [CWX17]. On the other hand, it is known that adaptivity does not help with testing monotonicity of functions $f : [n]^d \rightarrow \mathbb{R}$ with respect to the relative Hamming distance [CS14]. In the analytic setting, we ask the following question, whose resolution would most likely also require progress on Problem 8.1.

Question 8.3. *Does adaptivity help with L^p monotonicity testing of M -Lipschitz functions $f : [0, 1]^d \rightarrow \mathbb{R}$?*

8.2 Further directions

Expanding our scope from monotonicity testing to the broader landscape of property testing in the analytic setting, there are many interesting directions for future exploration. Many of the classic properties which have been widely studied by the property testing community over the past decades also have natural formulations in the analytic setting, and here we list a few of them. Specifically, we ask about query-efficient property testers for the following properties in the analytic setting. We believe that some of these properties may be especially interesting to study in the tolerant regime.

Convexity. Two natural settings for testing convexity of functions are the unit cube setting $f : [0, 1]^d \rightarrow \mathbb{R}$ under the uniform measure, and the setting of functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ under the Gaussian measure—in particular, the rotation-invariance of the Gaussian measure seems to be a natural fit for the rotation-invariance of the convexity property. We remark that, in the discrete setting of functions $f : [n]^d \rightarrow \mathbb{R}$, there is an exponential $\Omega((n/d)^{d/2})$ nonadaptive lower bound for testing convexity with respect to the relative Hamming distance [BBB20].

Unateness. A function $f : [n]^d \rightarrow \mathbb{R}$ is unate if it is either monotone or anti-monotone along each direction $i \in [d]$. In the case of Boolean or real-valued functions over the hypergrid $[n]^d$ or Boolean cube $\{0, 1\}^d$, it is known that the query complexity of unateness testing (with respect to the relative Hamming distance) is polynomial in d , $1/\varepsilon$, and $\log n$ [CWX17; CW19; BCPRS20].

Juntas. A function $f : \{0, 1\}^d \rightarrow \{0, 1\}$ on n variables is a k -junta if it depends on at most k input variables. The query complexity of k -junta testing is $\Theta(k \log k)$ for constant proximity parameter ε [Bla09; Sag18]. In another relevant line of work, [DMN21] defined a rotation-invariant notion of k -juntas, namely *linear k -juntas*, for the case of Boolean functions $f : \mathbb{R}^d \rightarrow \{\pm 1\}$ over Euclidean space endowed with the Gaussian measure. They showed that the class of linear k -juntas is not testable without further assumptions, but it becomes testable if one assumes that f satisfies a “bounded surface area” assumption. Here, we ask about testing k -juntas (or linear k -juntas) of *real-valued* $f : [0, 1]^d \rightarrow \mathbb{R}$ (or $f : \mathbb{R}^d \rightarrow \mathbb{R}$ under the Gaussian measure), under a Lipschitz regularity assumption.

The dynamical approach. Finally, on a more conceptual note, we may ask about further applications of the dynamical approach embodied by our proof of Theorem 1.4 in Chapter 7. In property testing, the idea of comparing the input object f to some “ideal” object f^* that satisfies the property of interest is very natural, and we showed that techniques from partial differential equations and optimal transport can be useful tools for reasoning about this comparison smoothly in time. Therefore, it is plausible that these tools may have something to say about other property testing problems of continuous nature. In particular, we could attempt to apply this perspective for testing geometric properties of sets in $\mathbb{R}^d$, e.g. convexity. Testing convexity of sets $S \subset \mathbb{R}^d$ with respect to the Gaussian measure is a notoriously challenging problem: the best known query complexity upper bound complexity is $2^{\tilde{O}(\sqrt{d}) \cdot \text{poly}(1/\varepsilon)}$ [CFSS17], whereas the best known lower bounds are $\text{poly}(d)$ bounds for either nonadaptive testers, or testers with one-sided error [CDNSW25]. Is there a natural dynamical process by which an initial set $S \subset \mathbb{R}^d$ converges over time to a convex set, and could the properties of such a process inspire new testers?

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APPENDICES

Appendix A

Nonadaptive lower bound from prior work on Hamming testing

We briefly explain how the $\Omega(d \log n)$ nonadaptive lower bound of [BRY14b] for Hamming testing monotonicity of functions $f : [n]^d \rightarrow \mathbb{R}$ (with sufficiently small constant ε) also applies to L^1 -testing of $(\ell^1, O(1))$ -Lipschitz functions. The construction of [BRY14b] relies on two main ingredients: *step functions* and *Walsh functions*.

Let $n = 2^\ell$ for simplicity. For each $i \in \{0, \dots, \ell\}$ the i -th step function $s_i : [2^\ell] \rightarrow [2^{\ell-i}]$ is given by

$$s_i(x) = \left\lfloor \frac{x-1}{2^i} \right\rfloor + 1.$$

In words, $s_i(x)$ increases by 1 after every 2^i consecutive elements (called a *block* of size 2^i).

The Walsh functions are defined as follows. For each $i \in [\ell]$, the function $w_i : [2^\ell] \rightarrow \{\pm 1\}$ is given by

$$w_i(x) = (-1)^{\text{bit}_i(x-1)},$$

where the operator bit_i extracts the i -th bit of its input (indexed from least to most significant). Then for each $S \subseteq [\ell]$, the function $w_S : [2^\ell] \rightarrow \{\pm 1\}$ is given by

$$w_S(x) = \prod_{i \in S} w_i(x).$$

These two types of functions are defined on the line, and they are extended into a multidimensional construction on the hypergrid as follows. Given a vector $\mathbf{i} \in [\ell]^d$, the

step function $s_{\mathbf{i}} : [2^\ell]^d \rightarrow [d2^\ell]$ is given by

$$s_{\mathbf{i}}(x_1, \dots, x_d) = \sum_{j=1}^d s_{i_j}(x_j),$$

and given a vector $\mathbf{S} = (\mathbf{S}_1, \dots, \mathbf{S}_d)$ of subsets of $[\ell]$, the Walsh function $w_{\mathbf{S}} : [2^\ell]^d \rightarrow \{\pm 1\}$ is given by

$$w_{\mathbf{S}}(x_1, \dots, x_d) = \prod_{j=1}^d w_{\mathbf{S}_j}(x_j).$$

Then, [BRY14b] use a communication complexity argument (namely a reduction from the AUGMENTINDEX problem) to show that (nonadaptive) Hamming testing monotonicity of functions $h_{\mathbf{i}, \mathbf{S}} : [2^\ell]^d \rightarrow \mathbb{N}$ of the form

$$h_{\mathbf{i}, \mathbf{S}}(x) = 2s_{\mathbf{i}}(x) + w_{\mathbf{S}}(x),$$

for appropriate choices of $\mathbf{i}$ and $\mathbf{S}$, requires at least $\Omega(d \log n)$ queries. Therefore, to show that (nonadaptive) L^1 testing monotonicity of $(\ell^1, O(1))$ -Lipschitz functions also requires at least this number of queries, it suffices to show that every such function $h = h_{\mathbf{i}, \mathbf{S}}$ satisfies

1. h is $(\ell^1, O(1))$ -Lipschitz; and
2. If $\text{dist}_H^{\text{mono}}(h) \geq \varepsilon$, then $\text{dist}_1^{\text{mono}}(h) \gtrsim \varepsilon$.

The first property follows from the definitions of the step and Walsh functions: let $x, y \in [2^\ell]^d$ be such that $\|x - y\|_1 = 1$. Then let $j \in [d]$ be the coordinate such that $|x_j - y_j| = 1$ and $x_k = y_k$ for $k \neq j$. Then

$$|h_{\mathbf{i}, \mathbf{S}}(x) - h_{\mathbf{i}, \mathbf{S}}(y)| \leq 2|s_{i_j}(x_j) - s_{i_j}(y_j)| + |w_{\mathbf{S}}(x) - w_{\mathbf{S}}(y)| \leq 2 + 2 = 4,$$

the first inequality because x and y agree on every coordinate except for j , and the second inequality because the step function s_{i_j} changes by at most 1 on adjacent inputs, and the Walsh functions only take values ± 1 . Thus the first property holds.

As for the second property, note that if $\text{dist}_H^{\text{mono}}(h) \geq \varepsilon$, then there exists a matching of the form $(x^i, y^i)_i$ where for each i we have $x^i \preceq y^i$ and $h(x^i) > h(y^i)$ (i.e. x^i, y^i form a violating pair), such that at least an ε -fraction of the points $[n]^d$ belong to this matching (see [FLNRRS02]). Now, for each such violating pair x^i, y^i , it follows that $h(x^i) - h(y^i) \geq 1$, since h is integer-valued. Therefore for any monotone function $h' : [n]^d \rightarrow \mathbb{R}$, it must be the case that $|h(x^i) - h'(x^i)| + |h(y^i) - h'(y^i)| \geq 1$. Since this is true for disjoint pairs x^i, y^i covering an ε -fraction of the domain, it follows that $\mathbb{E}[|h - h'|] \geq \varepsilon/2$ for any monotone function h' . Hence $\text{dist}_1^{\text{mono}}(h) \geq \varepsilon/2$, and we are done.

Appendix B

Missing proofs from Chapter 7

We require the following standard result from convex optimization.

Lemma B.1 (See e.g. [Bre11, Corollary 3.23]). *Let E be a reflexive Banach space and let $C \subseteq E$ be nonempty, closed and convex. Suppose $f : C \rightarrow (-\infty, +\infty]$ is convex, proper, lower semicontinuous, and coercive in the sense that for every sequence $(x_n)_{n \in \mathbb{N}}$ in C with $\|x_n\|_E \rightarrow \infty$, we have $f(x_n) \rightarrow +\infty$. Then there exists $x^* \in C$ satisfying $f(x^*) = \inf_{x \in C} f(x)$.*

Lemma B.2. *Define $\Omega := (0, 1)^d$. Suppose $f_n \rightharpoonup f$ weakly in $L^2(\Omega)$, and each f_n is monotone nondecreasing (resp. monotone nonincreasing). Then f is monotone nondecreasing (resp. monotone nonincreasing).*

Proof. Recall that for each $\varepsilon > 0$ and $g \in L^2(\Omega)$, $g^\varepsilon = \eta_\varepsilon * g$ is the mollification of g , defined on $\Omega_\varepsilon = (\varepsilon, 1 - \varepsilon)^d$.

Let $\varepsilon \in (0, 1/3)$. We claim that $f_n^\varepsilon \rightarrow f^\varepsilon$ pointwise in Ω_ε . Indeed, fix any $x \in \Omega_\varepsilon$ and define $\eta_{\varepsilon, x} \in C^\infty(\mathbb{R}^d)$ by $\eta_{\varepsilon, x}(y) := \eta_\varepsilon(x - y)$. Since $f_n \rightharpoonup f$ weakly in $L^2(\Omega)$, we obtain

$$f^\varepsilon(x) = \int_{\Omega} \eta_\varepsilon(x-y) f(y) dy = \langle \eta_{\varepsilon, x}, f \rangle = \lim_{n \rightarrow \infty} \langle \eta_{\varepsilon, x}, f_n \rangle = \lim_{n \rightarrow \infty} \int_{\Omega} \eta_\varepsilon(x-y) f_n(y) dy = \lim_{n \rightarrow \infty} f_n^\varepsilon(x).$$

Now, suppose without loss of generality that each f_n is monotone nondecreasing. Let $\varepsilon \in (0, 1/3)$. It is immediate that each f_n^ε is also monotone nondecreasing (in Ω_ε). Since $f_n^\varepsilon \rightarrow f^\varepsilon$ pointwise in Ω_ε , we conclude that f^ε is monotone nondecreasing as well.

The conclusion follows the fact that $f^\varepsilon \rightarrow f$ almost everywhere as $\varepsilon \rightarrow 0$. $\square$

Proposition 7.10 (Existence and uniqueness of optimal representation). *For all $u \in \mathcal{U}$, $\mathcal{R}^*(u)$ contains exactly one element.*

Proof of existence. Let $u \in \mathcal{U}$. Let $E := L^2(I) \oplus H^1(I)$, where $\oplus$ denotes the direct sum of Hilbert spaces, so that E is also a Hilbert space (and hence a reflexive Banach space). Note that $\mathcal{R}(u) \subseteq E$ by definition. Our goal is to show that $C = \mathcal{R}(u)$ and $f = \mathcal{D}^-$ satisfy the conditions of [Lemma B.1](#).

It is clear that $\mathcal{R}(u)$ is nonempty, because $u \in \mathcal{U}$. It is immediate to verify that $\mathcal{R}(u)$ is convex as well. We now check that $\mathcal{R}(u)$ is closed in E . Let $((u_n^{(1)}, u_n^{(2)}))_{n \in \mathbb{N}}$ be any sequence in $\mathcal{R}(u)$ that converges in E to some $(u^{(1)}, u^{(2)}) \in E$; we would like to show that $(u^{(1)}, u^{(2)}) \in \mathcal{R}(u)$. First, since $(u_n^{(1)}, u_n^{(2)}) \rightarrow (u^{(1)}, u^{(2)})$ in E , we have $u_n^{(1)} \rightarrow u^{(1)}$ in $L^2(I)$ and $u_n^{(2)} \rightarrow u^{(2)}$ in $H^1(I)$ and hence in $L^2(I)$, so that

$$\begin{aligned} & \|u - (u^{(1)} + u^{(2)})\|_{L^2(I)} \\ &= \lim_{n \rightarrow \infty} \|(u - (u_n^{(1)} + u_n^{(2)})) + (u_n^{(1)} - u^{(1)}) + (u_n^{(2)} - u^{(2)})\|_{L^2(I)} \\ &\leq \lim_{n \rightarrow \infty} \|u - (u_n^{(1)} + u_n^{(2)})\|_{L^2(I)} + \lim_{n \rightarrow \infty} \|u_n^{(1)} - u^{(1)}\|_{L^2(I)} + \lim_{n \rightarrow \infty} \|u_n^{(2)} - u^{(2)}\|_{L^2(I)} \\ &= 0, \end{aligned}$$

where the last equality used the fact that $u = u_n^{(1)} + u_n^{(2)}$ a.e. for every n (since $(u_n^{(1)}, u_n^{(2)}) \in \mathcal{R}(u)$) and the two convergence observations above. Thus $u = u^{(1)} + u^{(2)}$ in $L^2(I)$ and hence almost everywhere. Moreover, by [Lemma B.2](#), $u^{(1)}$ is nondecreasing and $u^{(2)}$ is nonincreasing. Finally, since $\int_I u_n^{(2)} dx = 0$ for each n and $u_n^{(2)} \rightarrow u^{(2)}$ in $L^2(I)$, it follows that $\int_I u^{(2)} dx = 0$ as well. We conclude that $(u^{(1)}, u^{(2)}) \in \mathcal{R}(u)$, and thus $\mathcal{R}(u)$ is closed.

Now, clearly $\mathcal{D}^-$ is proper since it is finitely valued by definition. It is also continuous (and thus lower semicontinuous) because for any $(u_n^{(1)}, u_n^{(2)}) \rightarrow (u^{(1)}, u^{(2)})$ in E , we have $u_n^{(2)} \rightarrow u^{(2)}$ in $H^1(I)$ and hence $\int_I (\partial_x u_n^{(2)})^2 dx \rightarrow \int_I (\partial_x u^{(2)})^2 dx$, so $\mathcal{D}^-(u_n^{(1)}, u_n^{(2)}) \rightarrow \mathcal{D}^-(u^{(1)}, u^{(2)})$. It is also immediate to check that $\mathcal{D}^-$ is convex.

Finally, we need to show that $\mathcal{D}^-$ is coercive in the sense of [Lemma B.1](#). Let $((u_n^{(1)}, u_n^{(2)}))_{n \in \mathbb{N}}$ be a sequence in $\mathcal{R}(u)$ such that $\|(u_n^{(1)}, u_n^{(2)})\|_E^2 = \|u_n^{(1)}\|_{L^2(I)}^2 + \|u_n^{(2)}\|_{H^1(I)}^2 \rightarrow \infty$. We claim that $\|\partial_x u_n^{(2)}\|_{L^2(I)} \rightarrow \infty$. Suppose for a contradiction that this false. There are two cases. First, suppose $\|u_n^{(2)}\|_{H^1(I)} \rightarrow \infty$. Then necessarily $\|u_n^{(2)}\|_{L^2(I)} \rightarrow \infty$, but since each $u_n^{(2)}$ has mean zero, the Poincaré-Wirtinger inequality gives that

$$\|\partial_x u_n^{(2)}\|_{L^2(I)} \geq \frac{1}{C} \|u_n^{(2)}\|_{L^2(I)}$$

for some constant $C > 0$. Thus $\|\partial_x u_n^{(2)}\|_{L^2(I)} \rightarrow \infty$, a contradiction. In the second case, $\|u_n^{(2)}\|_{H^1(I)}$ remains bounded, so we must have $\|u_n^{(1)}\|_{L^2(I)} \rightarrow \infty$. But since $u = u_n^{(1)} + u_n^{(2)}$ in $L^2(I)$, the reverse triangle inequality implies that $\|u_n^{(2)}\|_{L^2(I)} \geq \|u_n^{(1)}\|_{L^2(I)} - \|u\|_{L^2(I)} \rightarrow \infty$, so $\|u_n^{(2)}\|_{L^2(I)} \rightarrow \infty$. Then again Poincaré-Wirtinger implies that $\|\partial_x u_n^{(2)}\|_{L^2(I)} \rightarrow \infty$, so the claim holds. But since $\mathcal{D}^-(u_n^{(1)}, u_n^{(2)}) = \frac{1}{2}\|\partial_x u_n^{(2)}\|_{L^2(I)}^2$, we have $\mathcal{D}^-(u_n^{(1)}, u_n^{(2)}) \rightarrow +\infty$, so $\mathcal{D}^-$ is coercive. Thus $\mathcal{D}^-$ achieves its minimum on $\mathcal{R}(u)$ by [Lemma B.1](#), so $\mathcal{R}^*(u)$ is nonempty. $\square$

Proof of uniqueness. We show uniqueness using strict convexity. Suppose $(u^{(1)}, u^{(2)}), (v^{(1)}, v^{(2)}) \in \mathcal{R}^*(u)$, and suppose for a contradiction that $(u^{(1)}, u^{(2)}) \neq (v^{(1)}, v^{(2)})$ in E (in the notation from the previous part of the proof; note that equality in E is equivalent to equality as tuples in $L^2(I) \times H^1(I)$).

We first claim that $\partial_x u^{(2)} \neq \partial_x v^{(2)}$ in $L^2(I)$. Indeed, suppose $\partial_x u^{(2)} = \partial_x v^{(2)}$ in $L^2(I)$. Note that $u^{(2)} \neq v^{(2)}$ in $L^2(I)$, because otherwise we would have $u^{(2)} = v^{(2)}$ in $H^1(I)$ and $u^{(1)} = u - u^{(2)} = u - v^{(2)} = v^{(1)}$ in $L^2(I)$, and thus $(u^{(1)}, u^{(2)}) = (v^{(1)}, v^{(2)})$ in E , a contradiction. Now, since $\partial_x u^{(2)} = \partial_x v^{(2)}$ and $u^{(2)} \neq v^{(2)}$ in $L^2(I)$, we conclude that $u^{(2)} = v^{(2)} + C$ in $L^2(I)$ for some constant $C \neq 0$. But this contradicts the fact that $\int_I u^{(2)} dx = \int_I v^{(2)} dx = 0$, which holds by the definition of $\mathcal{R}(u)$. Thus $\partial_x u^{(2)} \neq \partial_x v^{(2)}$ in $L^2(I)$.

Define the function $f : L^2(I) \rightarrow \mathbb{R}_{\geq 0}$ by $f(w) := \frac{1}{2}\|w\|_{L^2(I)}^2$, which is strictly convex, and note that $\mathcal{D}^-(r^{(1)}, r^{(2)}) = f(\partial_x r^{(2)})$ for all $r \in \mathcal{R}(u)$. Now, the element $(z^{(1)}, z^{(2)}) := \frac{1}{2}(u^{(1)}, u^{(2)}) + \frac{1}{2}(v^{(1)}, v^{(2)})$ is in $\mathcal{R}(u)$ by convexity of that set, and satisfies $\mathcal{D}^-(z^{(1)}, z^{(2)}) = f(\frac{1}{2}\partial_x u^{(2)} + \frac{1}{2}\partial_x v^{(2)}) < \frac{1}{2}f(\partial_x u^{(2)}) + \frac{1}{2}f(\partial_x v^{(2)}) = \frac{1}{2}\mathcal{D}^-(u^{(1)}, u^{(2)}) + \frac{1}{2}\mathcal{D}^-(v^{(1)}, v^{(2)})$, contradicting the fact that $(u^{(1)}, u^{(2)}), (v^{(1)}, v^{(2)}) \in \mathcal{R}^*(u)$. Thus $(u^{(1)}, u^{(2)}) = (v^{(1)}, v^{(2)})$ in E , as needed. $\square$

Proposition 7.13. *The functional $\mathcal{E}^- : L^2(I) \rightarrow [0, +\infty]$ is convex, proper and lower semicontinuous.*

Proof. Properness is trivial, since e.g. $0 \in \mathcal{U}$ and $\mathcal{E}^-(0) = 0$. Convexity also follows easily from the convexity of $\mathcal{U}$ and $\mathcal{D}^-$.

It remains to show lower semicontinuity, and since $L^2(I)$ is a metric space, it suffices to show sequential lower semicontinuity. Let $u \in L^2(I)$, and let $(u_n)_{n \in \mathbb{N}}$ be a sequence in $L^2(I)$ converging to u in $L^2(I)$. We must show that

$$\mathcal{E}^-(u) \stackrel{?}{\leq} \liminf_{n \rightarrow \infty} \mathcal{E}^-(u_n). \quad (\text{B.1})$$

The only relevant case is when the RHS above is finite, so suppose there exists a subsequence $(u_{n_k})_{k \in \mathbb{N}}$ such that $\lim_{k \rightarrow \infty} \mathcal{E}^-(u_{n_k}) = A < +\infty$. By extracting a subsequence if necessary, we may assume that $\mathcal{E}^-(u_{n_k}) < +\infty$, and thus $u_{n_k} \in \mathcal{U}$, for every k .

We claim that $((u_{n_k} \uparrow, u_{n_k} \downarrow))_k$ is bounded as a sequence in $L^2(I) \oplus H^1(I)$. First, since $\lim_{k \rightarrow \infty} \mathcal{E}^-(u_{n_k}) = A$, we have that $(\int_I (\partial_x u_{n_k} \downarrow)^2 dx)_k$ is bounded. Then, since every $u_{n_k} \in \mathcal{U}$ and hence $\int_I u_{n_k} \downarrow dx = 0$, the Poincaré-Wirtinger inequality implies that

$$\int_I (u_{n_k} \downarrow)^2 dx \leq C \int_I (\partial_x u_{n_k} \downarrow)^2 dx$$

for some $C > 0$, and hence $(u_{n_k} \downarrow)_k$ is bounded in $H^1(I)$. Now, since $u_{n_k} \rightarrow u$ in $L^2(I)$, we have that $(\|u_{n_k}\|_{L^2(I)})_k$ is bounded, and since $(\|u_{n_k} \downarrow\|_{L^2(I)})_k$ is bounded as a consequence of boundedness in $H^1(I)$, it follows that $(\|u_{n_k} \uparrow\|_{L^2(I)})_k = (\|u_{n_k} - u_{n_k} \downarrow\|_{L^2(I)})_k$ is also bounded, that is, $(u_{n_k} \uparrow)_k$ is bounded in $L^2(I)$. Hence $((u_{n_k} \uparrow, u_{n_k} \downarrow))_k$ is bounded in $L^2(I) \oplus H^1(I)$ as claimed.

Since $L^2(I) \oplus H^1(I)$ is a Hilbert space, we conclude that $((u_{n_k} \uparrow, u_{n_k} \downarrow))_k$ has a weakly convergent subsequence, which we denote by $((u_{n_{k_m}} \uparrow, u_{n_{k_m}} \downarrow))_m$. Then there exist $v^{(1)} \in L^2(I)$ and $v^{(2)} \in H^1(I)$ such that $u_{n_{k_m}} \uparrow \rightharpoonup v^{(1)}$ weakly in $L^2(I)$ and $u_{n_{k_m}} \downarrow \rightharpoonup v^{(2)}$ weakly in $H^1(I)$. So letting $v := v^{(1)} + v^{(2)}$, we conclude that $u_{n_{k_m}} \rightharpoonup v$ weakly in $L^2(I)$; indeed, for every $f \in L^2(I)$, we have

$$\langle u_{n_{k_m}}, f \rangle = \langle u_{n_{k_m}} \uparrow + u_{n_{k_m}} \downarrow, f \rangle = \langle u_{n_{k_m}} \uparrow, f \rangle + \langle u_{n_{k_m}} \downarrow, f \rangle \rightarrow \langle v^{(1)}, f \rangle + \langle v^{(2)}, f \rangle = \langle v, f \rangle.$$

We now claim that $v \in \mathcal{U}$. Indeed, by [Lemma B.2](#), $v^{(1)}$ is nondecreasing and $v^{(2)}$ is nonincreasing, and moreover $\int_I v^{(2)} dx = 0$ by weak convergence; hence $(v^{(1)}, v^{(2)}) \in \mathcal{R}(v)$. Then, since $u_{n_{k_m}} \rightharpoonup v$ weakly in $L^2(I)$ and $u_{n_{k_m}} \rightarrow u$ in $L^2(I)$, we conclude that $u = v$, so $u \in \mathcal{U}$ with $(v^{(1)}, v^{(2)}) \in \mathcal{R}(u)$. In particular, this means that $\mathcal{E}^-(u) \leq \mathcal{D}^-(v^{(1)}, v^{(2)}) = \frac{1}{2} \int_I (\partial_x v^{(2)})^2 dx$.

Now, the fact that $u_{n_{k_m}} \downarrow \rightharpoonup v^{(2)}$ weakly in $H^1(I)$ implies that $\|v^{(2)}\|_{H^1(I)} \leq \liminf_{m \rightarrow \infty} \|u_{n_{k_m}} \downarrow\|_{H^1(I)}$. Moreover, since $H^1(I)$ embeds compactly into $L^2(I)$ by the Rellich-Kondrachov theorem (see e.g. [\[Bre11, Theorem 9.16\]](#)), we have that $u_{n_{k_m}} \downarrow \rightarrow v^{(2)}$

in $L^2(I)$, so $\|u_{n_{k_m}} \downarrow\|_{L^2(I)}^2 \rightarrow \|v^{(2)}\|_{L^2(I)}^2$. Therefore we obtain

$$\begin{aligned} \liminf_{m \rightarrow \infty} \mathcal{E}^-(u_{n_{k_m}}) &= \frac{1}{2} \liminf_{m \rightarrow \infty} \int_I (\partial_x u_{n_{k_m}} \downarrow)^2 dx = \frac{1}{2} \liminf_{m \rightarrow \infty} \left(\|u_{n_{k_m}} \downarrow\|_{H^1(I)}^2 - \|u_{n_{k_m}} \downarrow\|_{L^2(I)}^2 \right) \\ &= \left(\frac{1}{2} \liminf_{m \rightarrow \infty} \|u_{n_{k_m}} \downarrow\|_{H^1(I)}^2 \right) - \frac{1}{2} \|v^{(2)}\|_{L^2(I)}^2 \\ &\geq \frac{1}{2} \left(\|v^{(2)}\|_{H^1(I)}^2 - \|v^{(2)}\|_{L^2(I)}^2 \right) = \frac{1}{2} \int_I (\partial_x v^{(2)})^2 dx \\ &\geq \mathcal{E}^-(u), \end{aligned}$$

and thus (B.1) holds. $\square$

The following lemma is essentially standard, and below we present the proof sketched in [Fen].

Lemma B.3. *Let $\Omega \subset \mathbb{R}^d$ be an open set, let $u \in L^2(\Omega)$ and let $(u_n)_{n \in \mathbb{N}}$ be a bounded sequence in $L^2(\Omega)$ such that $u_n \rightarrow u$ in $L^2_{\text{loc}}(\Omega)$. Then $u_n \rightharpoonup u$ weakly in $L^2(\Omega)$.*

Proof. It suffices to show that every subsequence $(u_{n_k})_{k \in \mathbb{N}}$ has a subsequence that weakly converges to u . Fix any subsequence $(u_{n_k})_{k \in \mathbb{N}}$. First, since $u_{n_k} \rightarrow u$ in $L^2_{\text{loc}}(\Omega)$, we have that $\langle u_{n_k}, \phi \rangle \rightarrow \langle u, \phi \rangle$ for all $\phi \in C_c^\infty(\Omega)$. Moreover, since $L^2(\Omega)$ is a Hilbert space and $(u_{n_k})_{k \in \mathbb{N}}$ is bounded, let $(u_{n_{k_\ell}})_{\ell \in \mathbb{N}}$ be a subsequence such that $u_{n_{k_\ell}} \rightharpoonup w$ weakly in $L^2(\Omega)$ for some $w \in L^2(\Omega)$. Now, let $v \in L^2(\Omega)$. Since $C_c^\infty(\Omega)$ is dense in $L^2(\Omega)$, let $(\phi_m)_{m \in \mathbb{N}}$ be a sequence in $C_c^\infty(\Omega)$ with $\phi_m \rightarrow v$ in $L^2(\Omega)$. Then

$$\begin{aligned} \langle u, v \rangle &= \lim_{m \rightarrow \infty} \langle u, \phi_m \rangle && \text{(Since } \phi_m \rightarrow v \text{ in } L^2(\Omega)) \\ &= \lim_{m \rightarrow \infty} \lim_{k \rightarrow \infty} \langle u_{n_k}, \phi_m \rangle && \text{(As observed above)} \\ &= \lim_{m \rightarrow \infty} \lim_{\ell \rightarrow \infty} \langle u_{n_{k_\ell}}, \phi_m \rangle && \text{(Taking subsequence preserves the limit)} \\ &= \lim_{m \rightarrow \infty} \langle w, \phi_m \rangle && \text{(Since } u_{n_{k_\ell}} \rightharpoonup w \text{ weakly in } L^2(\Omega)) \\ &= \langle w, v \rangle && \text{(Since } \phi_m \rightarrow v \text{ in } L^2(\Omega)) \\ &= \lim_{\ell \rightarrow \infty} \langle u_{n_{k_\ell}}, v \rangle && \text{(Since } u_{n_{k_\ell}} \rightharpoonup w \text{ weakly in } L^2(\Omega)), \end{aligned}$$

so $u_{n_{k_\ell}} \rightharpoonup u$ weakly in $L^2(\Omega)$ as needed. $\square$

Lemma 7.29. *Let $J \subset \mathbb{R}$ be a finite, nonempty open interval. Let $f \in L^2(J)$ be nondecreasing and let $g \in H^1(J)$ be such that $g < 0$ in J and $g = 0$ on ∂J , i.e. the endpoints of J . Let $[a, b] \subset J$, let $\delta := f(b) - f(a)$ and let $\alpha := -\sup_{(a,b)} g$. Then $\langle f, \partial_x g \rangle_{L^2(J)} \geq \alpha \delta$.*

We first give a proof assuming $f \in L^\infty(J)$, and then generalize to $f \in L^2(J)$ via an approximation argument.

Proof assuming $f \in L^\infty(J)$. Without loss of generality, we may assume that $J = I$. Now, for each sufficiently small $\varepsilon > 0$, recall that $I_\varepsilon = (\varepsilon, 1 - \varepsilon)$ and let $f_\varepsilon \in C^\infty(I_\varepsilon)$ be the mollification of f . Let $f_\varepsilon^* : I \rightarrow \mathbb{R}$ be given by

$$f_\varepsilon^*(x) := \begin{cases} f_{\varepsilon/2}(\varepsilon) & \text{if } x \in (0, \varepsilon) \\ f_{\varepsilon/2}(x) & \text{if } x \in [\varepsilon, 1 - \varepsilon] \\ f_{\varepsilon/2}(1 - \varepsilon) & \text{if } x \in (1 - \varepsilon, 1). \end{cases}$$

Note that each $f_\varepsilon^* \in H^1(I)$, in particular because the piecewise definition is continuous and $f_{\varepsilon/2}$ is smooth on $[\varepsilon, 1 - \varepsilon]$. Also, each $f_{\varepsilon/2}$ is nondecreasing and hence so is each f_ε^* . Moreover, $f_\varepsilon^* \rightarrow f$ a.e. as $\varepsilon \rightarrow 0$ since this is true of $(f_\varepsilon)_{\varepsilon>0}$. Finally, we have $f_\varepsilon^* \rightarrow f$ in $L^2_{\text{loc}}(I)$ since $f_\varepsilon \rightarrow f$ in $L^2_{\text{loc}}(\Omega)$, and since $\|f_\varepsilon^*\|_{L^\infty(I)} \leq \|f_{\varepsilon/2}\|_{L^\infty(I)} \leq \|f\|_{L^\infty(I)}$, we have that $(f_\varepsilon^*)_{\varepsilon>0}$ is bounded in $L^2(I)$ and Lemma B.3 implies that $f_\varepsilon^* \rightharpoonup f$ weakly in $L^2(I)$ as $\varepsilon \rightarrow 0$.

Let $(a_n)_{n \in \mathbb{N}}, (b_n)_{n \in \mathbb{N}}$ be two sequences such that $a_n \uparrow a$, $b_n \downarrow b$, and moreover, for every $n \in \mathbb{N}$, $f_\varepsilon^*(a_n) \rightarrow f(a_n)$ and $f_\varepsilon^*(b_n) \rightarrow f(b_n)$ as $\varepsilon \rightarrow 0$; the existence of such sequences is guaranteed by the fact that $f_\varepsilon^* \rightarrow f$ almost everywhere. For each $n \in \mathbb{N}$, let $\delta_n := f(b_n) - f(a_n)$ and $\alpha_n := -\sup_{(a_n, b_n)} g$. Note that $\delta_n \geq \delta$ because f is nondecreasing while $a_n \leq a$ and $b \leq b_n$, and that $\alpha_n \rightarrow \alpha$ as $n \rightarrow \infty$ by the continuity of g .

For each $\varepsilon > 0$, integration by parts gives

$$\langle f_\varepsilon^*, \partial_x g \rangle = \int_I f_\varepsilon^* (\partial_x g) \, dx = f_\varepsilon^* g|_0^1 - \int_I (\partial_x f_\varepsilon^*) g \, dx.$$

Recall that $g = 0$ on $\{0, 1\}$. Moreover, since f_ε^* is nondecreasing while $g < 0$ in I , the integrand in the RHS above is nonpositive. Hence we can only make the RHS smaller by restricting the range of integration. Thus, fixing any $n \in \mathbb{N}$ and using the definition of α_n ,

$$\langle f_\varepsilon^*, \partial_x g \rangle \geq - \int_{(a_n, b_n)} (\partial_x f_\varepsilon^*) g \, dx \geq - \int_{(a_n, b_n)} (\partial_x f_\varepsilon^*) (-\alpha_n) \, dx = \alpha_n f_\varepsilon^*|_{a_n}^{b_n}.$$

Since $f_\varepsilon^* \rightharpoonup f$ weakly in $L^2(I)$, we obtain

$$\langle f, \partial_x g \rangle = \lim_{\varepsilon \rightarrow 0} \langle f_\varepsilon^*, \partial_x g \rangle \geq \lim_{\varepsilon \rightarrow 0} \alpha_n f_\varepsilon^*|_{a_n}^{b_n} = \alpha_n \delta_n \geq \alpha_n \delta,$$

the second equality by the choice of sequences $(a_n)_n, (b_n)_n$ and definition of δ_n . We conclude that

$$\langle f, \partial_x g \rangle \geq \lim_{n \rightarrow \infty} \alpha_n \delta = \alpha \delta. \quad \square$$

Proof for general $f \in L^2(J)$. Again let $J = I$ without loss of generality. We proceed by an approximation argument. Let $(a_n)_{n \in \mathbb{N}}$ be a strictly decreasing sequence satisfying $a_1 = a$ and $a_n \rightarrow 0$. Similarly, let $(b_n)_{n \in \mathbb{N}}$ be a strictly increasing sequence satisfying $b_1 = b$ and $b_n \rightarrow 1$. For each $n \in \mathbb{N}$, define $f_n \in L^\infty(I)$ by

$$f_n(x) := \begin{cases} \inf_{[a_n, b_n]} f & \text{if } x \in (0, a_n) \\ f(x) & \text{if } x \in [a_n, b_n] \\ \sup_{[a_n, b_n]} f & \text{if } x \in (b_n, 1). \end{cases}$$

Note that the infimum and supremum above are finite by virtue of the monotonicity of f and the observation that each $[a_n, b_n] \subset I$; thus we indeed have $f_n \in L^\infty(I)$. Moreover, each f_n is nondecreasing and, since $a_n \leq a < b \leq b_n$, we have $f_n(a) = f(a)$ and $f_n(b) = f(b)$. Thus the first part implies that $\langle f_n, \partial_x g \rangle \geq \alpha \delta$ for α as in that statement and $\delta = f_n(b) - f_n(a) = f(b) - f(a)$. Finally, we have $f_n \rightarrow f$ in $L^2(I)$; indeed, letting $c := \frac{a+b}{2}$ for convenience and using the monotonicity of f ,

$$\begin{aligned} \|f - f_n\|_{L^2(I)}^2 &= \int_{(0, a_n)} \left(f(x) - \inf_{[a_n, b_n]} f \right)^2 dx + \int_{(b_n, 1)} \left(f(x) - \sup_{[a_n, b_n]} f \right)^2 dx \\ &\leq \int_{(0, a_n)} (f(x) - f(c))^2 dx + \int_{(b_n, 1)} (f(x) - f(c))^2 dx \\ &\leq 2 \left[\|f\|_{L^2(0, a_n)}^2 + \|f\|_{L^2(b_n, 1)}^2 + (a_n + 1 - b_n) f(c)^2 \right] \rightarrow 0, \end{aligned}$$

the last step by the continuity of the functions $x \rightarrow \|f\|_{L^2(0, x)}^2$ and $x \rightarrow \|f\|_{L^2(x, 1)}^2$ and the fact that $a_n + (1 - b_n) \rightarrow 0$ as $n \rightarrow \infty$. Hence $\langle f_n, \partial_x g \rangle \rightarrow \langle f, \partial_x g \rangle$, and the conclusion follows. $\square$

Lemma 7.44 (From “almost Lipschitz” to Lipschitz). *Let $M \in \mathbb{R}_{\geq 0}$ and let $N \subset I$ be a measure zero set. Suppose $f : I \rightarrow \mathbb{R}$ satisfies $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in I \setminus N$. Then there exists a M -Lipschitz function $g : I \rightarrow \mathbb{R}$ such that $f = g$ in $I \setminus N$.*

Proof. Let $\bar{f} : I \rightarrow (-\infty, +\infty]$ be given by $\bar{f} := f$ in $I \setminus N$ and $\bar{f} := +\infty$ in N . Define $g : I \rightarrow \mathbb{R}$ by

$$g(x) := \begin{cases} f(x) & \text{if } x \in I \setminus N \\ \liminf_{z \rightarrow x} \bar{f}(z) & \text{otherwise.} \end{cases}$$

Note that g is real-valued because, when $x \in N$, we may find a sequence $(z_n)_{n \in \mathbb{N}} \subset I \setminus N$ such that $z_n \rightarrow x$ and then, by the definition of $\liminf$ by subsequential limits and the assumption on f ,

$$g(x) = \liminf_{z \rightarrow x} \bar{f}(z) \leq \lim_{n \rightarrow \infty} \bar{f}(z_n) = \lim_{n \rightarrow \infty} f(z_n) \leq \lim_{n \rightarrow \infty} [f(z_1) + M|z_n - z_1|] = f(z_1) + M|x - z_1| < +\infty.$$

Clearly $g = f$ in $I \setminus N$. We claim that g is M -Lipschitz. By the assumption on f , we do have $|g(x) - g(y)| \leq M|x - y|$ for all $x, y \in I \setminus N$. Now, let $x \in N$ and $y \in I \setminus N$. By the definition of g , fix a sequence $(z_n)_{n \in \mathbb{N}} \subset I \setminus N$ such that $z_n \rightarrow x$ and moreover $\lim_{n \rightarrow \infty} f(z_n) = g(x)$. Then

$$|g(x) - g(y)| = \left| \lim_{n \rightarrow \infty} f(z_n) - f(y) \right| = \lim_{n \rightarrow \infty} |f(z_n) - f(y)| \leq \lim_{n \rightarrow \infty} M|z_n - y| = M|x - y|,$$

as desired. The case when $x, y \in N$ follows by the triangle inequality: suppose $x < y$ without loss of generality, choose any $z \in (x, y) \setminus N$, which is possible since N has measure zero, and use the above to conclude that

$$|g(x) - g(y)| \leq |g(x) - g(z)| + |g(z) - g(y)| \leq M|x - z| + M|z - y| = M|x - y|. \quad \square$$